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X665 MLB DVT												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE		
												6			2017-08-14		
D	1	1	TABLE OF CONTENTS	DEV_MLB	11/06/2016	61	65	SSD CONN A	AMAN	02/10/2017	121	140	DEV SUPPORT	AMAN	06/16/2017	D	
	2	2	SYSTEM PCB SUMMARY	DEV_MLB	11/06/2016	62	66	SSD CONN B	AMAN	02/10/2017	122	142	PMU DEV SUPPORT	AMAN	06/16/2017		
	3	3	BOM CONFIGURATION	DEV_MLB	11/06/2016	63	67	SPI ROM	DEV_MLB	11/06/2016	123	145	SOC DEV SUPPORT	AMAN	06/16/2017		
	4	4	BOM TABLES			64	68	LIFEBOAT	AMAN	07/25/2017	124	146	DEBUG HEADERS 1	AMAN	06/16/2017		
	5	6	BOM ALTERNATES			65	69	PCH LEVEL SHIFTERS	DEV_MLB	11/06/2016	125	147	DEBUG HEADERS 2	AMAN	06/16/2017		
	6	9	PD PARTS	DEV_MLB	11/06/2016	66	70	PCH CLOCKS, JTAG	AITKEN	03/06/2017	126	148	DEBUG SOC U.FL CONNECTORS	AMAN	06/16/2017		
	7	10	CPU MEMORY CHANNEL 0	DEV_MLB	11/06/2016	67	71	PCH DMI, PM, MISC	AITKEN	03/10/2017	127	150	AUDIO JACK CODEC	JAMES	05/12/2017		
	8	11	CPU MEMORY CHANNEL 1	DEV_MLB	11/06/2016	68	72	PCH PCIE, USB	DEV_MLB	11/06/2016	128	151	AUDIO LEFT SPKR AMP	JAMES	05/12/2017		
	9	12	CPU MEMORY CHANNEL 2	DEV_MLB	11/06/2016	69	73	PCH GPIO	AITKEN	03/10/2017	129	152	AUDIO RIGHT SPKR AMP	JAMES	05/12/2017		
	10	13	CPU MEMORY CHANNEL 3	DEV_MLB	11/06/2016	70	74	PCH POWER, DECOUPLING	AITKEN	03/08/2017	130	153	AUDIO JACK CONNECTOR	JAMES	05/12/2017		
C	11	14	CPU PCIE			71	75	PCH GND, RTC, ME	SUNIL	02/04/2017	131	154	AUDIO MIC CONNECTOR	JAMES	05/12/2017	C	
	12	15	CPU DMI, MISC	DEV_MLB	11/06/2016	72	76	PCH SUPPORT	AITKEN	03/10/2017	132	155	AUDIO MISC 1	JAMES	05/12/2017		
	13	16	CPU SVID, BCLK, JTAG	DEV_MLB	11/07/2016	73	77	XDP	DEV_MLB	11/06/2016	133	156	AUDIO MISC 2	JAMES	05/12/2017		
	14	17	CPU VCCIO, VDDQ, VCCSA	DEV_MLB	11/06/2016	74	78	PMIC BUCKS AND SWS	AMAN	07/24/2017	134	160	WIFI/BT MODULE 1	METE	05/04/2017		
	15	18	CPU VCCIN	DEV_MLB	11/06/2016	75	79	PMIC LDOS	AMAN	07/24/2017	135	161	WIFI/BT MODULE 2	METE	05/04/2017		
	16	19	CPU GND	DEV_MLB	11/06/2016	76	80	PMIC GPIOS & CONTROL	AMAN	07/24/2017	136	162	WIFI/BT MODULE 3	METE	05/04/2017		
	17	20	CPU SUPPORT	DEV_MLB	11/06/2016	77	81	PMIC SUPPORT	AMAN	06/16/2017	137	164	WIFI/BT SUPPORT 1	METE	05/04/2017		
	18	21	DDR4 DIMM CONN CH0	DEV_MLB	11/06/2016	78	82	POWER FETS	DEV_MLB	11/06/2016	138	165	WIFI/BT SUPPORT 2	METE	05/10/2017		
	19	22	DDR4 DIMM CONN CH1	DEV_MLB	11/06/2016	79	83	VR P3V3 G3W, P5V G3S USBC	DEV_MLB	11/06/2016	139	180	POWER INPUT FROM AC/DC 1	AMAN	06/16/2017		
	20	23	DDR4 DIMM CONN CH2	DEV_MLB	11/06/2016	80	84	VR ENET 1V2, 2V1	DEV_MLB	11/06/2016	140	181	POWER INPUT FROM AC/DC 2	AMAN	06/16/2017		
B	21	24	DDR4 DIMM CONN CH3	DEV_MLB	11/06/2016	81	85	VR ENET 0V85	DEV_MLB	11/06/2016	141	182	POWER INPUT FROM AC/DC 3	AMAN	06/22/2017	B	
	22	25	DDR4 CA VREF, VTT	DEV_MLB	11/06/2016	82	86	VR P3V3 G3H, P5V G3S	DEV_MLB	11/06/2016	142	190	POWER SEQUENCING S3	DEV_MLB	11/06/2016		
	23	27	USB-C (T) CONNECTORS	DEV_MLB	11/06/2016	83	87	PSU PRESENCE G3H ENABLE	DEV_MLB	11/06/2016	143	191	POWER SEQUENCING S0 1	DEV_MLB	11/06/2016		
	24	28	USB-C (T) HIGH SPEED 1	DEV_MLB	11/06/2016	84	88	VR 5.75V CPUVR	SUNIL	02/04/2017	144	192	POWER SEQUENCING S0 2	DEV_MLB	11/06/2016		
	25	29	USB-C (T) HIGH SPEED 2	DEV_MLB	11/06/2016	85	90	VR CPU CORE & VCCSA CTRL	SUNIL	02/04/2017	145	197	RESETS	DEV_MLB	11/06/2016		
	26	30	USB-C (T) SUPPORT	DEV_MLB	11/07/2016	86	91	VR CPU CORE PHASES 1-3	SUNIL	02/04/2017	146	200	SENSORS THERMAL 1	BOYLE	03/01/2017		
	27	31	USB-C (T) PORT CONTROLLER A	DEV_MLB	11/07/2016	87	92	VR CPU CORE PHASES 4-5	SUNIL	02/04/2017	147	201	SENSORS THERMAL 2	BOYLE	03/01/2017		
	28	32	USB-C (T) PORT CONTROLLER B	DEV_MLB	11/06/2016	88	94	VR CPU VCCIO	SUNIL	02/04/2017	148	202	SENSORS THERMAL 3	BOYLE	03/01/2017		
	29	33	USB-C (X) CONNECTORS	DEV_MLB	11/06/2016	89	95	VR CPU VCCSA	SUNIL	02/04/2017	149	206	SENSORS POWER HIGHSIDE 1	BOYLE	03/17/2017		
	30	34	USB-C (X) HIGH SPEED 1			90	96	VR CPU VPP & VTT CH 0,1	SUNIL	02/04/2017	150	207	SENSORS POWER HIGHSIDE 2	BOYLE	03/17/2017		
A	31	35	USB-C (X) HIGH SPEED 2			91	97	VR CPU VPP & VTT CH 2,3	SUNIL	02/04/2017	151	208	SENSORS POWER LOWSIDE 1	BOYLE	03/17/2017	A	
	32	36	USB-C (X) SUPPORT			92	98	VR CPU VDDQ CH 0,1	SUNIL	02/04/2017	152	209	SENSORS POWER LOWSIDE 2	BOYLE	03/17/2017		
	33	37	USB-C (X) PORT CONTROLLER A	DEV_MLB	11/06/2016	93	99	VR CPU VDDQ CH 2,3	SUNIL	02/04/2017	153	211	LEDS POR	DEV_MLB	11/06/2016		
	34	38	USB-C (X) PORT CONTROLLER B	DEV_MLB	11/06/2016	94	100	GPU PCIE, STRAPS, JTAG	DEV_MLB	11/06/2016	154	212	LEDS DEBUG 1	AMAN	02/16/2017		
	35	39	SOC GPIO/SEP/USB/DDR/TEST	AMAN	07/25/2017	95	101	GPU PCIE AC CAPS	DEV_MLB	11/06/2016	155	213	LEDS DEBUG 2	AMAN	01/20/2017		
	36	40	SOC AOP/AON/SMC	AMAN	07/25/2017	96	102	GPU DISPLAYPORT	DEV_MLB	11/06/2016	156	215	LEDS SUPPORT	AMAN	01/20/2017		
	37	41	SOC ISP/I2C/UART/SPI/I2S	AMAN	07/24/2017	97	103	GPU GPIOS	DEV_MLB	11/06/2016	157	220	PSU RESISTANCE MEASUREMENT				
	38	42	SOC PCIE	AMAN	07/24/2017	98	104	GPU CLK GEN	DEV_MLB	11/06/2016	158	221	TEST PADS FOR SOC AREA				
	39	43	SOC POWER 1	AMAN	07/24/2017	99	105	GPU MEMORY AND IO PWR	DEV_MLB	11/06/2016	159	230	DFU TEST POINTS	ANDRES	02/13/2017		
	40	44	SOC POWER 2	AMAN	07/24/2017	100	106	GPU CORE PWR	DEV_MLB	11/06/2016	160	233	SVT TEST PROPERTIES	DEV_MLB	11/06/2016		
												161	235	FCT TEST PROPERTIES	DEV_MLB	11/06/2016	
												162	240	ICT NO TEST 1			
												163	241	ICT NO TEST 2			
												164	242	ICT NO TEST 3			
												165	243	ICT NO TEST 4			
												166	244	ICT NO TEST 5			
												167	248	EE NO TEST			
												168	250	DESENSE 1			
												169	251	DESENSE 2			
												170	252	DESENSE 3			
												171	255	EMC			
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	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00133</td><td>1</td><td>1C,SP1 SERIAL FLASH,256KTS,3-VT,05008</td><td>U2890</td><td>CRITICAL</td><td>TBT_T_ROM:BLANK</td></tr><tr><td>341S00738</td><td>1</td><td>1C,T29,AA1,VTR0,P0 DEV,8665</td><td>U2890</td><td>CRITICAL</td><td>TBT_T_ROM:POC</td></tr><tr><td>341S00795</td><td>1</td><td>1C,T29,AA,V15.4,PROTD 0,8665</td><td>U2890</td><td>CRITICAL</td><td>TBT_T_ROM:P0_V15P4</td></tr><tr><td>341S00846</td><td>1</td><td>1C,T29,AA1-V,V15.10,PROTD1,8665</td><td>U2890</td><td>CRITICAL</td><td>TBT_T_ROM:PIA_V15P10</td></tr><tr><td>341S00902</td><td>1</td><td>1C,T29,AA1-V,V15.10,EVT,8665</td><td>U2890</td><td>CRITICAL</td><td>TBT_T_ROM:EVT_V15P10</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S00133	1	1C,SP1 SERIAL FLASH,256KTS,3-VT,05008	U2890	CRITICAL	TBT_T_ROM:BLANK	341S00738	1	1C,T29,AA1,VTR0,P0 DEV,8665	U2890	CRITICAL	TBT_T_ROM:POC	341S00795	1	1C,T29,AA,V15.4,PROTD 0,8665	U2890	CRITICAL	TBT_T_ROM:P0_V15P4	341S00846	1	1C,T29,AA1-V,V15.10,PROTD1,8665	U2890	CRITICAL	TBT_T_ROM:PIA_V15P10	341S00902	1	1C,T29,AA1-V,V15.10,EVT,8665	U2890	CRITICAL	TBT_T_ROM:EVT_V15P10																																				
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341S00738	1	1C,T29,AA1,VTR0,P0 DEV,8665	U2890	CRITICAL	TBT_T_ROM:POC																																																																																			
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341S00846	1	1C,T29,AA1-V,V15.10,PROTD1,8665	U2890	CRITICAL	TBT_T_ROM:PIA_V15P10																																																																																			
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	335S00133	1	1C,SP1 SERIAL FLASH,256KTS,3-VT,05008	U3490	CRITICAL	TBT_X_ROM:BLANK																																																																																		
341S00738	1	1C,T29,AA1,VTR0,P0 DEV,8665	U3490	CRITICAL	TBT_X_ROM:POC																																																																																			
341S00795	1	1C,T29,AA,V15.4,PROTD 0,8665	U3490	CRITICAL	TBT_X_ROM:P0_V15P4																																																																																			
341S00847	1	1C,T29,AA2-V,V15.10,PROTD1,8665	U3490	CRITICAL	TBT_X_ROM:PIA_V15P10																																																																																			
341S00903	1	1C,T29,AA2-V,V15.10,EVT,8665	U3490	CRITICAL	TBT_X_ROM:EVT_V15P10																																																																																			
SD READER ROM																																																																																								
<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00024</td><td>1</td><td>1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600</td><td>U6230</td><td>CRITICAL</td><td>SD_ROM:BLANK_WINBOUND</td></tr><tr><td>335S00282</td><td>1</td><td>1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600</td><td>U6230</td><td>CRITICAL</td><td>SD_ROM:BLANK_MACRONIX</td></tr><tr><td>341S00733</td><td>1</td><td>1C,ROM,SDCARD CTRL GL3227A,V001,POC,8665</td><td>U6230</td><td>CRITICAL</td><td>SD_ROM:POC</td></tr><tr><td>341S00783</td><td>1</td><td>FW,SDCARD CTRL GL3227A,V002,POC,8665</td><td>U6230</td><td>CRITICAL</td><td>SD_ROM:P0_V002</td></tr><tr><td>341S00897</td><td>1</td><td>1C,ROM,SDCARD CTL GL3227A,VAP03,EVT,8665</td><td>U6230</td><td>CRITICAL</td><td>SD_ROM:EVT_VAP03</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S00024	1	1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600	U6230	CRITICAL	SD_ROM:BLANK_WINBOUND	335S00282	1	1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600	U6230	CRITICAL	SD_ROM:BLANK_MACRONIX	341S00733	1	1C,ROM,SDCARD CTRL GL3227A,V001,POC,8665	U6230	CRITICAL	SD_ROM:POC	341S00783	1	FW,SDCARD CTRL GL3227A,V002,POC,8665	U6230	CRITICAL	SD_ROM:P0_V002	341S00897	1	1C,ROM,SDCARD CTL GL3227A,VAP03,EVT,8665	U6230	CRITICAL	SD_ROM:EVT_VAP03																																					
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335S00024	1	1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600	U6230	CRITICAL	SD_ROM:BLANK_WINBOUND																																																																																			
335S00282	1	1C,SERIAL FLASH,256KTS,3V,8-050W,243F,600	U6230	CRITICAL	SD_ROM:BLANK_MACRONIX																																																																																			
341S00733	1	1C,ROM,SDCARD CTRL GL3227A,V001,POC,8665	U6230	CRITICAL	SD_ROM:POC																																																																																			
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341S00897	1	1C,ROM,SDCARD CTL GL3227A,VAP03,EVT,8665	U6230	CRITICAL	SD_ROM:EVT_VAP03																																																																																			
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<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00269</td><td>1</td><td>1C,SP1 SERIAL FLASH,128KTS,3V,800C8</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:BLANK_ADESTO</td></tr><tr><td>341S00751</td><td>1</td><td>1C,ETHERNET,VTR0,PROTD 0,8665</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:POC</td></tr><tr><td>341S00793</td><td>1</td><td>1C,ETHERNET,VTR0,PROTD 0,8665</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:P0</td></tr><tr><td>341S00817</td><td>1</td><td>1C,ETHERNET,V2.7-24,PROTD1,8665</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:P1_V2P9P24</td></tr><tr><td>341S00864</td><td>1</td><td>1C,ETHERNET,V2.7-24,M01 NORM01,P10,8665</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:PIB_M01_NORM01</td></tr><tr><td>341S00901</td><td>1</td><td>1C,ETHERNET,V2.7-24,M01 NORM01,EVT,8665</td><td>U6090</td><td>CRITICAL</td><td>ETHERNET_ROM:EVT_M01_NORM01</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S00269	1	1C,SP1 SERIAL FLASH,128KTS,3V,800C8	U6090	CRITICAL	ETHERNET_ROM:BLANK_ADESTO	341S00751	1	1C,ETHERNET,VTR0,PROTD 0,8665	U6090	CRITICAL	ETHERNET_ROM:POC	341S00793	1	1C,ETHERNET,VTR0,PROTD 0,8665	U6090	CRITICAL	ETHERNET_ROM:P0	341S00817	1	1C,ETHERNET,V2.7-24,PROTD1,8665	U6090	CRITICAL	ETHERNET_ROM:P1_V2P9P24	341S00864	1	1C,ETHERNET,V2.7-24,M01 NORM01,P10,8665	U6090	CRITICAL	ETHERNET_ROM:PIB_M01_NORM01	341S00901	1	1C,ETHERNET,V2.7-24,M01 NORM01,EVT,8665	U6090	CRITICAL	ETHERNET_ROM:EVT_M01_NORM01																															
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335S00269	1	1C,SP1 SERIAL FLASH,128KTS,3V,800C8	U6090	CRITICAL	ETHERNET_ROM:BLANK_ADESTO																																																																																			
341S00751	1	1C,ETHERNET,VTR0,PROTD 0,8665	U6090	CRITICAL	ETHERNET_ROM:POC																																																																																			
341S00793	1	1C,ETHERNET,VTR0,PROTD 0,8665	U6090	CRITICAL	ETHERNET_ROM:P0																																																																																			
341S00817	1	1C,ETHERNET,V2.7-24,PROTD1,8665	U6090	CRITICAL	ETHERNET_ROM:P1_V2P9P24																																																																																			
341S00864	1	1C,ETHERNET,V2.7-24,M01 NORM01,P10,8665	U6090	CRITICAL	ETHERNET_ROM:PIB_M01_NORM01																																																																																			
341S00901	1	1C,ETHERNET,V2.7-24,M01 NORM01,EVT,8665	U6090	CRITICAL	ETHERNET_ROM:EVT_M01_NORM01																																																																																			
B	BT SERIAL FLASH																																																																																							
	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00255</td><td>1</td><td>1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98</td><td>UG450</td><td>CRITICAL</td><td>BT_FLASH:BLANK_ADESTO</td></tr><tr><td>335S00256</td><td>1</td><td>1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98</td><td>UG450</td><td>CRITICAL</td><td>BT_FLASH:BLANK_WINBOUND</td></tr><tr><td>341S00726</td><td>1</td><td>1C,BT,VTR0,POC,8665</td><td>UG450</td><td>CRITICAL</td><td>BT_FLASH:POC</td></tr><tr><td>341S00844</td><td>1</td><td>1C,BT,VTR0,POC,8665</td><td>UG450</td><td>CRITICAL</td><td>BT_FLASH:PIB</td></tr><tr><td>341S00892</td><td>1</td><td>1C,BT FLASH,VTR0,EVT,8665</td><td>UG450</td><td>CRITICAL</td><td>BT_FLASH:EVT</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S00255	1	1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98	UG450	CRITICAL	BT_FLASH:BLANK_ADESTO	335S00256	1	1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98	UG450	CRITICAL	BT_FLASH:BLANK_WINBOUND	341S00726	1	1C,BT,VTR0,POC,8665	UG450	CRITICAL	BT_FLASH:POC	341S00844	1	1C,BT,VTR0,POC,8665	UG450	CRITICAL	BT_FLASH:PIB	341S00892	1	1C,BT FLASH,VTR0,EVT,8665	UG450	CRITICAL	BT_FLASH:EVT																																				
	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION																																																																																		
	335S00255	1	1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98	UG450	CRITICAL	BT_FLASH:BLANK_ADESTO																																																																																		
335S00256	1	1C,SP1 SERIAL FLASH,256KTS,1.8V,0P98	UG450	CRITICAL	BT_FLASH:BLANK_WINBOUND																																																																																			
341S00726	1	1C,BT,VTR0,POC,8665	UG450	CRITICAL	BT_FLASH:POC																																																																																			
341S00844	1	1C,BT,VTR0,POC,8665	UG450	CRITICAL	BT_FLASH:PIB																																																																																			
341S00892	1	1C,BT FLASH,VTR0,EVT,8665	UG450	CRITICAL	BT_FLASH:EVT																																																																																			
WLAN SERIAL EEPROM																																																																																								
<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00214</td><td>1</td><td>1C,EEPROM,64K,WM8E,14K,1.8V,0P98</td><td>UG410</td><td>CRITICAL</td><td>WLAN_EEPROM:BLANK_ONSEMI</td></tr><tr><td>335S00216</td><td>1</td><td>1C,EEPROM,64K,WM8E,14K,BITS,1.7V,0P98</td><td>UG410</td><td>CRITICAL</td><td>WLAN_EEPROM:BLANK_ROHM</td></tr><tr><td>341S00725</td><td>1</td><td>1C,WIFI ROM,VTR0,POC,8665</td><td>UG410</td><td>CRITICAL</td><td>WLAN_EEPROM:POC</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S00214	1	1C,EEPROM,64K,WM8E,14K,1.8V,0P98	UG410	CRITICAL	WLAN_EEPROM:BLANK_ONSEMI	335S00216	1	1C,EEPROM,64K,WM8E,14K,BITS,1.7V,0P98	UG410	CRITICAL	WLAN_EEPROM:BLANK_ROHM	341S00725	1	1C,WIFI ROM,VTR0,POC,8665	UG410	CRITICAL	WLAN_EEPROM:POC																																																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION																																																																																			
335S00214	1	1C,EEPROM,64K,WM8E,14K,1.8V,0P98	UG410	CRITICAL	WLAN_EEPROM:BLANK_ONSEMI																																																																																			
335S00216	1	1C,EEPROM,64K,WM8E,14K,BITS,1.7V,0P98	UG410	CRITICAL	WLAN_EEPROM:BLANK_ROHM																																																																																			
341S00725	1	1C,WIFI ROM,VTR0,POC,8665	UG410	CRITICAL	WLAN_EEPROM:POC																																																																																			
A	GPU BOOT ROM																																																																																							
	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S0399</td><td>1</td><td>1C,FLASH,SERIAL,EVT,1K X 8,3-VT,MLP9,1F</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:BLANK</td></tr><tr><td>341S00729</td><td>1</td><td>VR006_ROM,V00A10 GPU,V001,POC,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:POC</td></tr><tr><td>341S00792</td><td>1</td><td>1C,VR006_ROM,V00A10 GPU,V002,P0,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:P0_V002</td></tr><tr><td>341S00863</td><td>1</td><td>1C,VR006_ROM,V00A10 A0-XLA GPU,P10,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:PIB_A0_XLA</td></tr><tr><td>341S00862</td><td>1</td><td>1C,VR006_ROM,V00A10 A0-XTA GPU,P10,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:PIB_A0_XTA</td></tr><tr><td>341S00888</td><td>1</td><td>1C,VR006_ROM,V00A10 XLA GPU,A1,P1C,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:P1C_A1_XLA</td></tr><tr><td>341S00887</td><td>1</td><td>1C,VR006_ROM,V00A10 XTA GPU,A1,P1C,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:P1C_A1_XTA</td></tr><tr><td>341S00899</td><td>1</td><td>1C,VR006_ROM,V00A10 XLA GPU,A1,EVT,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:EVT_A1_XLA</td></tr><tr><td>341S00898</td><td>1</td><td>1C,VR006_ROM,V00A10 XTA GPU,A1,EVT,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:EVT_A1_XTA</td></tr><tr><td>341S00943</td><td>1</td><td>1C,VR006_ROM,V00A10 XTA GPU(V15)DVT,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:DVT_A1_XTA</td></tr><tr><td>341S00944</td><td>1</td><td>1C,VR006_ROM,V00A10 XLA GPU(V15)DVT,8665</td><td>UA920</td><td>CRITICAL</td><td>GPU_BOOT_ROM:DVT_A1_XLA</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	335S0399	1	1C,FLASH,SERIAL,EVT,1K X 8,3-VT,MLP9,1F	UA920	CRITICAL	GPU_BOOT_ROM:BLANK	341S00729	1	VR006_ROM,V00A10 GPU,V001,POC,8665	UA920	CRITICAL	GPU_BOOT_ROM:POC	341S00792	1	1C,VR006_ROM,V00A10 GPU,V002,P0,8665	UA920	CRITICAL	GPU_BOOT_ROM:P0_V002	341S00863	1	1C,VR006_ROM,V00A10 A0-XLA GPU,P10,8665	UA920	CRITICAL	GPU_BOOT_ROM:PIB_A0_XLA	341S00862	1	1C,VR006_ROM,V00A10 A0-XTA GPU,P10,8665	UA920	CRITICAL	GPU_BOOT_ROM:PIB_A0_XTA	341S00888	1	1C,VR006_ROM,V00A10 XLA GPU,A1,P1C,8665	UA920	CRITICAL	GPU_BOOT_ROM:P1C_A1_XLA	341S00887	1	1C,VR006_ROM,V00A10 XTA GPU,A1,P1C,8665	UA920	CRITICAL	GPU_BOOT_ROM:P1C_A1_XTA	341S00899	1	1C,VR006_ROM,V00A10 XLA GPU,A1,EVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:EVT_A1_XLA	341S00898	1	1C,VR006_ROM,V00A10 XTA GPU,A1,EVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:EVT_A1_XTA	341S00943	1	1C,VR006_ROM,V00A10 XTA GPU(V15)DVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XTA	341S00944	1	1C,VR006_ROM,V00A10 XLA GPU(V15)DVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XLA
	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION																																																																																		
	335S0399	1	1C,FLASH,SERIAL,EVT,1K X 8,3-VT,MLP9,1F	UA920	CRITICAL	GPU_BOOT_ROM:BLANK																																																																																		
341S00729	1	VR006_ROM,V00A10 GPU,V001,POC,8665	UA920	CRITICAL	GPU_BOOT_ROM:POC																																																																																			
341S00792	1	1C,VR006_ROM,V00A10 GPU,V002,P0,8665	UA920	CRITICAL	GPU_BOOT_ROM:P0_V002																																																																																			
341S00863	1	1C,VR006_ROM,V00A10 A0-XLA GPU,P10,8665	UA920	CRITICAL	GPU_BOOT_ROM:PIB_A0_XLA																																																																																			
341S00862	1	1C,VR006_ROM,V00A10 A0-XTA GPU,P10,8665	UA920	CRITICAL	GPU_BOOT_ROM:PIB_A0_XTA																																																																																			
341S00888	1	1C,VR006_ROM,V00A10 XLA GPU,A1,P1C,8665	UA920	CRITICAL	GPU_BOOT_ROM:P1C_A1_XLA																																																																																			
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341S00898	1	1C,VR006_ROM,V00A10 XTA GPU,A1,EVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:EVT_A1_XTA																																																																																			
341S00943	1	1C,VR006_ROM,V00A10 XTA GPU(V15)DVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XTA																																																																																			
341S00944	1	1C,VR006_ROM,V00A10 XLA GPU(V15)DVT,8665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XLA																																																																																			
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SILICON

CPU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00415	1	CPU,SKYW,AM95,10,ESP,2.4,8C,130W,LGA2066	CPU	CRITICAL	CPU_X86:08C_EVT1
337S00360	1	CPU,SKYW,QLRX,10,ES,2.4,10C,140W,LGA2066	CPU	CRITICAL	CPU_X86:10C_EVT1
337S00435	1	CPU,SKYW,QMW7,Q6,8C,3.2,120W,BGA2066	CPU	CRITICAL	CPU_X86:08C_3P2GHZ_QMWZ
337S00434	1	CPU,SKYW,QMW7,Q6,10C,3.0,120W,BGA2066	CPU	CRITICAL	CPU_X86:10C_3P0GHZ_QMW7
337S00453	1	CPU,SKYW,QMH8,Q6,2.3,18C,140W,LGA2066	CPU	CRITICAL	CPU_X86:18C_2P3GHZ_QMH8

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-06198	1	CONN,CPU,SOCKET,R4,LGA2066,SKYLAKE W	U1000	CRITICAL	CPU_SOCKET:POC
511S00004	1	CONN,SOCKET R4 LGA2066	U1000	CRITICAL	CPU_SOCKET:POR_P0

PCH

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00359	1	1C,PCB-H,XBL,SSKU02,QL19,ES2,A0	U7000	CRITICAL	PCH:ES2_MAF_QLL9
337S00390	1	1C,PCB-H,XBL,SSKU02,QM36,SAF,ES2,A0	U7000	CRITICAL	PCH:ES2_SAF_QM36
337S00436	1	1C,PCB-H,XBL,SSKU02,QML1,SAF,Q6,BGA885	U7000	CRITICAL	PCH:Q6_SAF_QML1

GPU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-11669	1	VEGA10 SDLE ADAPTER (VDDC)	UA000	CRITICAL	GPU:SDLE_VDDC
998-11670	1	VEGA10 SDLE ADAPTER (VDDCI/MVDD)	UA000	CRITICAL	GPU:SDLE_VDDCI_MVDD
337S00452	1	1C,GPU,AMD,VEGA10,XLA,A1,1F,PS,BGA2013	UA000	CRITICAL	GPU:XLA_08_SEC
337S00451	1	1C,GPU,AMD,VEGA10,XTA,A1,1F,PS,BGA2013	UA000	CRITICAL	GPU:XTA_16_SEC

TBT X,T

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00254	2	1C,TBT,ALPINE RIDGE,SL28H,P0Q,C1,CP337	U2800,U3400	CRITICAL	TBT_AR:C1

ACE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00961	4	1C,C03215,ACT,CPU,USB PWR SW,SL28H,BGA96	U3100,U3200,U3700,U3800	CRITICAL	ACE:C0

SOC VR

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00268	1	1C,PMU,CALPE,D2249A0,OTF-BG,CSP324	U7800	CRITICAL	PMU:A0_B

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00372	1	POP,SOC,G18MAL7AR+2G 200M,H,B0,CSF1406	U3900	CRITICAL	SOC:B0_2G_MI_SCK

SILICON

SD CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-07938	1	1C,GL3227A,USB3 3.0 TO SD CARD HEADER 4.0	U6200	CRITICAL	SD:POC
338S00312	1	1C,GL3227A,USB3 3.0 TO SD CARD HEADER,QFN48	U6200	CRITICAL	SD:P0_A2

ETHERNET CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-08198	1	ETHERNET,LOGAME-T,PCIe,MAC+PHY	U5800	CRITICAL	ENET:A0
998-09963	1	1C,ETHERNET CONTROLLER,AQ107,A0,BGA224	U5800	CRITICAL	ENET:B0
338S00332	1	1C,ETHERNET CONTROLLER,AQ107,A1,BGA224	U5800	CRITICAL	ENET:B1

CPU CORE VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01192	1	1C,PXE1110-PMB001,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM001
353S01239	1	1C,PXE1110-PMB002,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM002
353S01376	1	1C,PXE1110-PMB003,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM003

CPU VCCIO VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01193	1	1C,PXE1110-PMA001,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PM001
353S01240	1	1C,PXE1110-PMA002,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PM002
353S01377	1	1C,PXE1110-PMA003,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PM003

CPU VDDQ 01 VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01194	1	1C,PXE1110-PMB001,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB001
353S01241	1	1C,PXE1110-PMB002,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB002
353S01309	1	1C,PXE1110-PMB003,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB003
353S01378	1	1C,PXE1110-PMB004,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB004

CPU VDDQ 23 VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01195	1	1C,PXE1110-PMC001,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC001
353S01242	1	1C,PXE1110-PMC002,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC002
353S01310	1	1C,PXE1110-PMC003,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC003
353S01379	1	1C,PXE1110-PMC004,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC004

GPU CORE VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00997	1	1C,1R35217-4199,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:P0_V4199
353S01246	1	1C,1R35217-4207,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:P1C_V4207
353S01380	1	1C,1R35217-4210,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:EVT_V4210
353S01469	1	1C,1R35217-3078,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:DVT_V3078

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COMPONENT ALTERNATES TRACKED IN <RDAR://24198640>																	
SYSTEM EE																	
19780588	19780591	ALT_CNN	ALL	DAVID <RDAR://31128006>	XTAL, 24.000MHZ, 20PPM, 9.5PF, 600MH, 1.6X1.2	377800103	37780057	ALT_CNN	ALL	DAVID <RDAR://31309975>	SUPPR, TRANS, VARISTOR, 6.8V, 100PF, 0402	126-00012	126-0306	ALT_CNN	ALL	LINDA KANAS 02/08/2017 EMAIL	CAP, AL EL, 100UF, 20%, 35V, 0.3A, 6.3X11, BLK
35384376	35383384	ALT_CNN	ALL	DAVID <RDAR://3120375>	IC, LMV331, COMP, OPEN COL, 2.7V, 600A, 18CT0	377800104	37780068	ALT_CNN	ALL	DAVID <RDAR://31312281>	SUPPR, TRANS, VARISTOR, 27V, 47PF, 0402	152800740	15281668	ALT_CNN	ALL	LINDA KANAS 02/08/2017 EMAIL	IND, MOD, 330H, 9.4A TYP, 39MOHM, 17X17X7MM
376800972	376800075	ALT_CNN	ALL	DAVID <RDAR://31218436>	MOSFET, N-CH, 30V, 100MA, 13.0OHM, VMT3, HF	371800127	371800096	ALT_CNN	ALL	DAVID <RDAR://31355983>	DIODE, XENHR, 3.0V, 5A, 250MH, SOD882-2	13881078	13880810	ALT_CNN	ALL	LINDA KANAS 02/08/2017 EMAIL	CAP, CER, 2.2UF, 10%, 100V, X7R, 1206, SPT, MXTA
107800112	107800006	ALT_CNN	ALL	DAVID <RDAR://31218480>	THERMISTOR, WTC, 47K OHM, 14, 4073K, 0603	353800636	35384037	ALT_CNN	ALL	DAVID <RDAR://30842514>	IC, WX30V42, DIFF USB SW, DPDT, QFN10, 0.33MH	124-00003	126-00013	ALT_CNN	ALL	BLC OUTPUT CAPS	CAP, AL-PLY, 180UF, 20, 100V, 105C, 3A, 10X13, BK
311800112	31180437	ALT_CNN	ALL	DAVID <RDAR://31156795>	IC, 74LVC2G126, DUAL, 3-STATE BUFFER, DFN8	311800138	31180436	ALT_CNN	ALL	DAVID <RDAR://31847473>	IC, 74AVC47245, XCVR, 4BIT, DUALSPELY, QFN16	37180748	37180731	ALT_CNN	ALL	BLC INPUT DIODE	DIODE, SCHOTTKY, 30V, 1A, 800-323
353800711	35382073	ALT_CNN	ALL	DAVID <RDAR://31174431>	IC, NC6210, CURRENT SENSE, 200V/V, SC70-6	378800074	378800072	ALT_CNN	ALL	DAVID <RDAR://31561551>	LED, GREEN, 568NM, 4.3MCD@1MA, 0402	37180782	37180783	ALT_CNN	ALL	BLC OUTPUT DIODE	DIODE, RECT, STPS8H100G, 100V, 8A, T0263
371800089	371800085	ALT_CNN	ALL	DAVID <RDAR://31174773>	DIODE, SCHOTTKY, 40V, 2A, LOW VF, 0603	378800073	378800067	ALT_CNN	ALL	DAVID <RDAR://31561520>	LED, RED, 621NM, 2.5MCD@1MA, 1X0.6X0.25MM	37680845	37681067	ALT_CNN	ALL	BLC POWER FET + 12V POWER FET	MOSFET, P-CH, 30V, 9.4MOHM, 23.7A, 5X6 DFN
37180684	37180495	ALT_CNN	ALL	DAVID <RDAR://31174773>	DIODE, DUAL, SCHOTTKY, 30V, 200MA, SOT-363	31180595	311800155	ALT_CNN	ALL	DAVID <RDAR://31204779>	IX, XCVR, DL-BIT/DL-SUPPLY, 2ST-OUT-QFN-10	37681204	37681106	ALT_CNN	ALL	BLC LED SINK FET	MOSFET, NCH, 100V, 12.5A, 24MOHM, 3X3 DFN
37280186	37280185	ALT_CNN	ALL	DAVID <RDAR://31174982>	XSTR, NPN, 65V, 100MA, 3P SOT883B	371800046	371800082	ALT_CNN	ALL	DAVID <RDAR://31847479>	DIODE, SCHOTTKY, 30V, 2A, LOW VF, 0603	74080145	74080146	ALT_CNN	ALL	BLC FUSE	FUSE 6A 32V SMD 0603
376800015	37680610	ALT_CNN	ALL	DAVID <RDAR://31195440>	XSTR, FET, DUAL N-CH, 60V, 3.3OHM, SOT-363	155800067	15580581	ALT_CNN	ALL	DAVID <RDAR://31230500>	FERR BD, 240 OHM, 25%, 420MA, 0.31 DCR, 0201	10780359	10780229	ALT_CNN	ALL	BLC ROOSTY SENSE RES	RES, FILM, 2W, .03 OHM, 1%, 2512, SMD
376800001	37680659	ALT_CNN	ALL	DAVID <RDAR://31196048>	MOSFET, P-CH, 20V, 4.2A, 60MOHM@4.5V, SOT23	31180596	31180593	ALT_CNN	ALL	DAVID <RDAR://31234608>	IC, VOLT-LEVEL TRANS, W/CONFIG GATE, SOT886	37681184	37681183	ALT_CNN	ALL	BLC ROOSTY FET	MOSFET, NCH, 100V, 30A, 31.5MOHM, 3X3 MLP
376800146	37681061	ALT_CNN	ALL	DAVID <RDAR://31197441>	FET, P-CH, 20V, 0.1A, 3.8MOHM, DFN1006	37780178	377800031	ALT_CNN	ALL	DAVID <RDAR://30715965>	SUPPR, TRANS, BIDIR, 5.5V, 0.2PF, 0201	37681256	37681073	ALT_CNN	ALL	BLC SHORT PROTECTION FETS	XSTR, FET, N-CH, 150V, 1.4A, 425MOHM, S80T-6L
311800116	31180324	ALT_CNN	ALL	DAVID <RDAR://31244248>	IC, 74LVC2G34, DUAL BUFFER GATE, DFN6	197800129	197800130	ALT_CNN	ALL	DAVID <RDAR://31239624>	OSC, XTAL, 50MHZ, 50PPM, 3.3V, 5032	15580831	15580797	ALT_CNN	ALL	BLC PERISTE BEADS	FLTR, BMT, FERR BD, 600 OHM, 100MHZ, 0603
311800133	311800130	ALT_CNN	ALL	DAVID <RDAR://312073450>	IC, 74AVC1745, XCVR, DL SUPPLY, 1BIT, DFN-6	376800246	37680865	ALT_CNN	ALL	LEWIS <RDAR://31232531>	NFET, 20V, 4A, 0.030OHM, SOT23	13880805	13880834	ALT_CNN	ALL	LINDA KANAS 02/24/2017 EMAIL	CAP, CER, 100F, 10%, 25V, X7R, 1206, SOPT
353801041	353801042	ALT_CNN	ALL	DAVID <RDAR://31207542>	IC, TS3312, V-REF, 1.25V, 0.15%, 5MA, QFN8	197800118	197800120	ALT_CNN	ALL	LEWIS <RDAR://31231847>	XTAL, 24MHZ, 30PPM, 9.5PF, 60 OHM MAX, 1612	N&V					
371800095	37180567	ALT_CNN	ALL	DAVID <RDAR://312318051>	DIODE, SWITCH, 100V, 500MA, SOD323, 2P	155800363	155800367	ALT_CNN	ALL	JOHN <RDAR://312886387>	CHOKE, COMMON MODE, 670OHM, 20%, 400MA, 0805-4	138800077	138800035	ALT_CNN	ALL	AKASH BUILD MATRIX REV 3.0	CAP, X6S, 200F, 20%, 2.5V, B2TTL, TY, 0402
372800008	37280173	ALT_CNN	ALL	DAVID <RDAR://31232978>	XSTR, NPN, 50V, 100MA, 100MHV, HFE=200, CST3	155800364	155800367	ALT_CNN	ALL	JOHN <RDAR://312886387>	CHOKE, COMMON MODE, 900OHM, 50%, 300MA, 0805-4	138800169	138800168	ALT_CNN	ALL	AKASH BUILD MATRIX REV 3.0	CAP, X6S, 430F, 20%, 2.5V, NU, 0603
376800224	37681128	ALT_CNN	ALL	DAVID <RDAR://31240125>	NFET, 30V, 590MA, 670MOHM, SOT883B	157800024	157800026	ALT_CNN	ALL	JOHN <RDAR://312886430>	XPMB, LAN, 10G, 1CH, 120UH, 4532-8	138800111	138800036	ALT_CNN	ALL	AKASH BUILD MATRIX REV 3.0	CAP, X6S, 200F, 20%, 2.5V, NU, 0402
311800141	31180653	ALT_CNN	ALL	LEE <RDAR://313054341>	IC, PIGUL85V9, I2C LEVEL TRANSLATOR, DFN-8	197800143	197800142	ALT_CNN	ALL	ALAN <RDAR://31847271>	XTAL, 27MHZ, 10PPM, 12PF, 400OHM, 2520	138800188	138800043	ALT_CNN	ALL	DAVID <RDAR://31343496>	CAP, X6S, 100F, 20%, 6.3V, 0402, NU
138800013	13880772	ALT_CNN	ALL	DAVID <RDAR://312073933>	CAP, CER, 2.2UF, 10V, X6S, 0402	10280880	10280879	ALT_CNN	ALL	DAVID <RDAR://31305391>	RES, 0.010OHM, 1%, 100PPM, 1/2W, 1206, BLK, CYN	WIRELESS					
31180649	31180541	ALT_CNN	ALL	DAVID <RDAR://31235049>	IC, SINGLE, 2-INPUT AND GATE, 5P SOT553	DC-DC				339800411	339800404	ALT_CNN	ALL	MIKE EMAIL ON 05/16/2017	MODULE, WIFI/BT, HARPOON, USI, ES7.1, LGA324		
13880714	13880713	ALT_CNN	ALL	DAVID <RDAR://31281224>	CAP, CER, 100F, 6.3V, X5R, 7MM, 0402, SAMSUNG	152800573	15281778	ALT_CNN	ALL	SURESH <RDAR://31260481>	IND, 0.105UH, 20%, 46A, 0.35MOHM, 6.8X6.8X5MM	339800425	339800404	ALT_CNN	ALL	MIKE EMAIL ON 05/16/2017	MODULE, WIFI/BT, HARPOON, M, ES7.3, LGA385
15280098	15580137	ALT_CNN	ALL	DAVID <RDAR://312812705>	IND, CHIP EMFILL, 1000OHM, 200MA, 10OHM DCR	152800685	15281778	ALT_CNN	ALL	LEWIS <RDAR://312677597>	IND, 0.105UH, 20%, 46A, 0.35MOHM, 6.7X6.8X5MM	339800426	339800404	ALT_CNN	ALL	MIKE EMAIL ON 05/16/2017	MODULE, WIFI/BT, HARPOON, USI, ES7.3, LGA385
116800007	116800006	ALT_CNN	ALL	DAVID <RDAR://31281280>	RES, MTL FILM, 0 OHM, TIGHT TOL, 3A, 0402	15281684	15281870	ALT_CNN	ALL	SURESH <RDAR://312193171>	IND, PWR, MLD, 2.2UH, 20%, 105MO, 2.0A, 2520	339800427	339800404	ALT_CNN	ALL	MIKE EMAIL ON 05/16/2017	MODULE, WIFI/BT, HARPOON, M, ES7.5, LGA385
31180426	311800007	ALT_CNN	ALL	DAVID <RDAR://312604120>	IC, 74LVC1G07, SMDL BUFFER W/OD OUT, SOT891	152800408	152800329	ALT_CNN	ALL	SURESH <RDAR://312878048>	IND, MLD, 1.5UH, 20%, 13A, 0.145MO, 8.44X8X2.4	339800428	339800404	ALT_CNN	ALL	MIKE EMAIL ON 05/16/2017	MODULE, WIFI/BT/HARPOON/USI/ES7.5, LGA385
107800015	107800011	ALT_CNN	ALL	DAVID <RDAR://31267426>	RES, MTL FILM, 0.00075OHM, 1%, 4-L, 0612, BLK	12880364	12880264	ALT_CNN	ALL	VICTORY HARPOON 01/20/2017	CAP, TANT, POLY, 680UF, 20%, 16V, 50MOHM, D2E, BLK	SOC					
107800101	107800005	ALT_CNN	ALL	DAVID <RDAR://31267411>													

D

C

B

A

SHIELD FENCE AR

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-11526	1	SHIELD FENCE,MLB,AR,X664	SHIELD_FENCE_AR	CRITICAL	

SHIELD CAN RJ45

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-10607	1	SHIELD,CAN,RJ45,X664	SHIELD_CAN_RJ45	CRITICAL	

SHIELD CAN HARPOON

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-10608	1	SHIELD CAN,HARPOON,X664	SHIELD_HPN_CAN	CRITICAL	

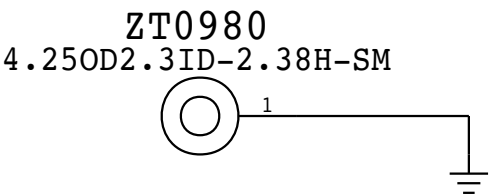
SHIELD FENCE HARPOON

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-12956	1	SHIELD,FENCE,WIFI,X664	SHIELD_HPN_FENCE	CRITICAL	

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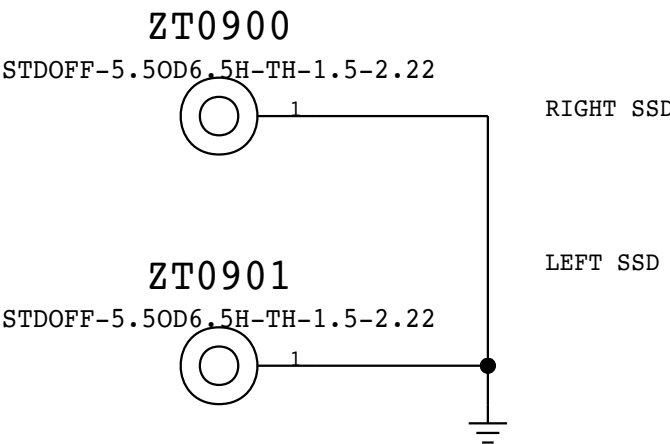
AUDIO BOSS

(860-00773)



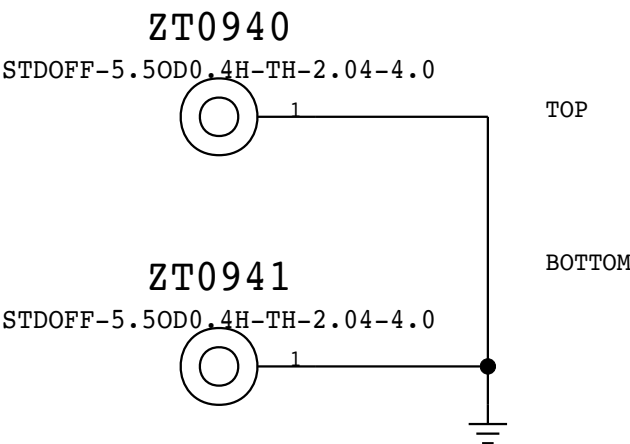
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2.35MM HOLE, 5.8MM PAD, WITH BOSS INCLUDED (860-00739)



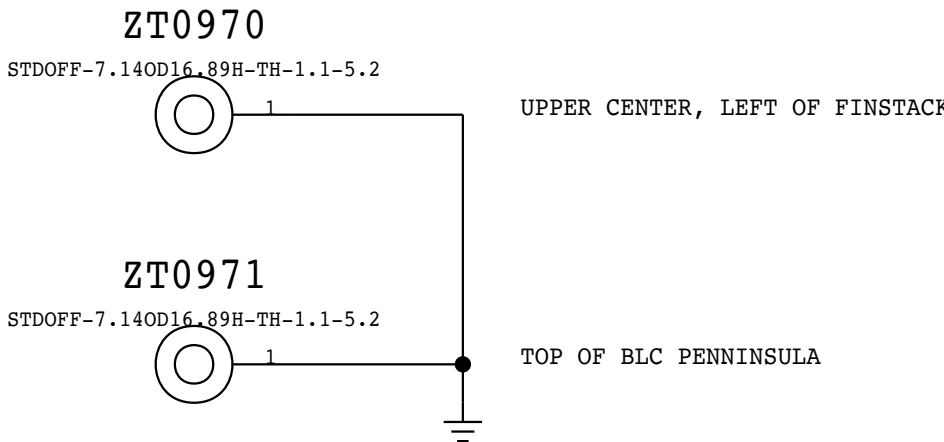
WIRELESS STRAIN RELIEF STANDOFFS

4.11MM HOLE, 6.00MM PAD, WITH BOSS INCLUDED (860-00713)



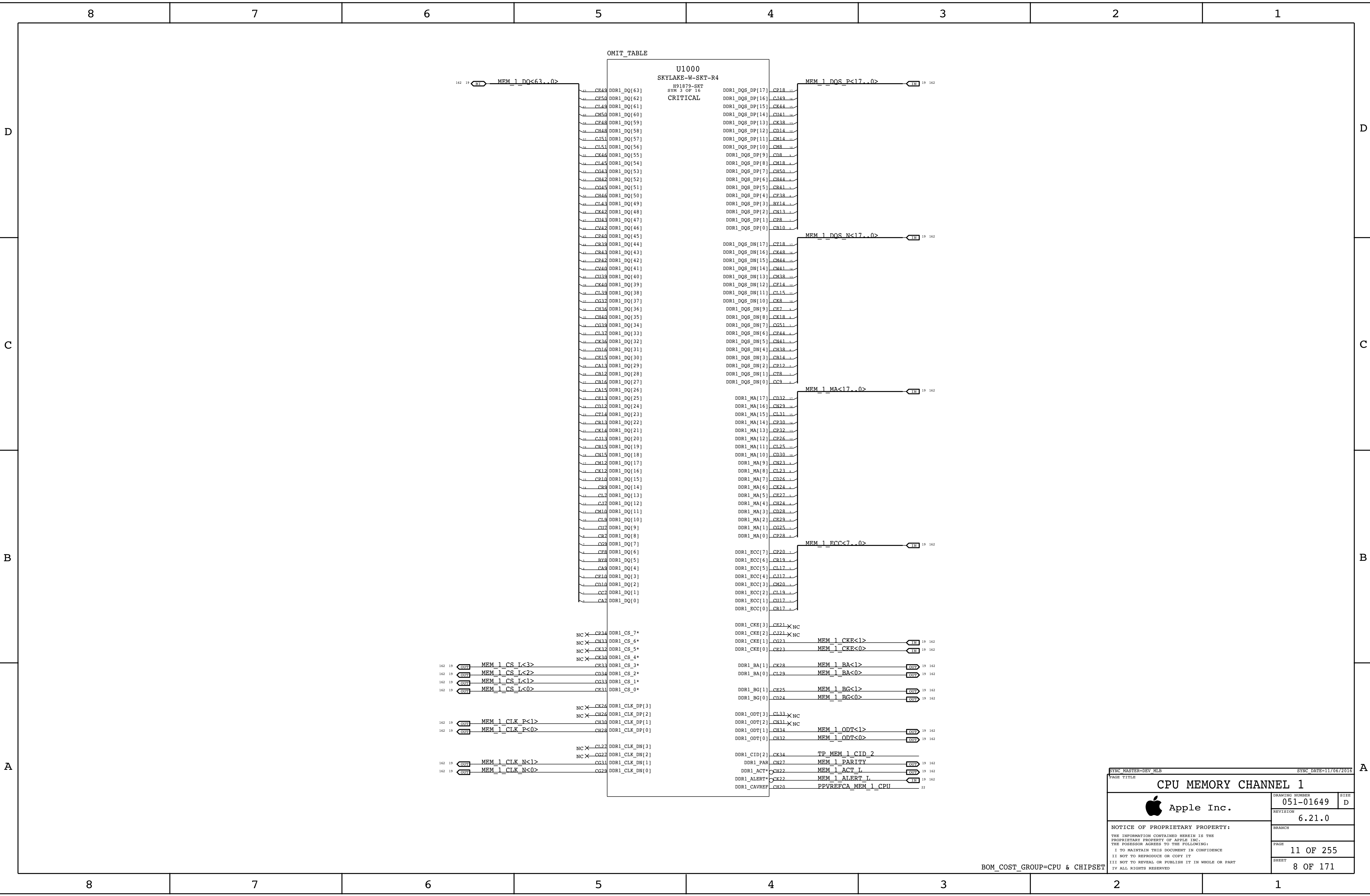
MLB STANDOFFS

5.38MM HOLE, 8.5MM PAD, WITH BOSS INCLUDED (860-00708)



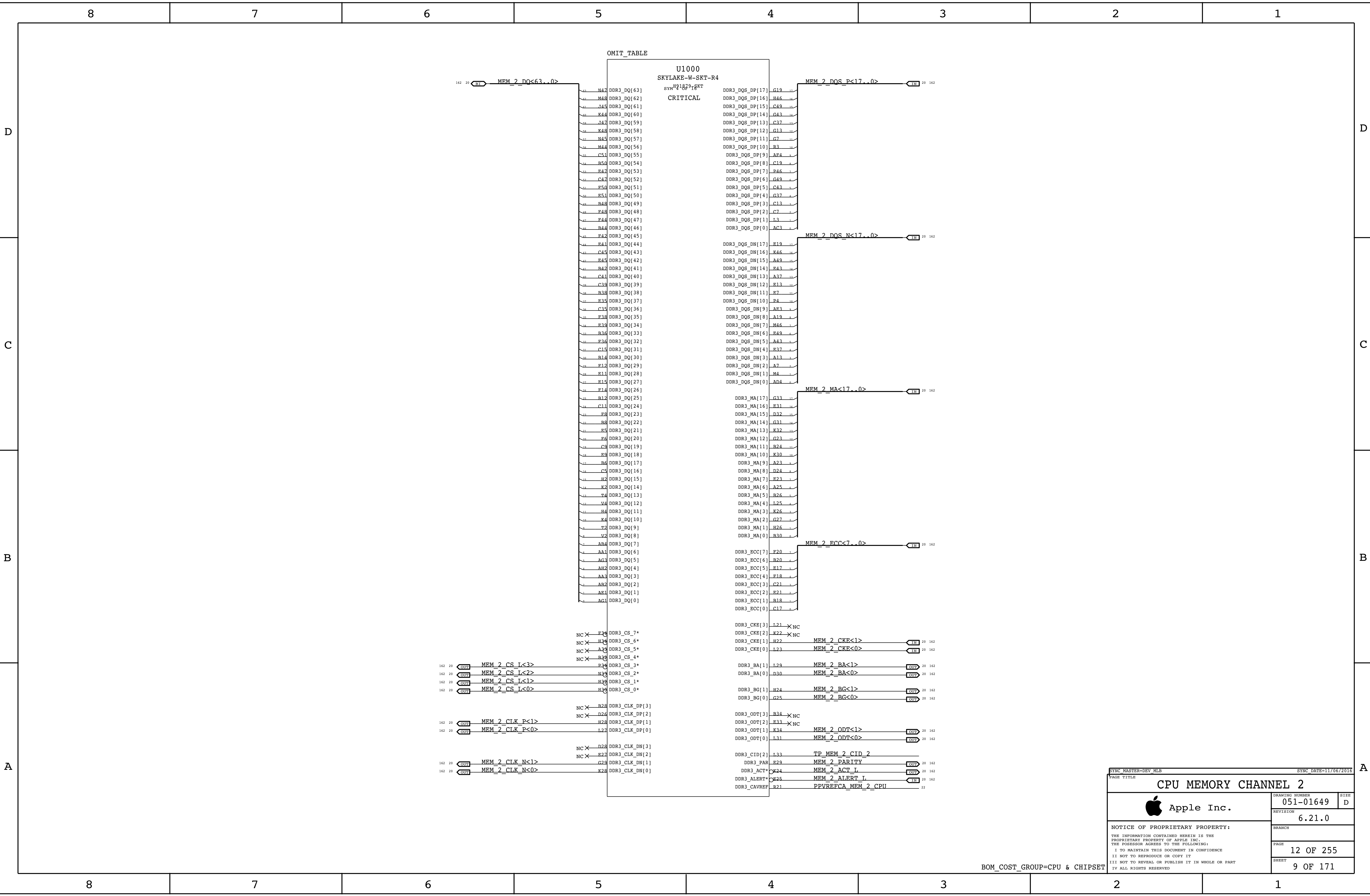
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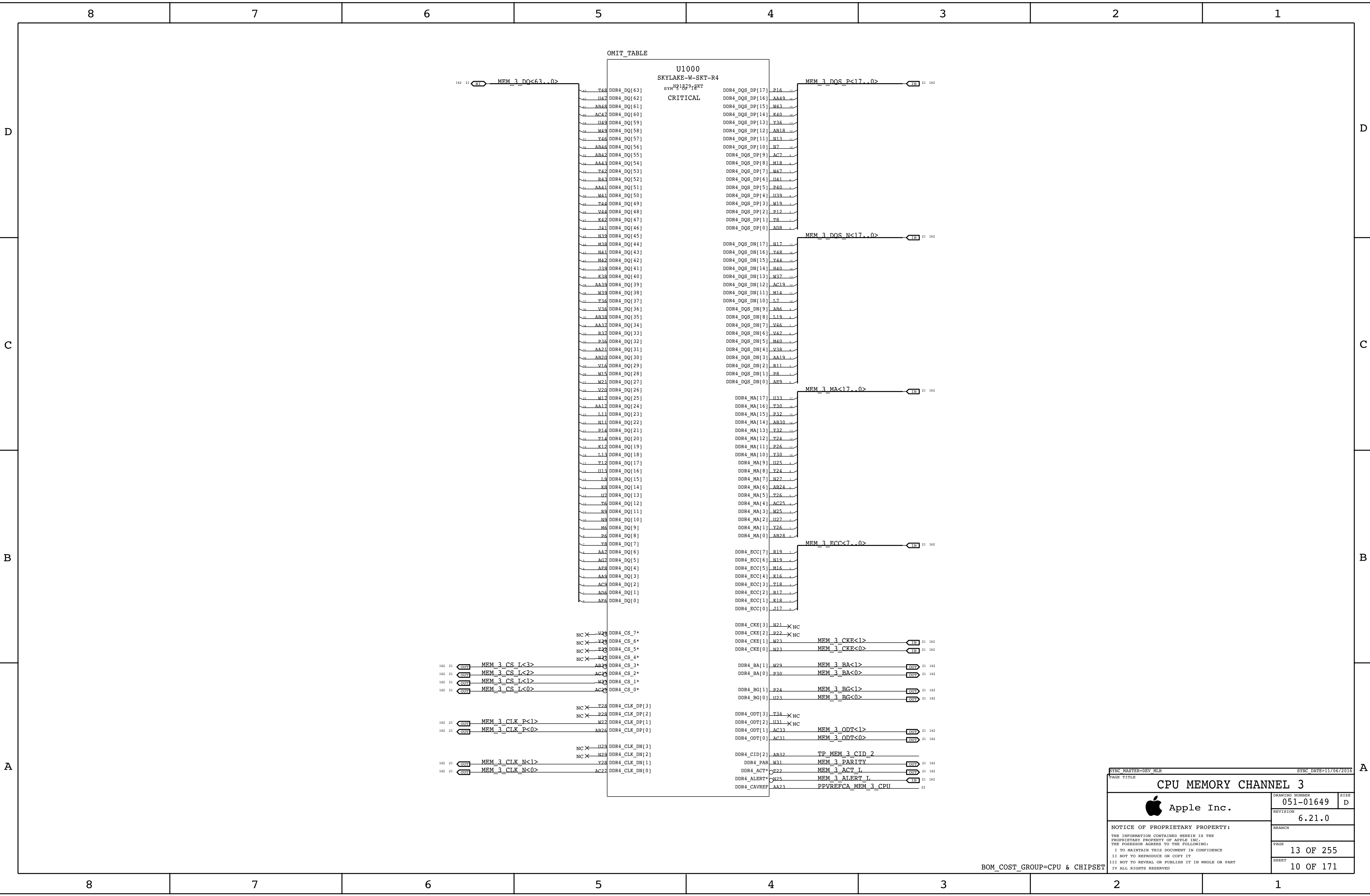
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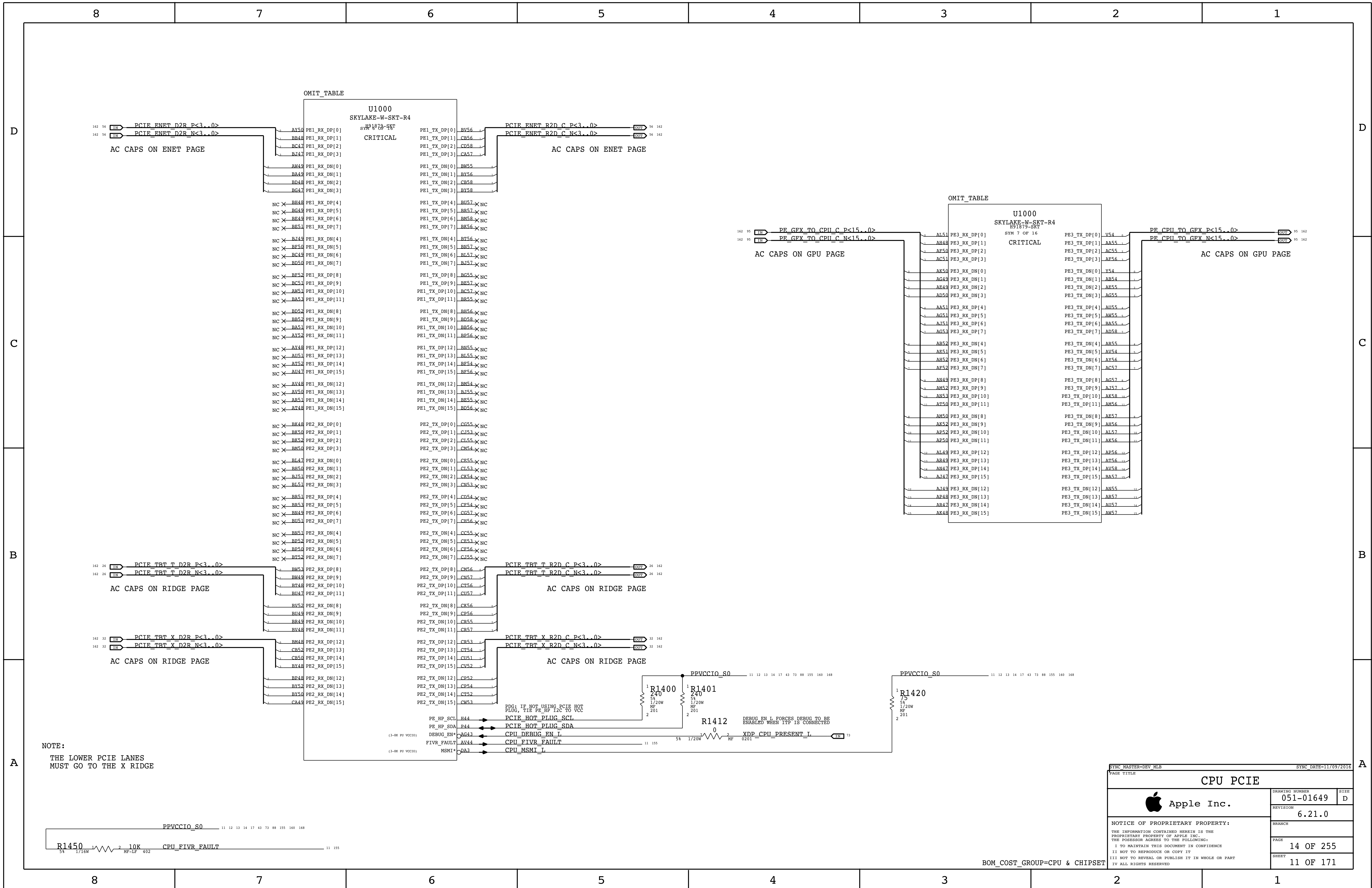
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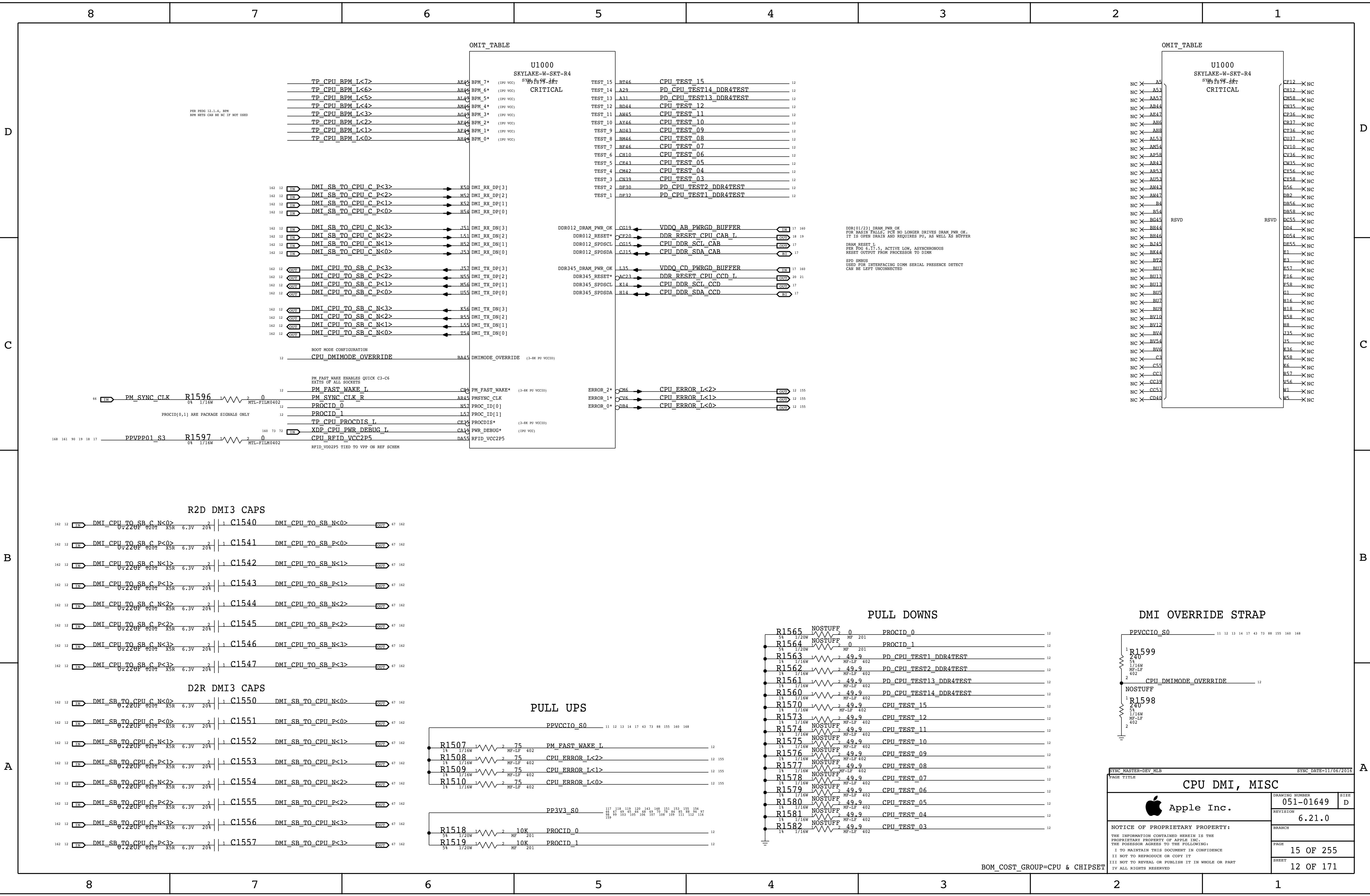
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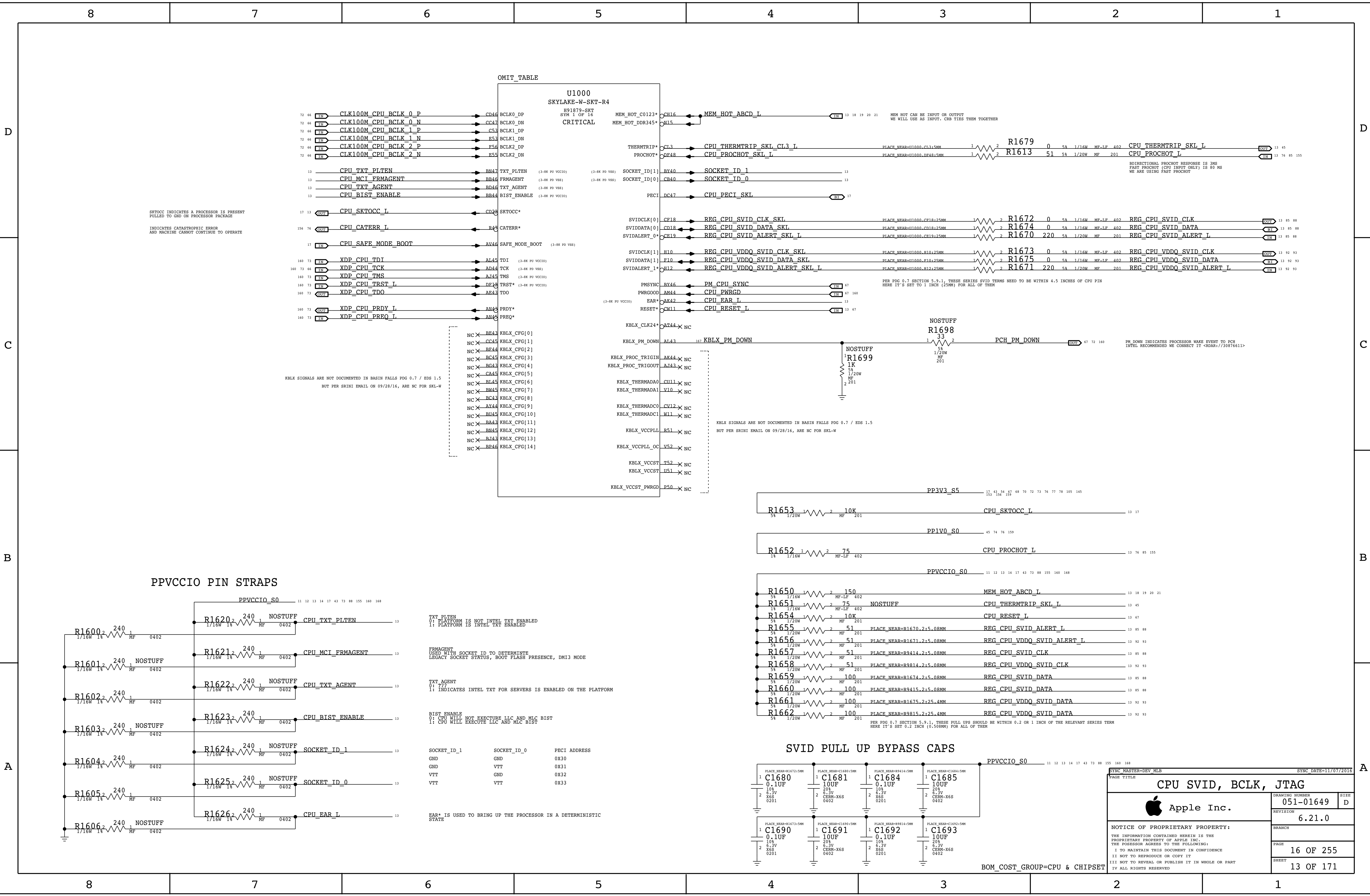


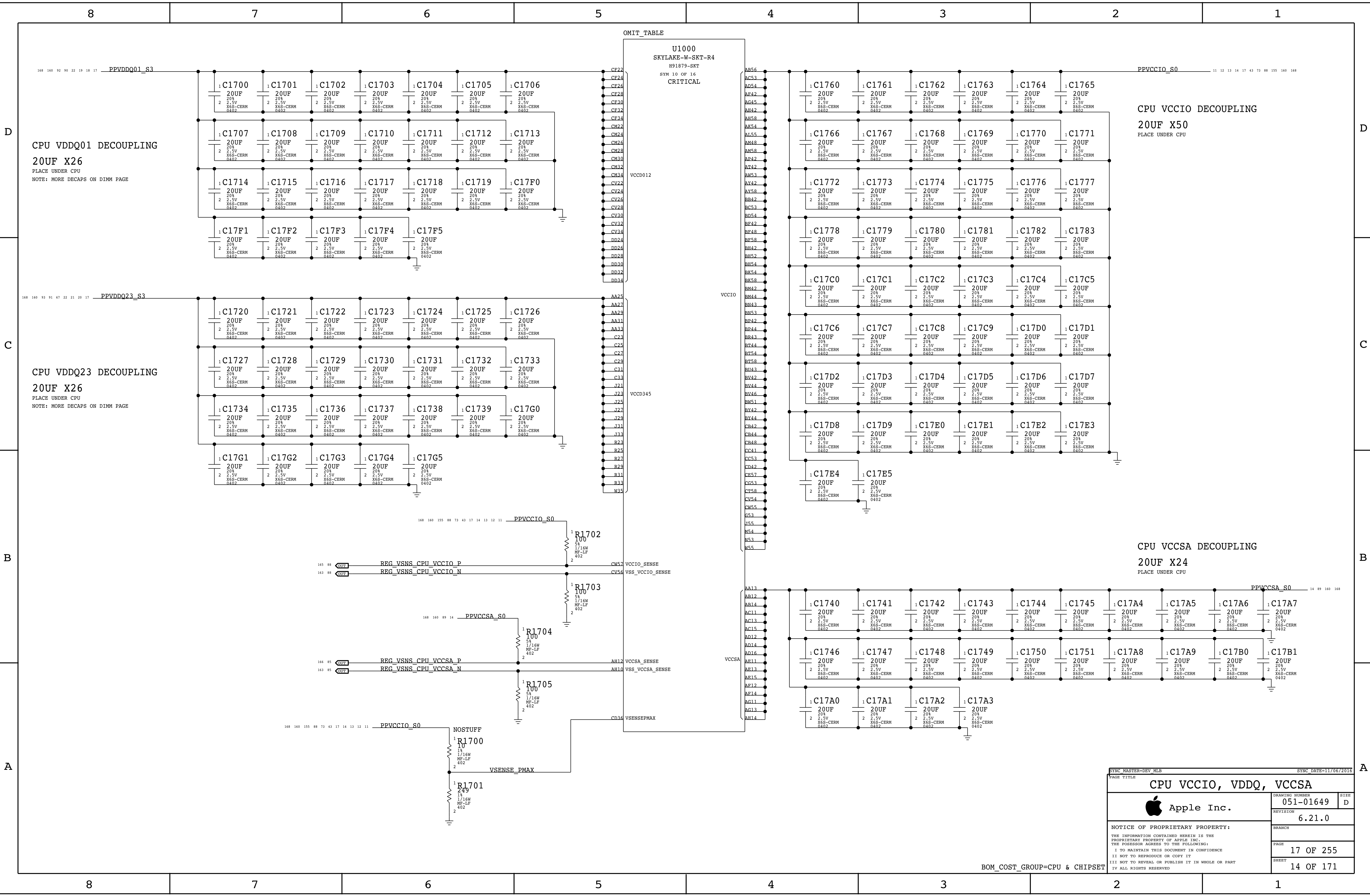
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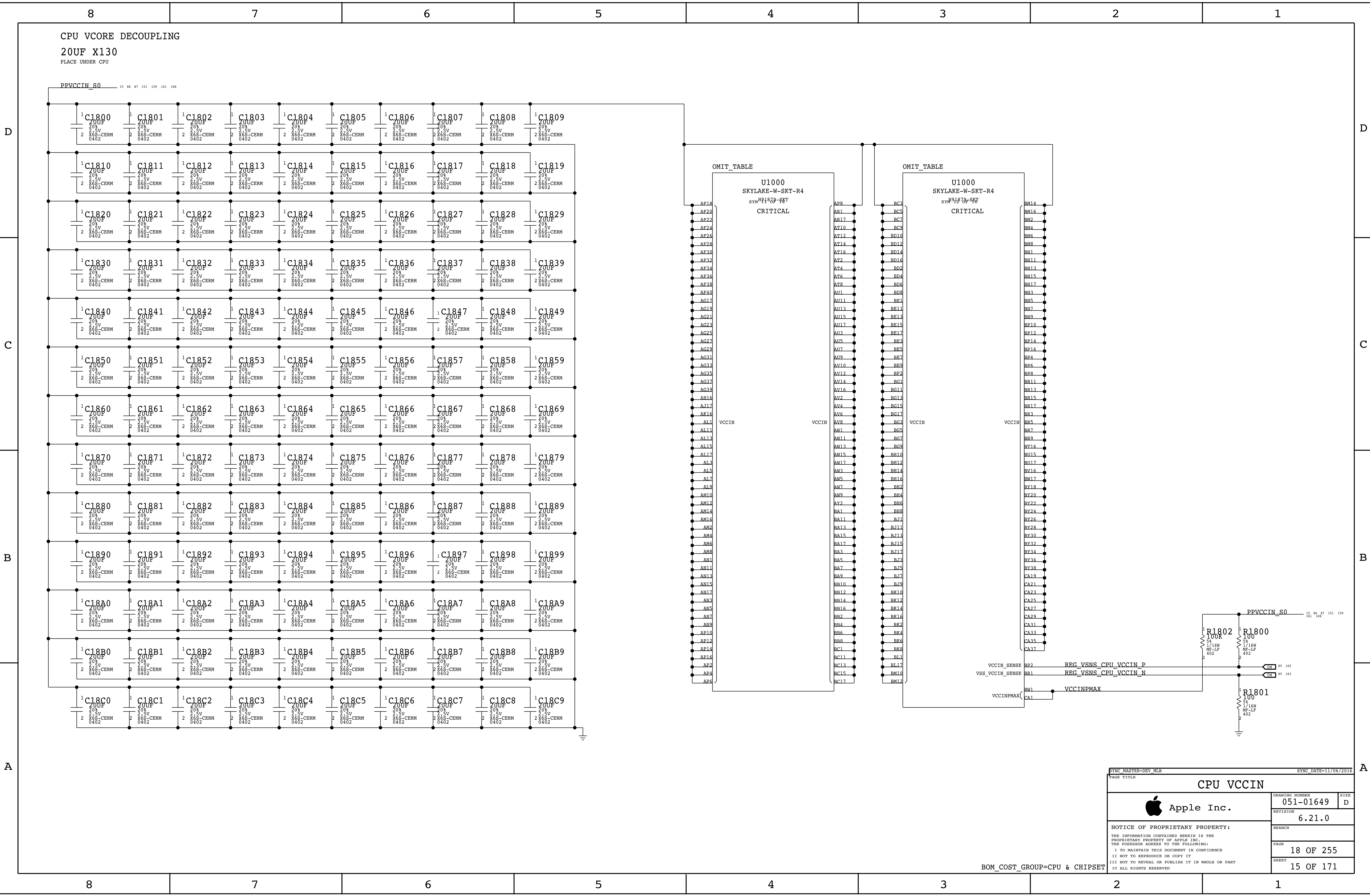






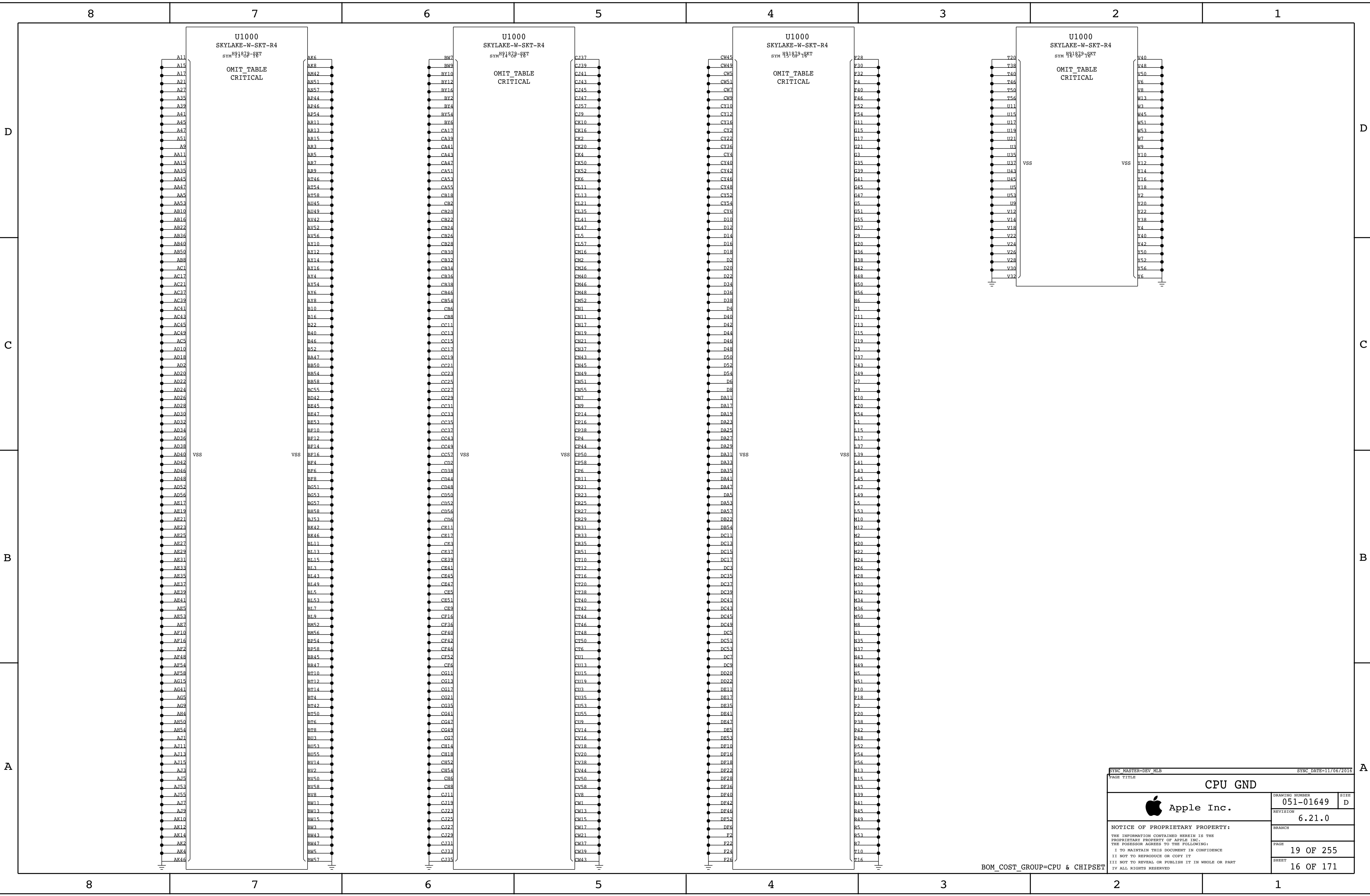
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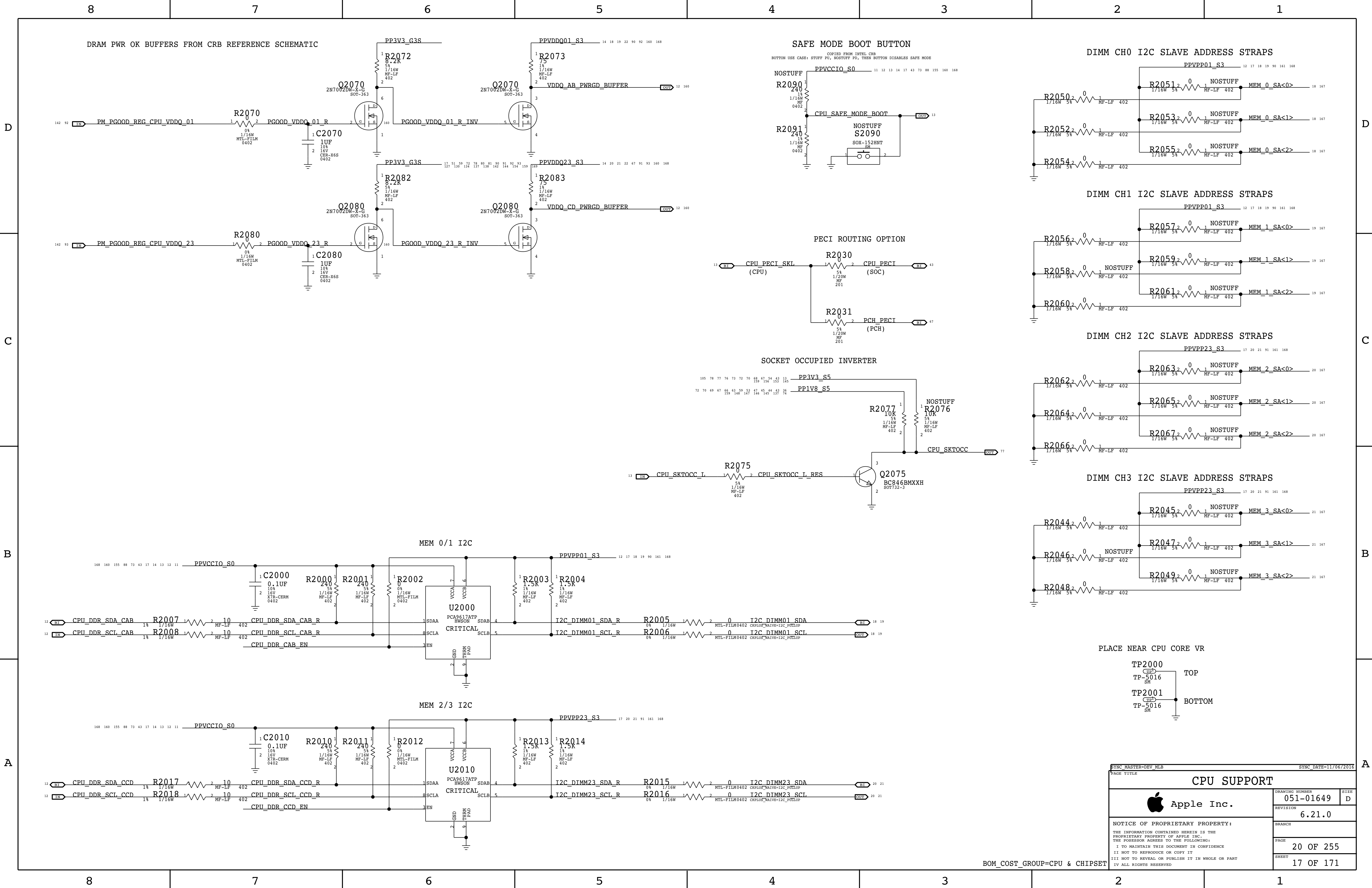
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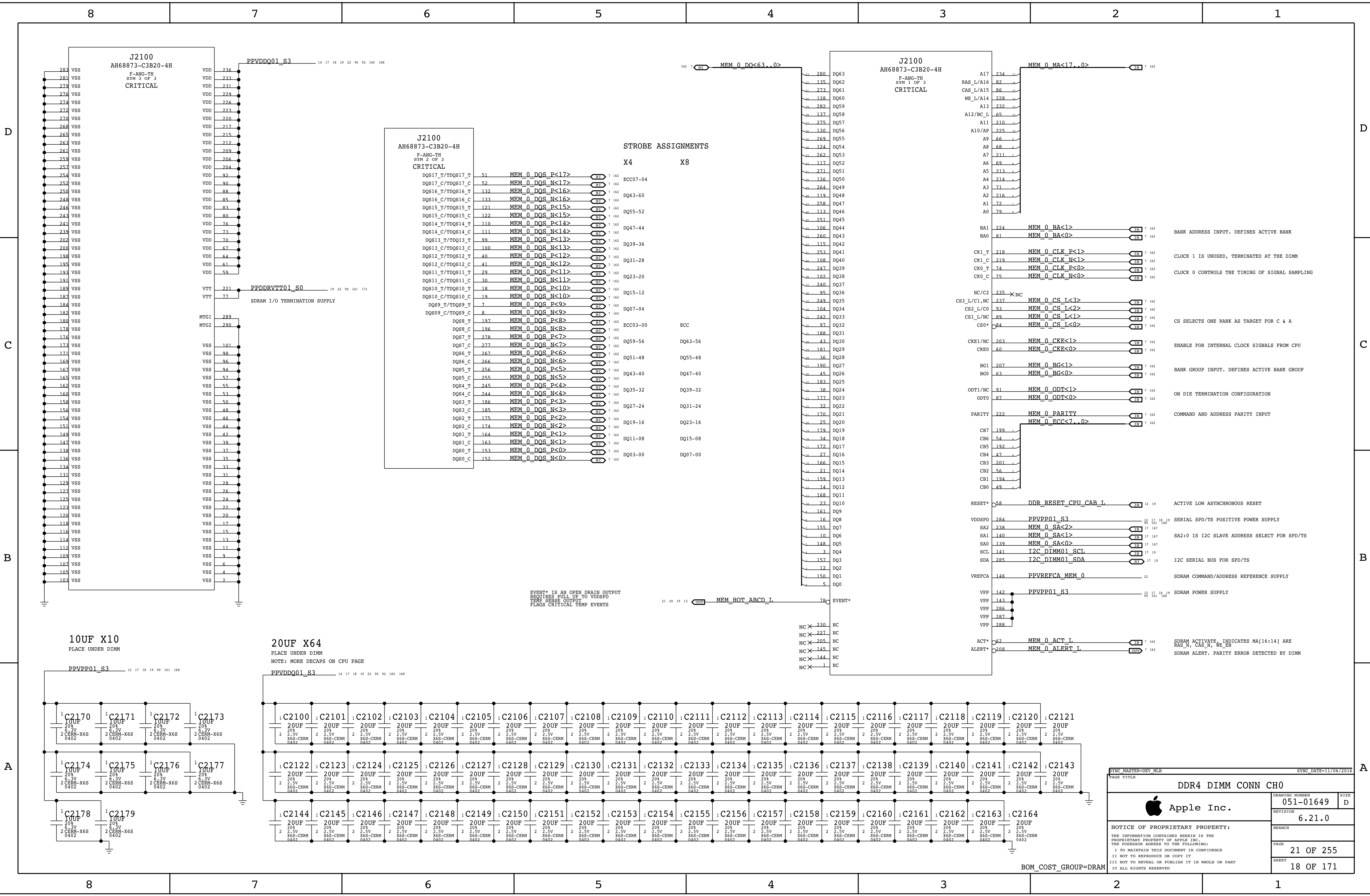
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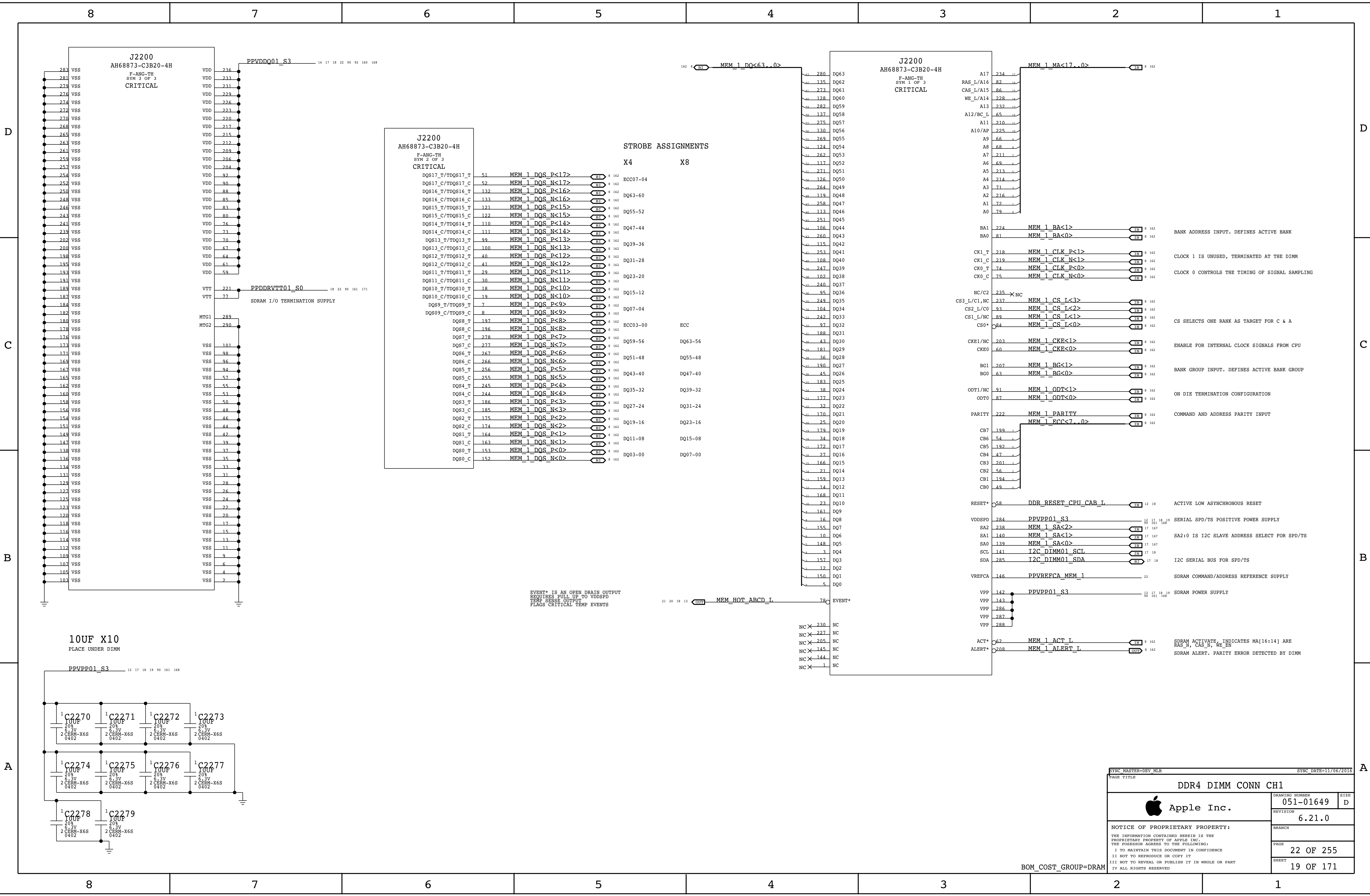
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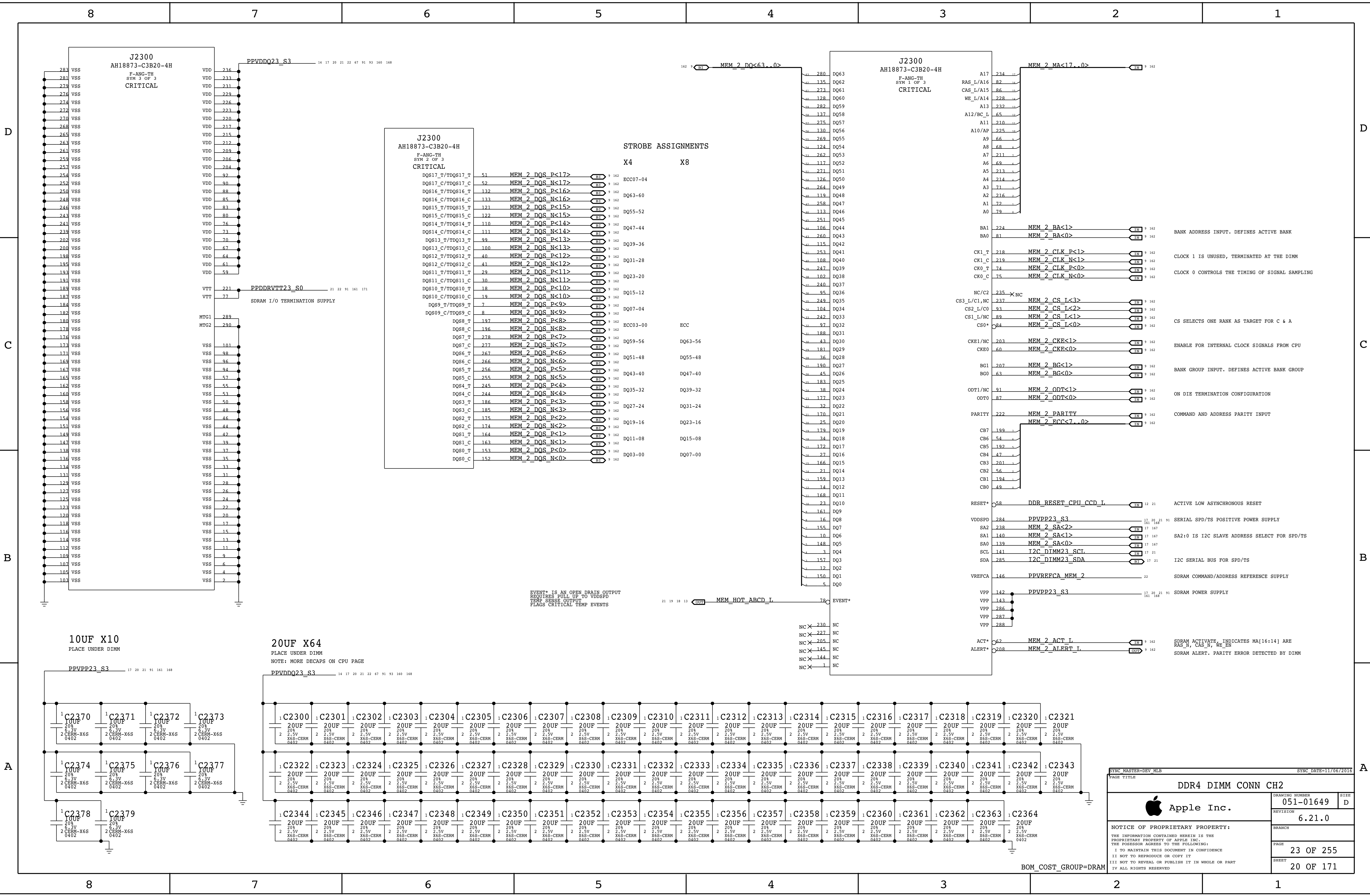


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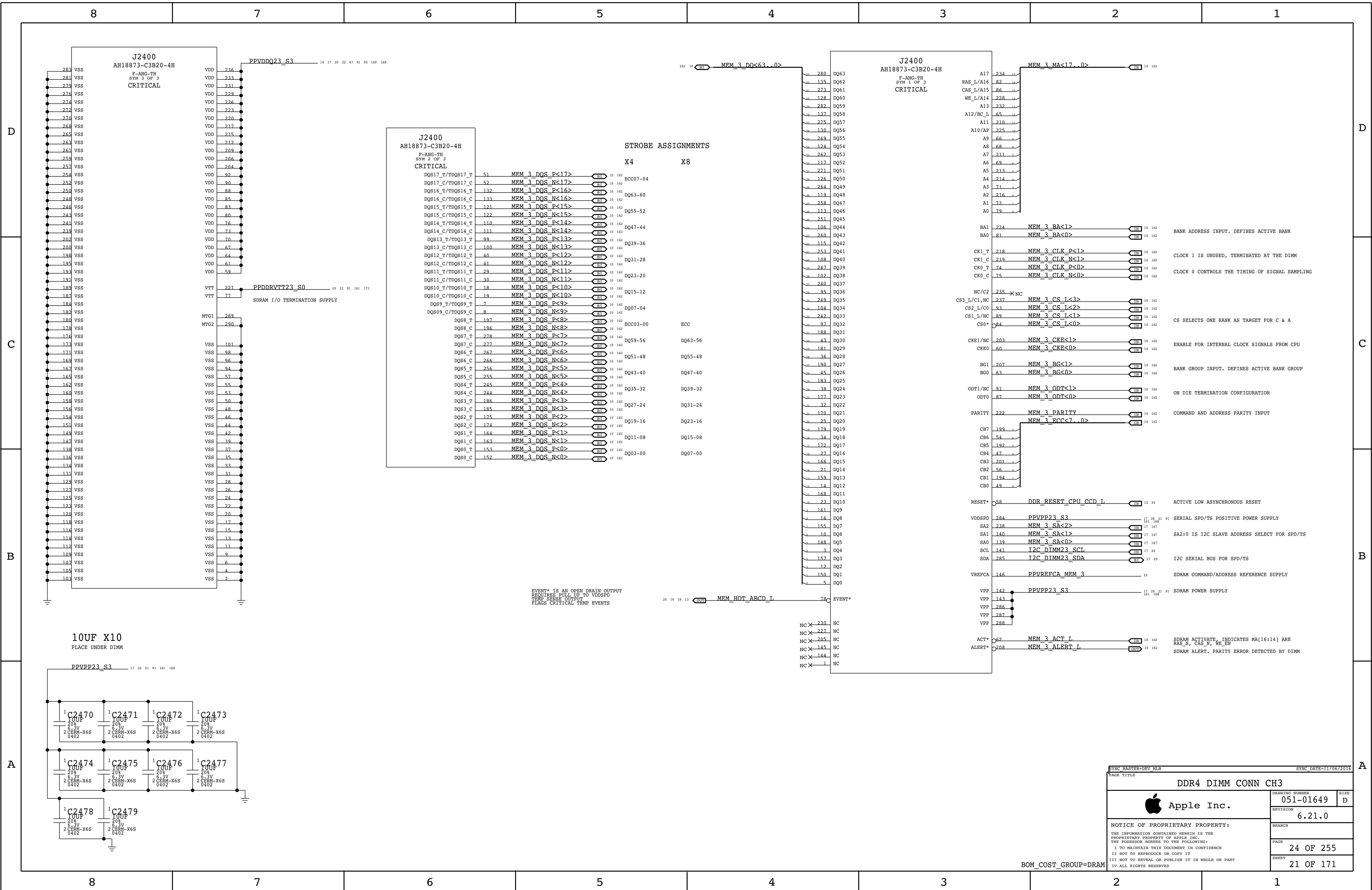
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DQS3_T	186	MEM_2_DQS_P<3>	9 162
DQS3_C	185	MEM_2_DQS_N<3>	9 162
DQS2_T	175	MEM_2_DQS_P<2>	9 162
DQS2_C	174	MEM_2_DQS_N<2>	9 162
DQS1_T	164	MEM_2_DQS_P<1>	9 162
DQS1_C	163	MEM_2_DQS_N<1>	9 162
DQS0_T	153	MEM_2_DQS_P<0>	9 162
DQS0_C	152	MEM_2_DQS_N<0>	9 162

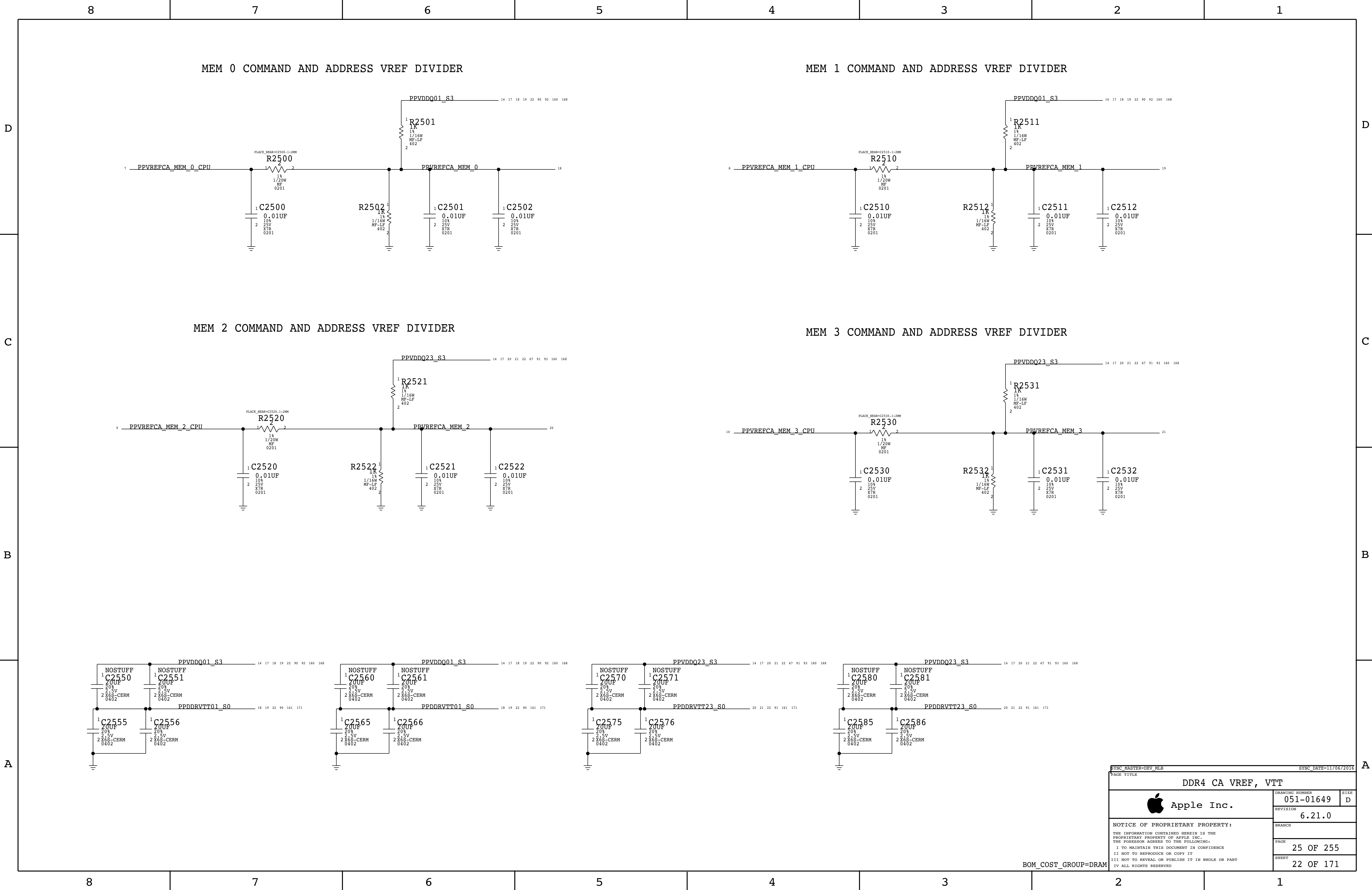
STROBE ASSIGNMENTS

X4	X8
ECC07-04	
DQ63-60	
DQ55-52	
DQ47-44	
DQ39-36	
DQ31-28	
DQ23-20	
DQ15-12	
DQ07-04	
ECC03-00	ECC
DQ59-56	DQ63-56
DQ51-48	DQ55-48
DQ43-40	DQ47-40
DQ35-32	DQ39-32
DQ27-24	DQ31-24
DQ19-16	DQ23-16
DQ11-08	DQ15-08
DQ03-00	DQ07-00

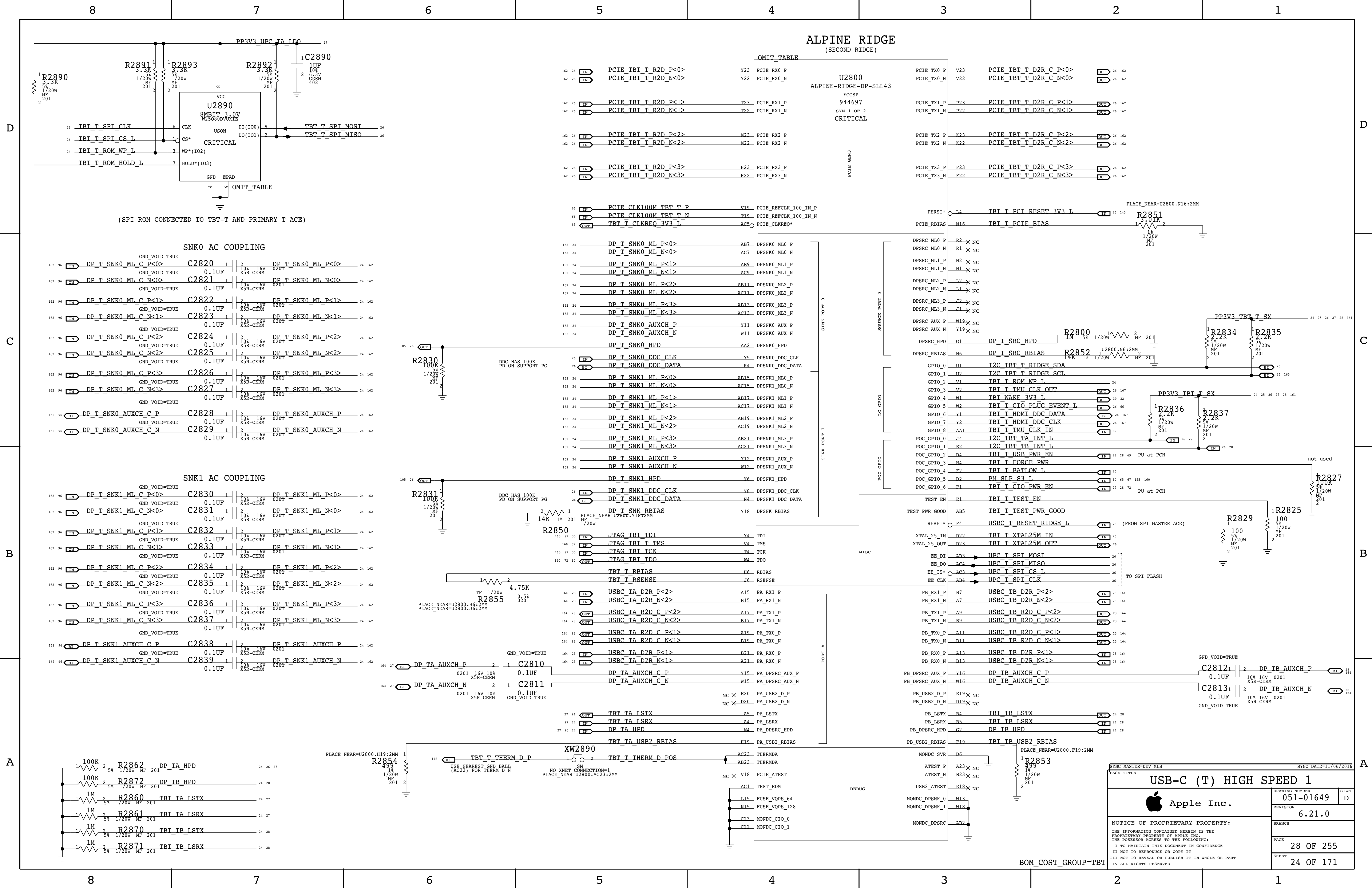
EVENT* IS AN OPEN DRAIN OUTPUT
REQUIRES PULL UP TO VDDSPD
TEMP CRITICAL TEMP EVENTS
TEMP CRITICAL TEMP EVENTS

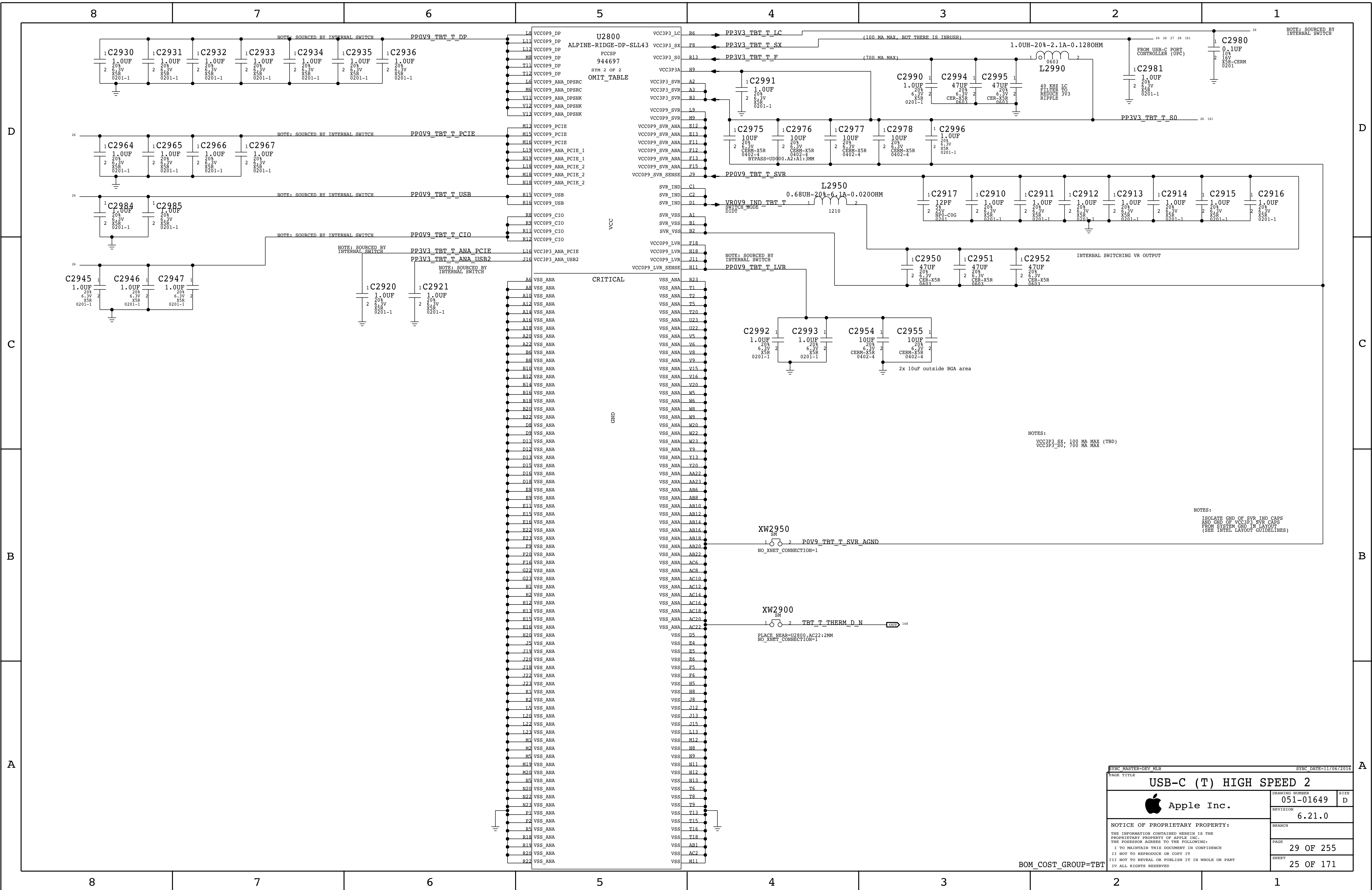
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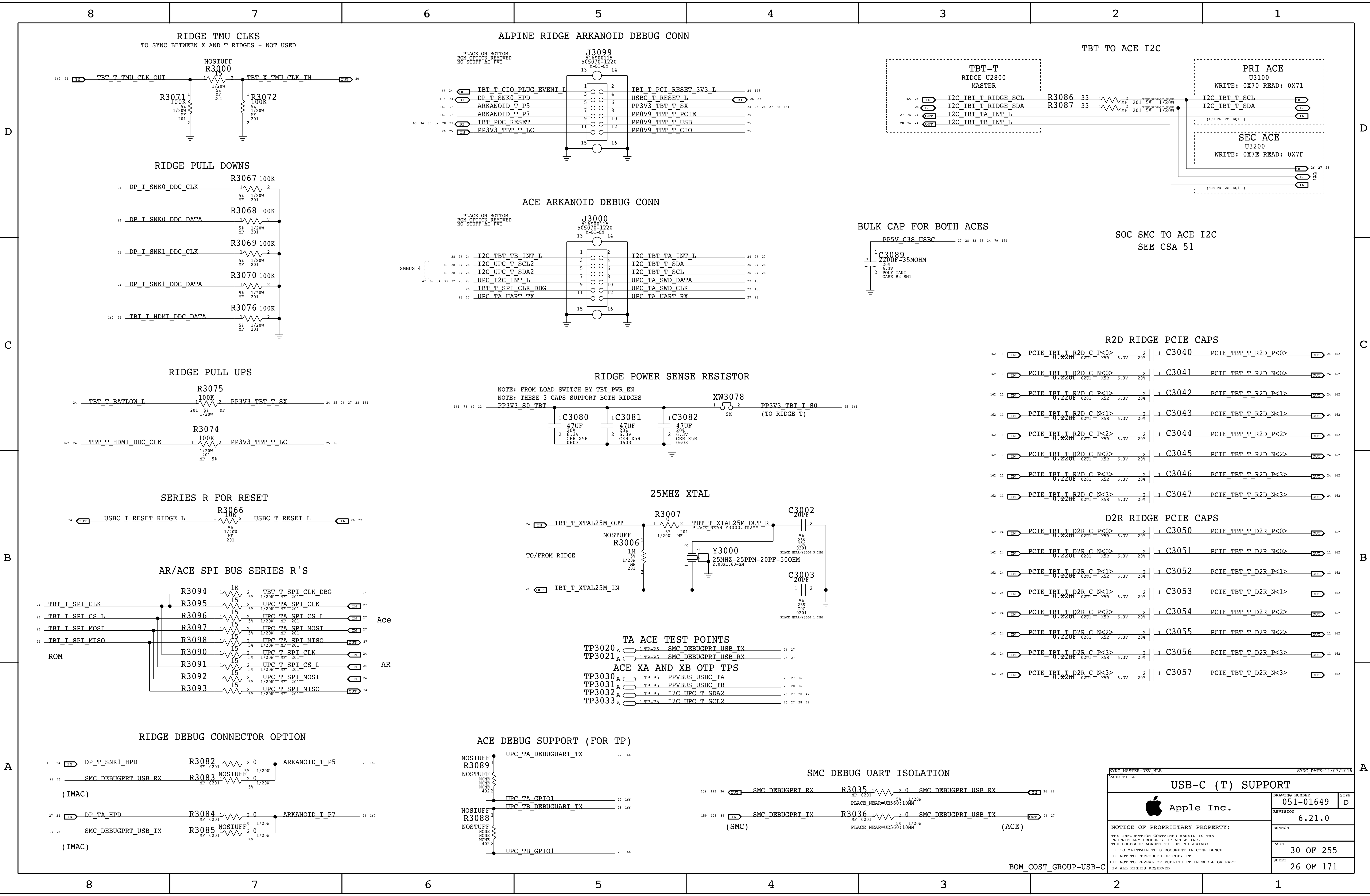




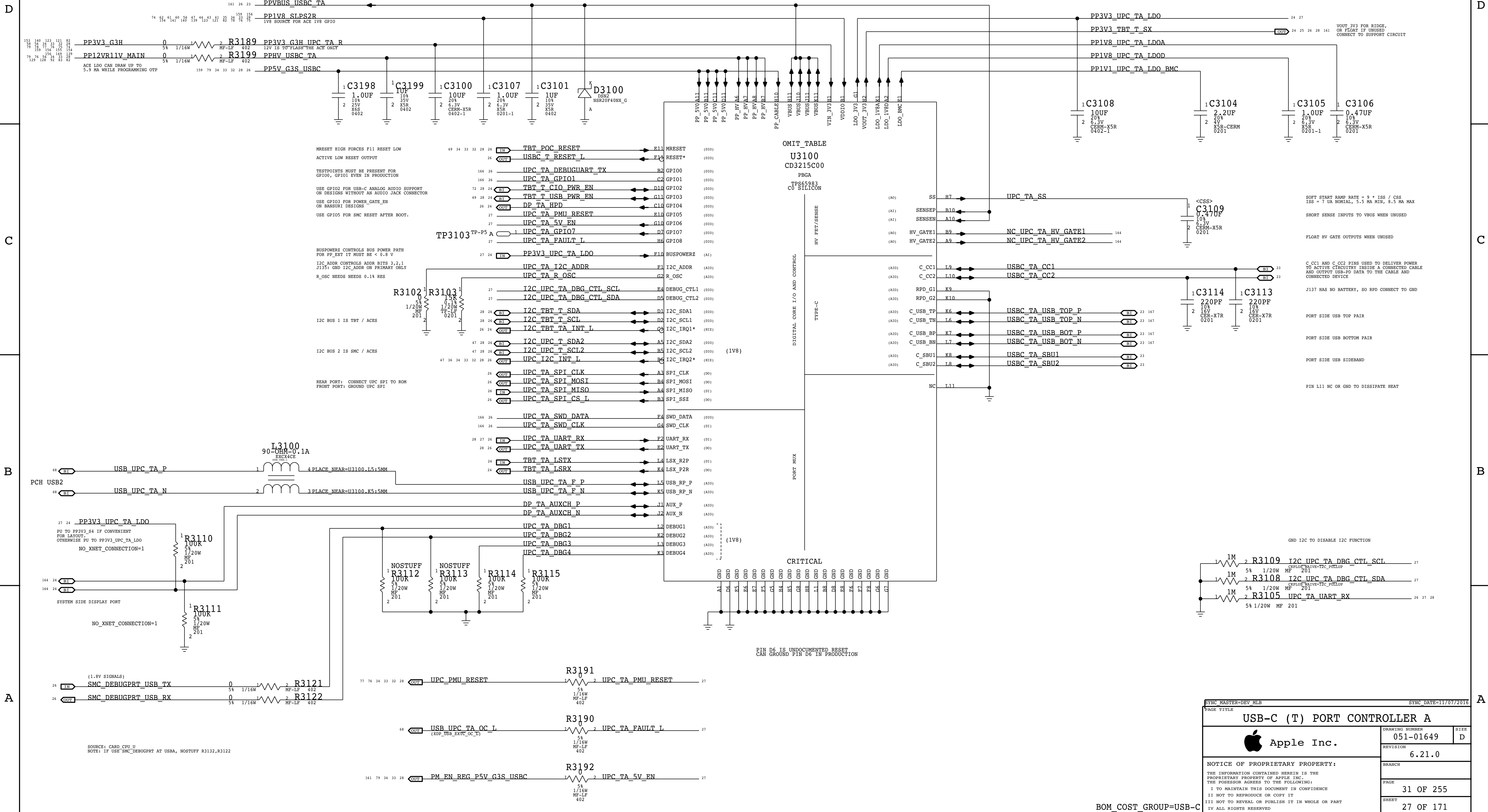




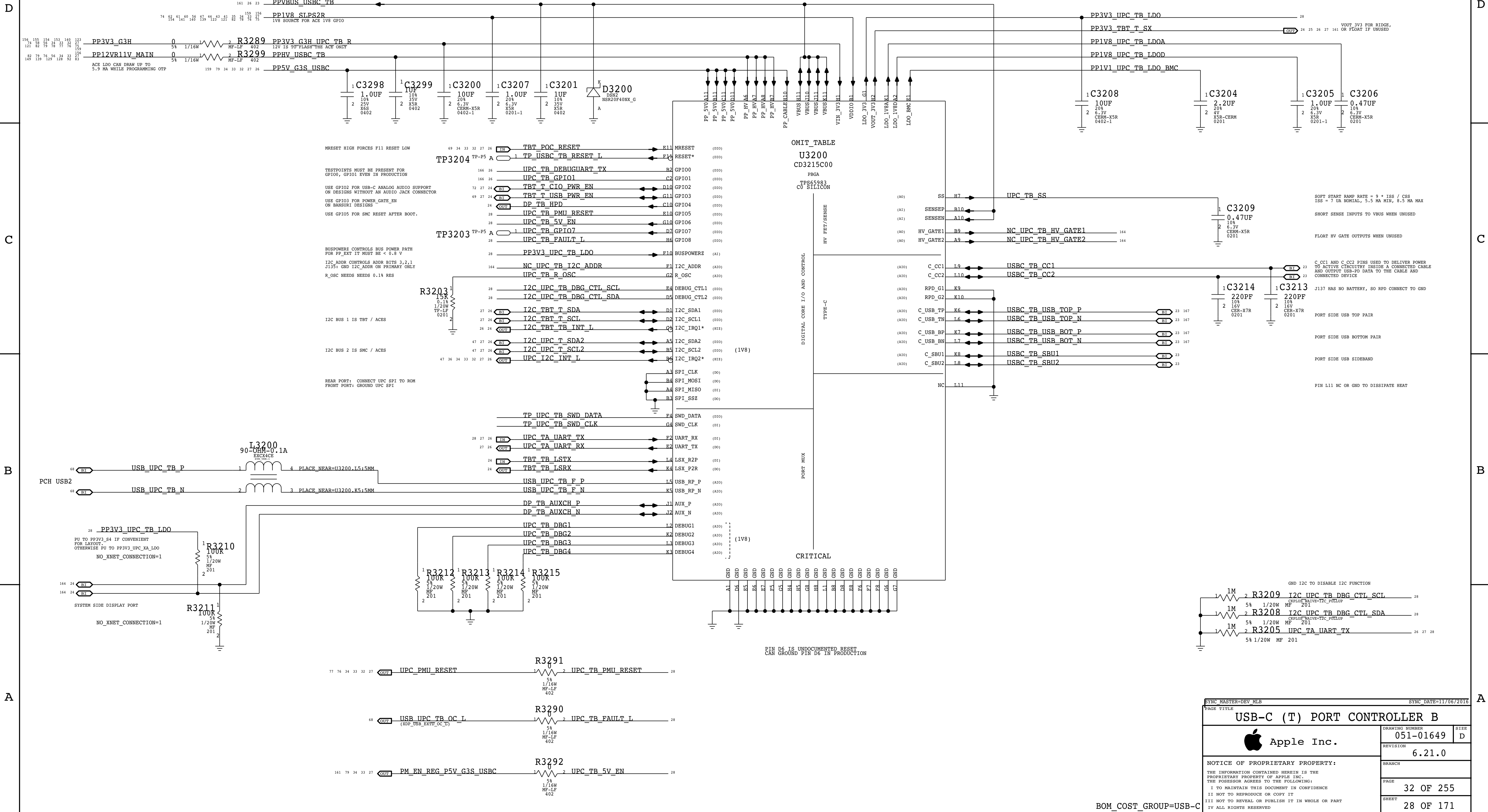




8	7	6	5	4	3	2	1
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T SECONDARY ACE USB-C PORT CONTROLLER (UPC)



J137 SUPPORTS 5V @ 3A
J137 HAS A QUAD-PACK USB-C CONNECTOR

D

D

C

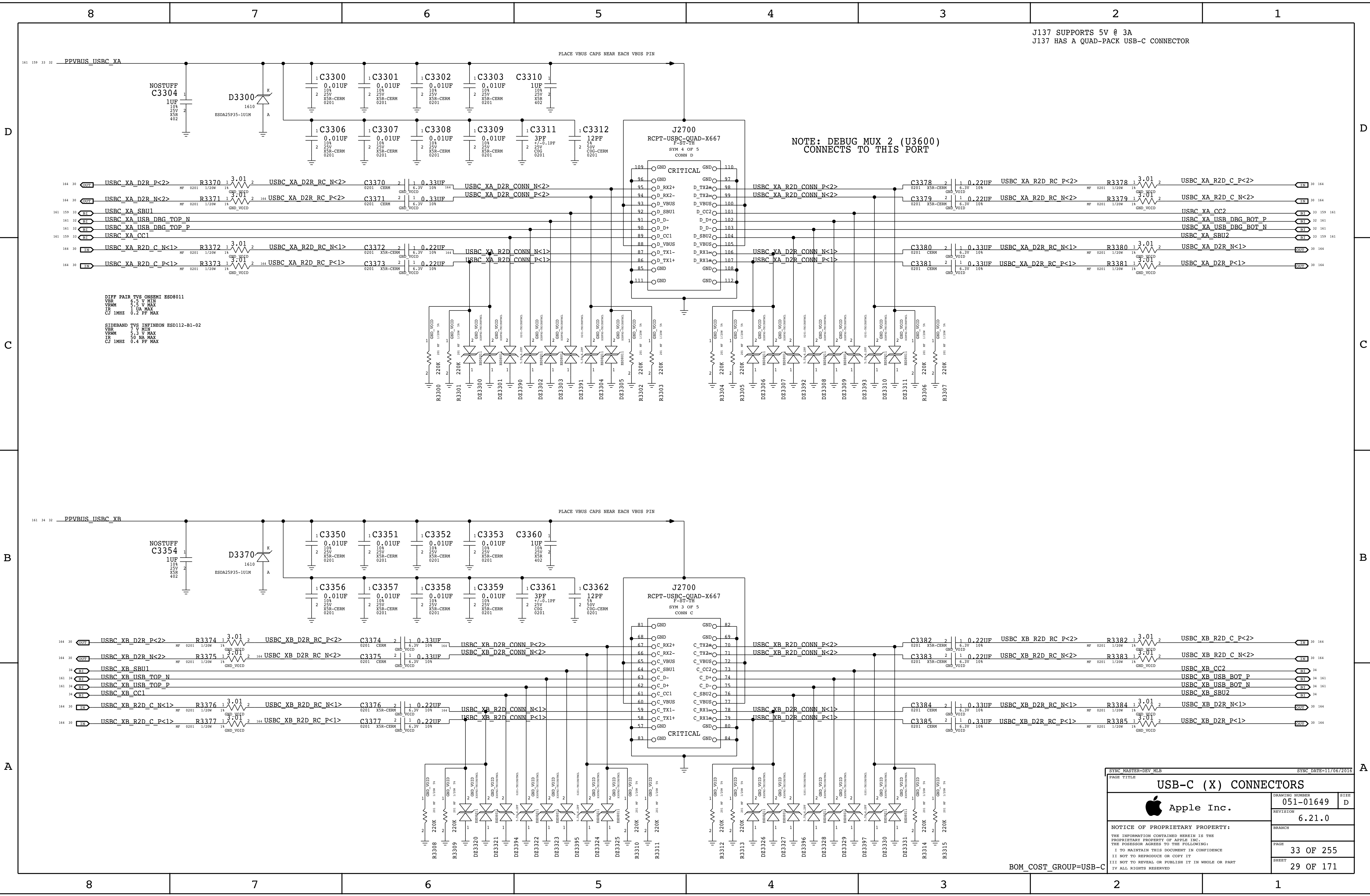
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B

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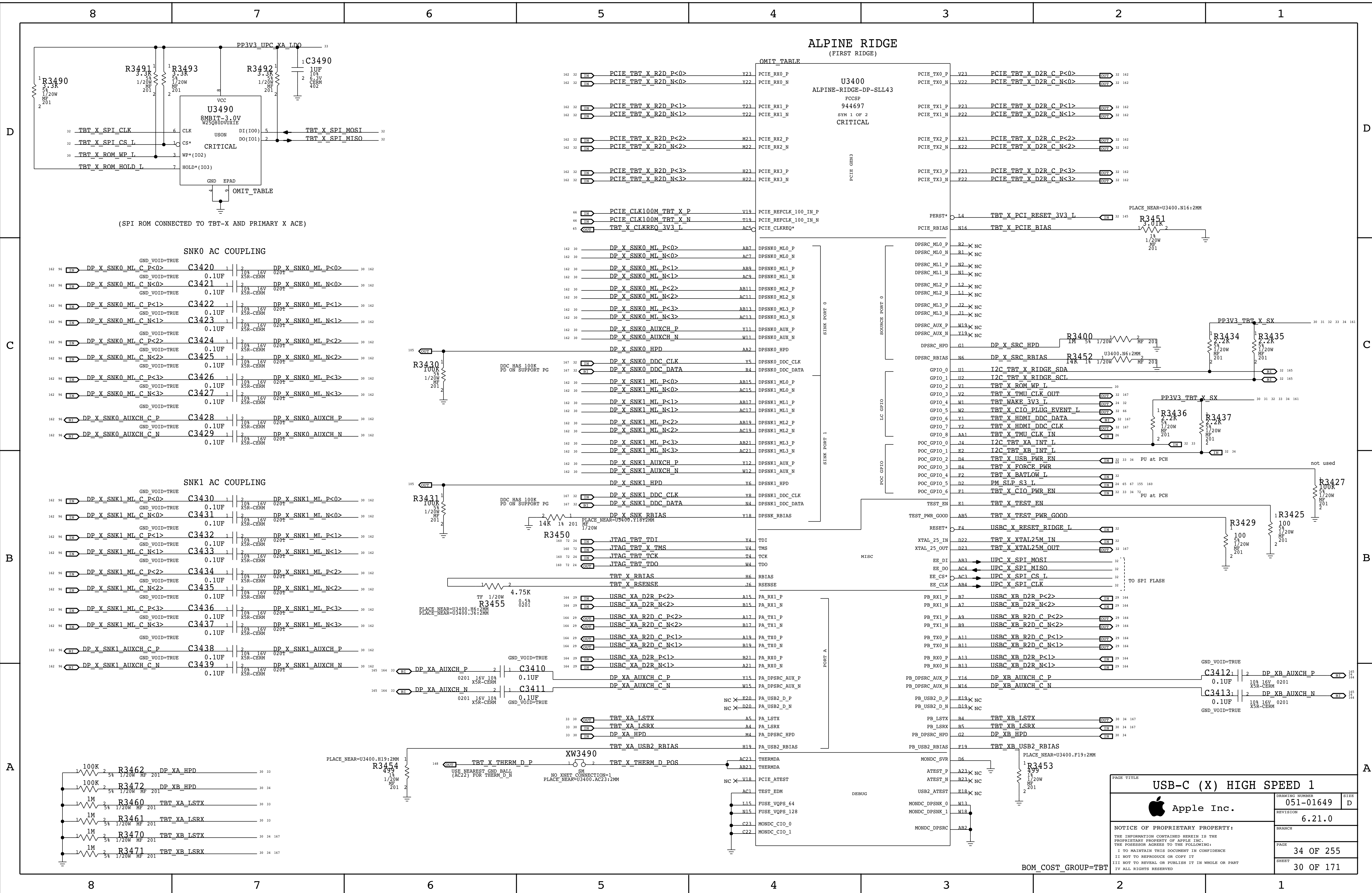
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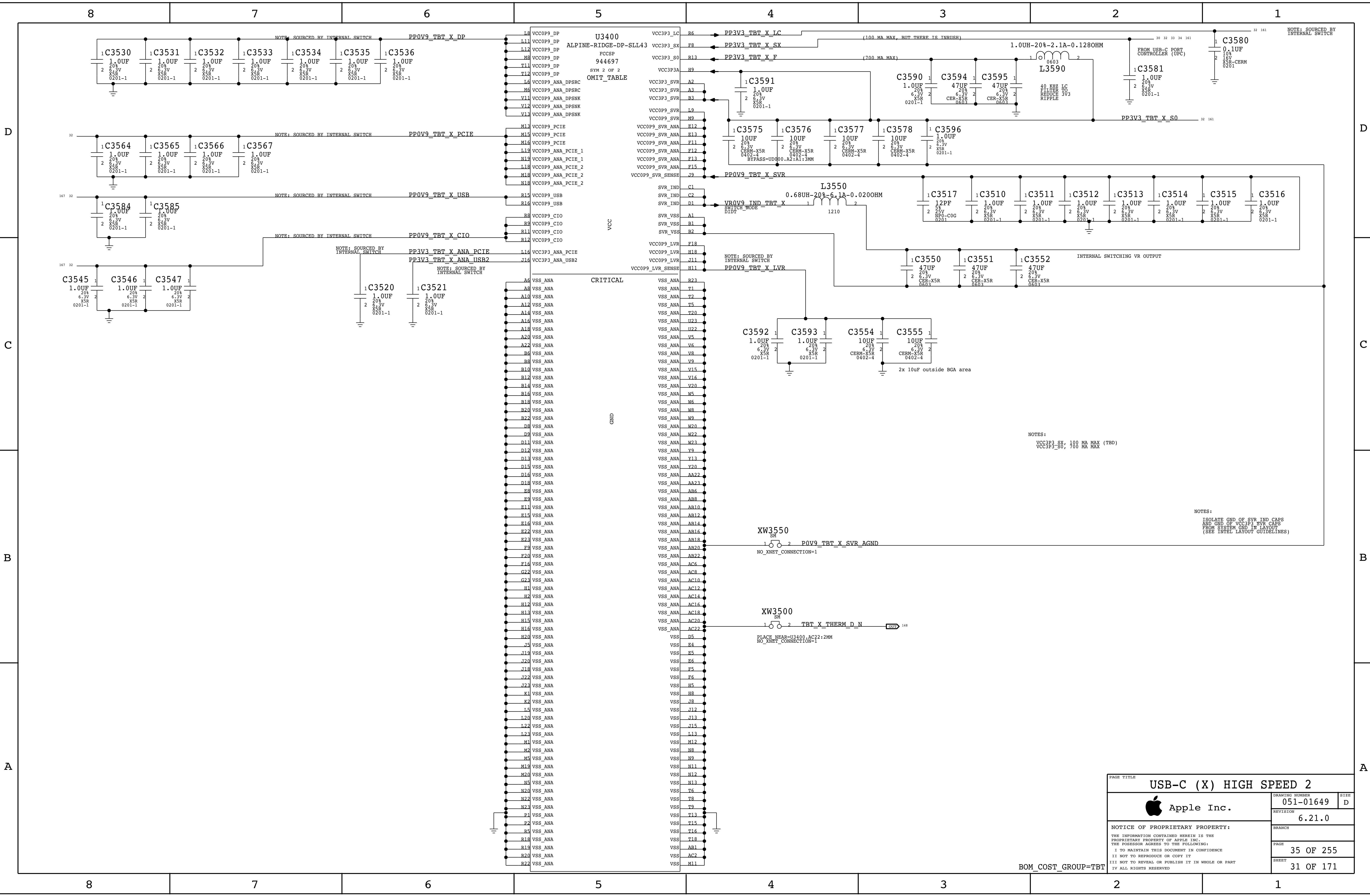
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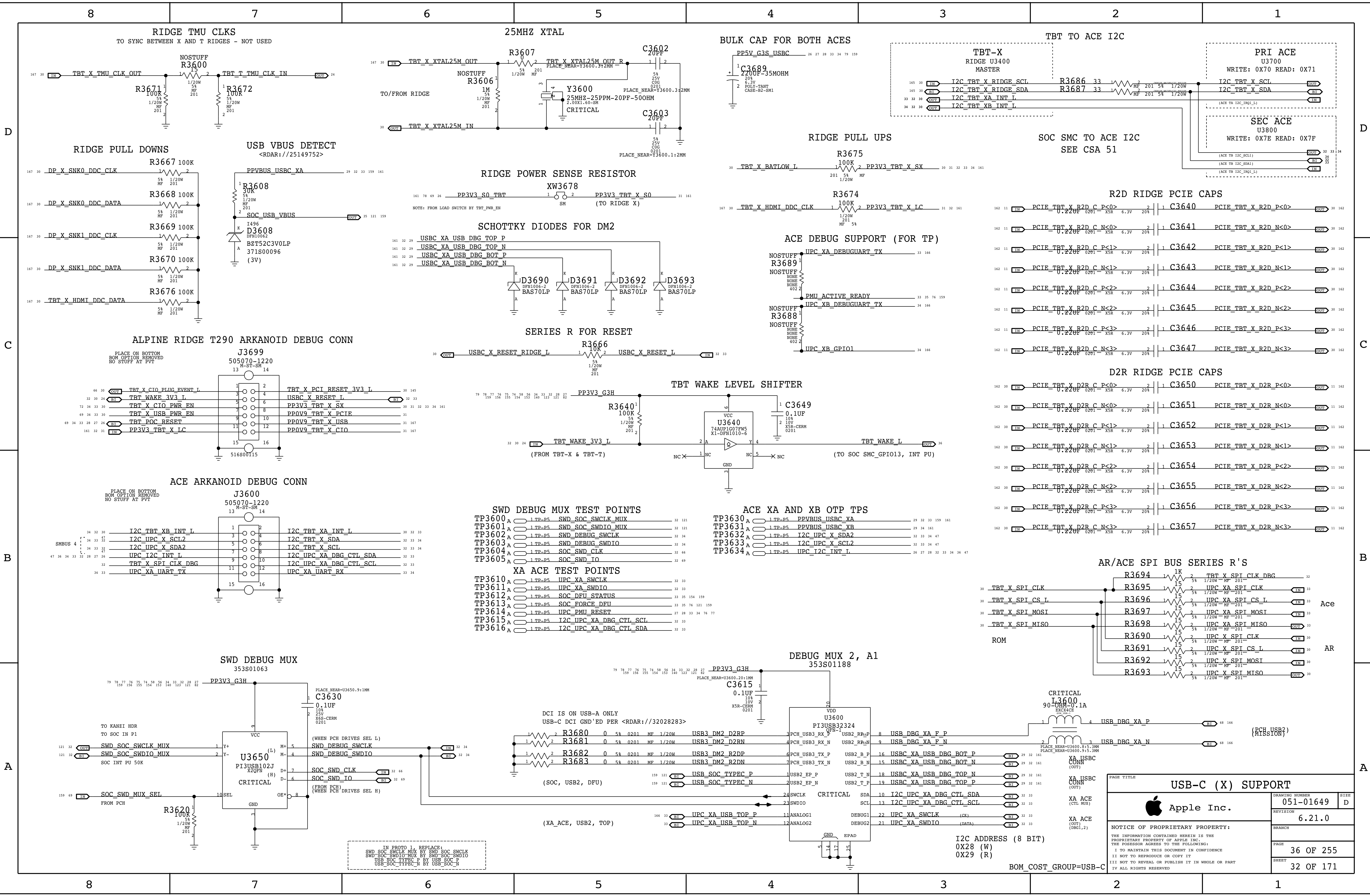
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	BRANCH		
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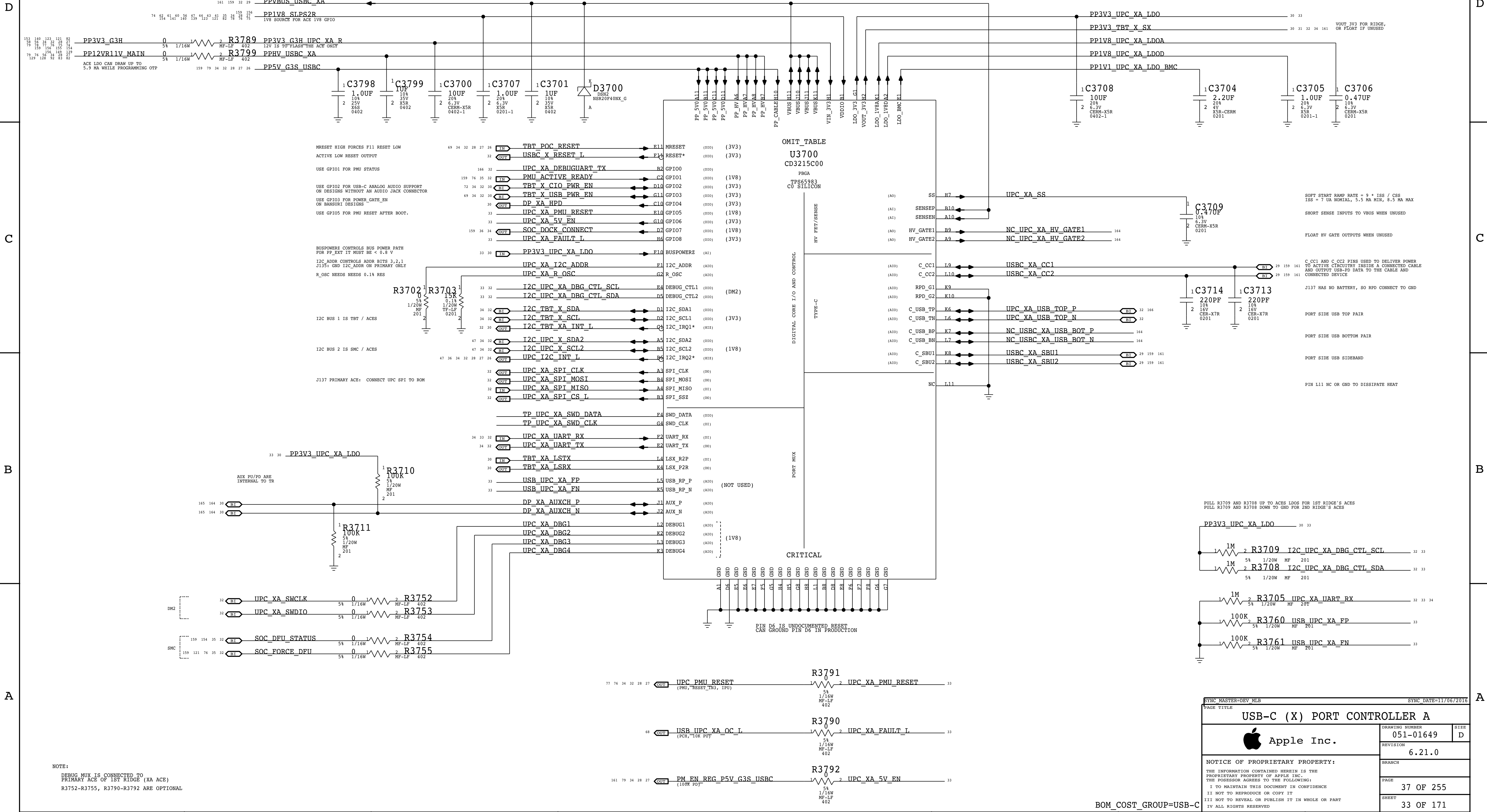
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BOM_COST_GROUP=TBT



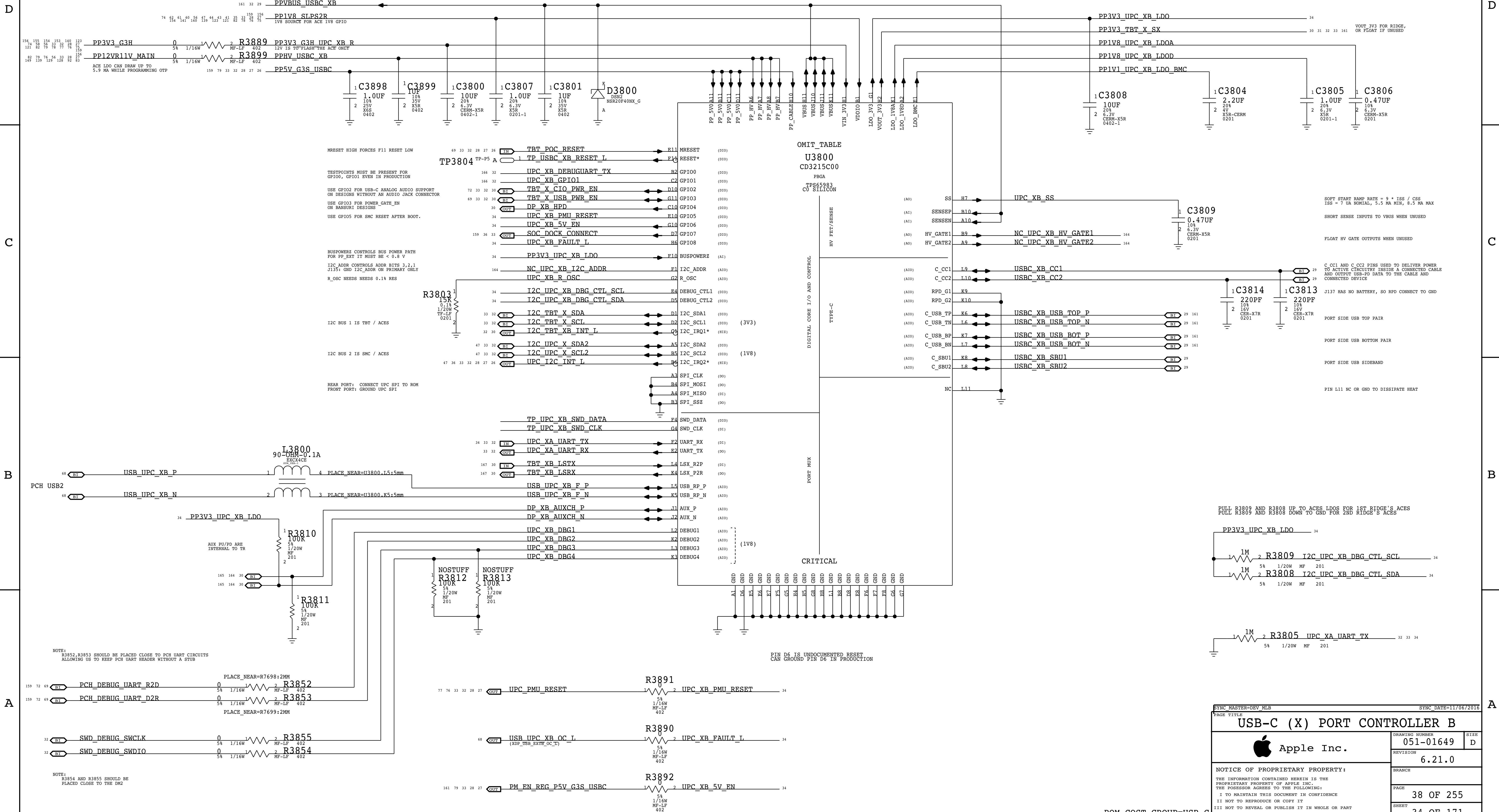
X PRIMARY ACE USB-C PORT CONTROLLER (UPC)

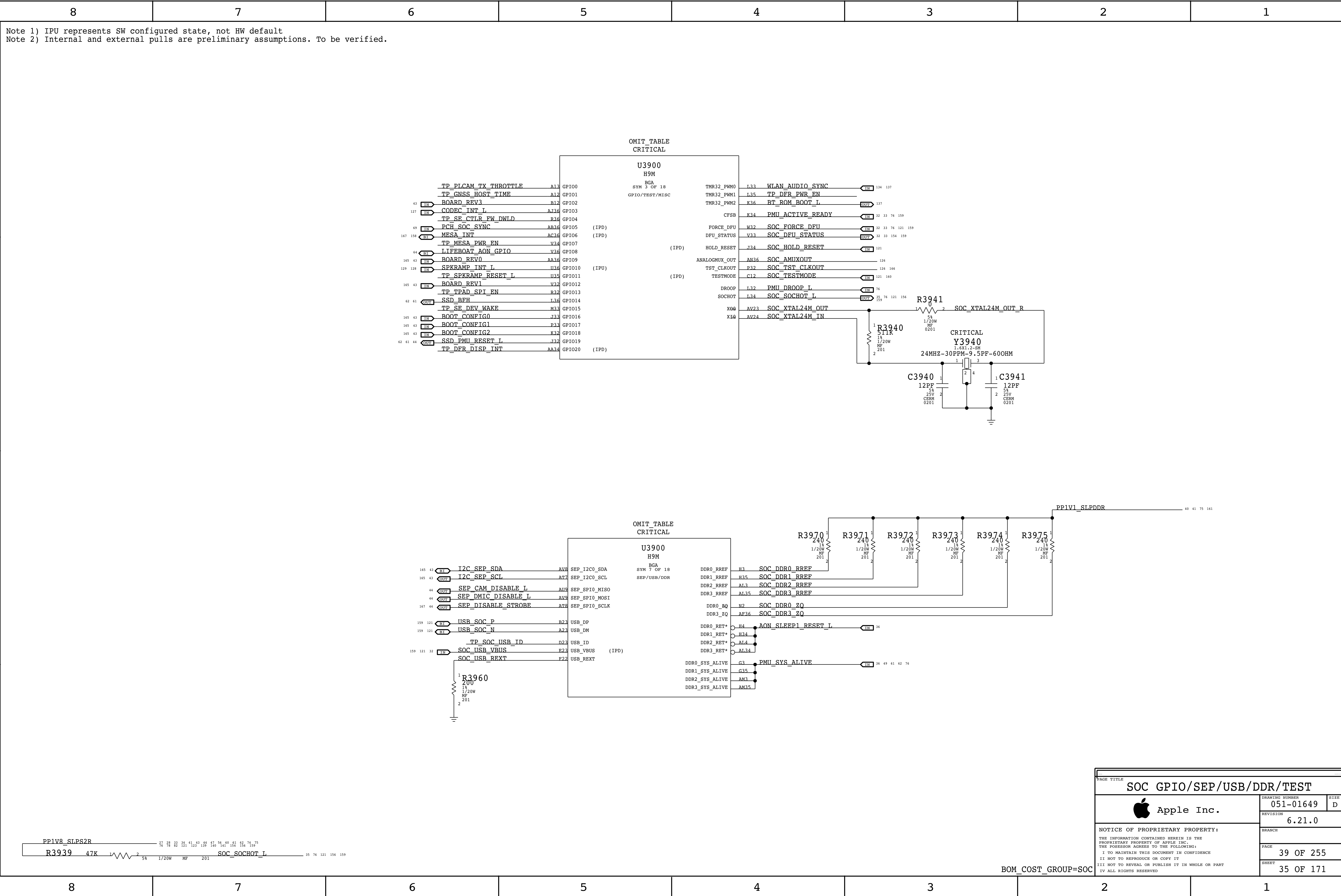
J137 USBC SUPPORTS 5V @ 3A
PP12V IS FOR PROGRAMMING ACE ONLY

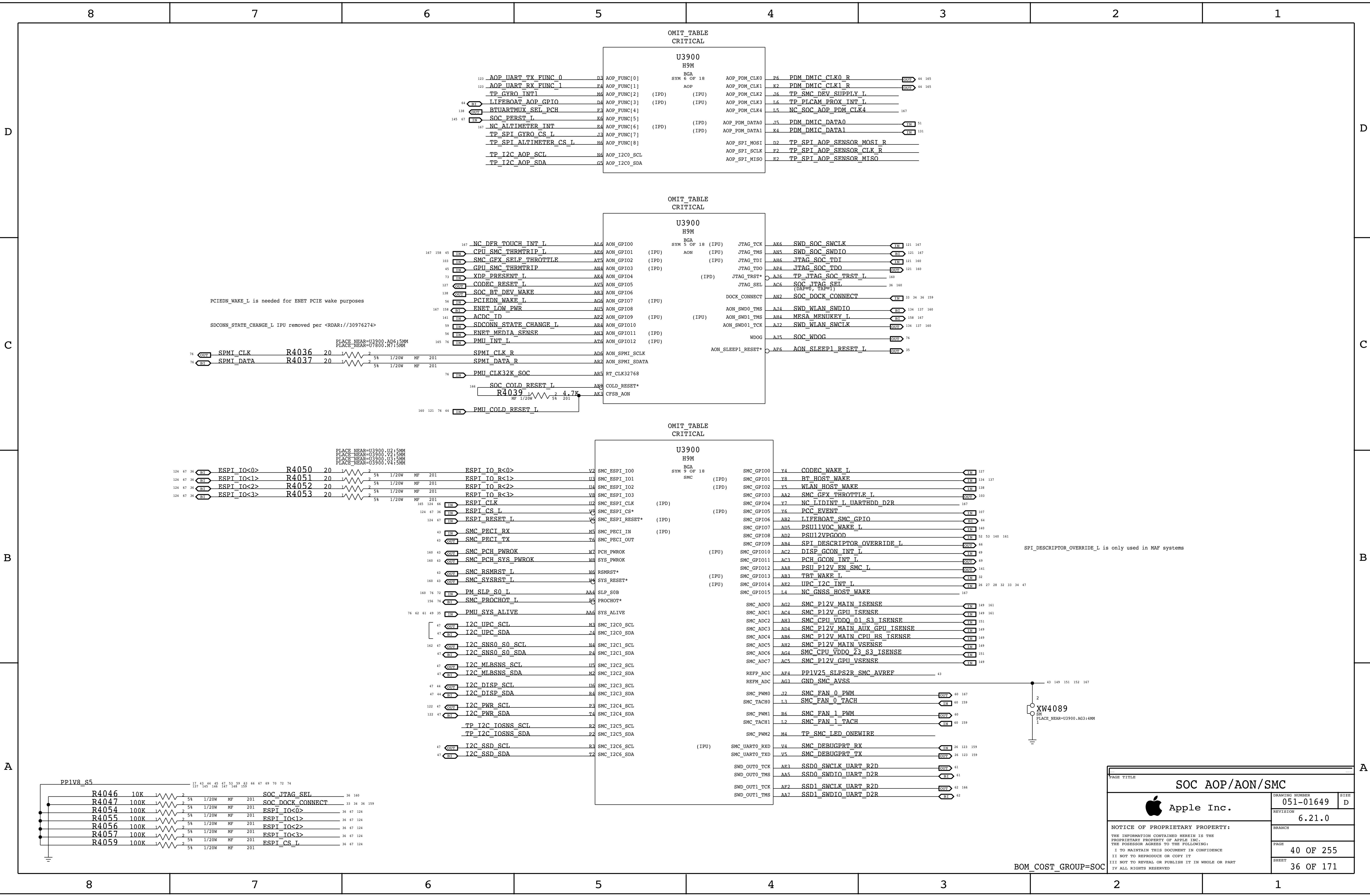


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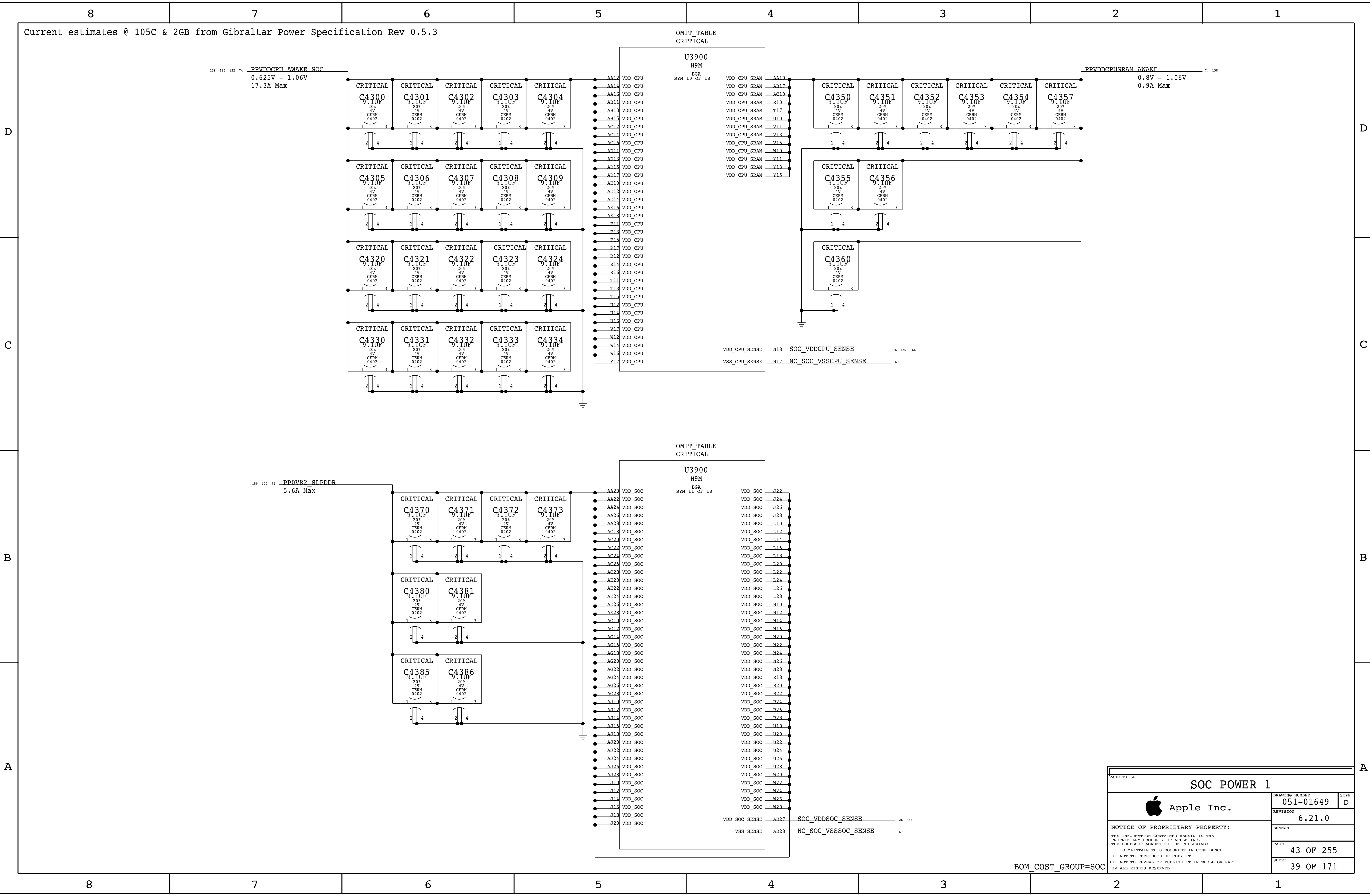
X SECONDARY ACE USB-C PORT CONTROLLER (UPC)





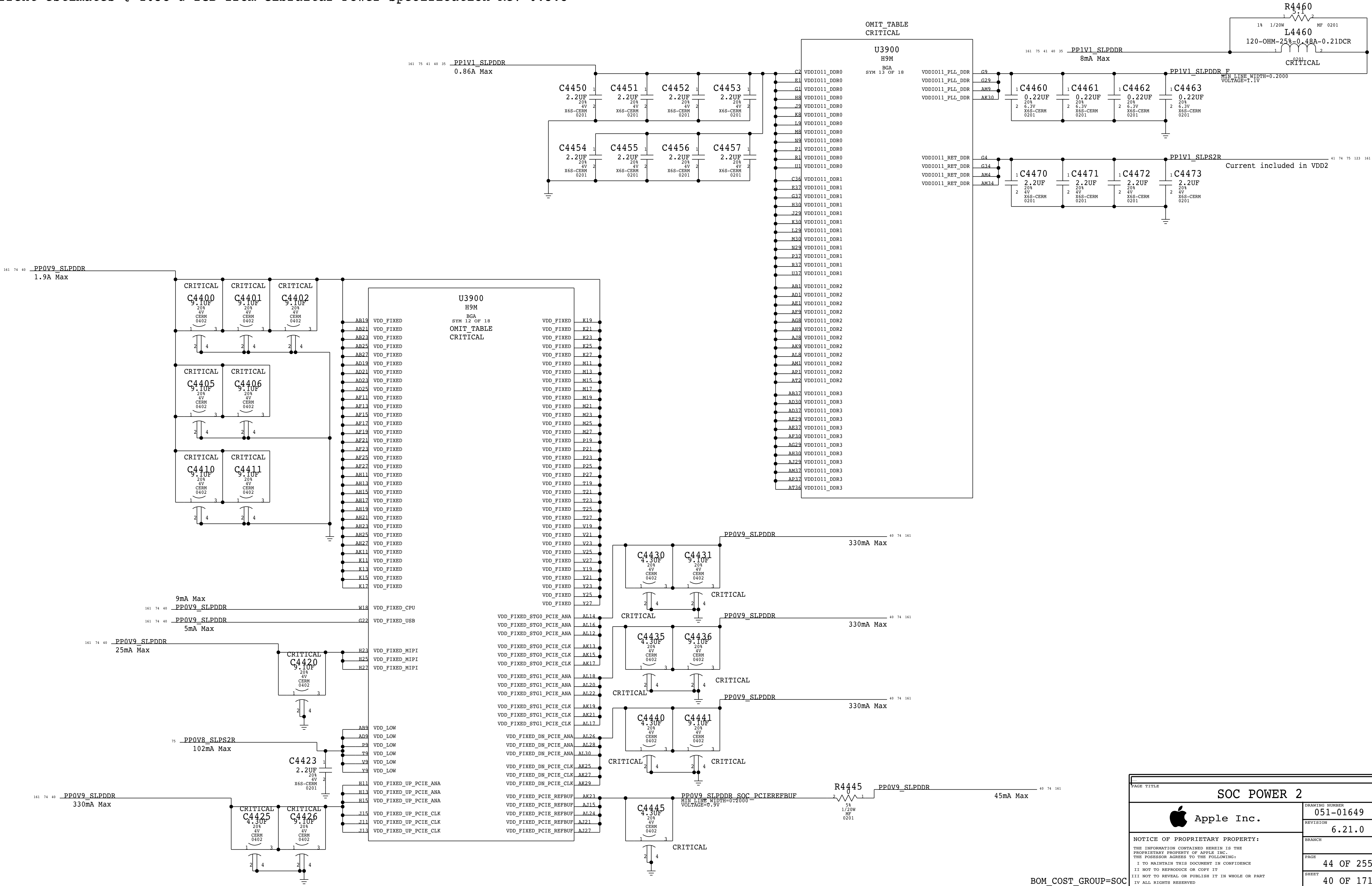


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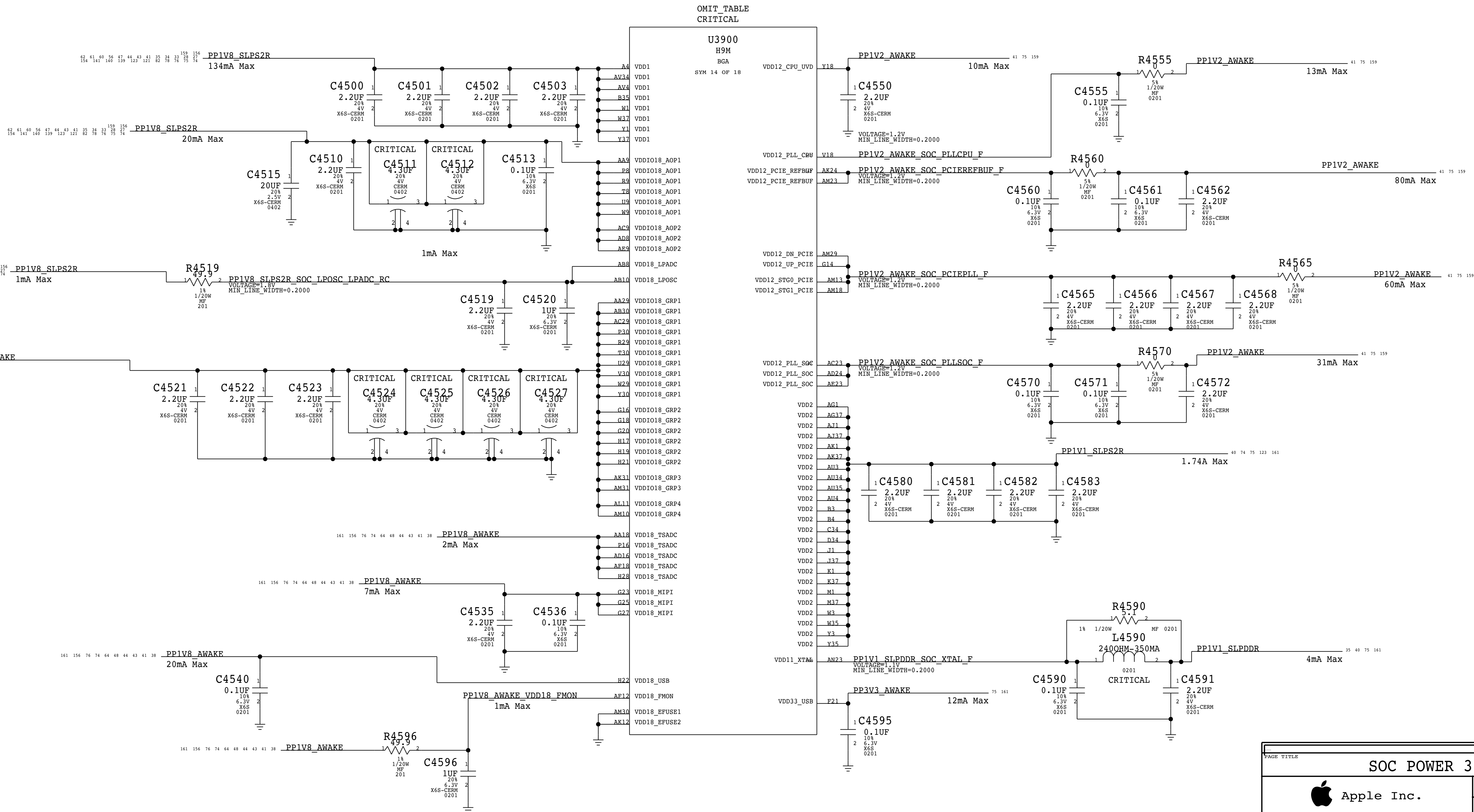
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Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3

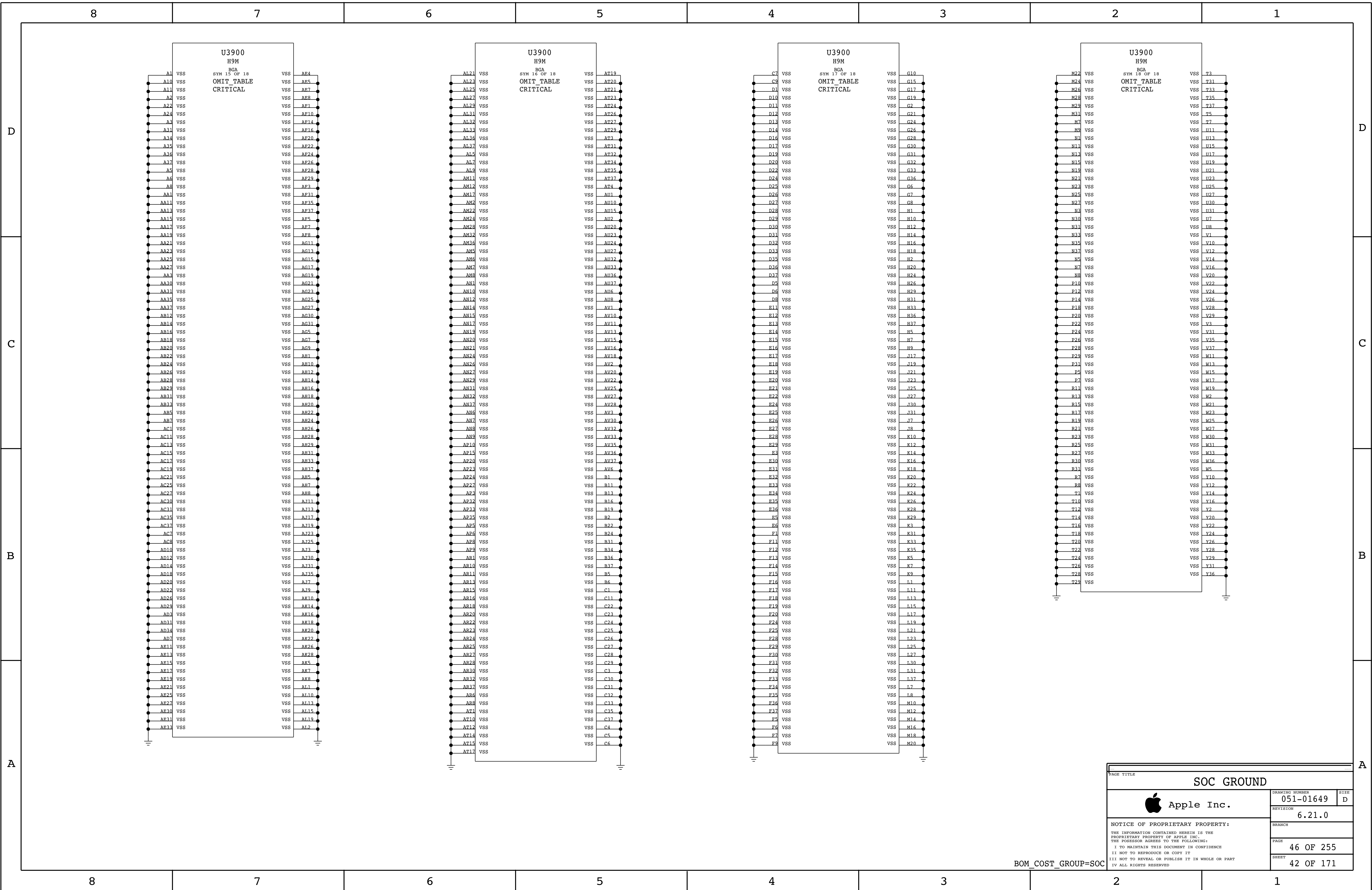


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SOC POWER 3			
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	BRANCH		
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	SHEET		
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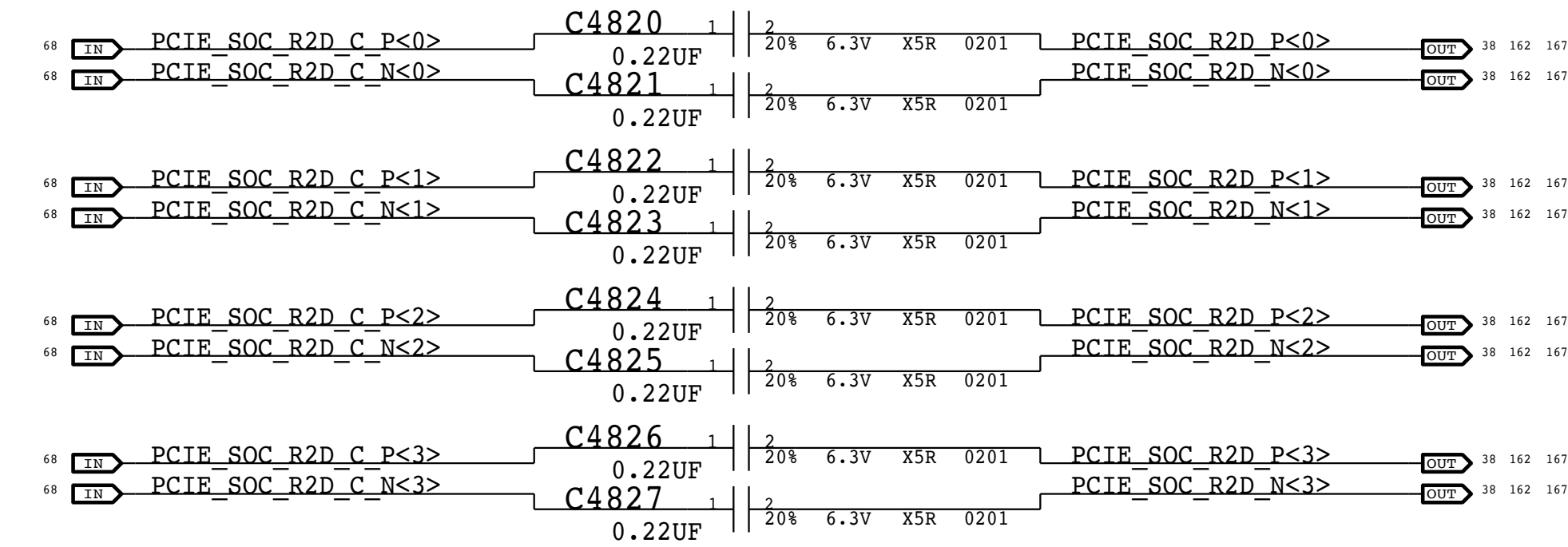
SMC ADC Assignments

Desktop Assignments

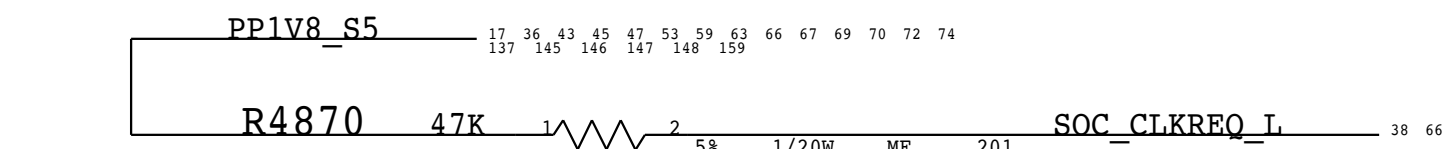
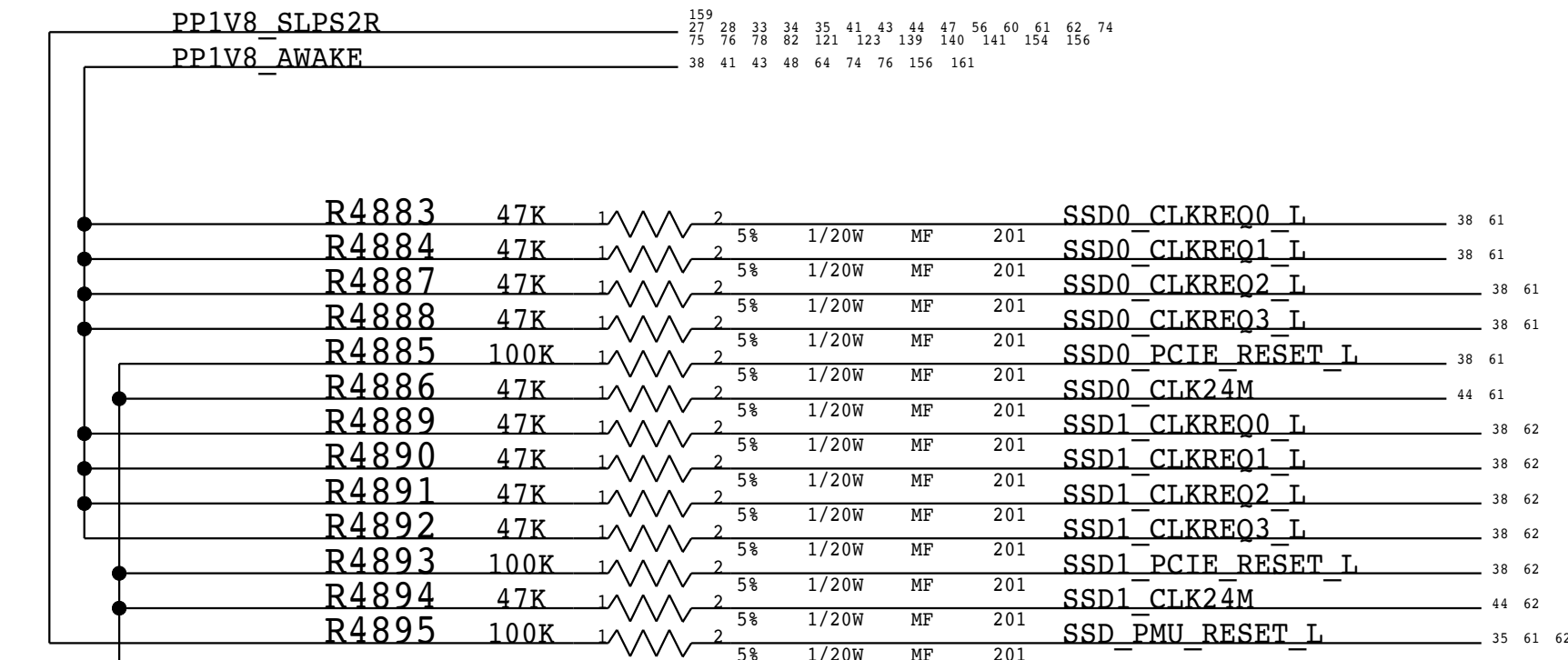
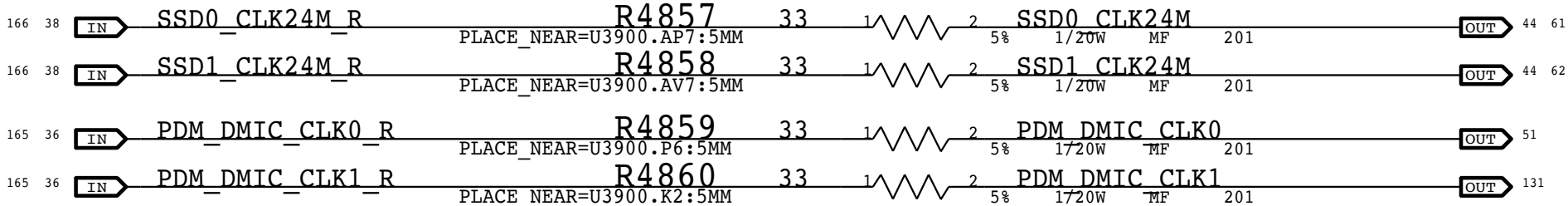
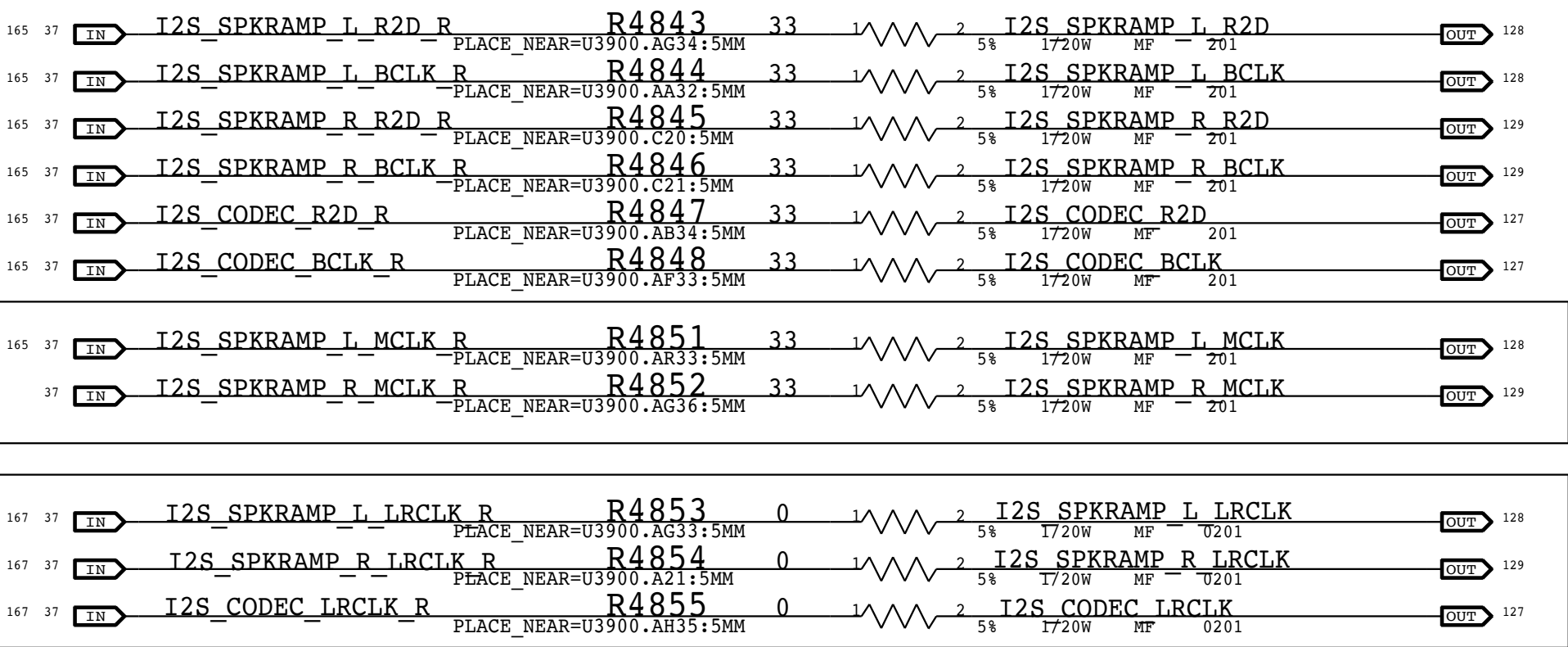
GPU_AUX	ADC0
GPU_CORE	ADC1
12V_VSNS	ADC2
12V_ISNS	ADC3
Same	ADC4-7

PCIe Up R2D AC Caps

(All Caps)

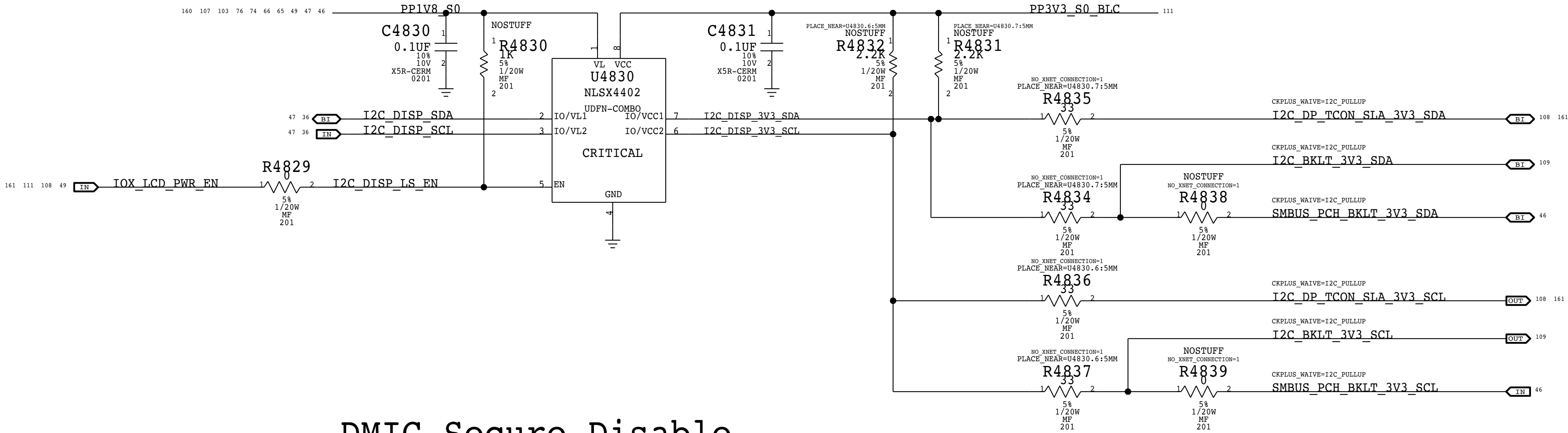


GPIO Source Termination

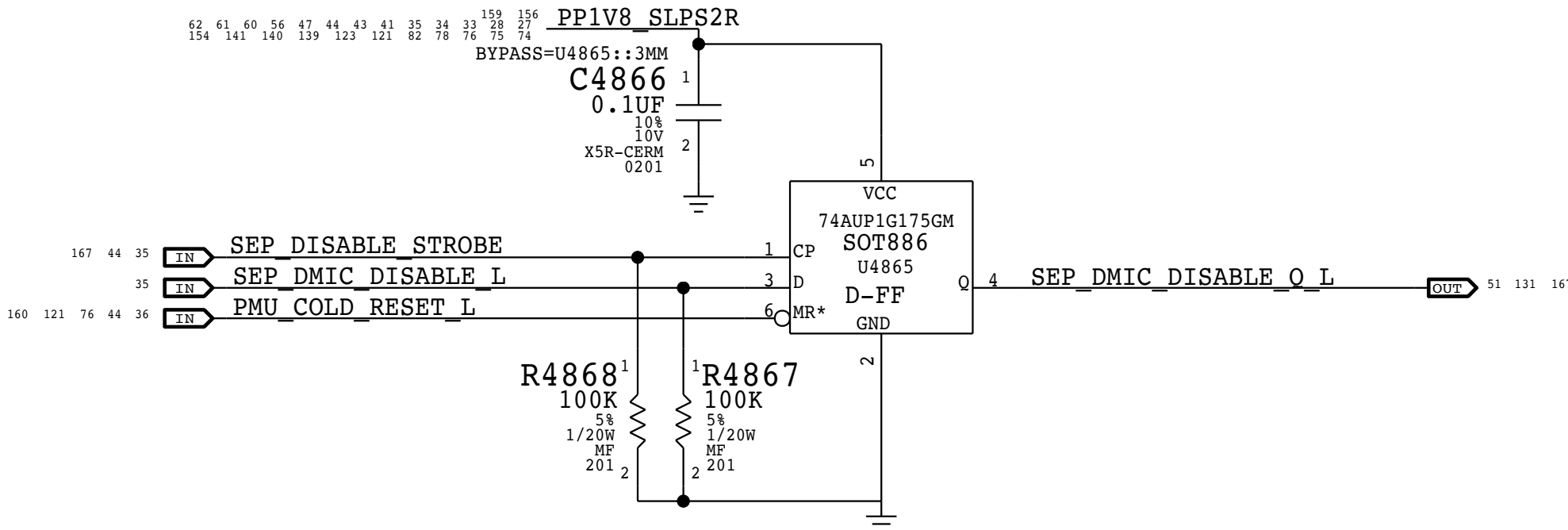


SMC I2C "3" LEVEL SHIFTING

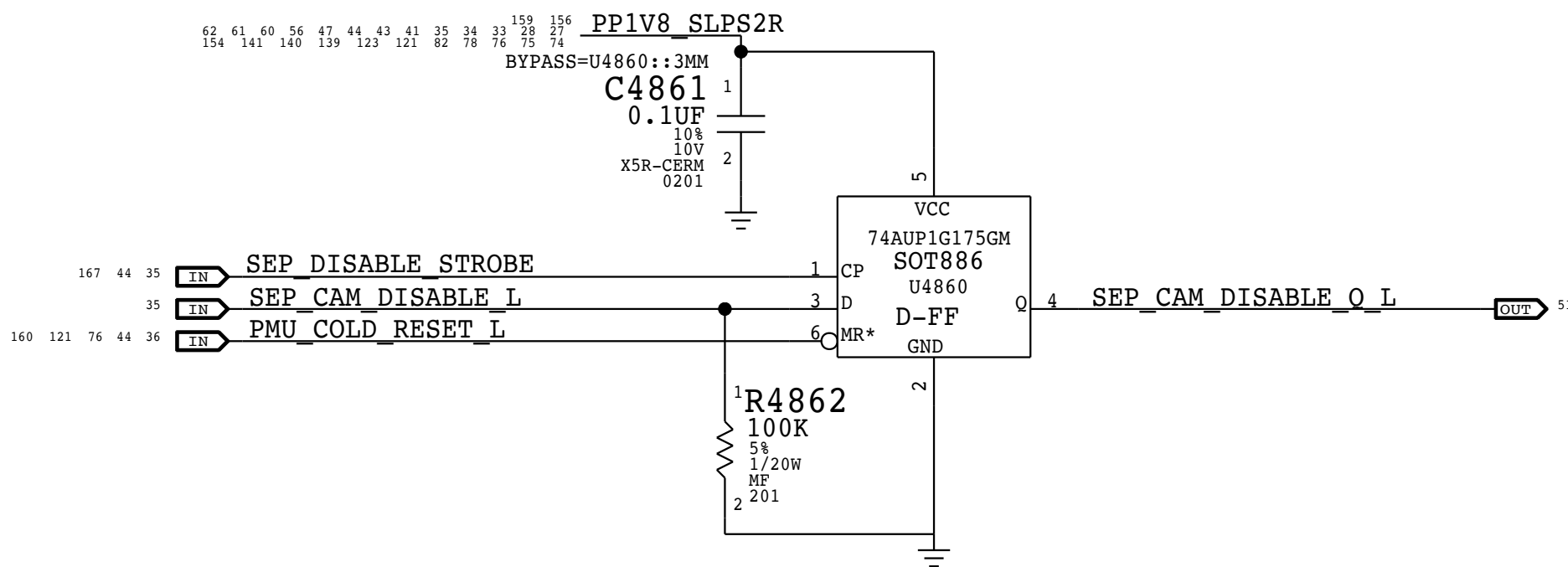
NOTE: LEVEL SHIFTER CONNECTIONS NEED TO BE FIXED PENDING ON-SEMI'S REPLY
Added options to connect PCH to BLC



DMIC Secure Disable



FTCAM Secure Disable



SOC PROJECT SUPPORT 1



Apple Inc.

DRAWING NUMBER

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BOM_COST_GROUP=SOC

PLATFORM THERMTRIP SUPPORT

INTENTION IS PMU GETS CPU/GPU THERMTRIP SO THAT IN THE CASE SMC HAS ISSUES, CALPE CAN STILL BRING DOWN THE SYSTEM GRACEFULLY
CALPE HAS AN INTERNAL DEBOUNCE, SMC DOESN'T, SO SMC VERSIONS OF THERMTRIP ARE MASKED
PCH GETS A COMBINED CPU/GPU THERMTRIP. SO IF EITHER ASSERTS, PCH SHOULD BRING DOWN THE SYSTEM GRACEFULLY

PMU

TO PMIC RESET IN5
PROGRAMMABLE 2MS OR 10MS DEBOUNCE
3V2, 200K IPD, CAN TAKE 1V8 TOO

SMC

1V8
ROM GPIO[3]
IPD

PCH

CPU VCCIO
PCH THERMTRIP* PIN AH4

PMU

TO PMIC RESET IN4
PROGRAMMABLE 2MS OR 10MS DEBOUNCE
BUCK8 1V0
NO IPU OR IPD

SMC

1V8
IPU TO AON DOMAIN (SLPS2R DOMAIN)

PAGE TITLE		
PLATFORM THERMTRIP SUPPORT		
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	SHEET	45 OF 171

BOM_COST_GROUP=PLATFORM POWER

D

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B

A

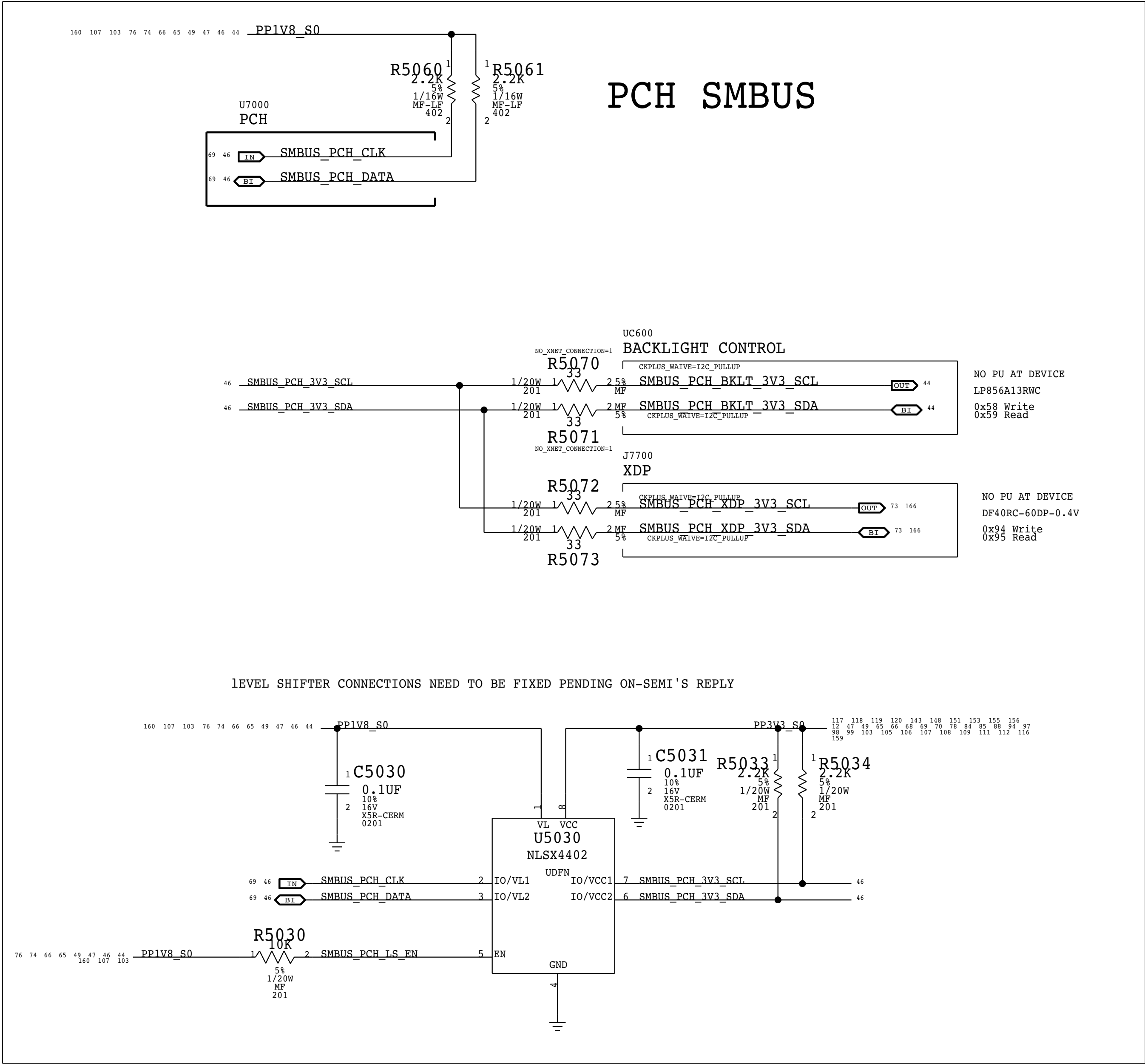
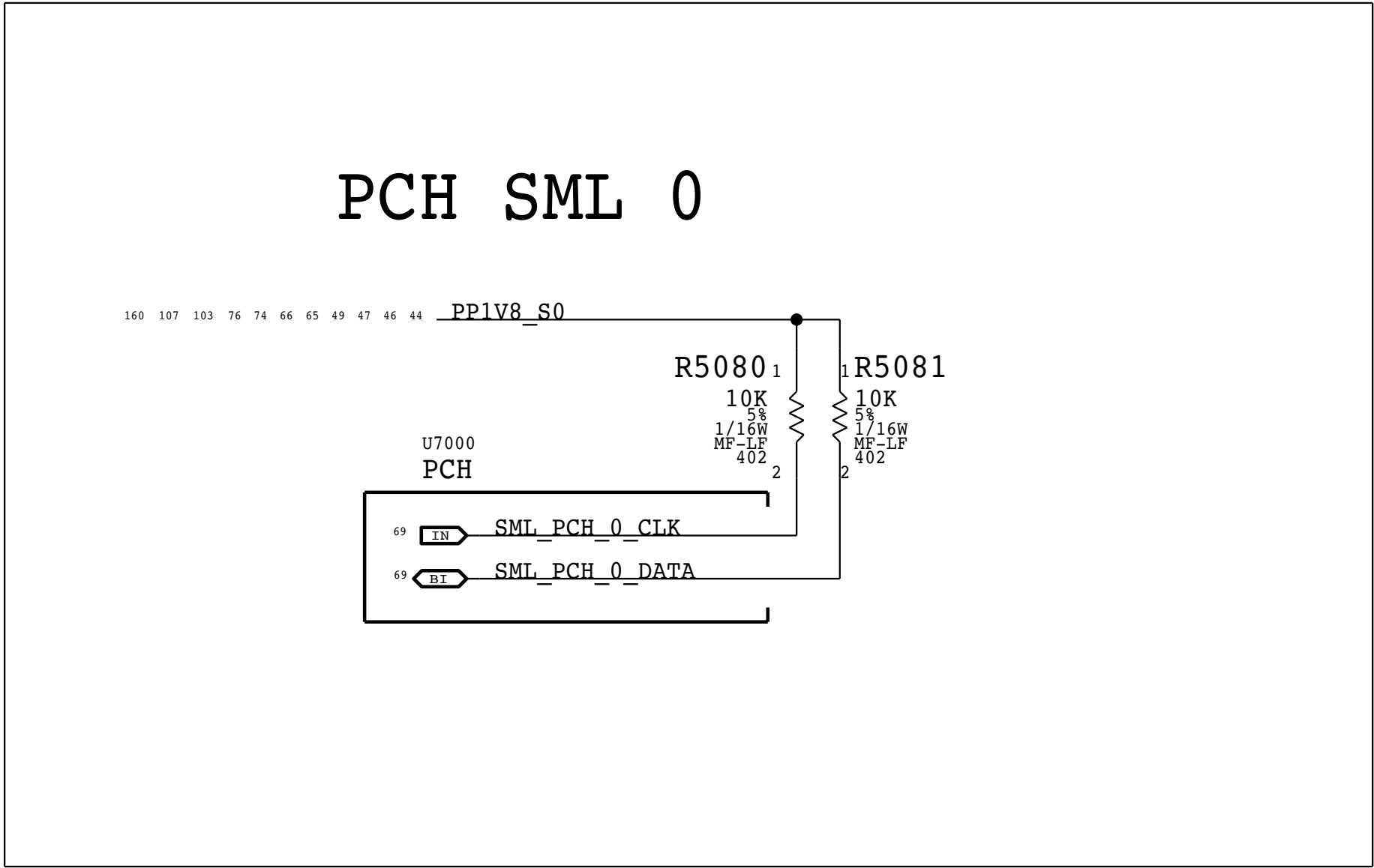
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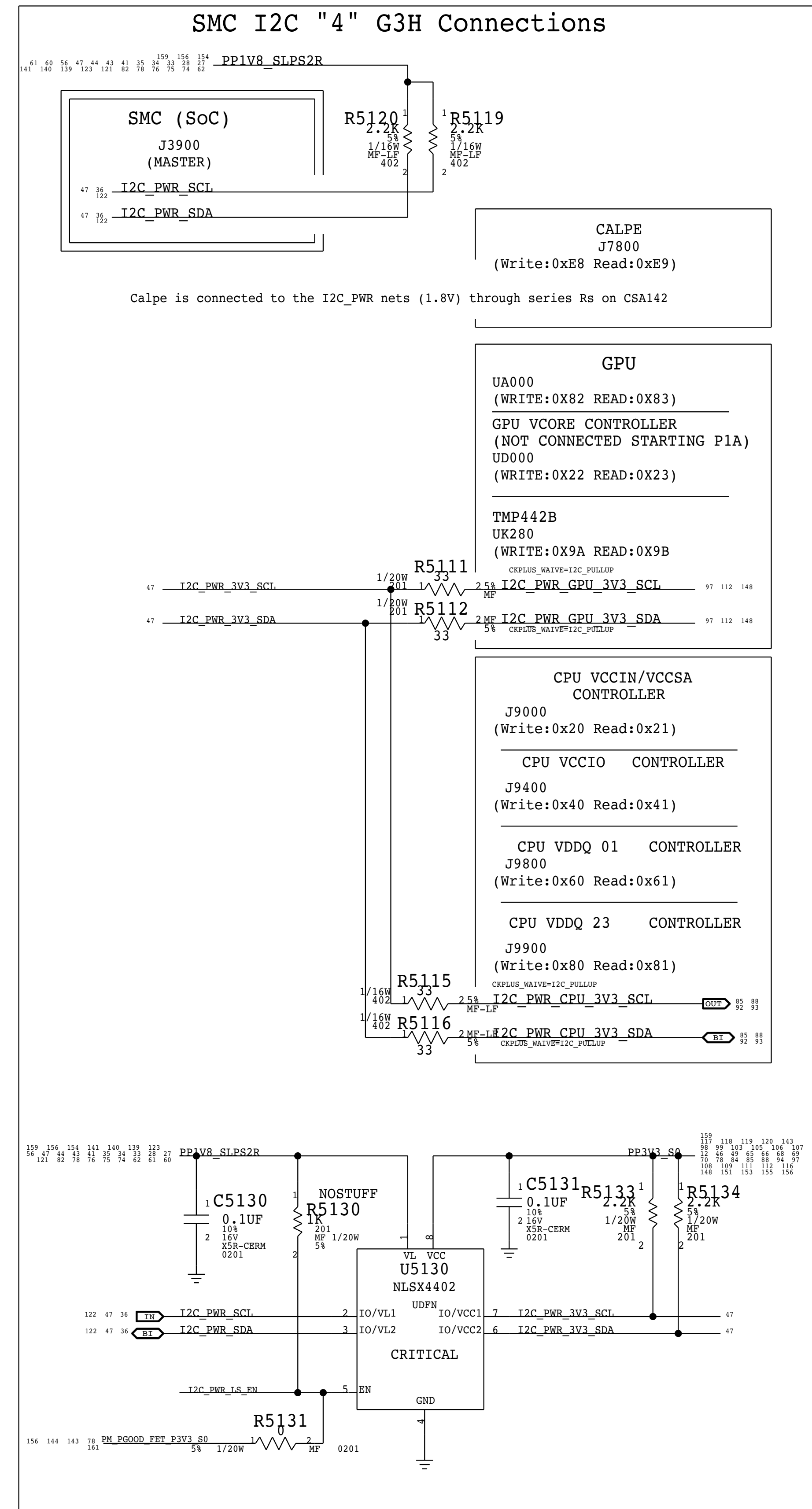
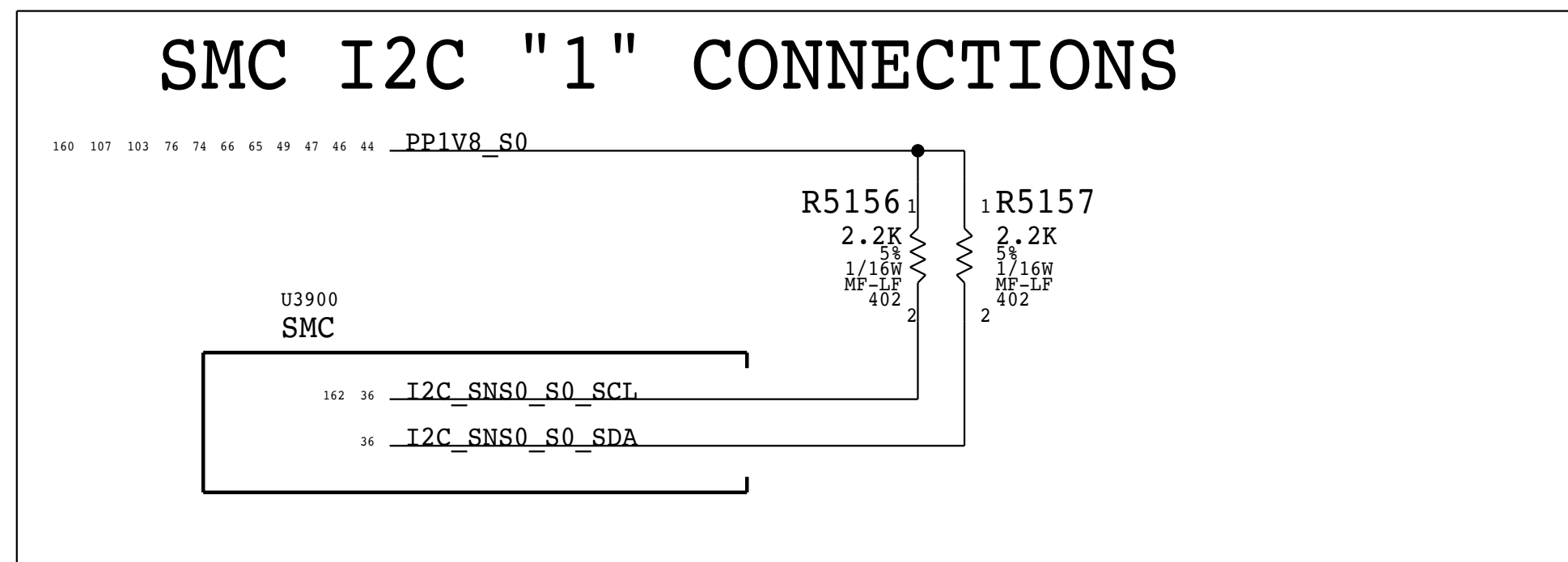
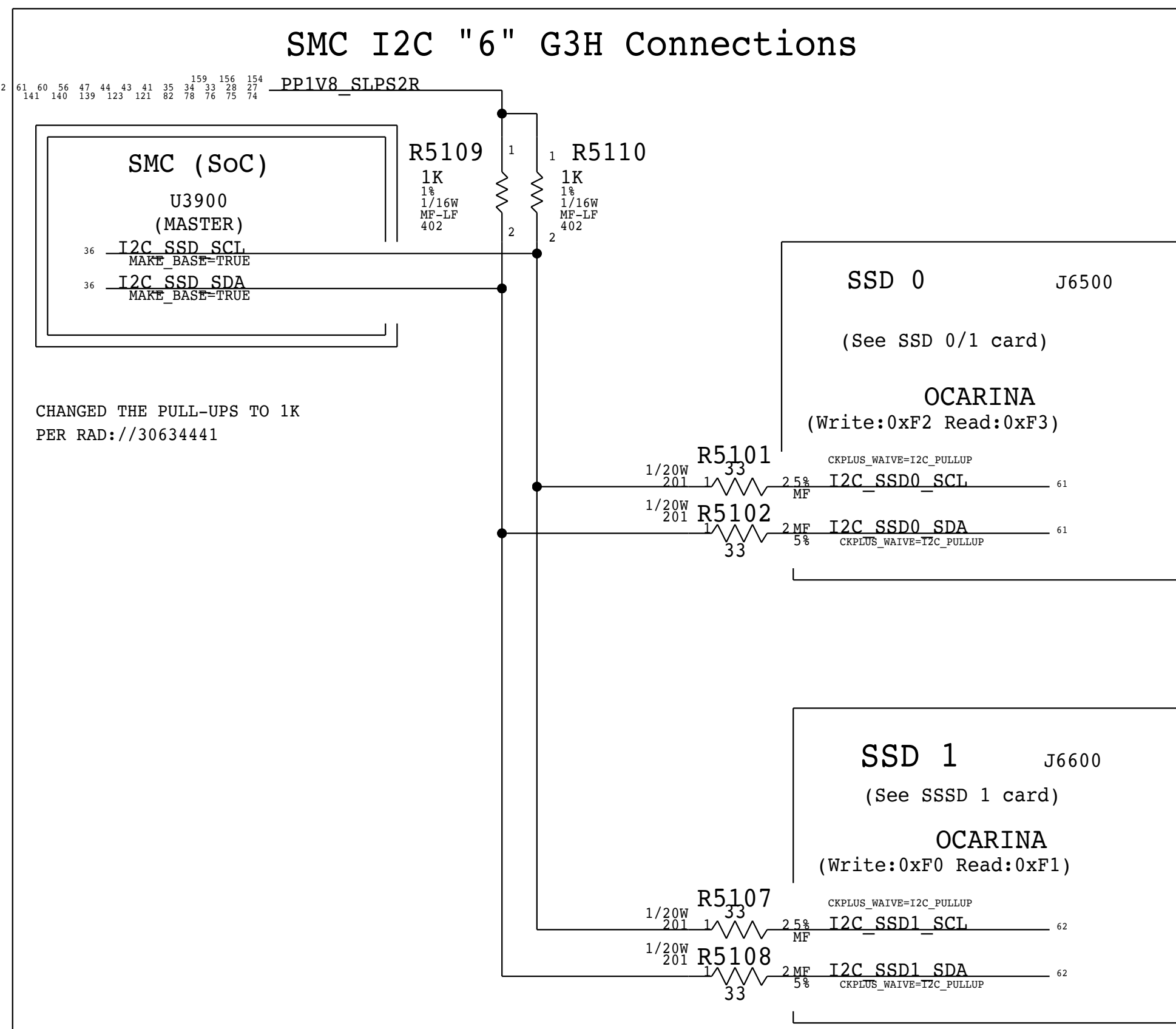
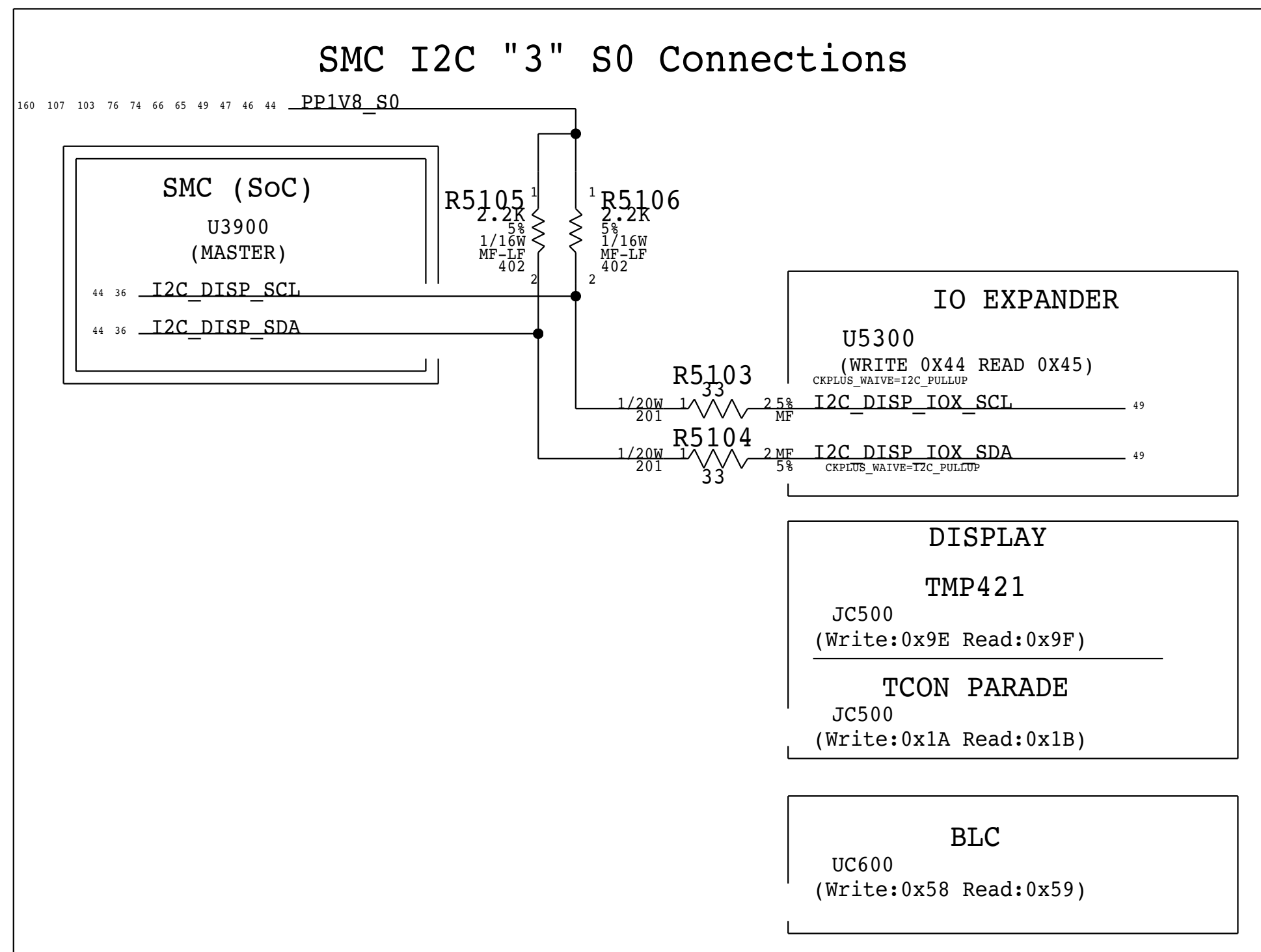
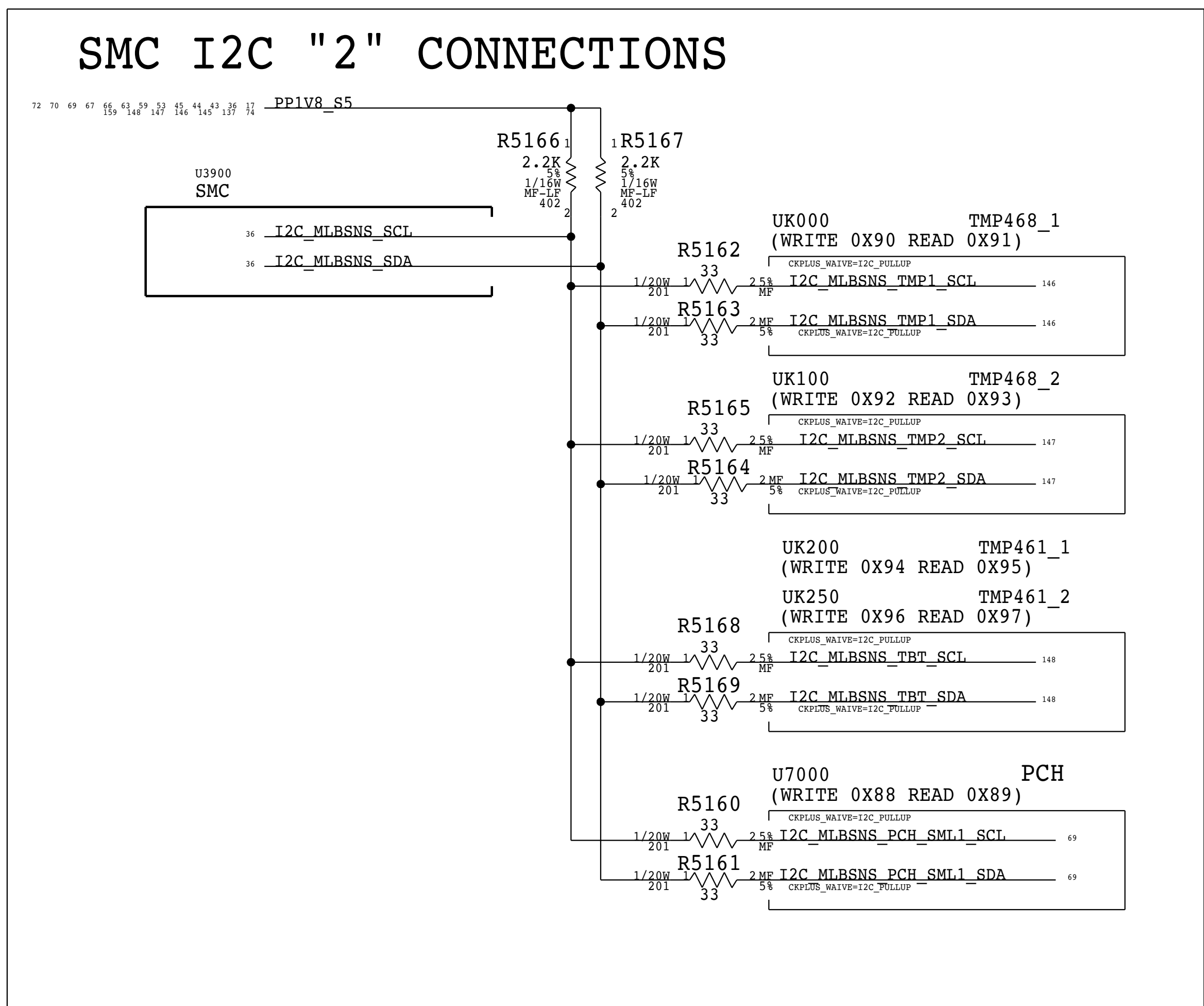
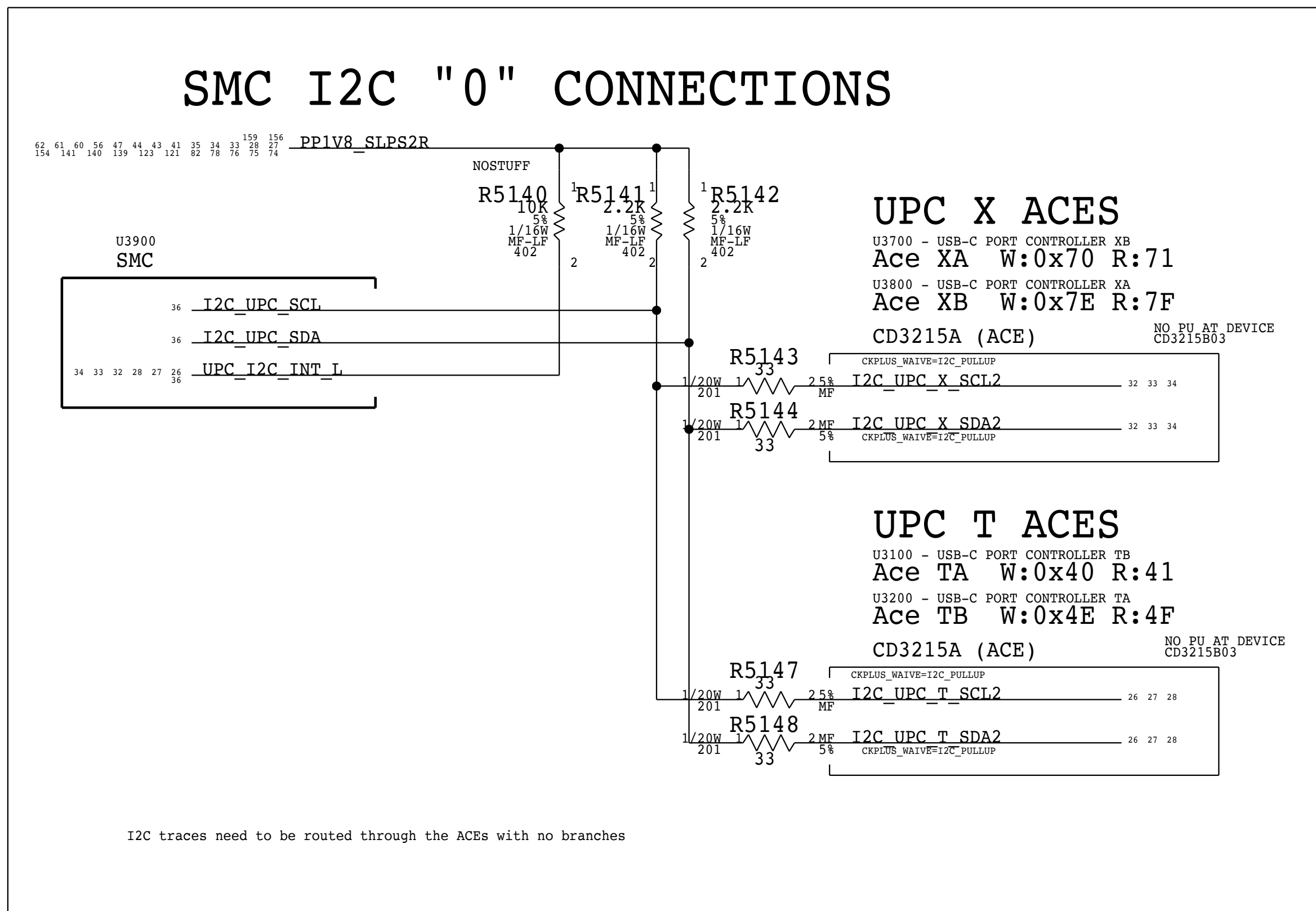
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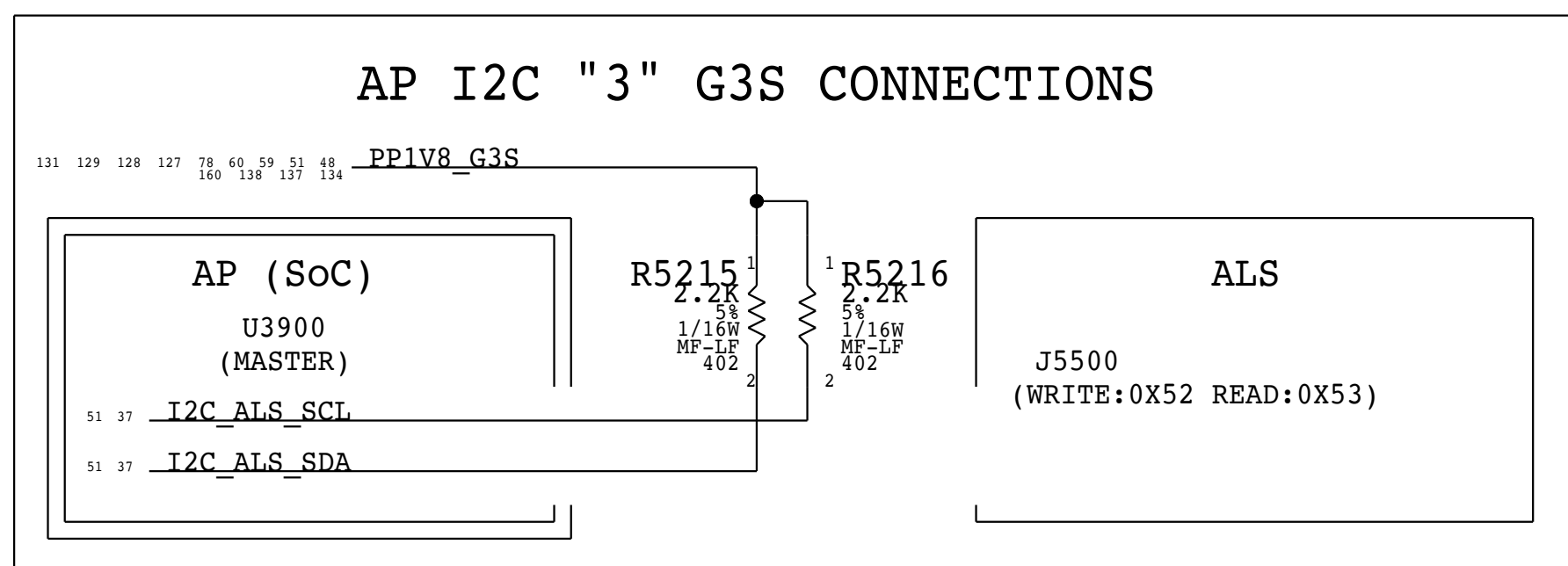
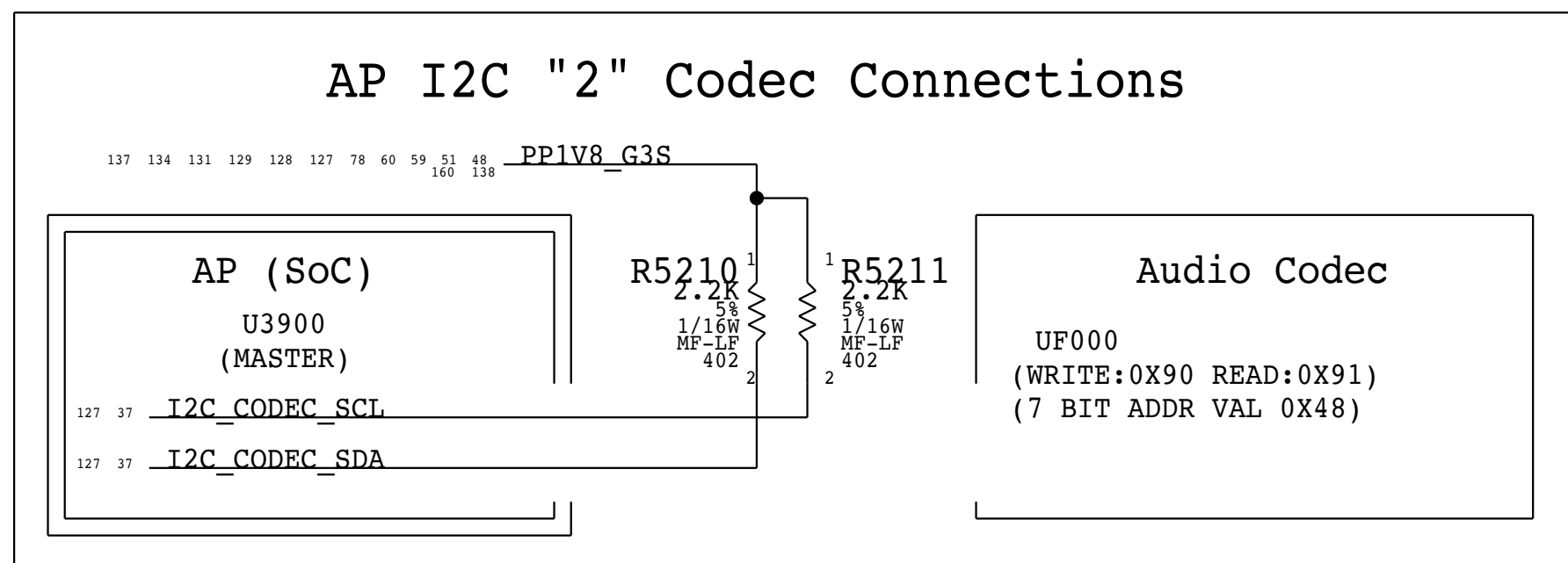
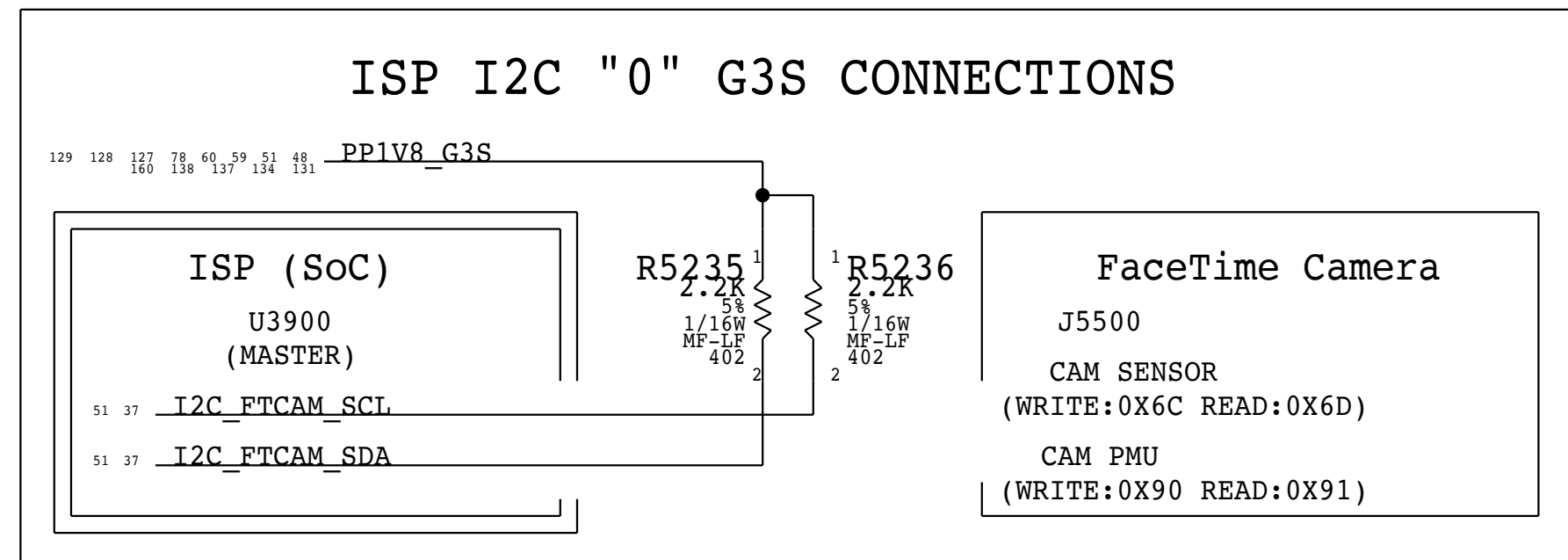
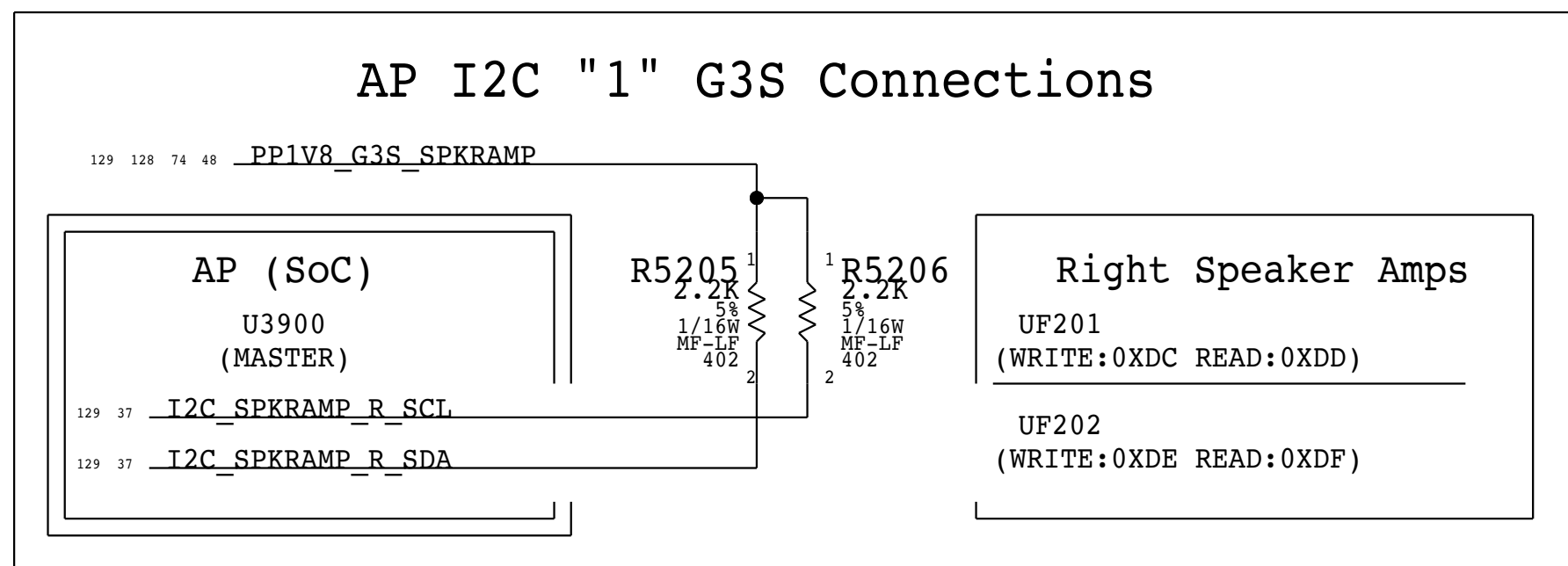
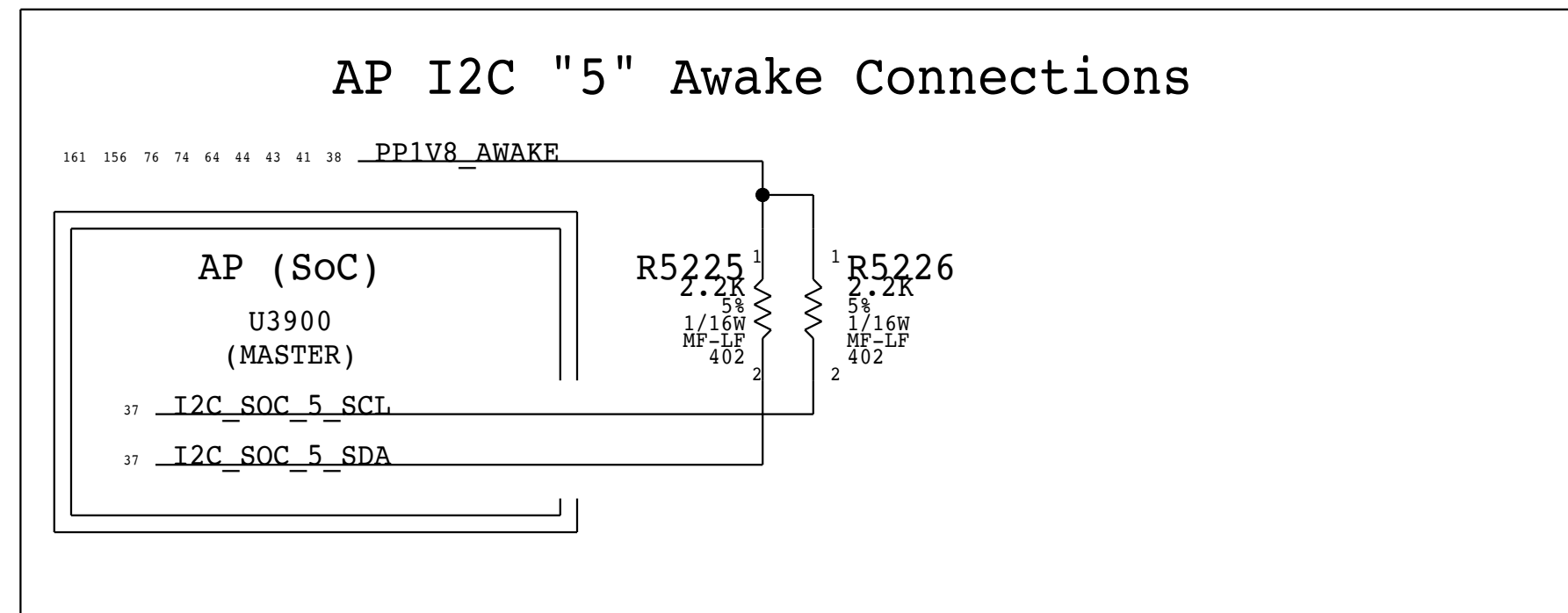
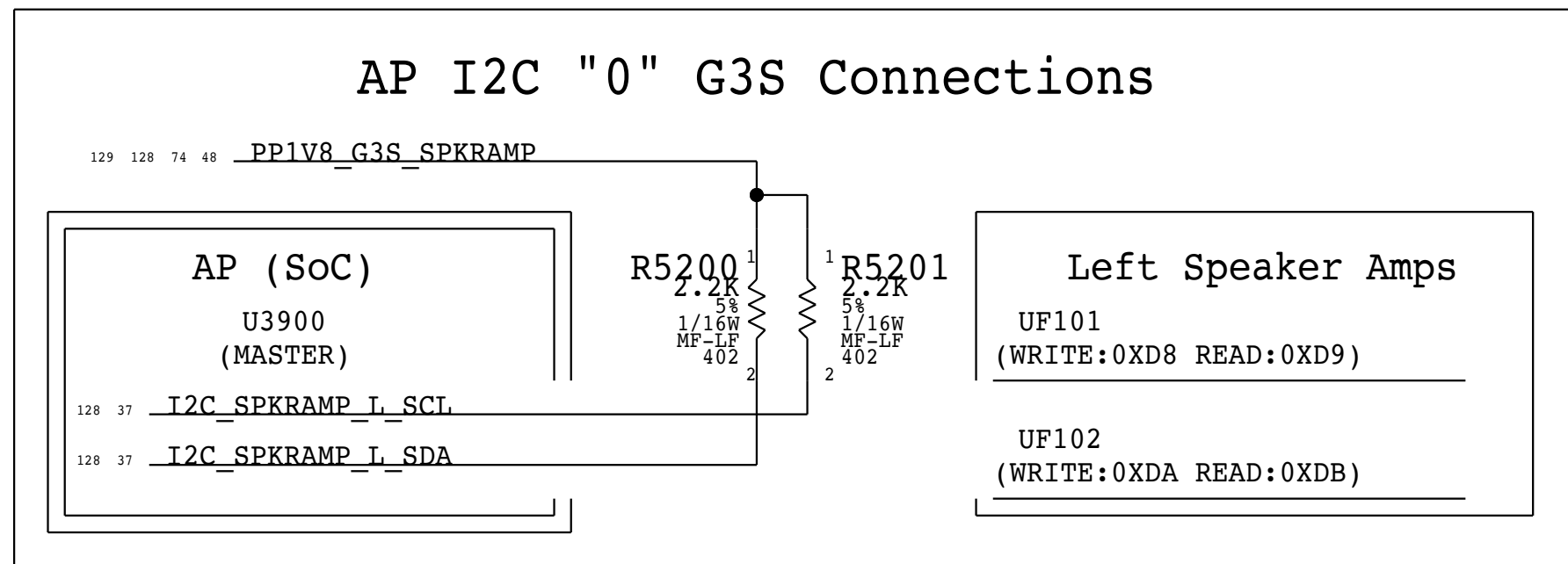
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BOM_COST_GROUP=SOC



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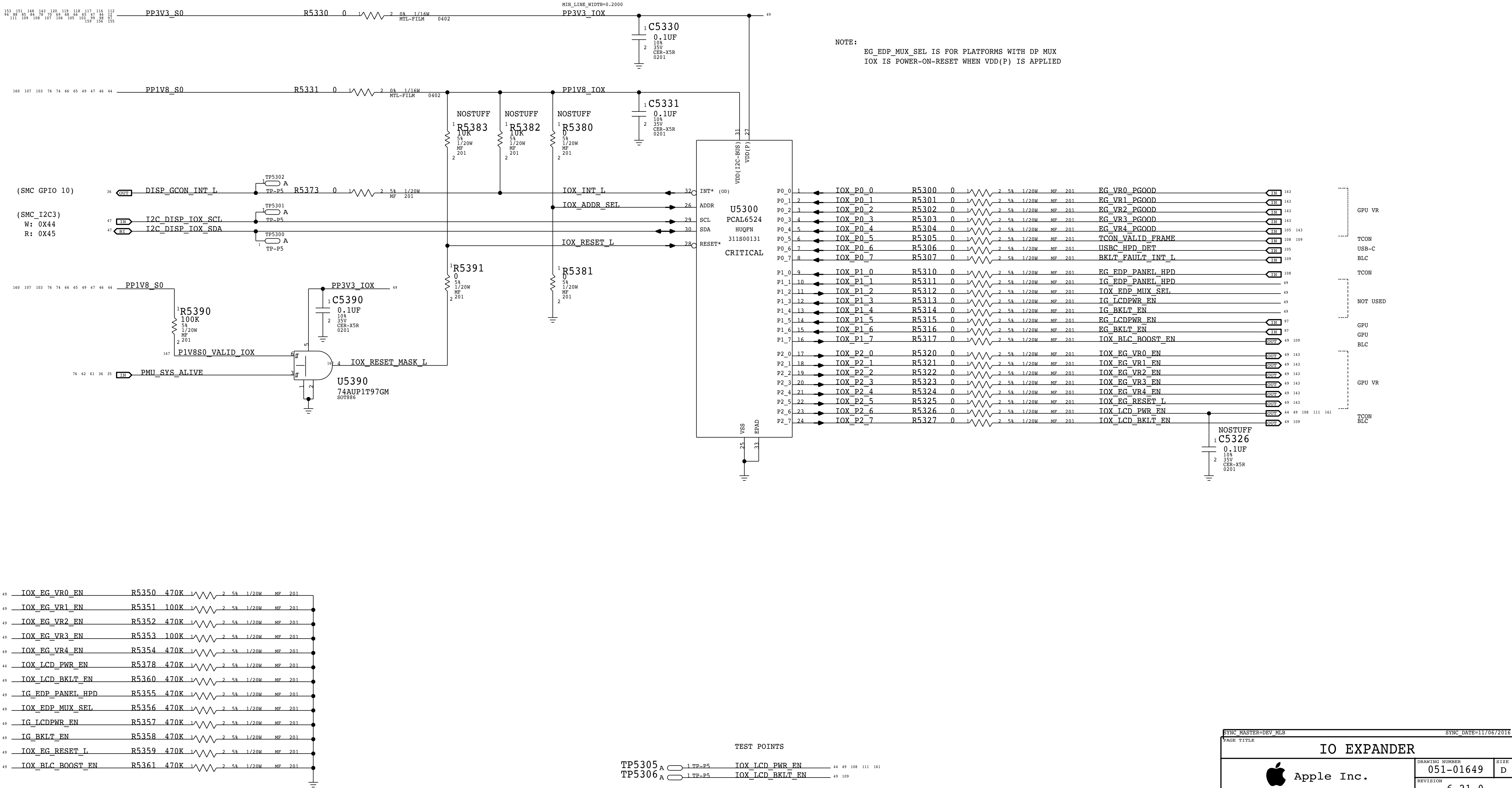
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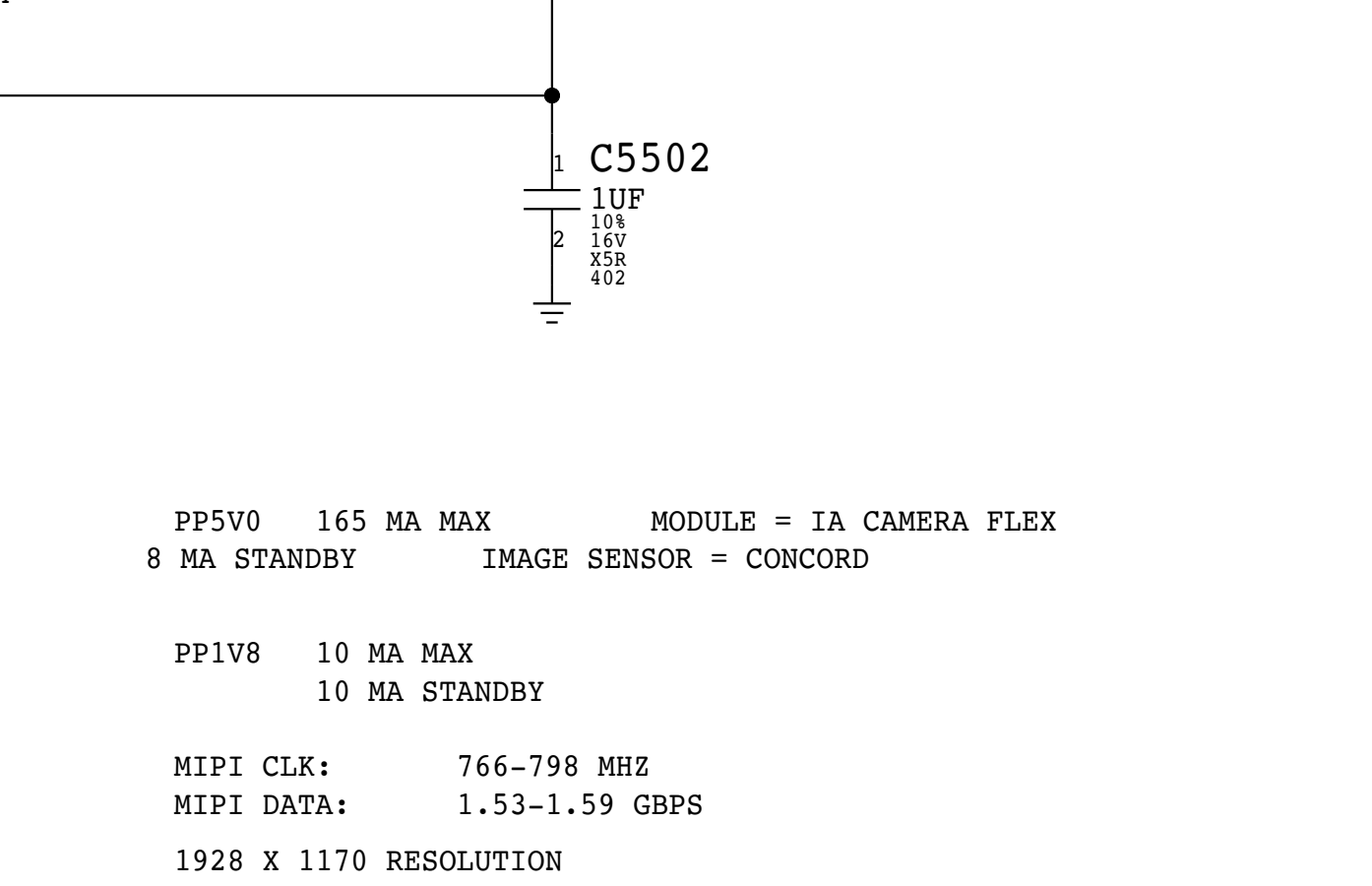
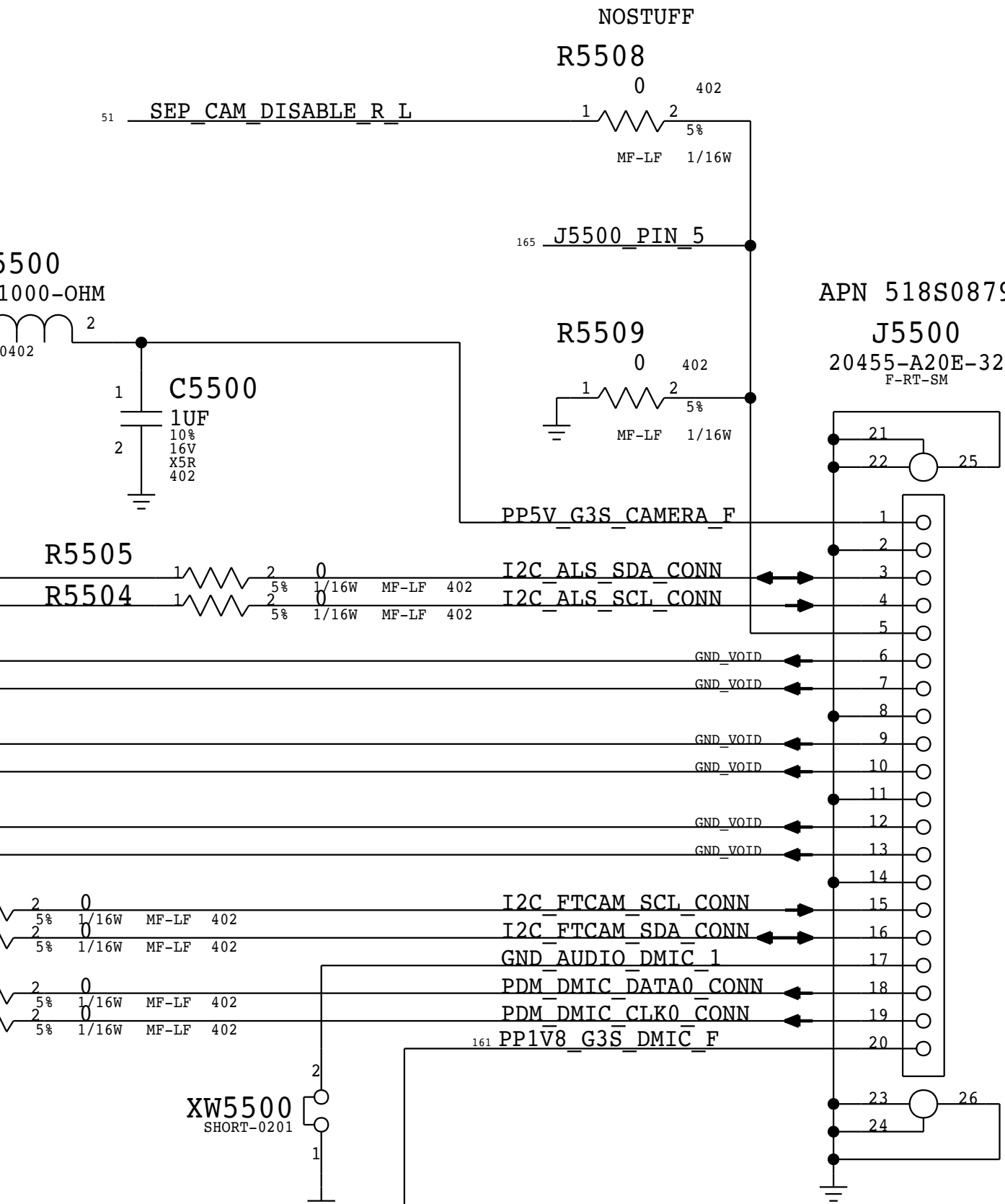
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2.2K PU TO I1V8_S0

CAM MIPI = SOC MIPI0C

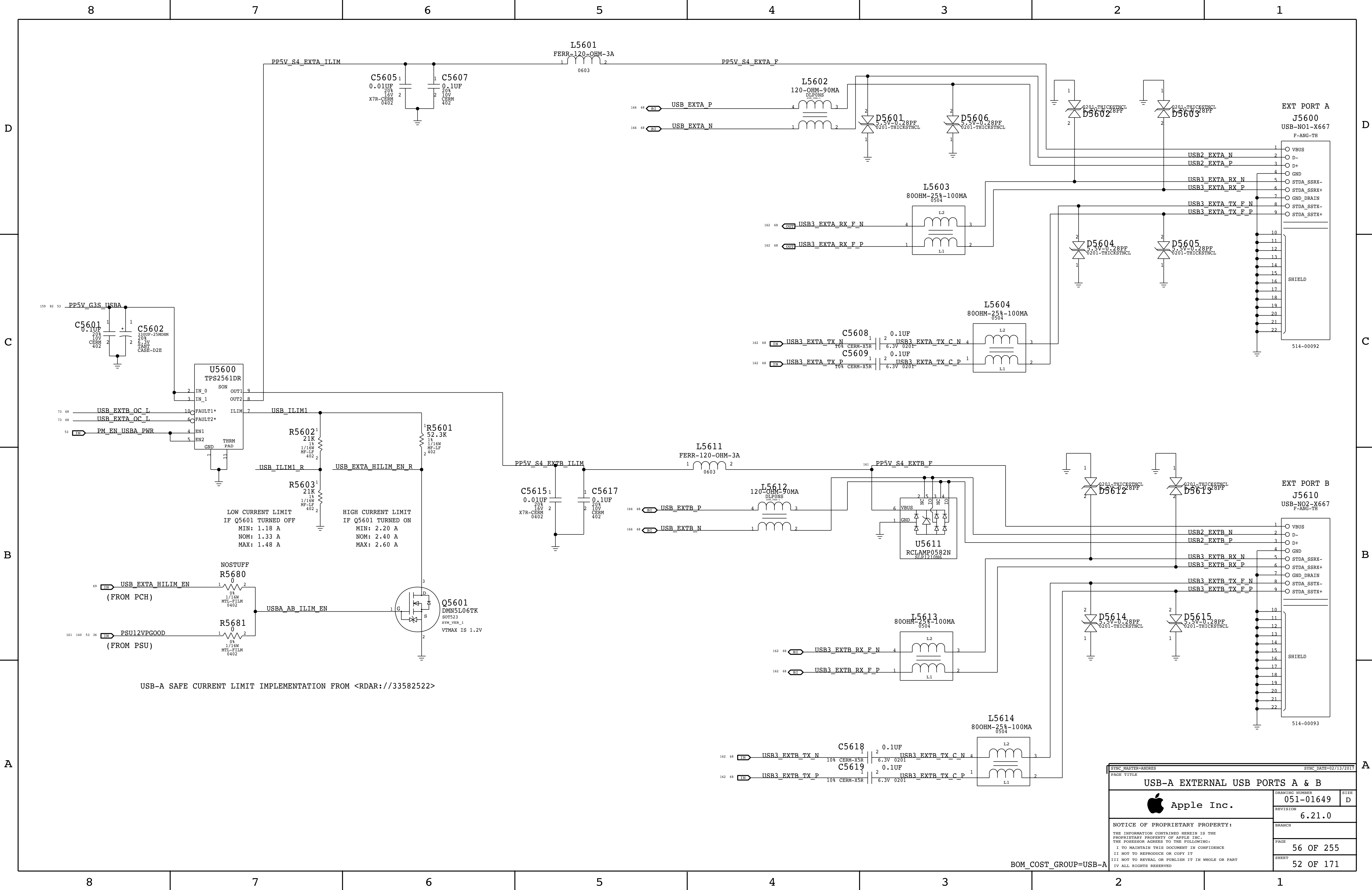
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2.2K PU TO I1V8_S0

DMIC = SOC AOP
I1V8 G3S



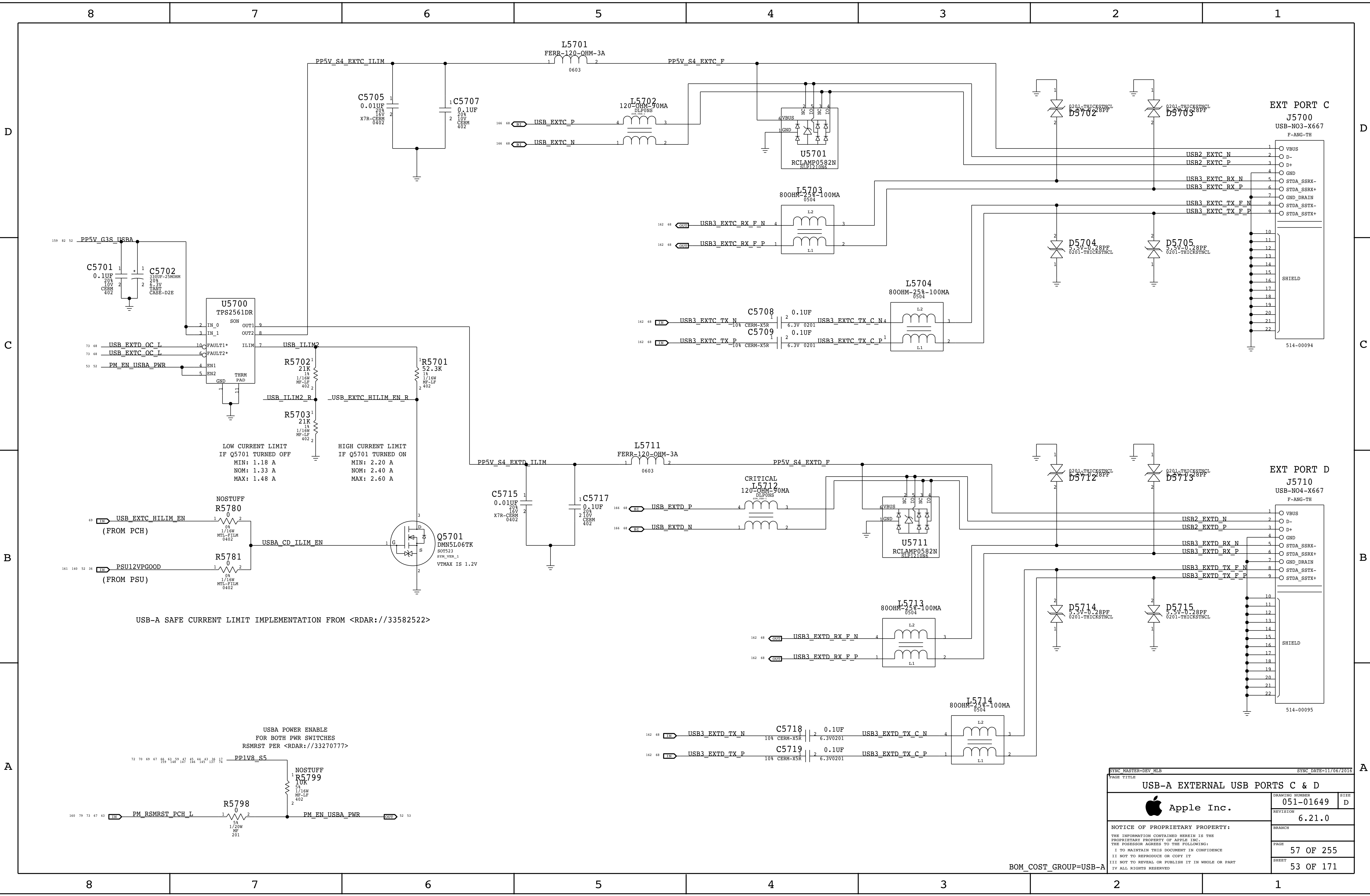
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BOM_COST_GROUP=CAMERA

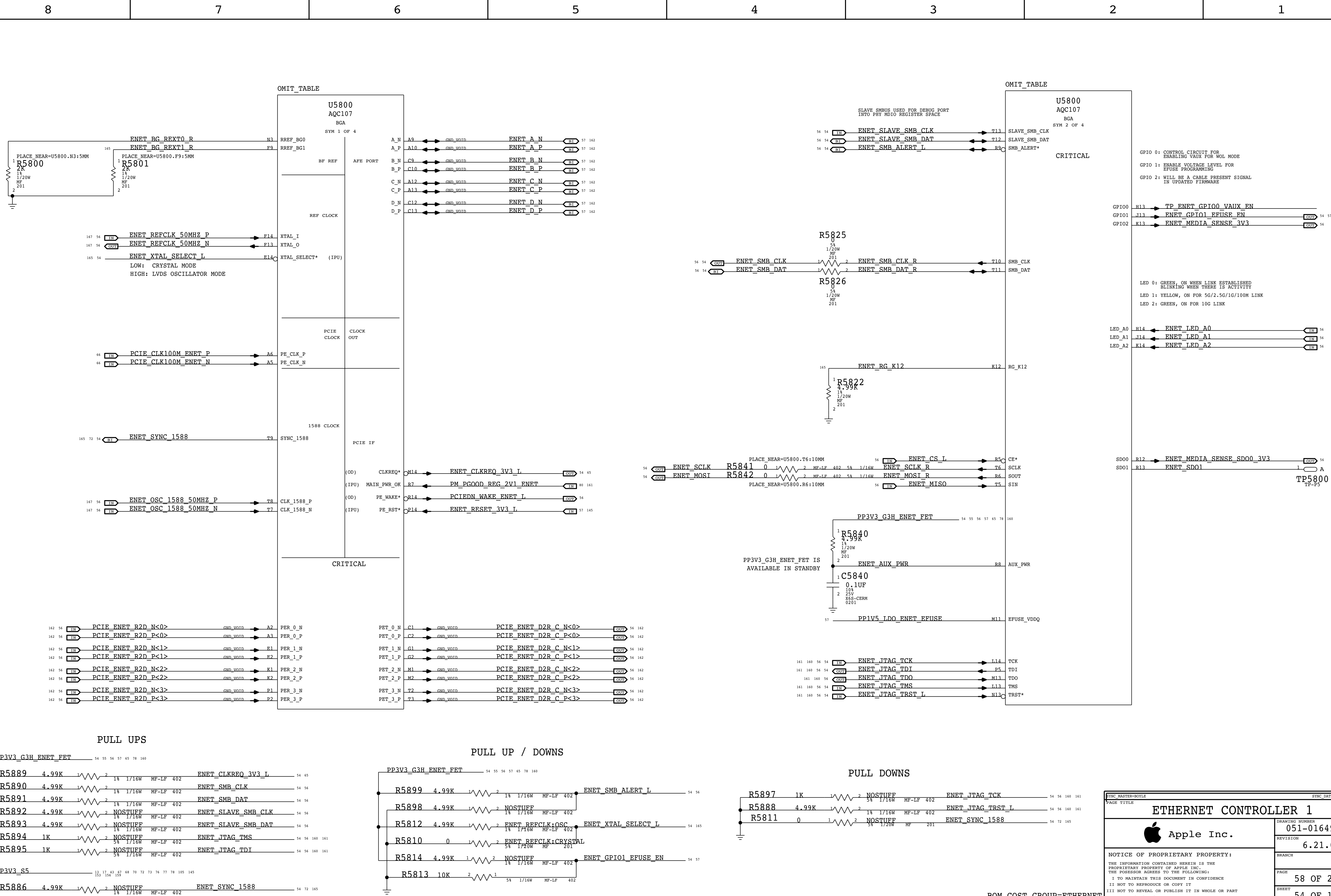


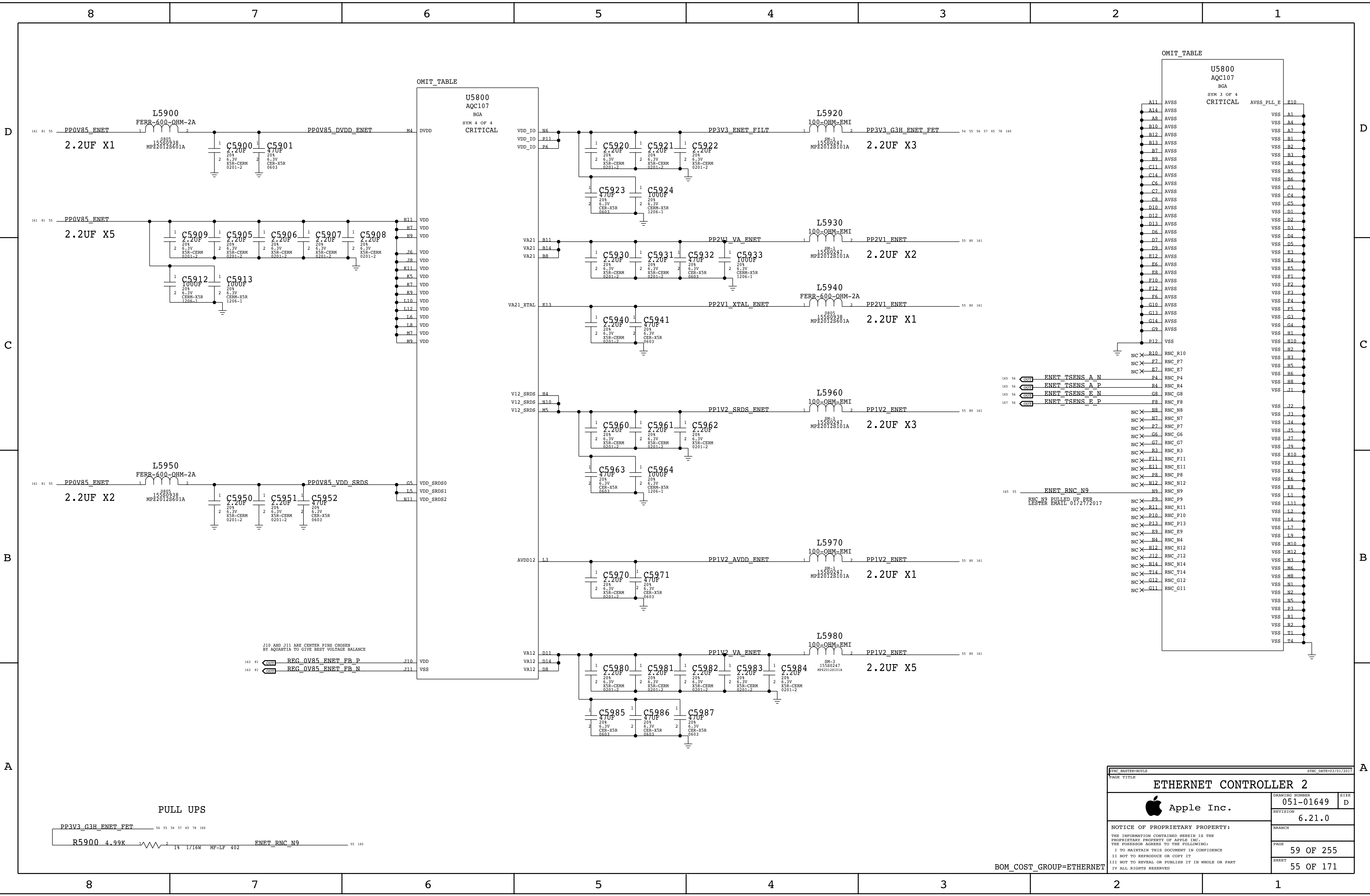
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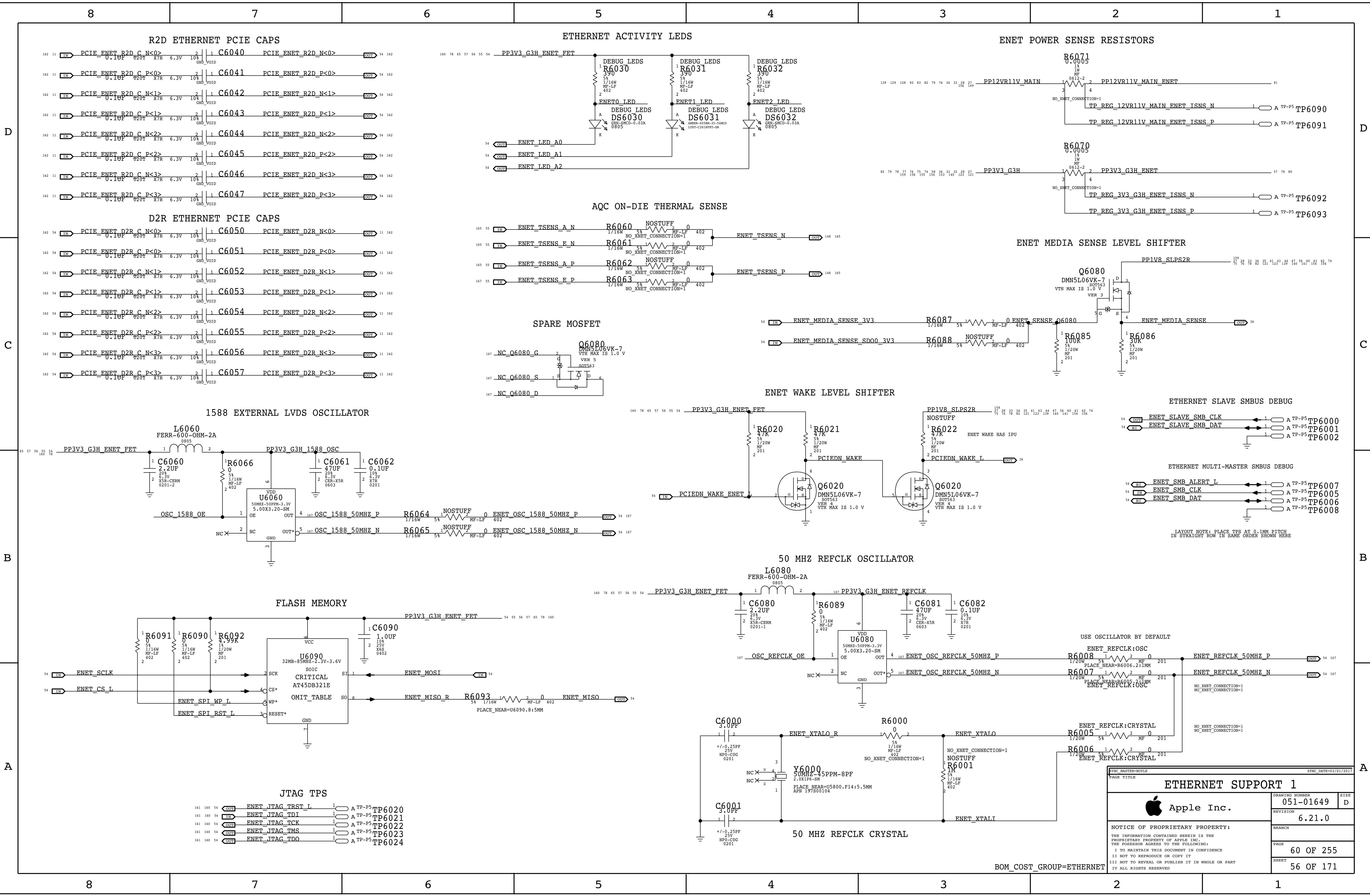
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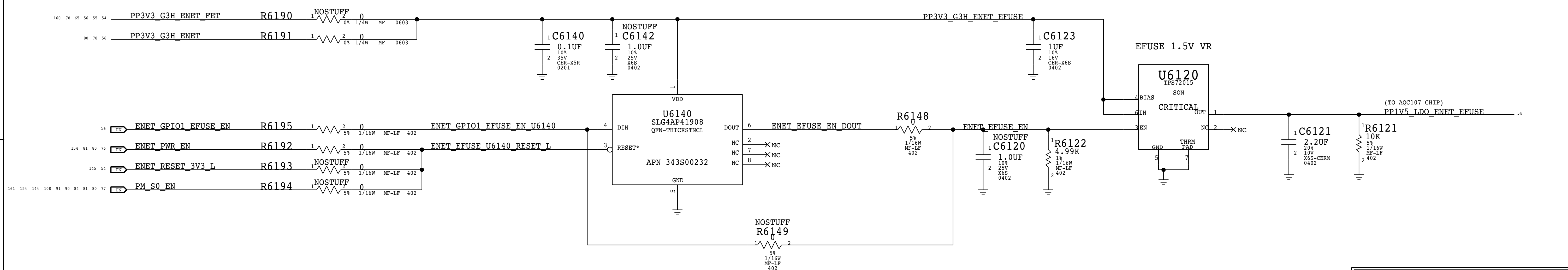
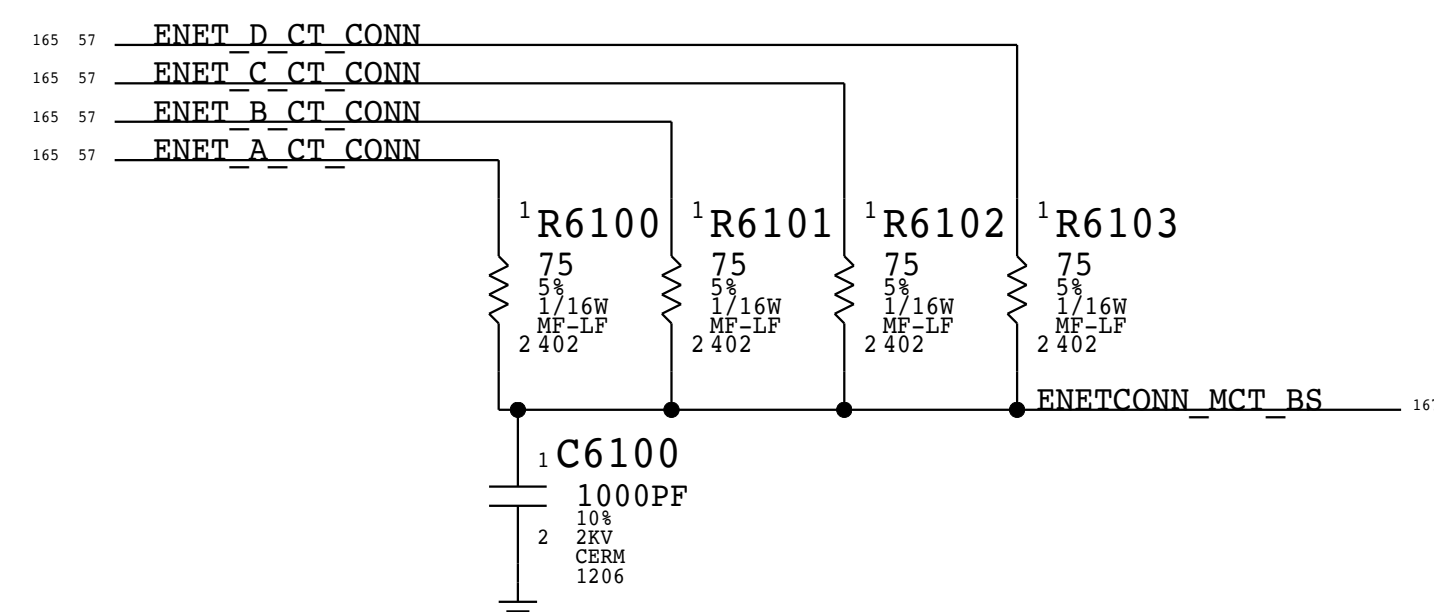
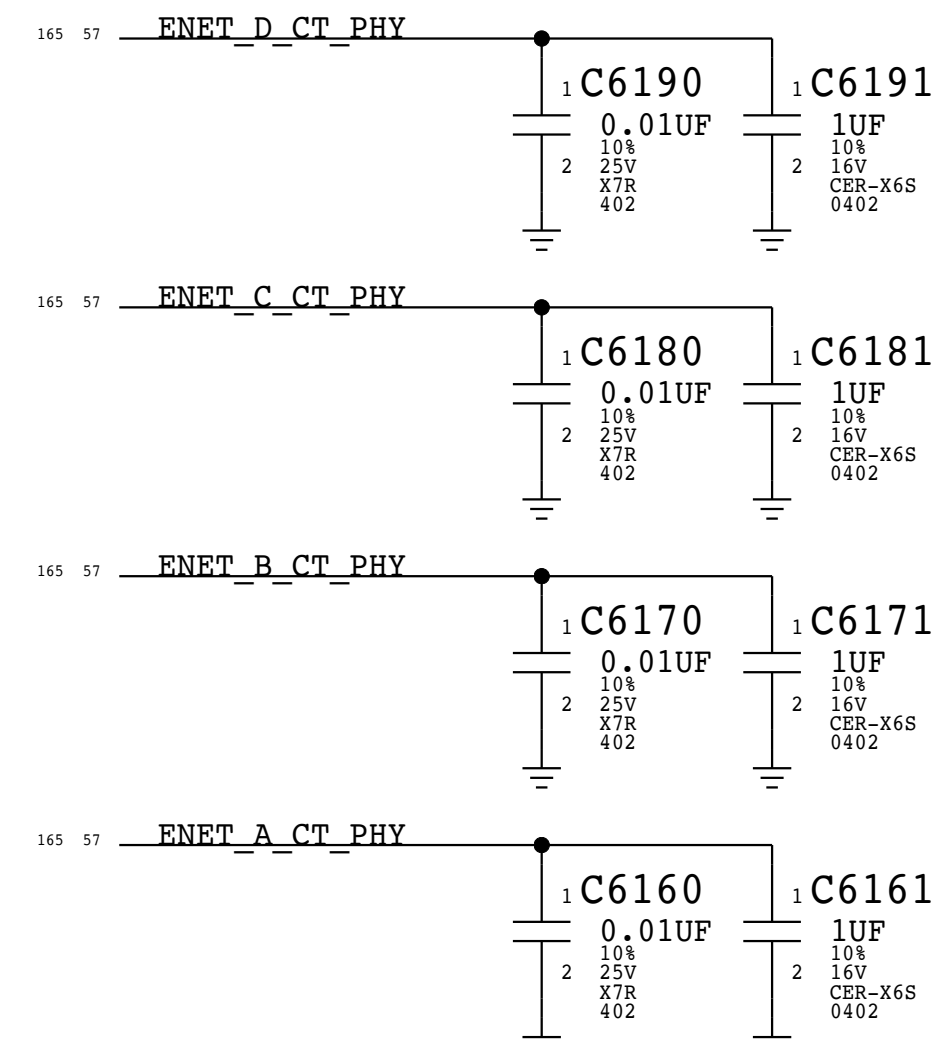
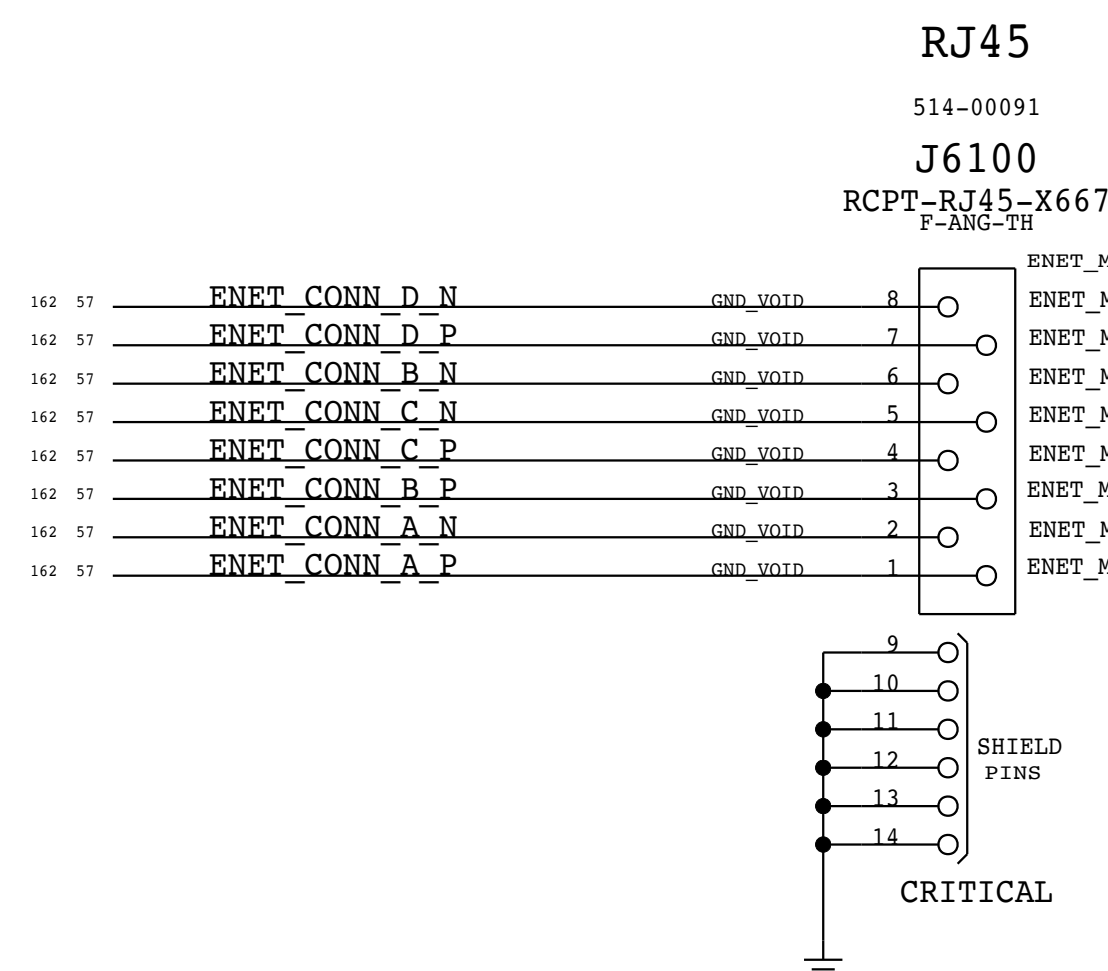
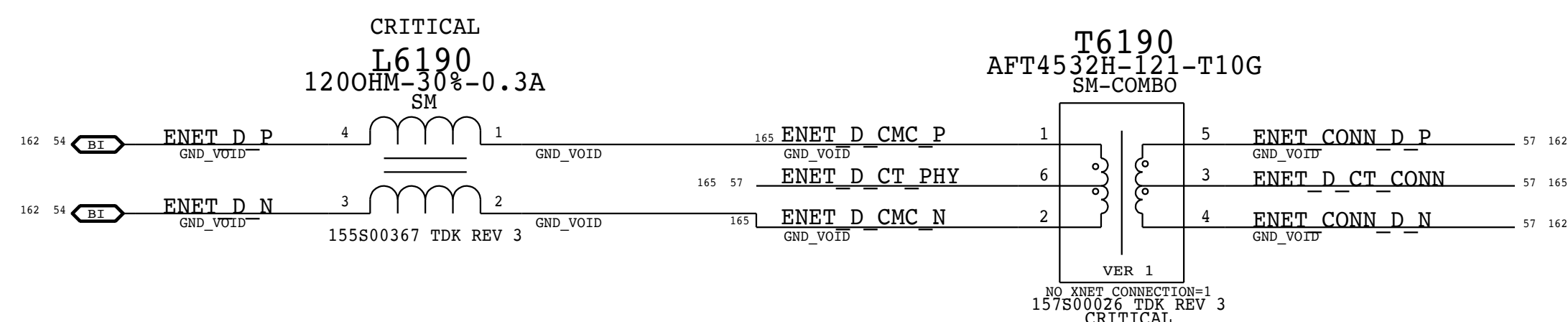
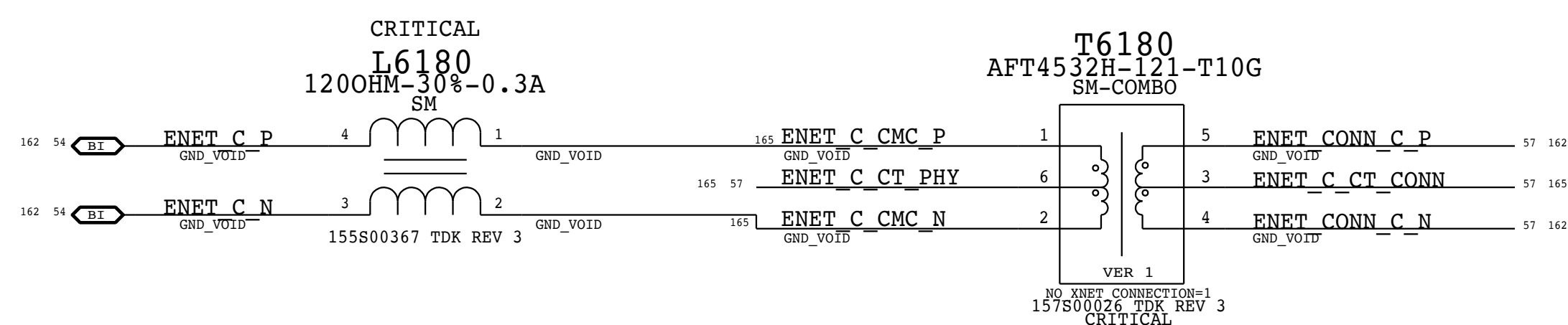
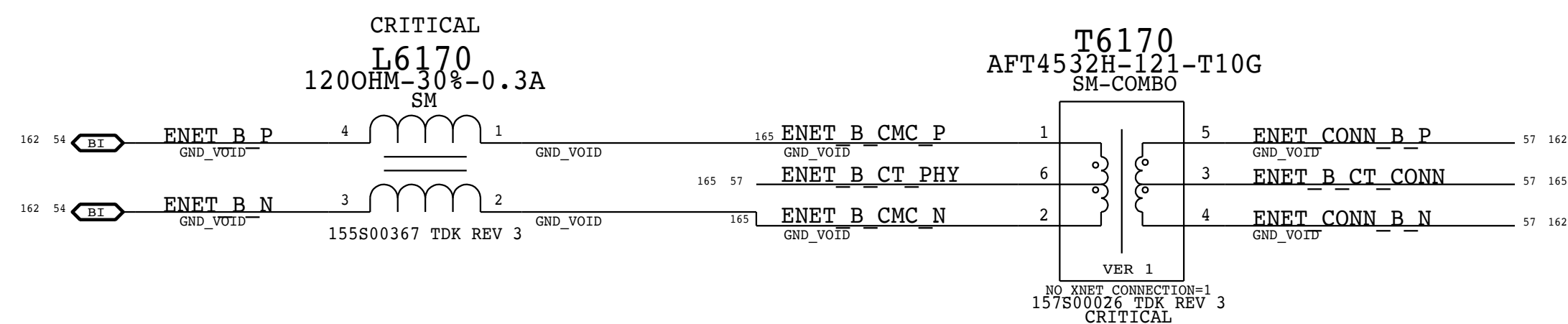
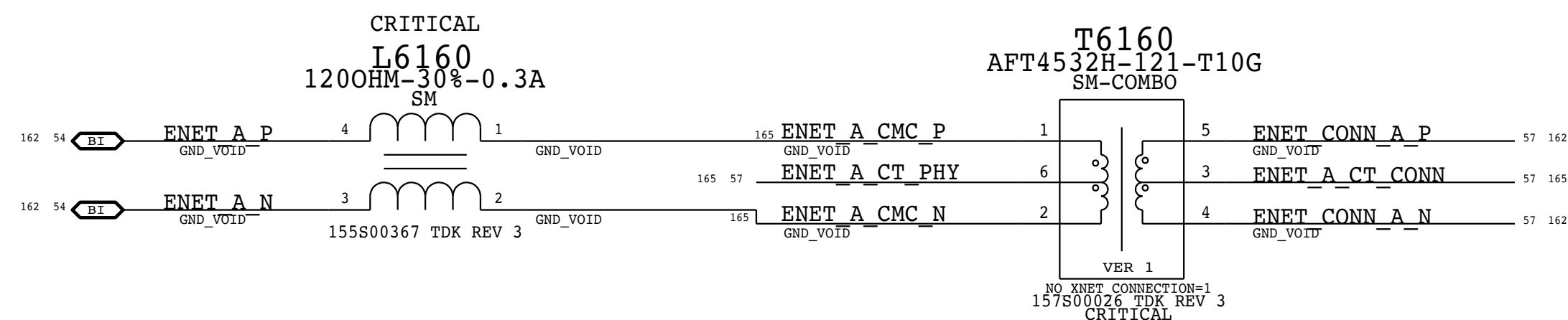






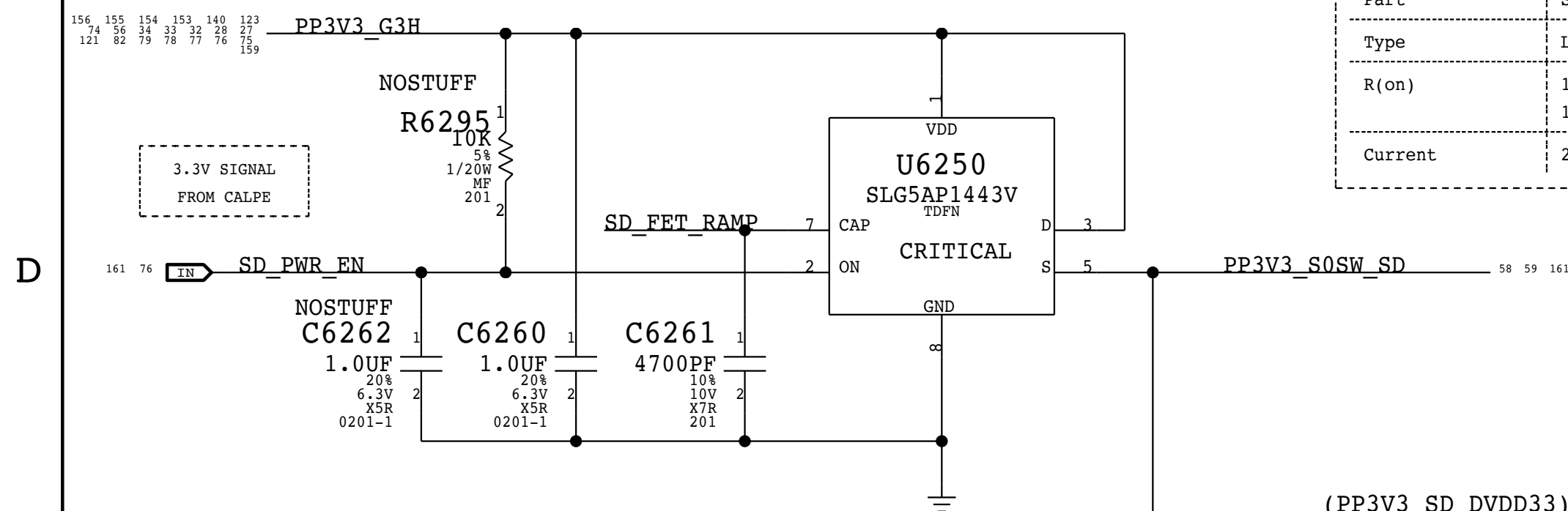


TO RJ45



SYNCH MASTER-BOYLE		SYNCH DATE-03/01/2017	
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ETHERNET SUPPORT 2			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-01649		D
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3.3V SD CARD LOAD SWITCH



U6250		EPD: 1.05A
Part	SLG5AP1438V	
Type	Load Switch	
R(on)	15 mOhm Typ 17 mOhm Max	
Current	2.5A	

NOTE:

HOST PROVIDES DVDD33 AND AVDD33
GL3227 GENERATES 1.2V USING INTERNAL
LDO OR DCDC. DCDC IS ENABLED IN FW

FOR USH-I: 3.3V @ 900MA

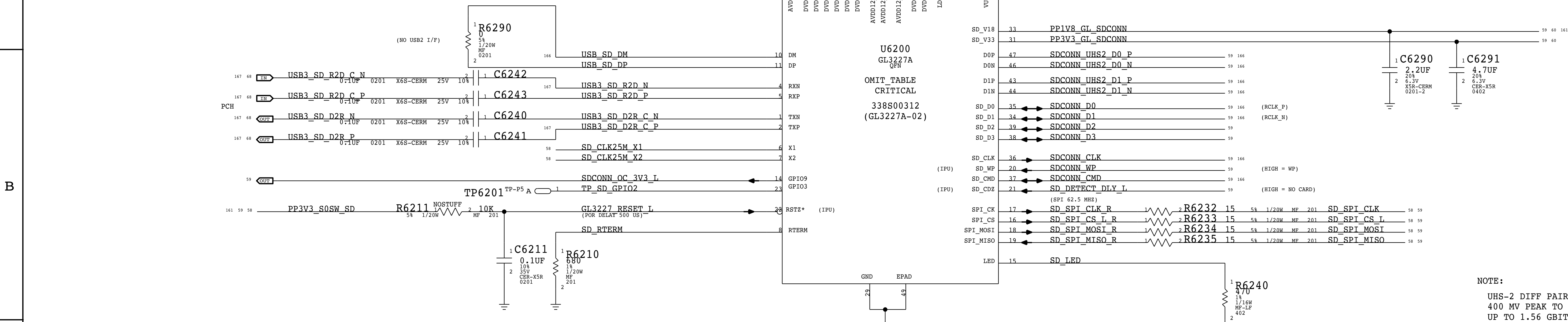
FOR USH-II: 3.3V @ 700MA

1.2V SVR OPERATES @ 1.6 MHZ

GL3227 PROVIDES POWER TO SD CARD

FOR UHS-I: SD_V33 @ 800MA

FOR UHS-II: SD_V33 @ 400MA
SD_V18 @ 200MA MAX



NOTE:

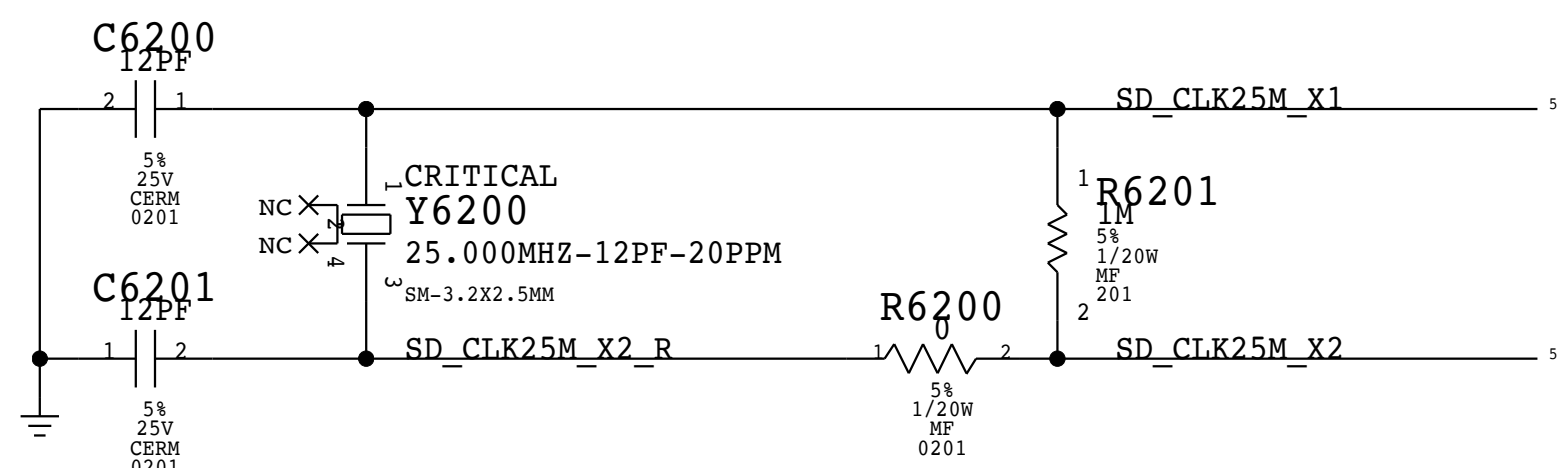
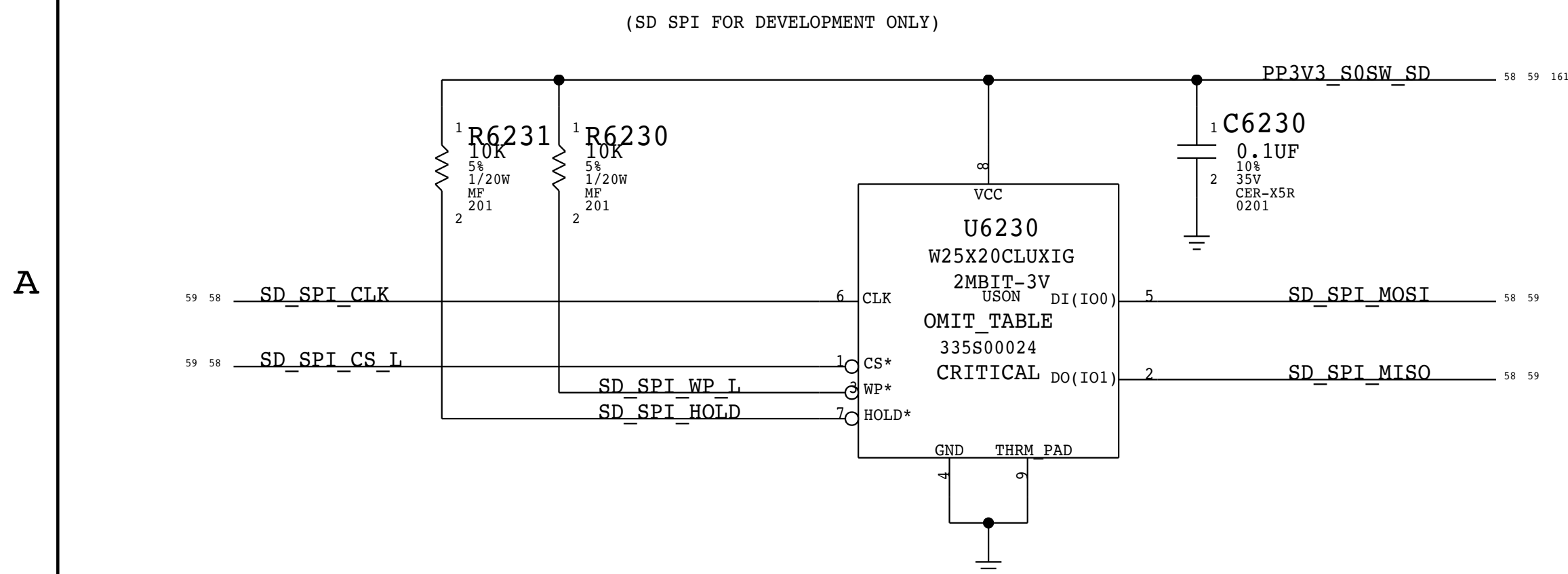
UHS-2 DIFF PAIRS 100 OHM
400 MV PEAK TO PEAK
UP TO 1.56 GBIT/S

UHS-2 DIFF CLK UP TO 52 MHZ

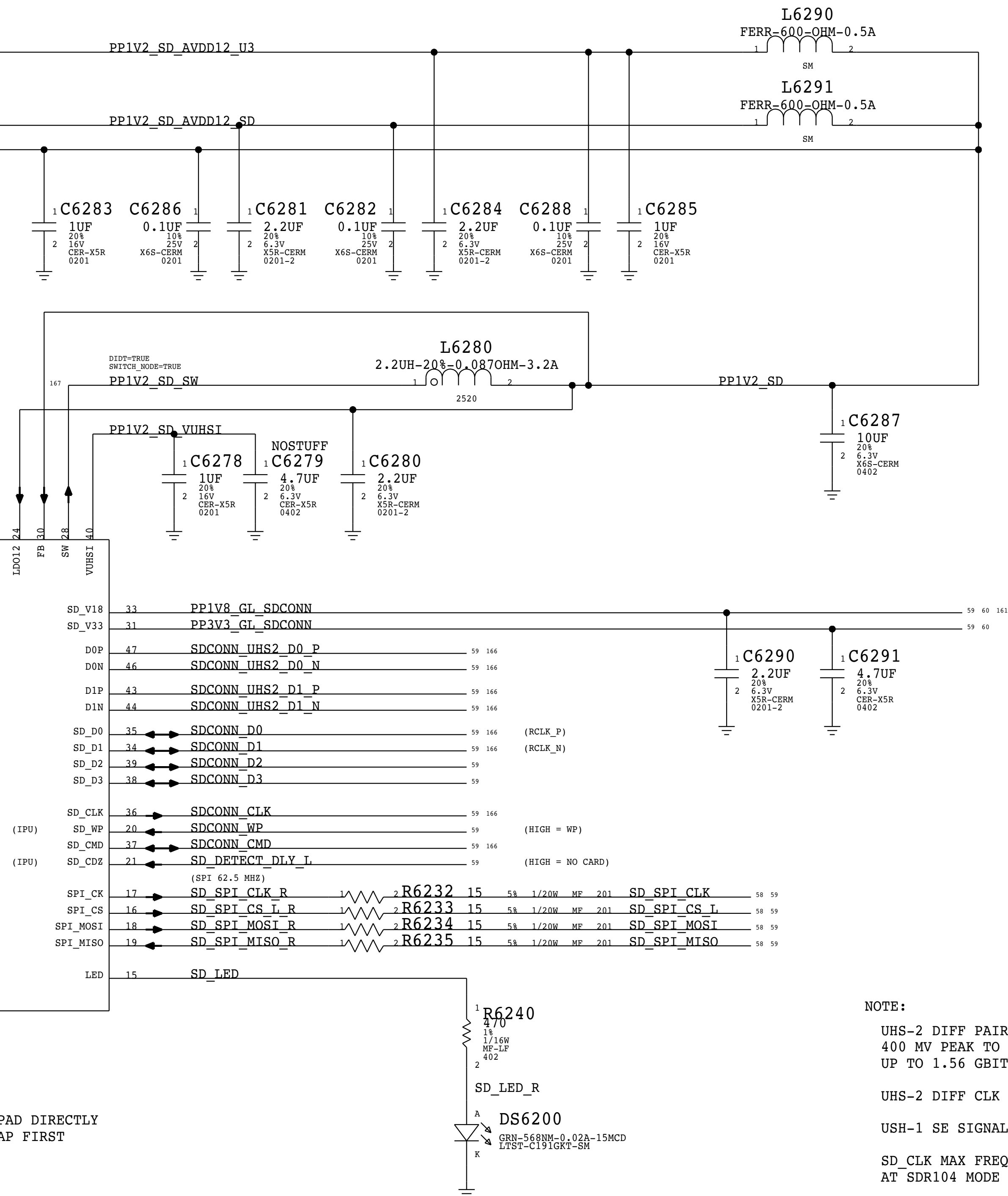
USH-1 SE SIGNALS 50-OHM

```
SD_CLK MAX FREQ IS 208 MHZ
AT SDR104 MODE
```

LED IS ON WHEN POWER ON
BLINKING DURING R/W OPERATIONS



PIN 29 IS SVR GND
DO NOT CONNECT PIN 29 TO EPAD DIRECTLY
CONNECT PIN 29 TO PIN 27 CAP FIRST
THEN TO MAIN GND



NOTE:

UHS-2 DIFF PAIRS 100 OHM
400 MV PEAK TO PEAK
UP TO 1.56 GBIT/S

UHS-2 DIFF CLK UP TO 52 MHZ

USH-1 SE SIGNALS 50-OHM

```
SD_CLK MAX FREQ IS 208 MHZ
AT SDR104 MODE
```

LED IS ON WHEN POWER ON
BLINKING DURING R/W OPERATIONS

SYNC MASTER=DEV MLB		SYNC DATE=11/06/2016	
PAGE TITLE			
SD CONTROLLER UHS-II			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-01649	D
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		PAGE	62 OF 255
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BOM_COST_GROUP=SD CARD

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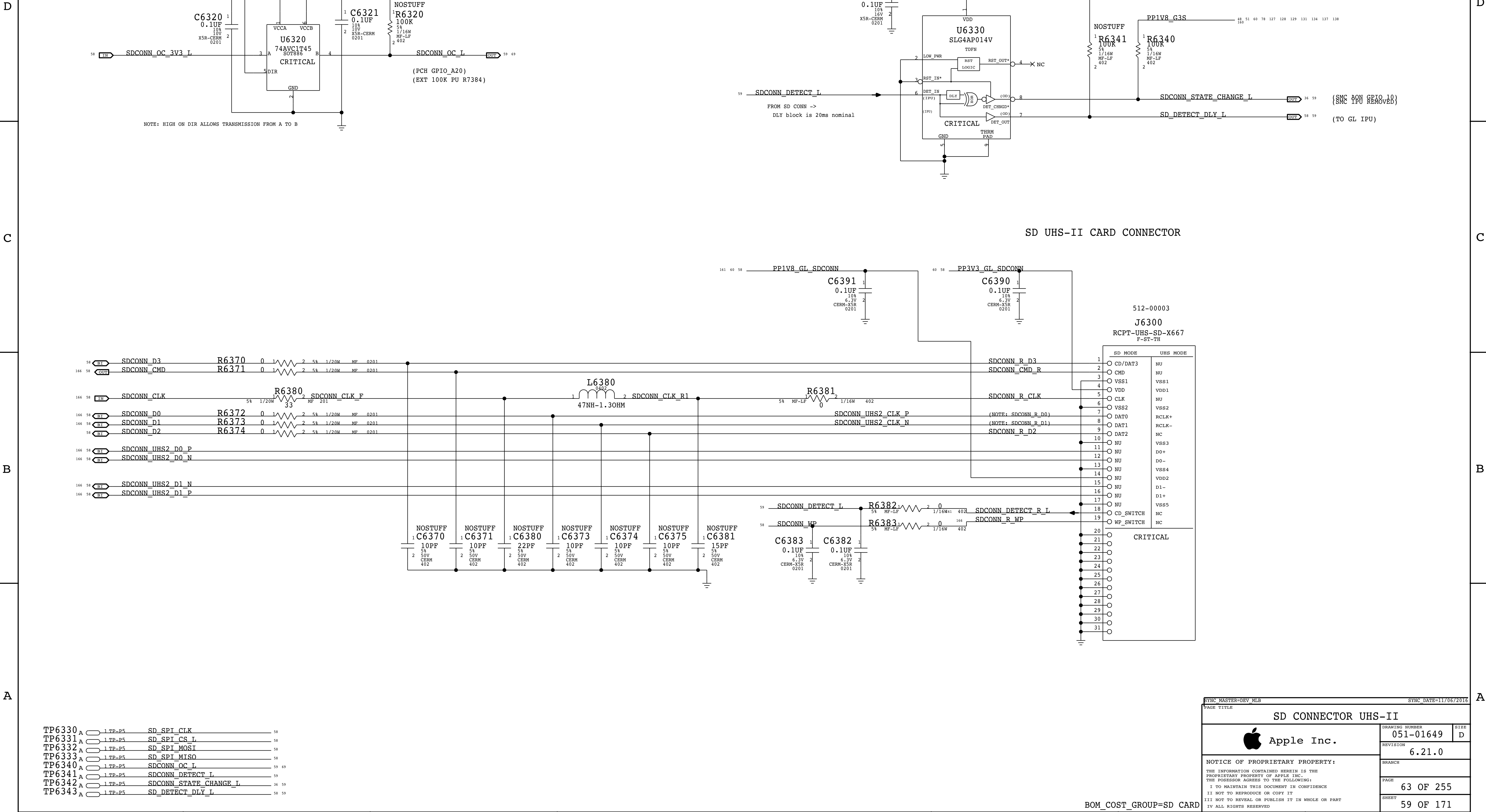
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TP6330

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TP-P5

SD_SPT_CLK

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TP6331

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TP-P5

SD_SPT_CS_L

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TP6332

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TP-P5

SD_SPT_MOSI

58

TP6333

A

1

TP-P5

SD_SPT_MISO

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TP6340

A

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TP-P5

SDCONN_OC_L

59 69

TP6341

A

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TP-P5

SDCONN_DETECT_L

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TP6342

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TP-P5

SDCONN_STATE_CHANGE_L

36 59

TP6343

A

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TP-P5

SD_DETECT_DLY_L

58 59

TP6330

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TP-P5

SD_SPT_CLK

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TP6331

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TP-P5

SD_SPT_CS_L

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SDCONN_OC_L

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SD_DETECT_DLY_L

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TP6330

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SD_SPT_CLK

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SDCONN_OC_L

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SDCONN_STATE_CHANGE_L

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TP6343

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TP-P5

SD_DETECT_DLY_L

58 59

87654321

SD CONNECTOR UHS-II

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TP6330

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TP-P5

SD_SPT_CLK

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TP6331

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SD_SPT_CS_L

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87654321

SD CONNECTOR UHS-II

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TP6330

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TP-P5

SD_SPT_CLK

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TP6331

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SD_SPT_CS_L

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SDCONN_OC_L

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TP-P5

SDCONN_DETECT_L

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SDCONN_STATE_CHANGE_L

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SD_DETECT_DLY_L

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SD_SPT_CLK

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TP6331

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SD_SPT_CS_L

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SD_SPT_MOSI

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SDCONN_OC_L

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SDCONN_STATE_CHANGE_L

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SD_DETECT_DLY_L

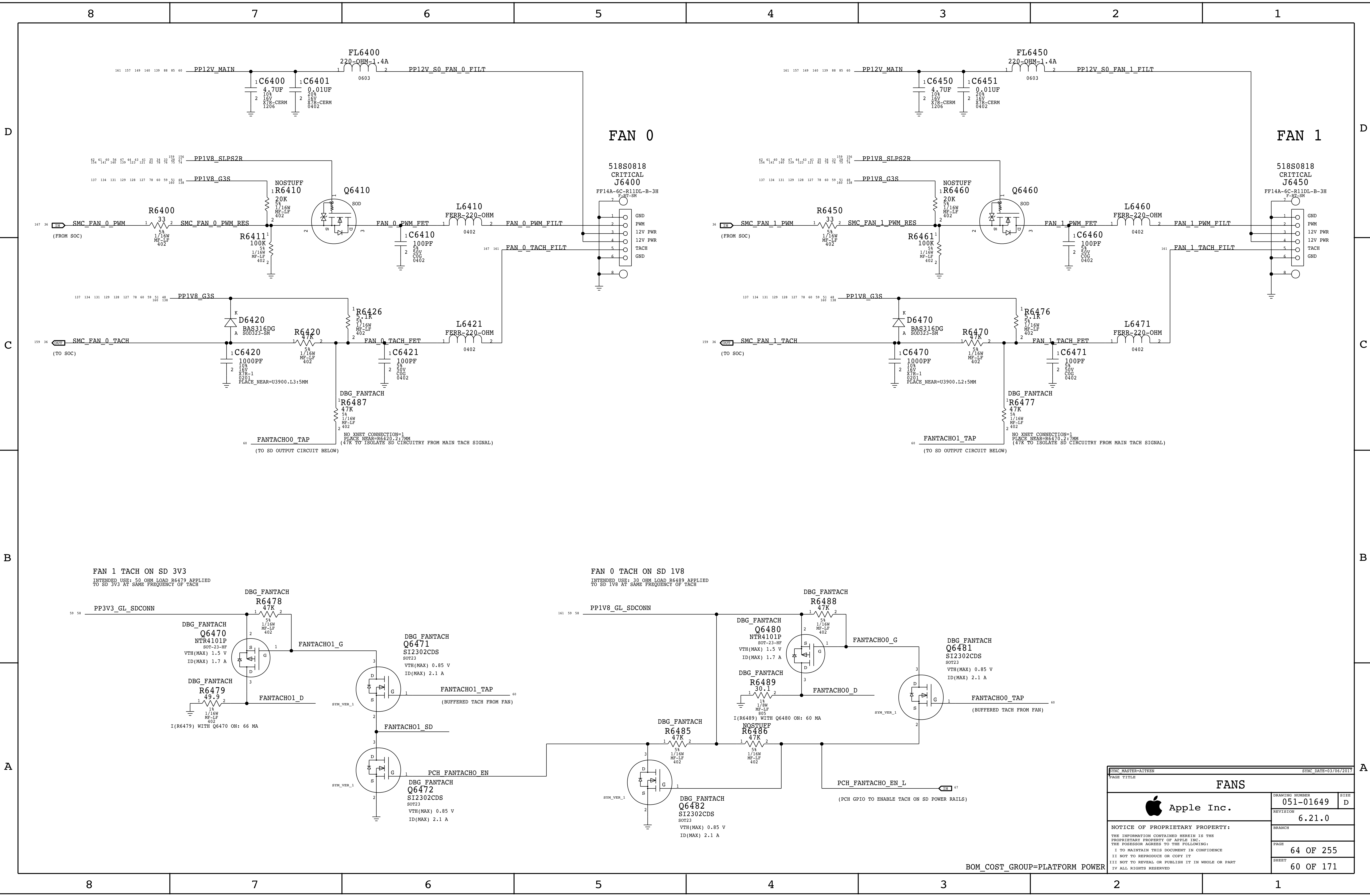
58 59

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			REVISION	6.21.0	
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			PAGE	64 OF 255	
			SHEET	60 OF 171	

BOM_COST_GROUP=PLATFORM POWER

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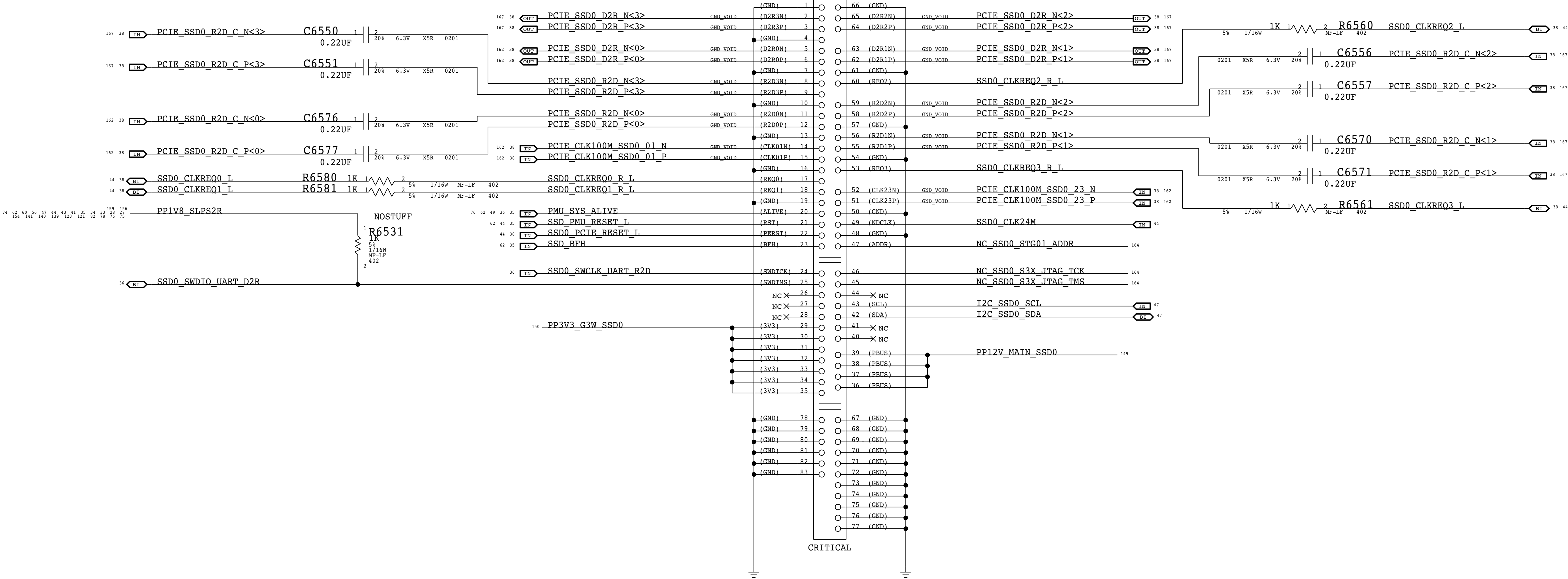
SSD GS5 CONNECTOR A

APN:514800120

J6500

SSD-J137

F-RT-SM



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SSD CONN A		
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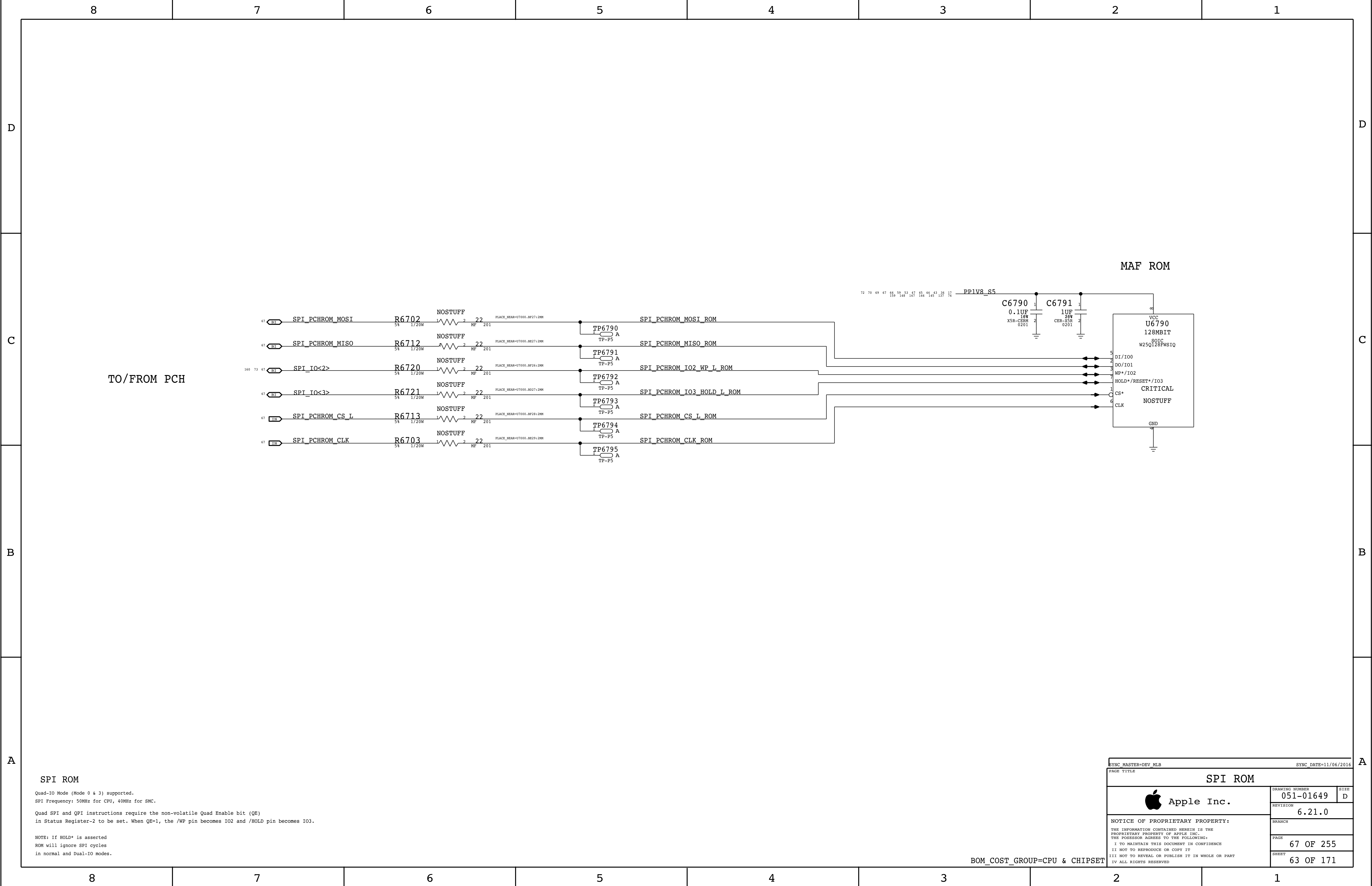
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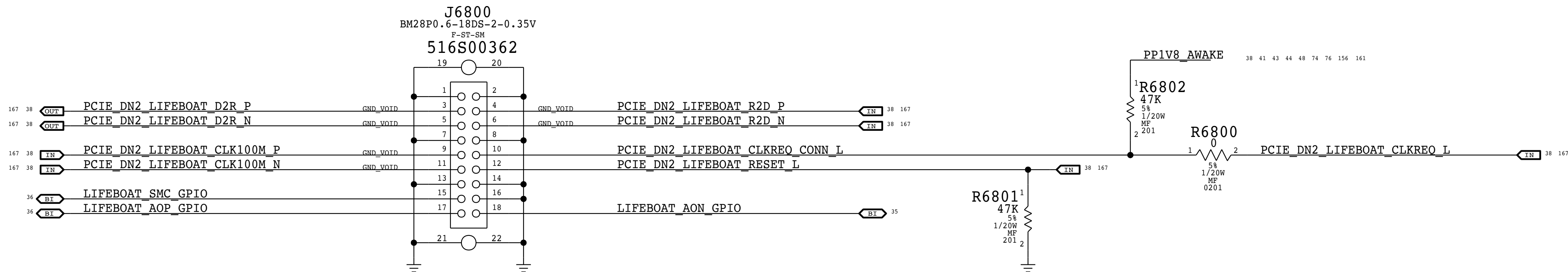
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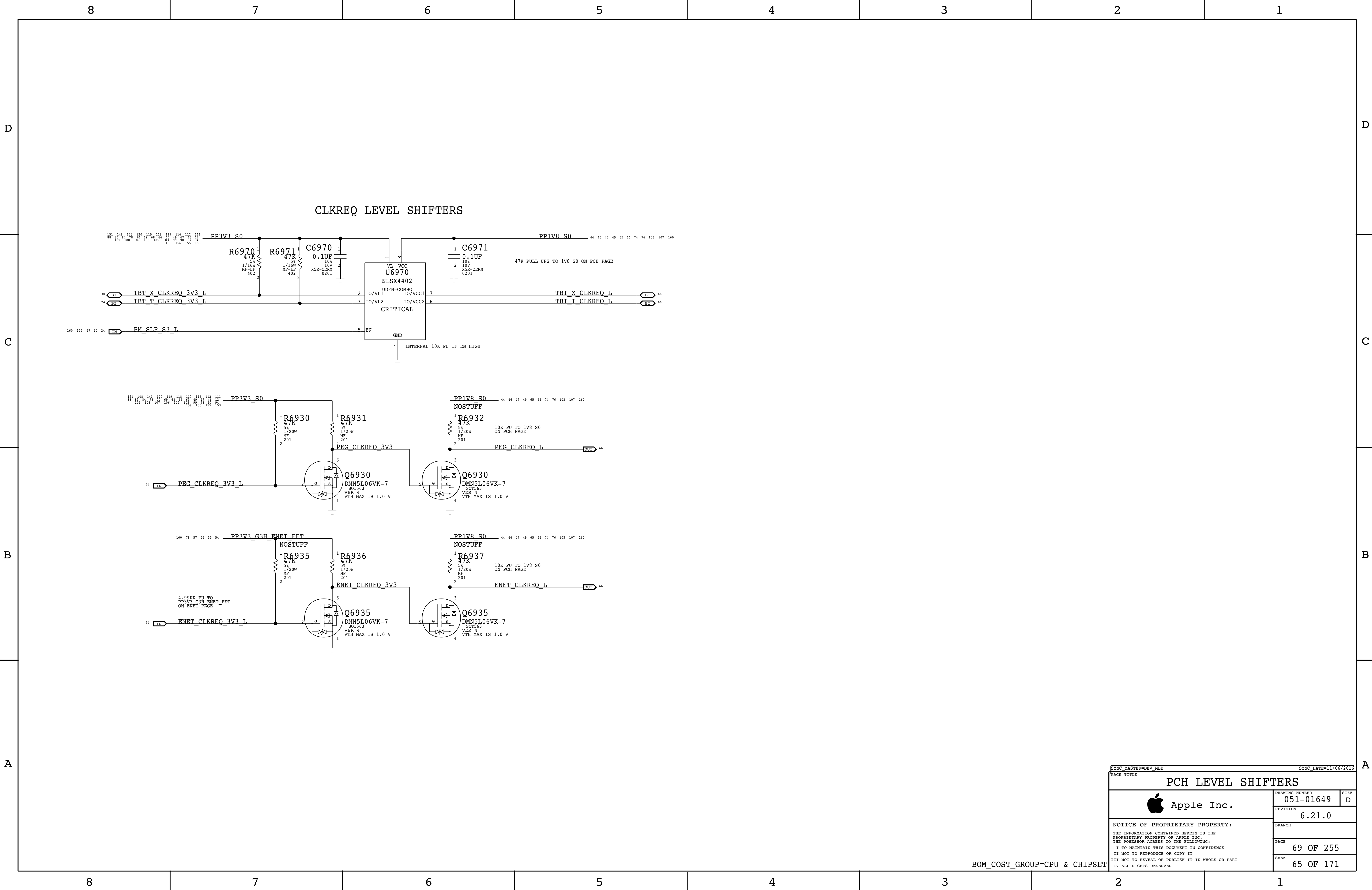
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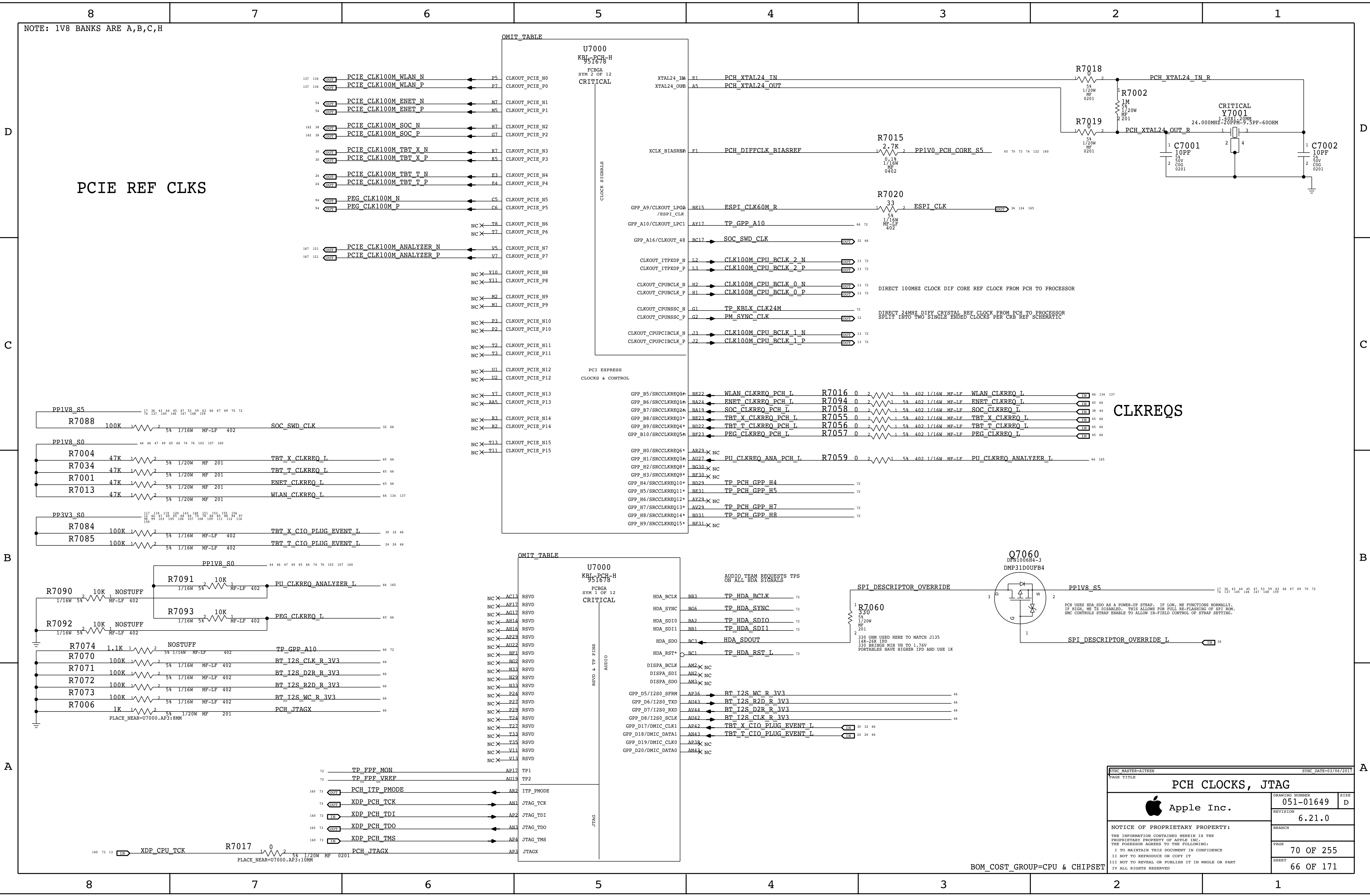


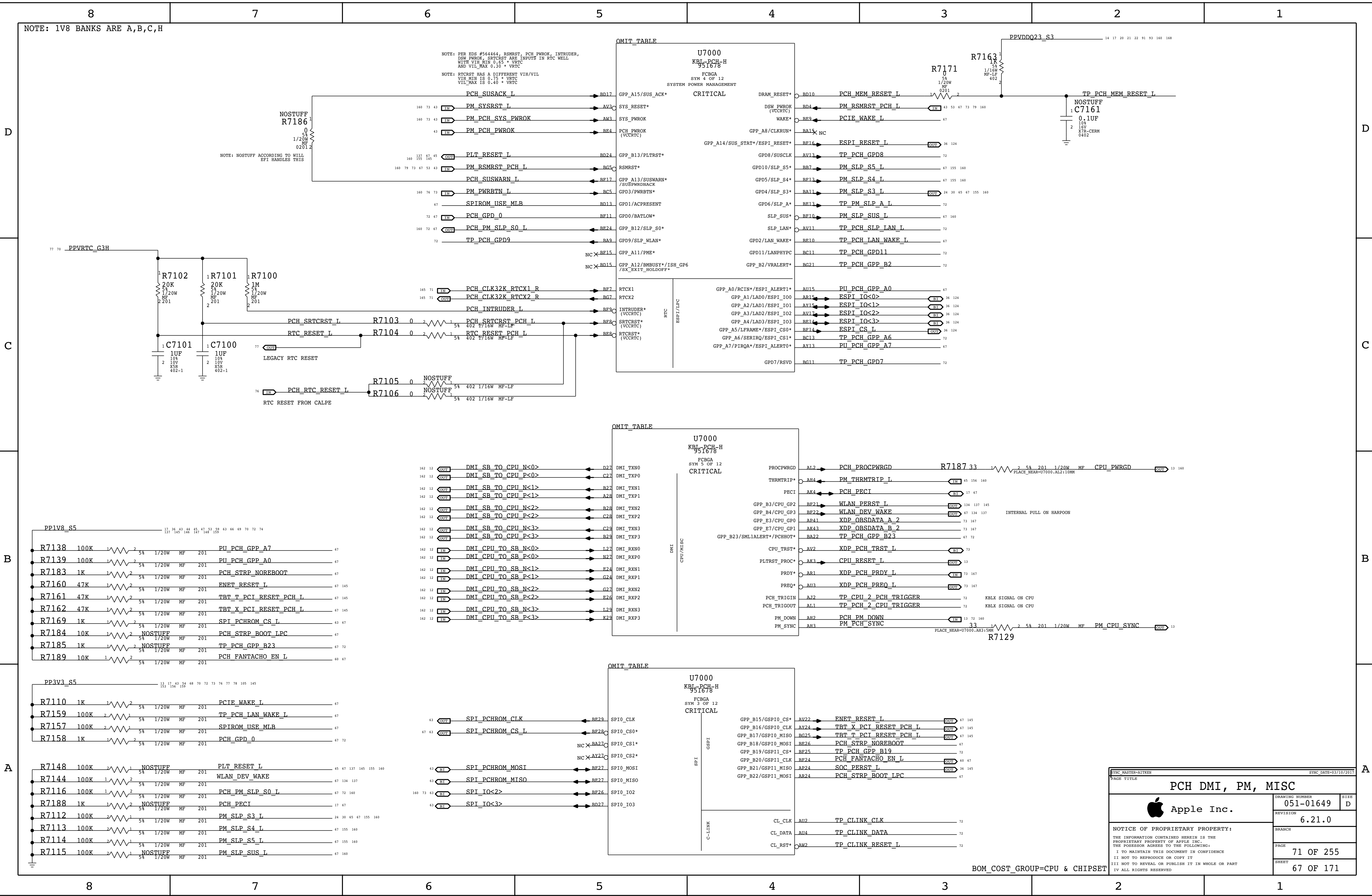
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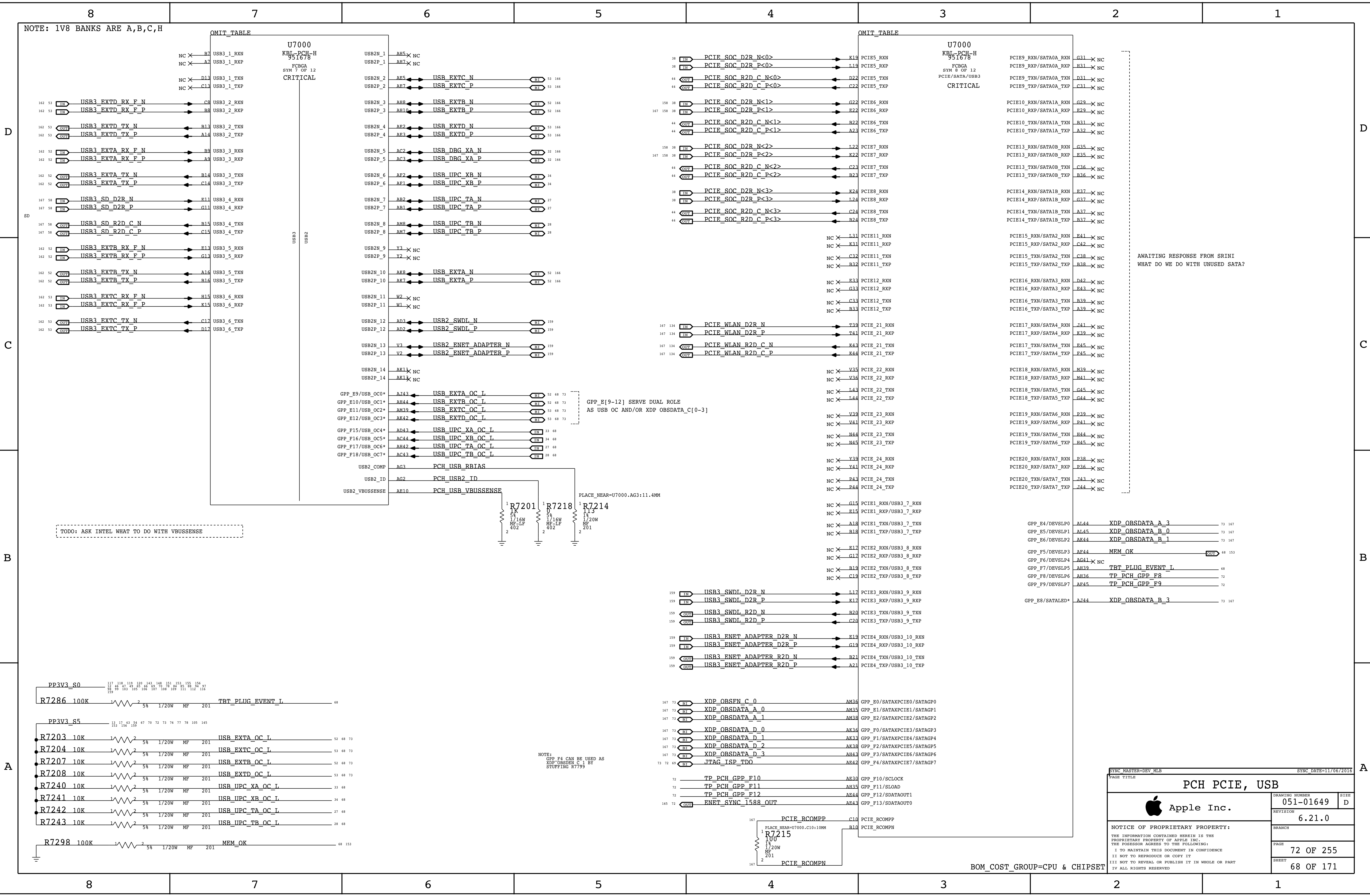
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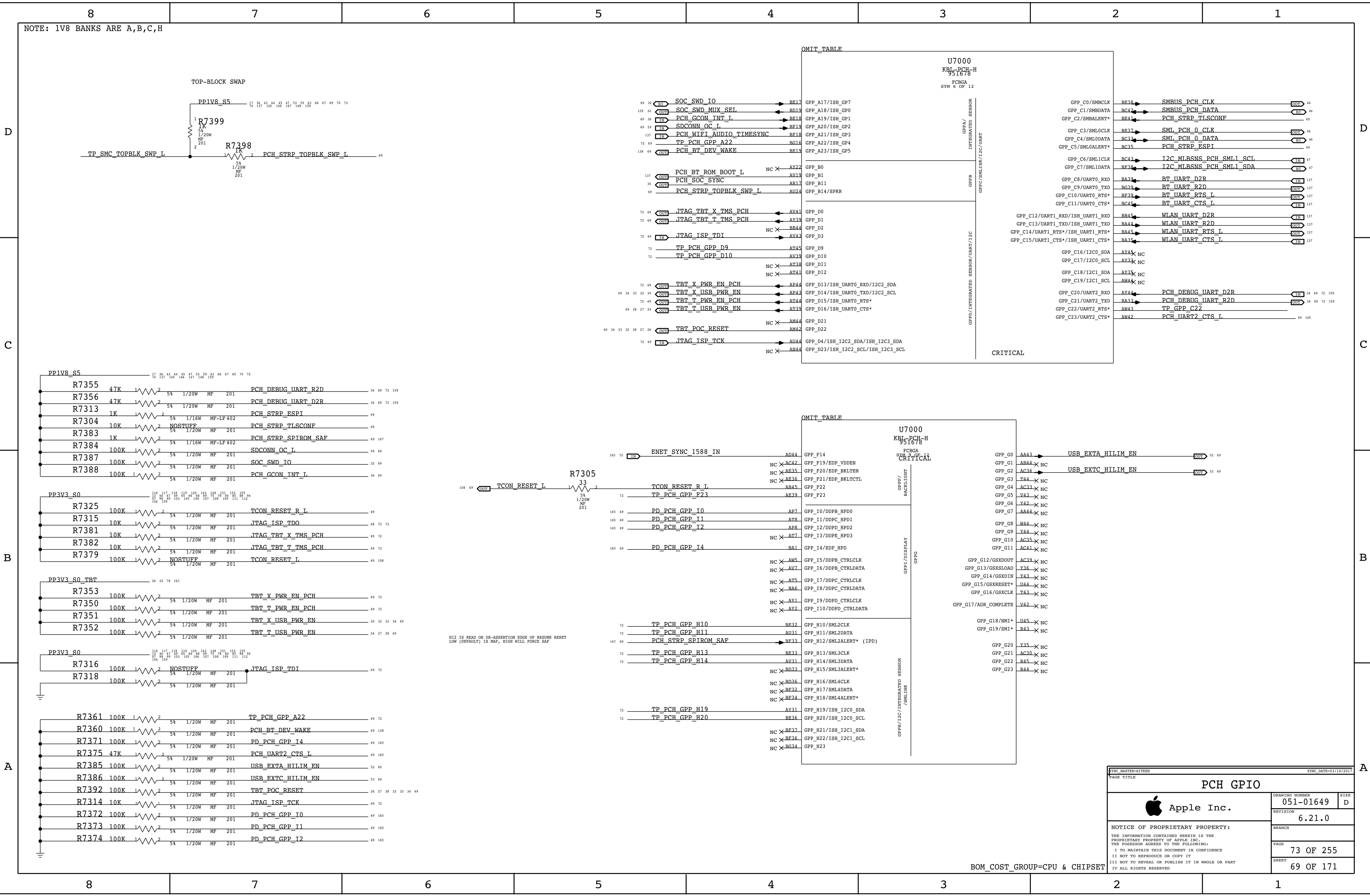
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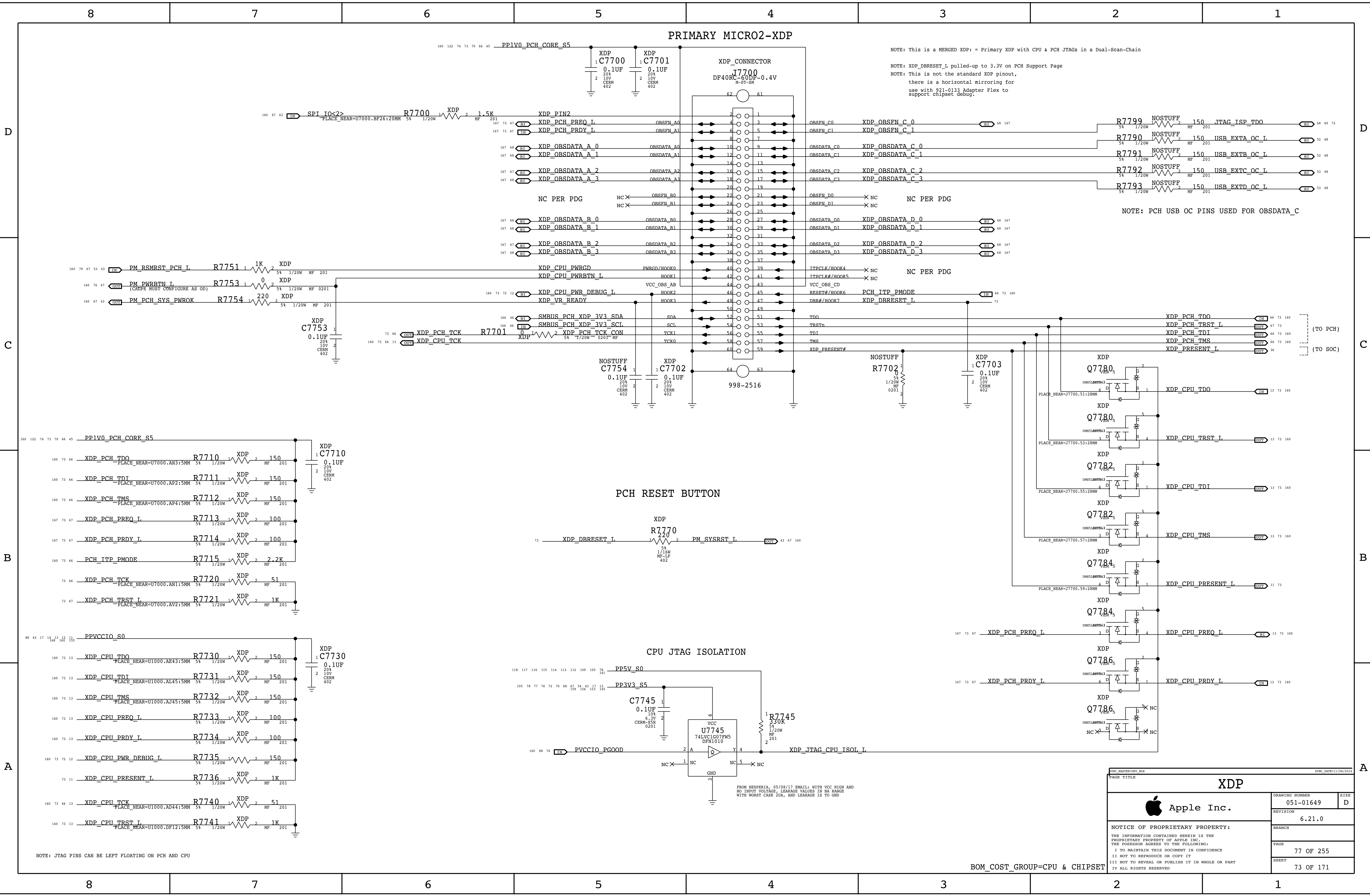
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NOTE: This is a MERGED XDP: = Primary XDP with CPU & PCH JTAGs in a Dual-Scan-Chain

NOTE: XDP_DBRESET_L pulled-up to 3.3V on PCH Support Page

NOTE: This is not the standard XDP pinout, there is a horizontal mirroring for use with 921-0133 Adapter Flex to support chipset debug.

NOTE: PCH USB OC PINS USED FOR OBSDATA_C

PCH RESET BUTTON

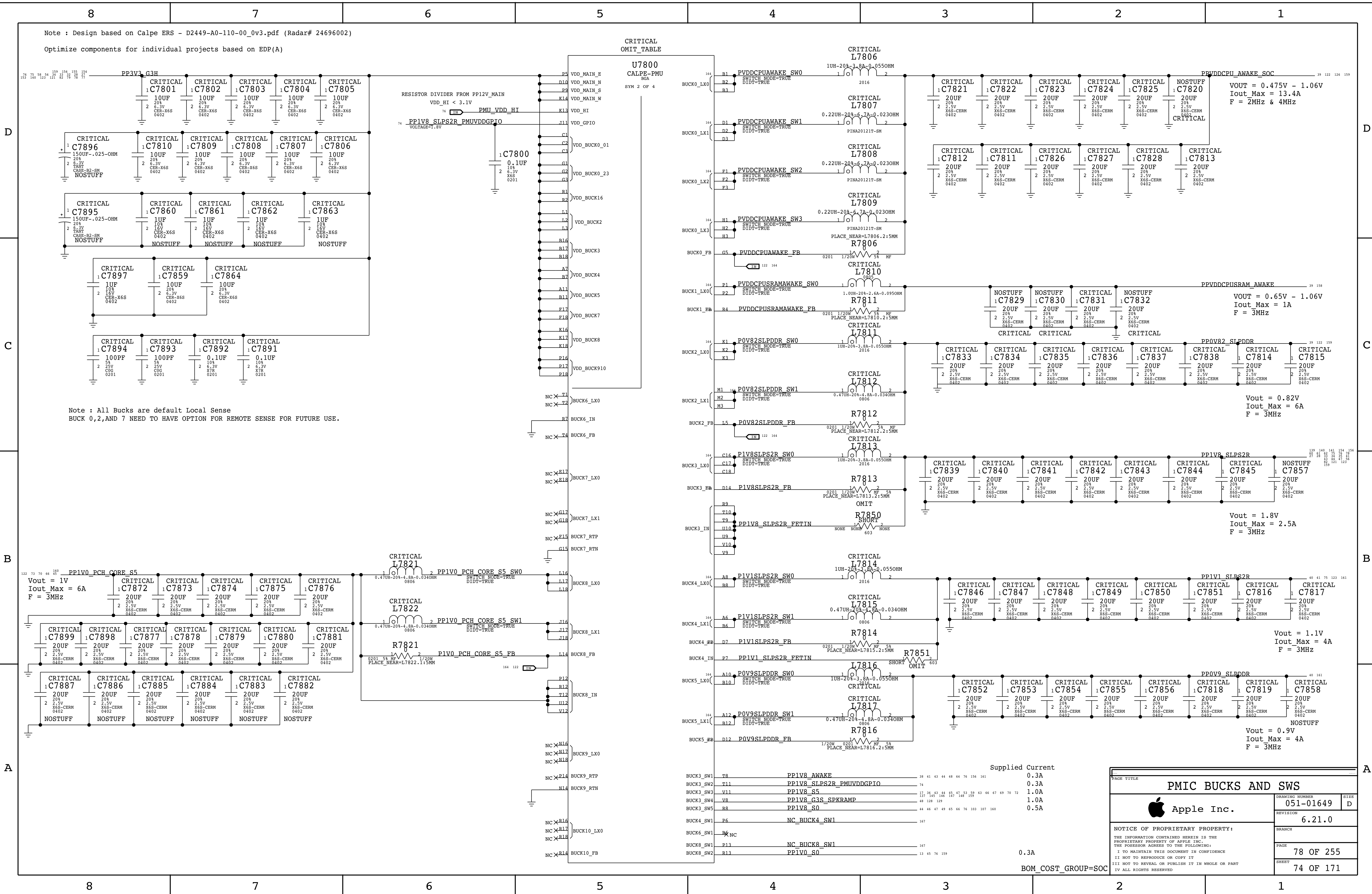
CPU JTAG ISOLATION

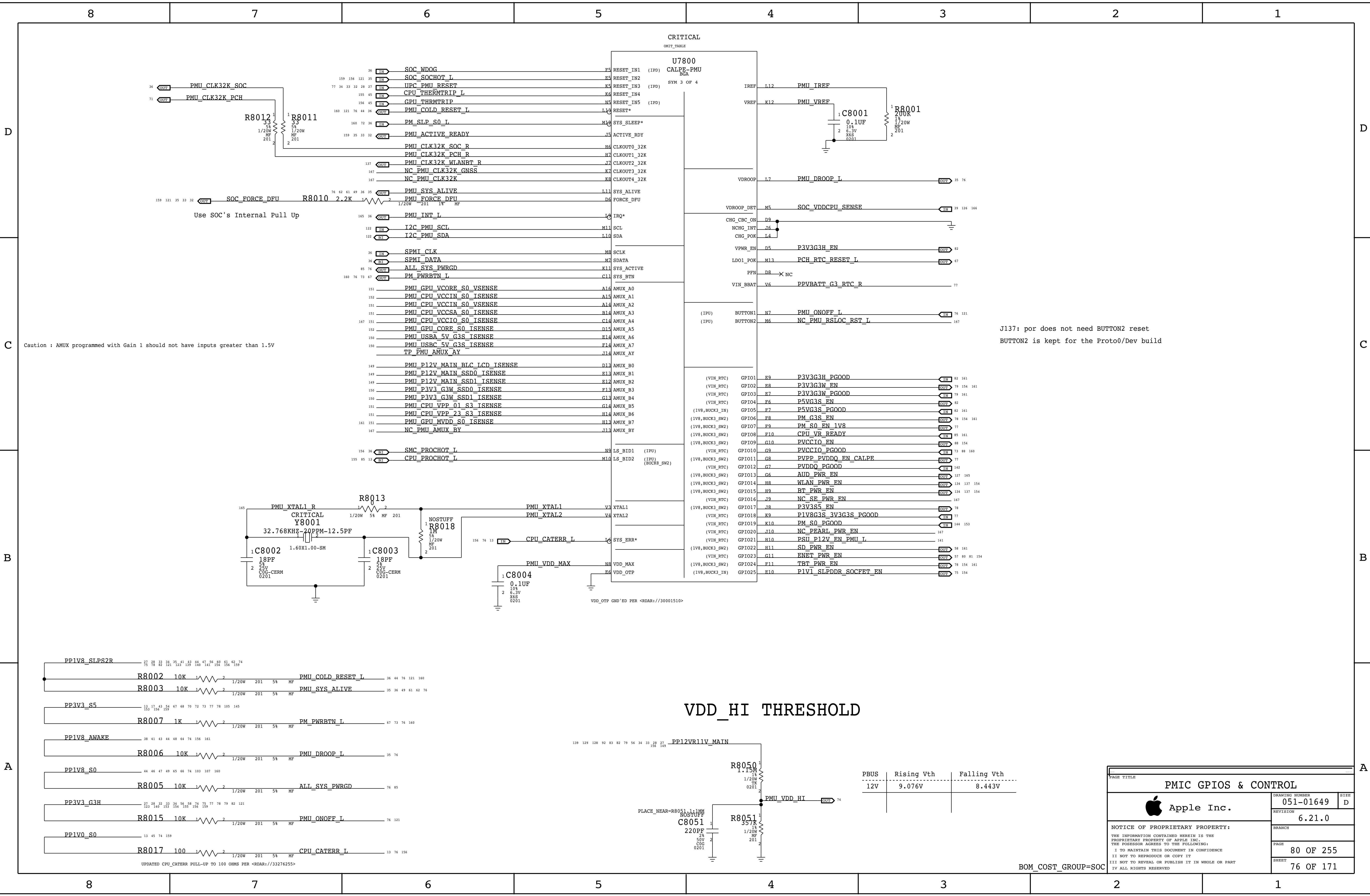
NOTE: JTAG PINS CAN BE LEFT FLOATING ON PCH AND CPU

FROM NEXPERIA, 05/08/17 EMAIL: WITH VCC HIGH AND NO INPUT VOLTAGE, LEAKAGE VALUES IN NA RANGE WITH WORST CASE ZIN, AND LEAKAGE IS TO GND

BOM_COST_GROUP=CPU & CHIPSET

XDP		
 Apple Inc.	DRAWING NUMBER	051-01649
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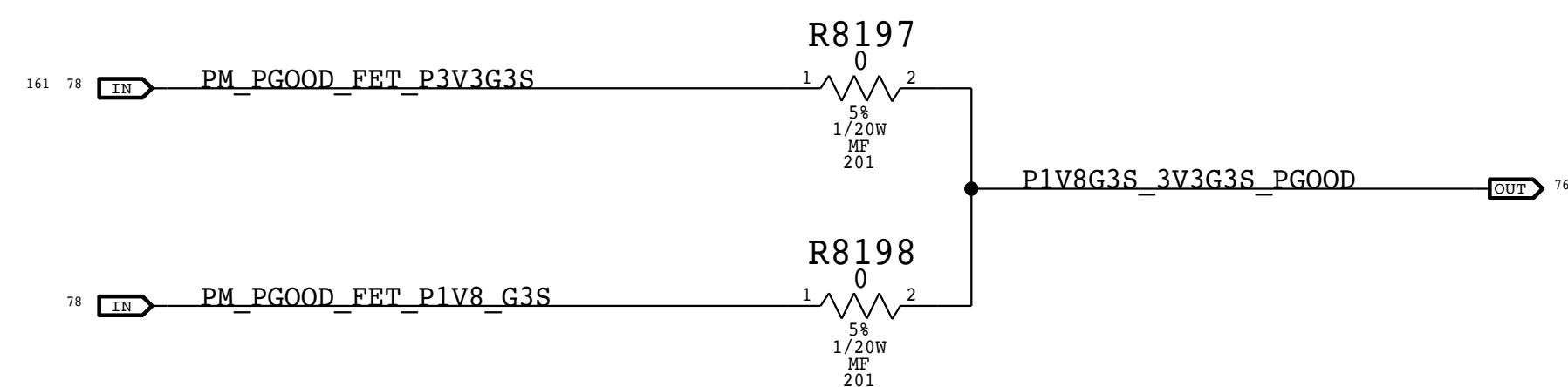
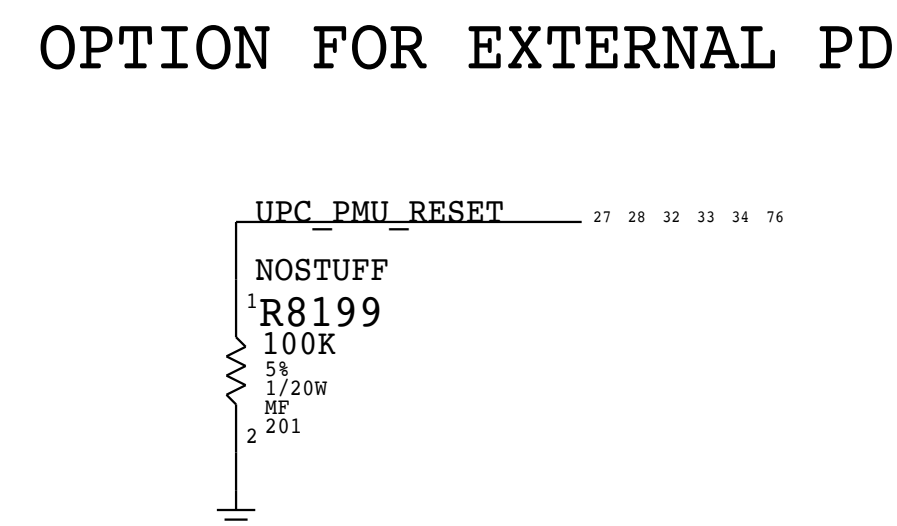
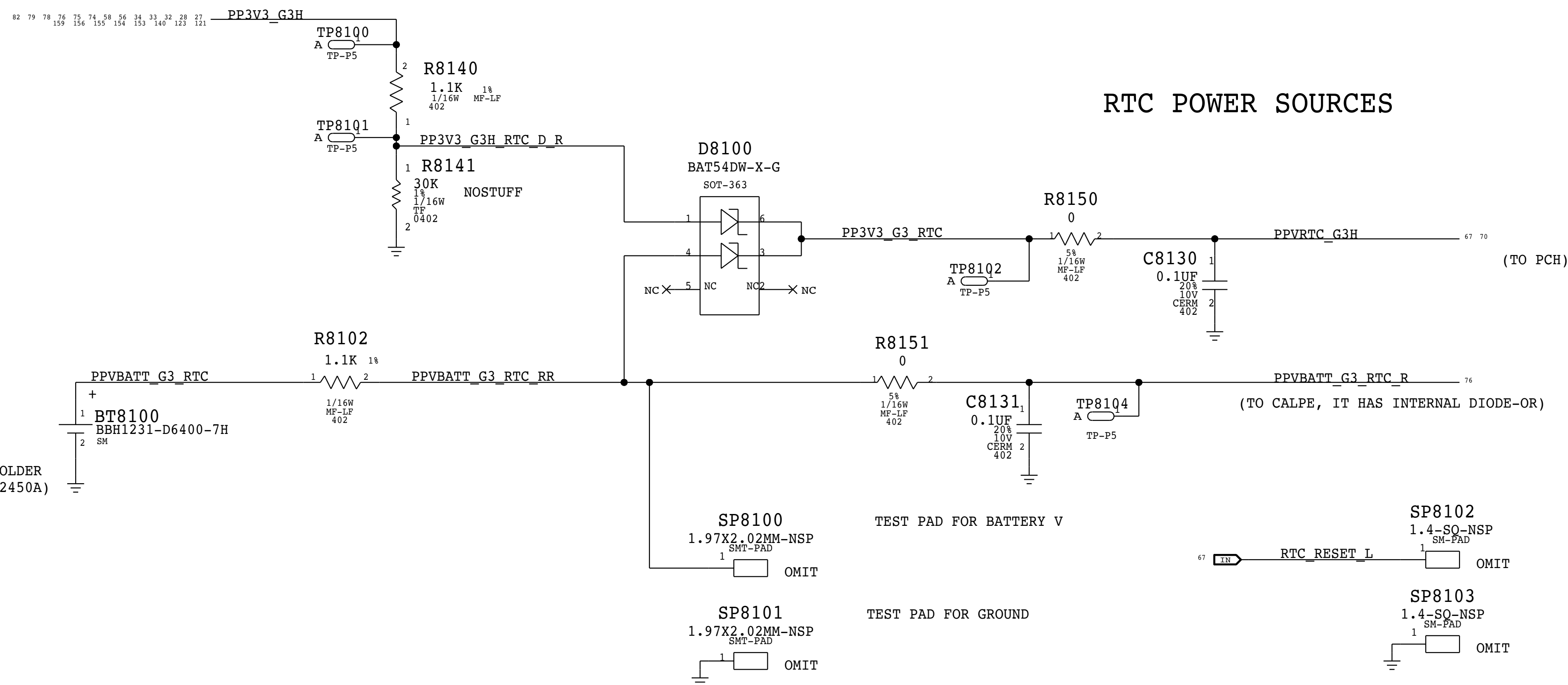
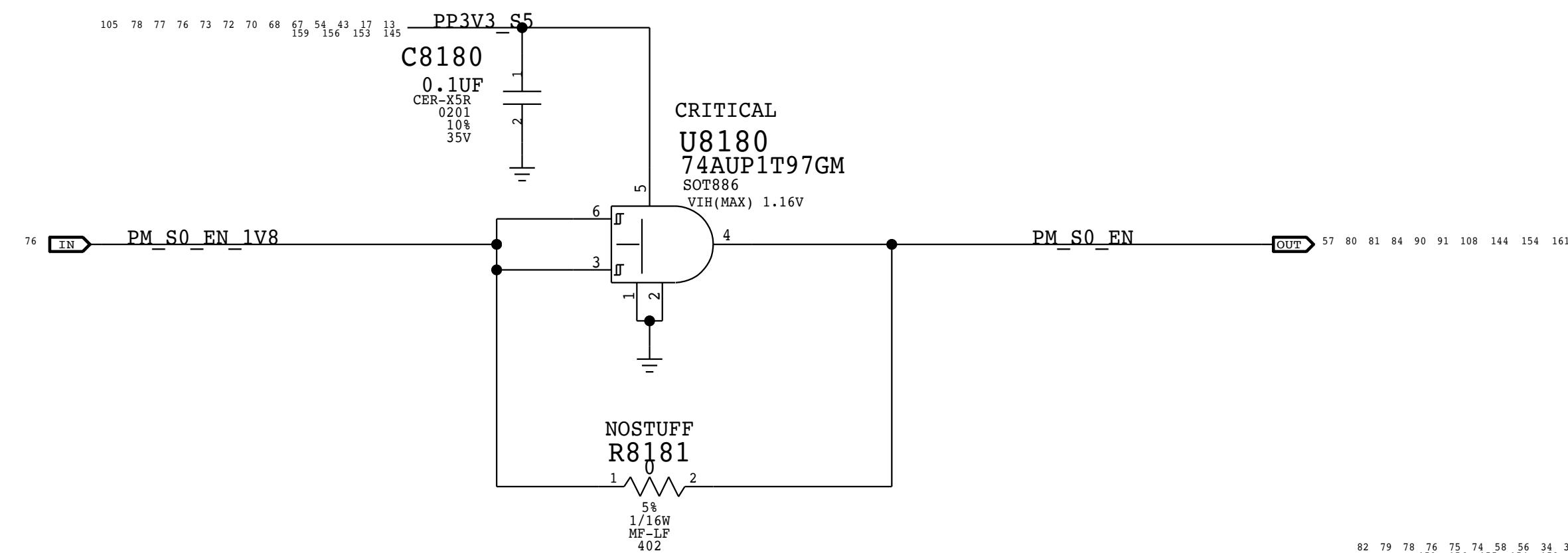
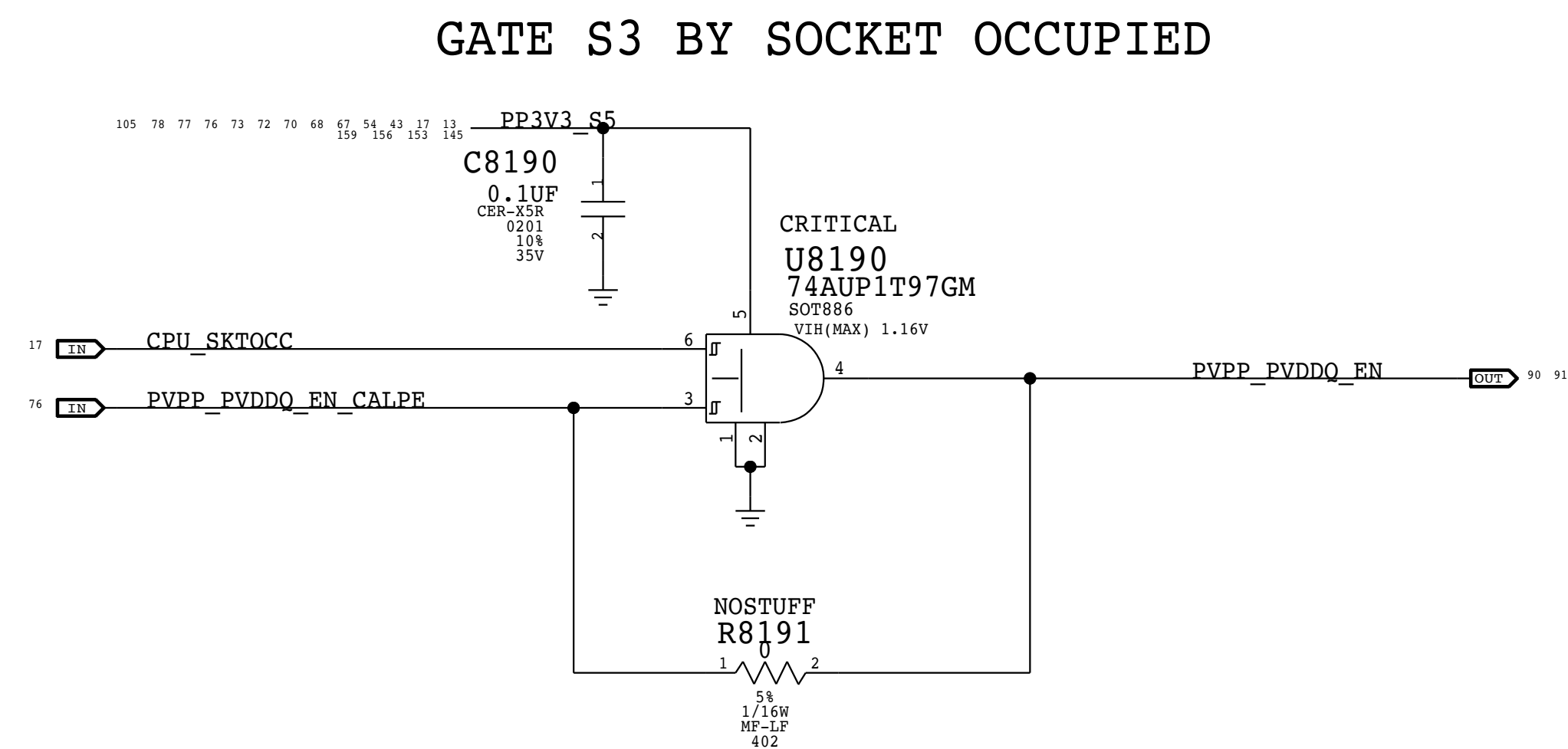
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VDD_HI THRESHOLD

PBUS	Rising Vth	Falling Vth
12V	9.076V	8.443V

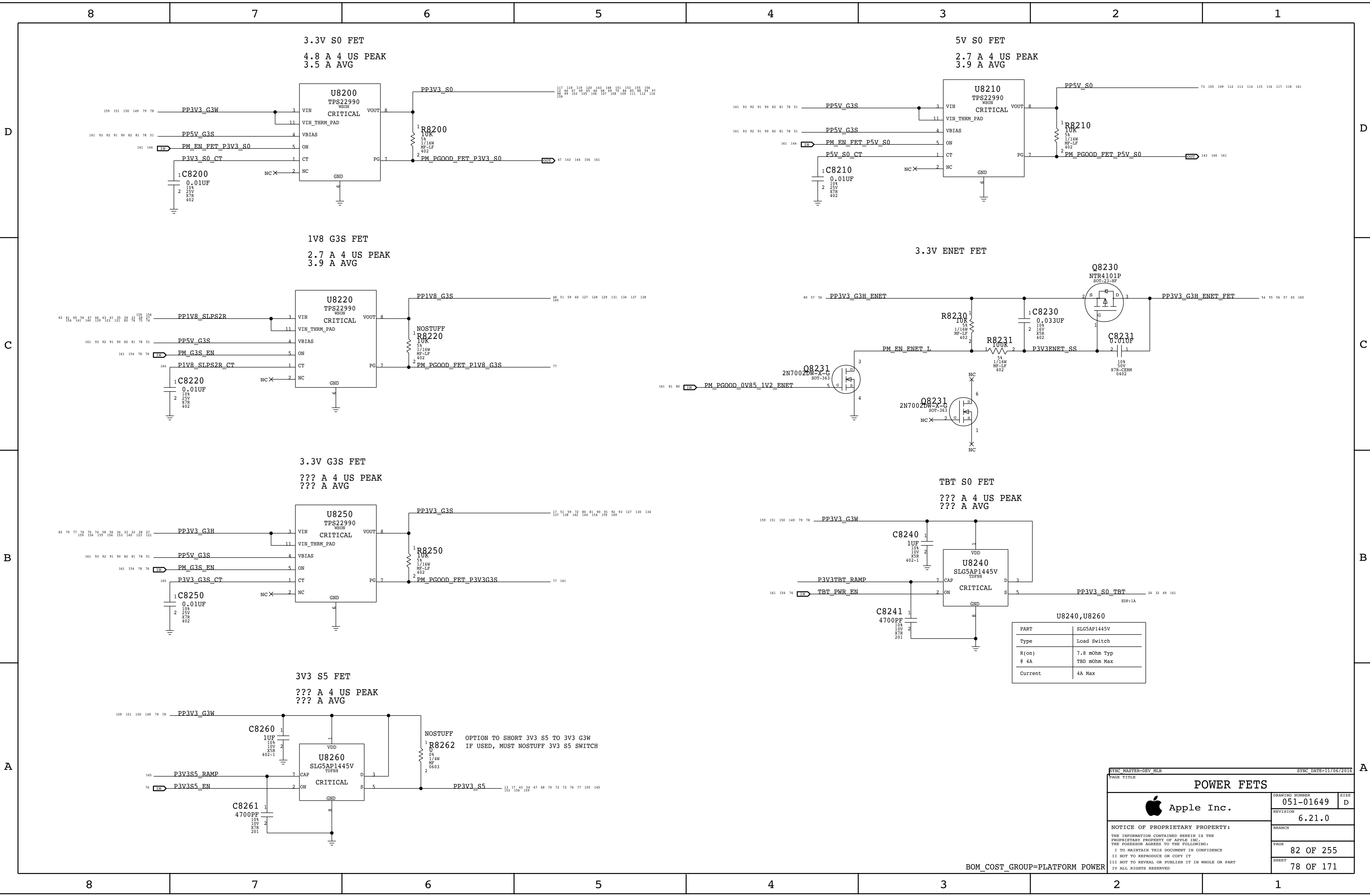
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PMIC GPIOs & CONTROL		
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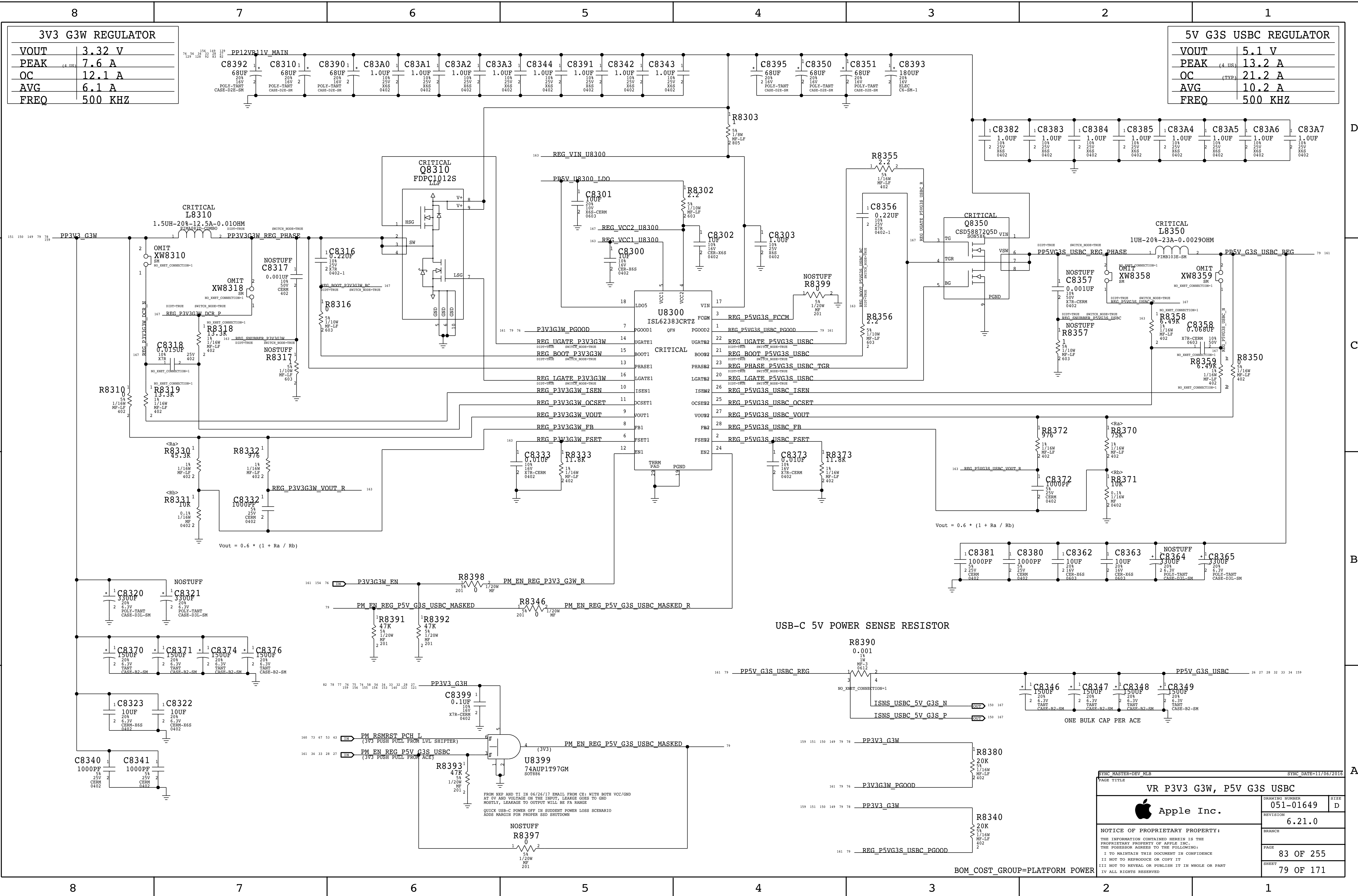
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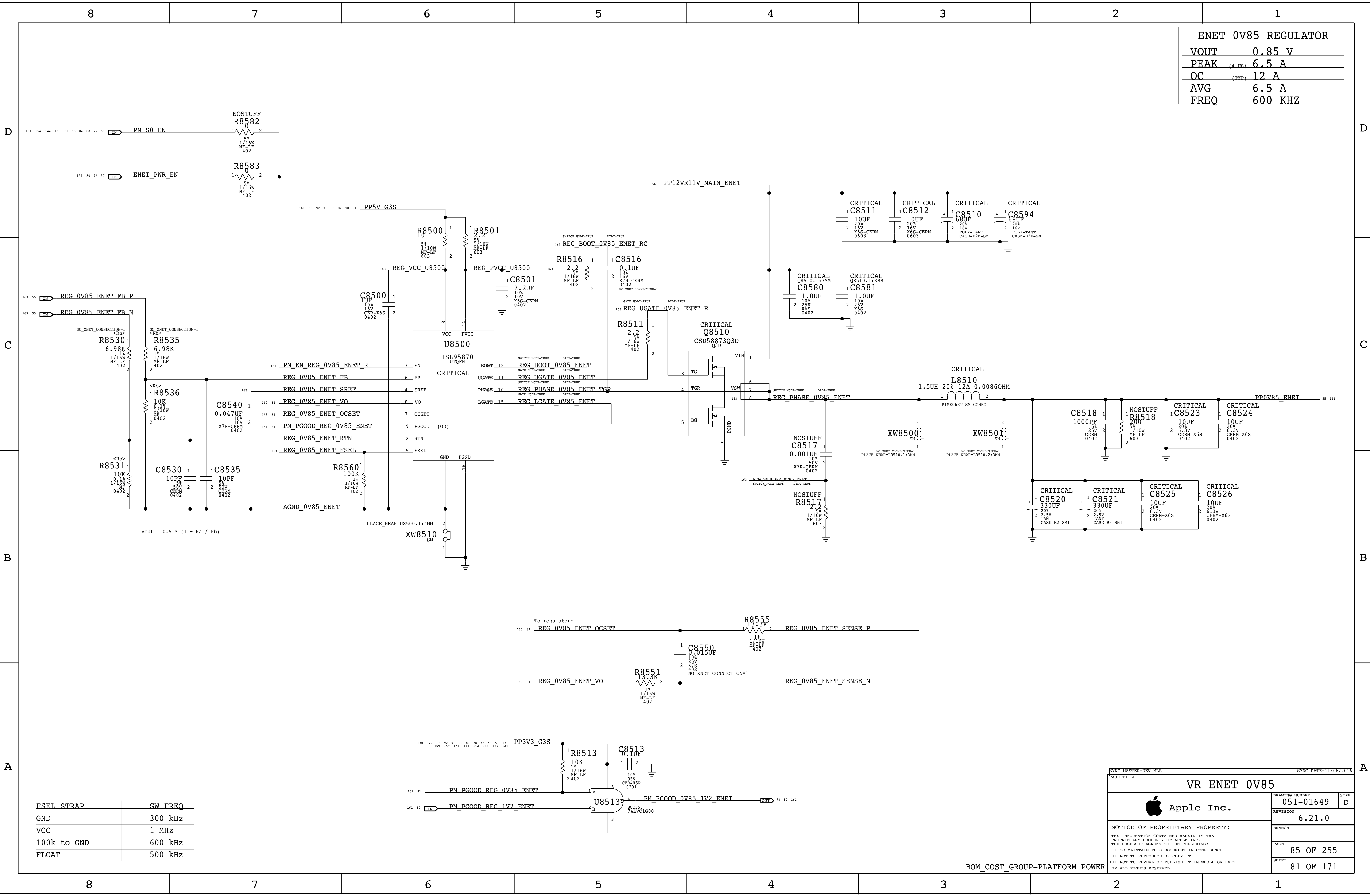


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		SHEET		77 OF 171	

BOM_COST_GROUP=SOC





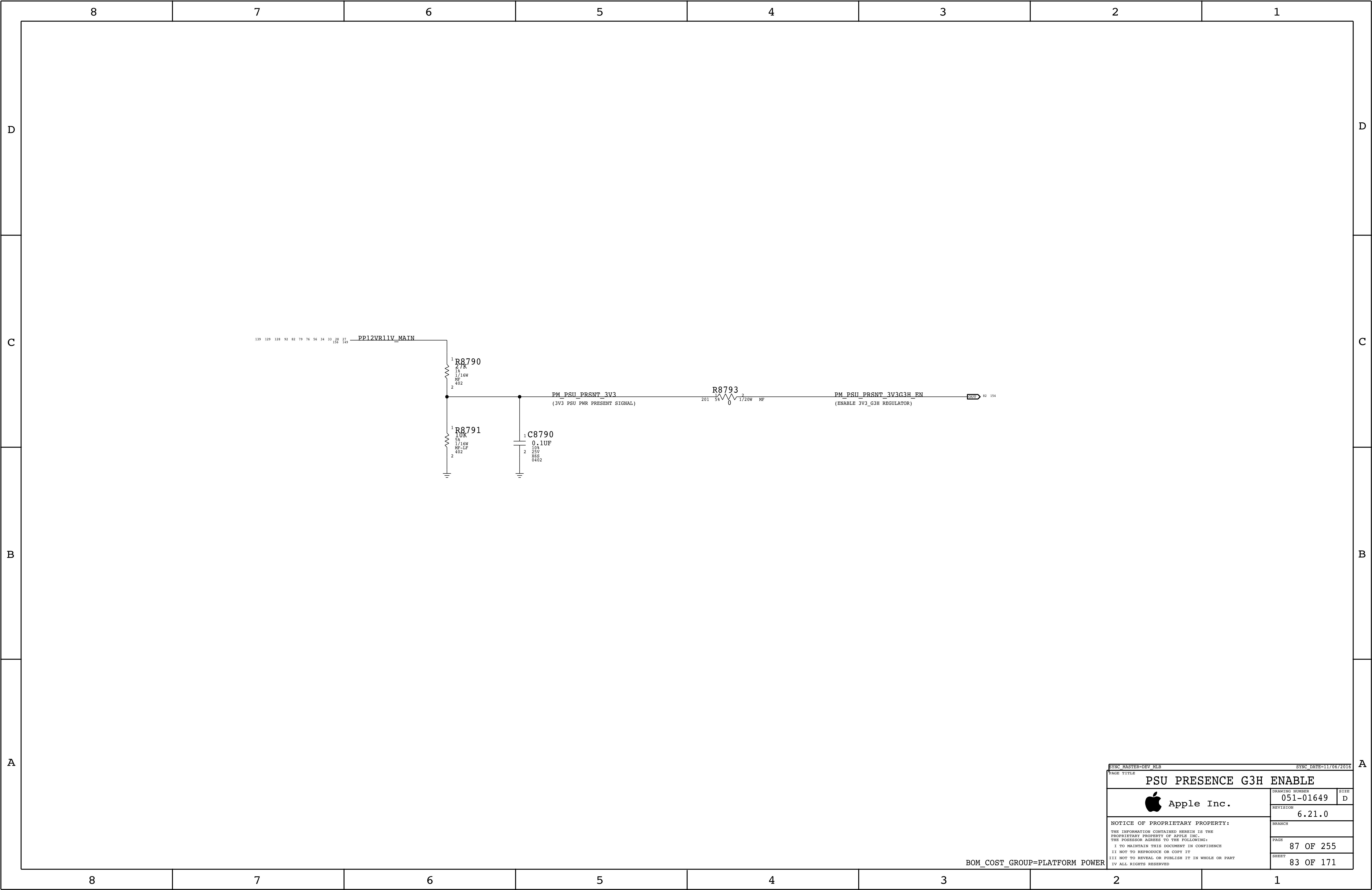


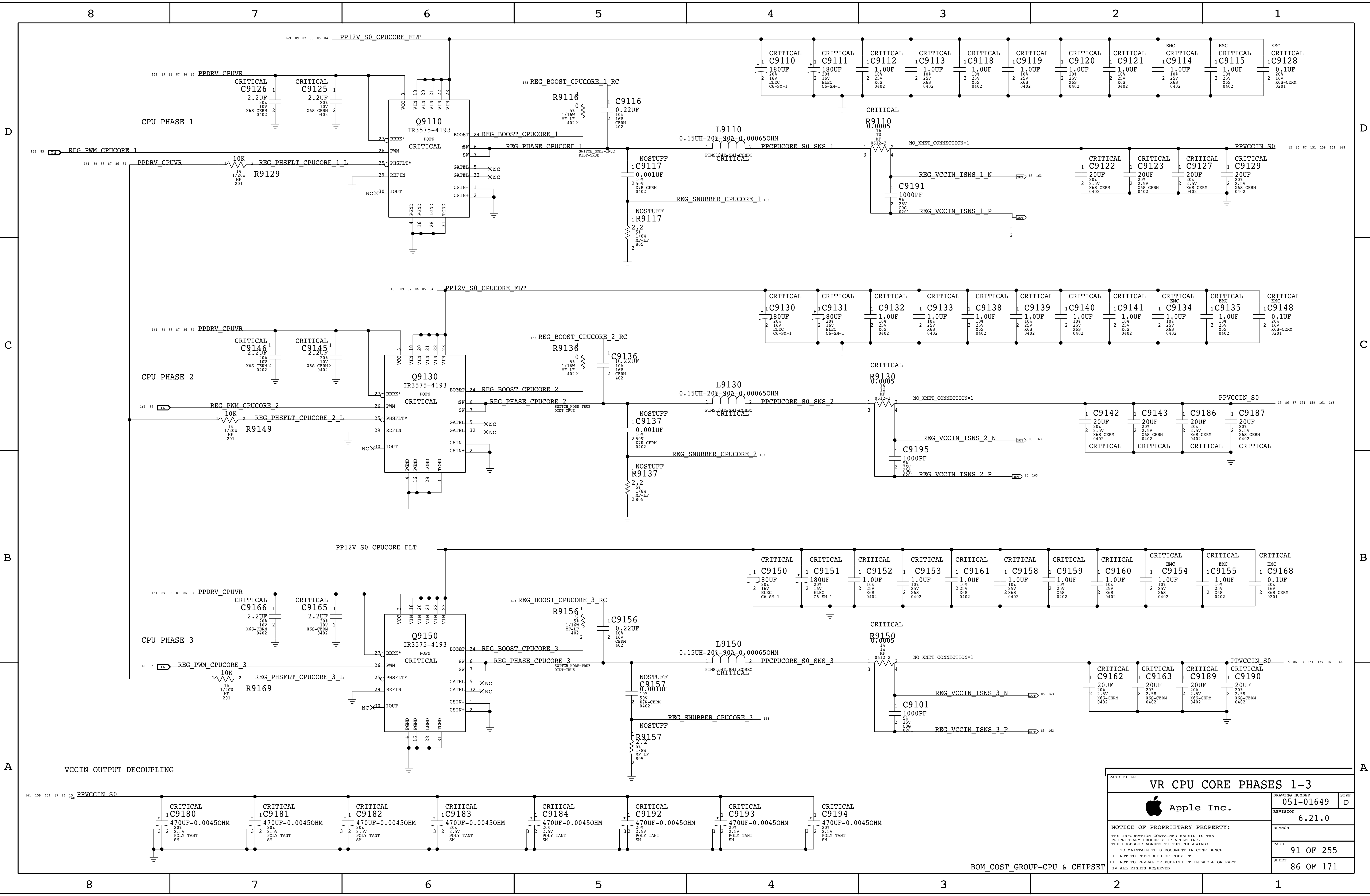
ENET 0V85 REGULATOR	
VOUT	0.85 V
PEAK	6.5 A
OC	12 A
AVG	6.5 A
FREQ	600 KHz

FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz

VR ENET 0V85		
	DRAWING NUMBER	051-01649
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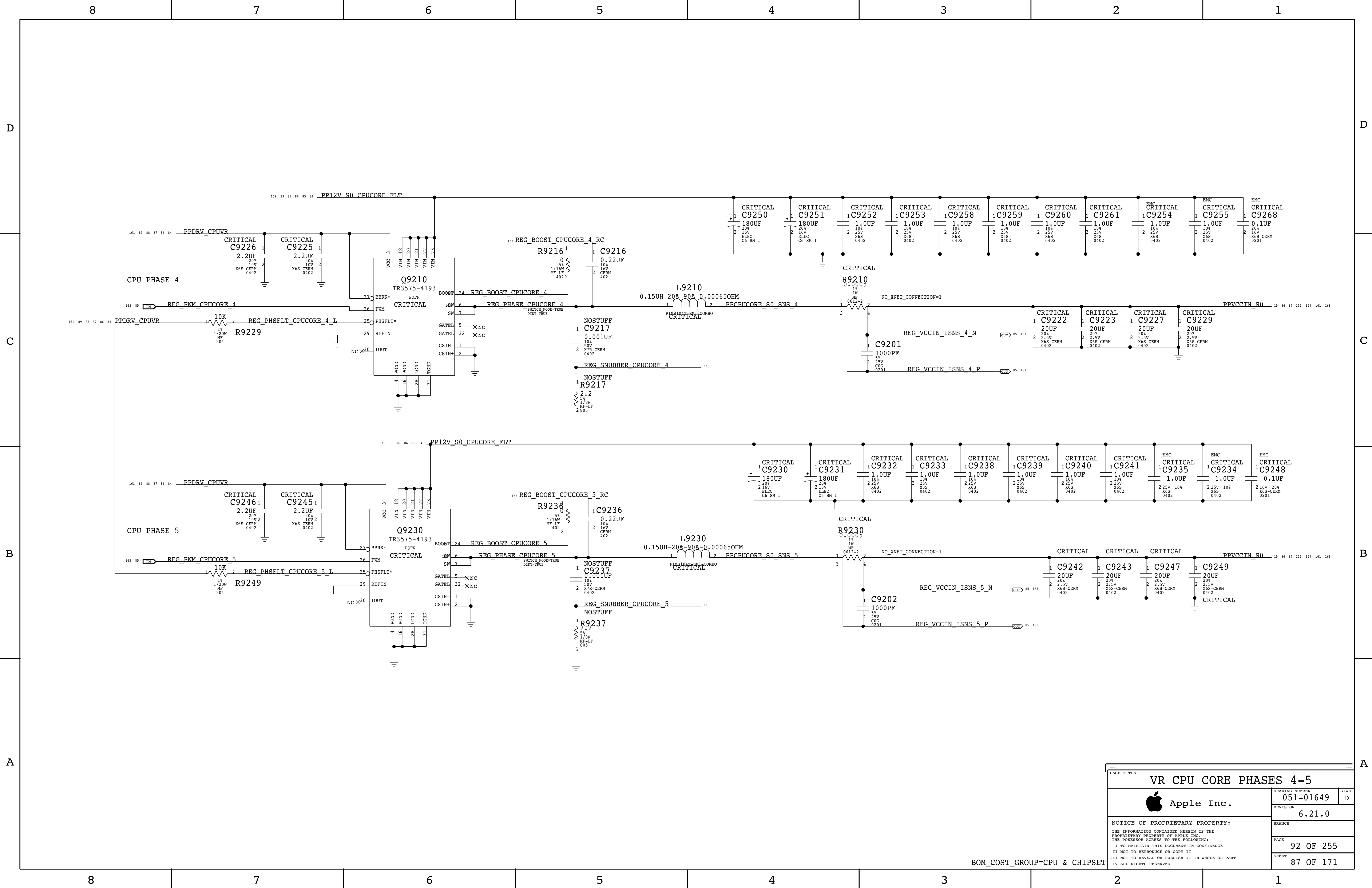
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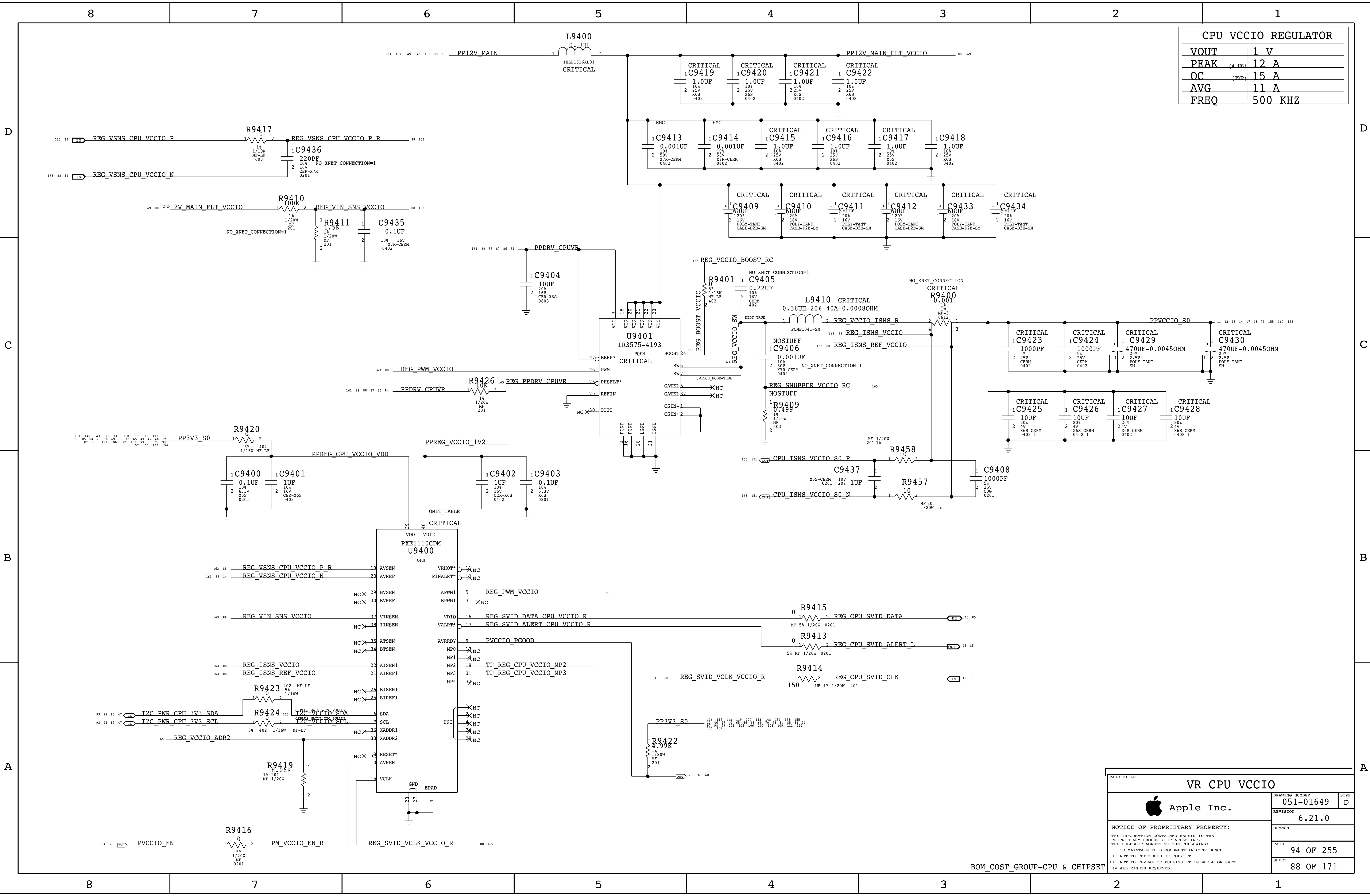
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VR CPU CORE PHASES 1-3		
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BOM_COST_GROUP=CPU & CHIPSET



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VR CPU CORE PHASES 4-5		
 Apple Inc.	DRAWING NUMBER	051-01649
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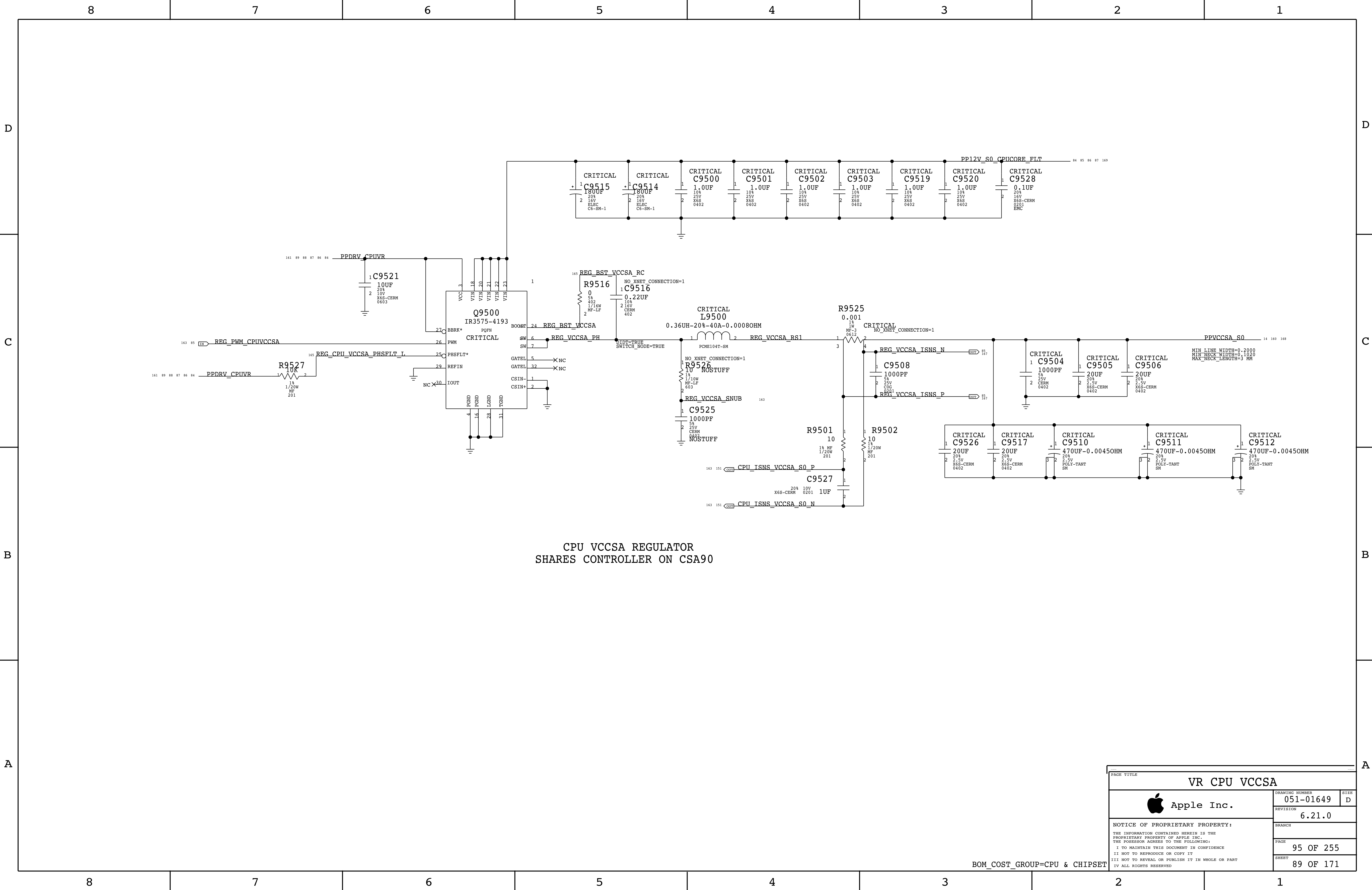
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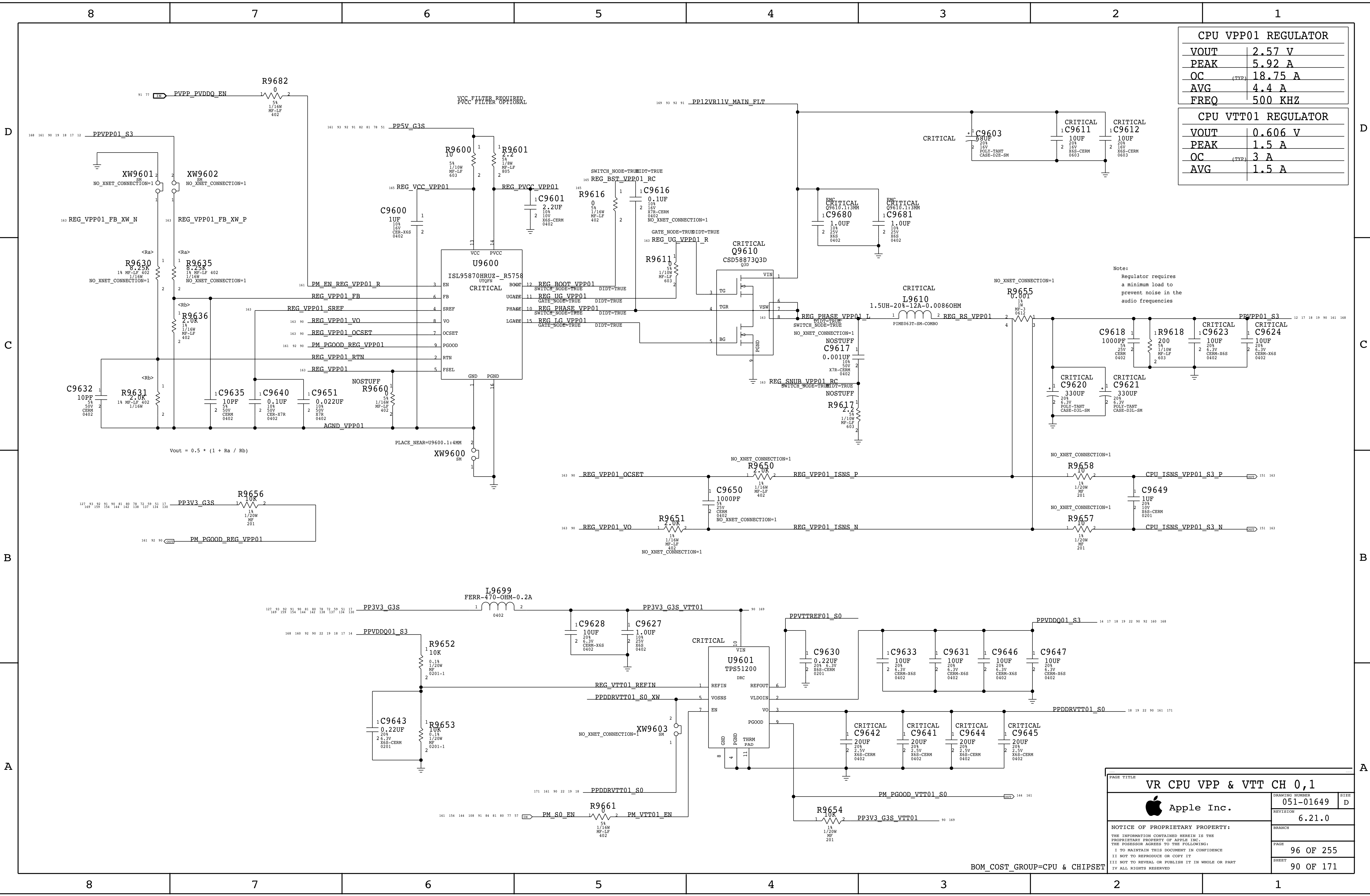


CPU VCCIO REGULATOR	
VOUT	1 V
PEAK	(4 US) 12 A
OC	(TYP) 15 A
AVG	11 A
FREQ	500 KHZ

VR CPU VCCIO		
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BOM_COST_GROUP=CPU & CHIPSET





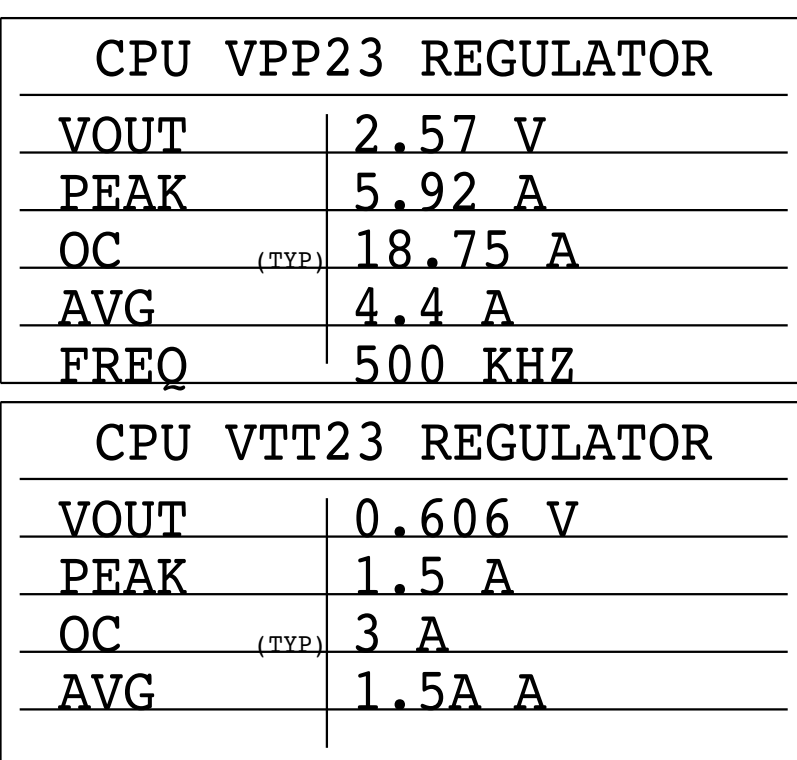
CPU VPP01 REGULATOR	
VOUT	2.57 V
PEAK	5.92 A
OC	18.75 A (TYP)
AVG	4.4 A
FREQ	500 KHZ

CPU VTT01 REGULATOR	
VOUT	0.606 V
PEAK	1.5 A
OC	3 A (TYP)
AVG	1.5 A

Note:
Regulator requires
a minimum load to
prevent noise in the
audio frequencies

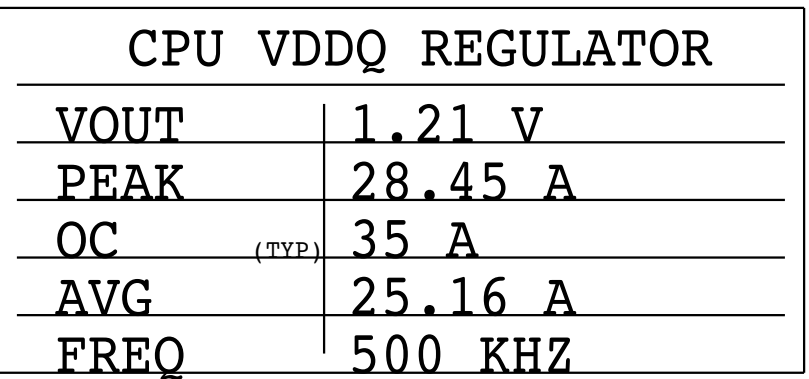
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VR CPU VPP & VTT CH 0,1		
	DRAWING NUMBER	051-01649
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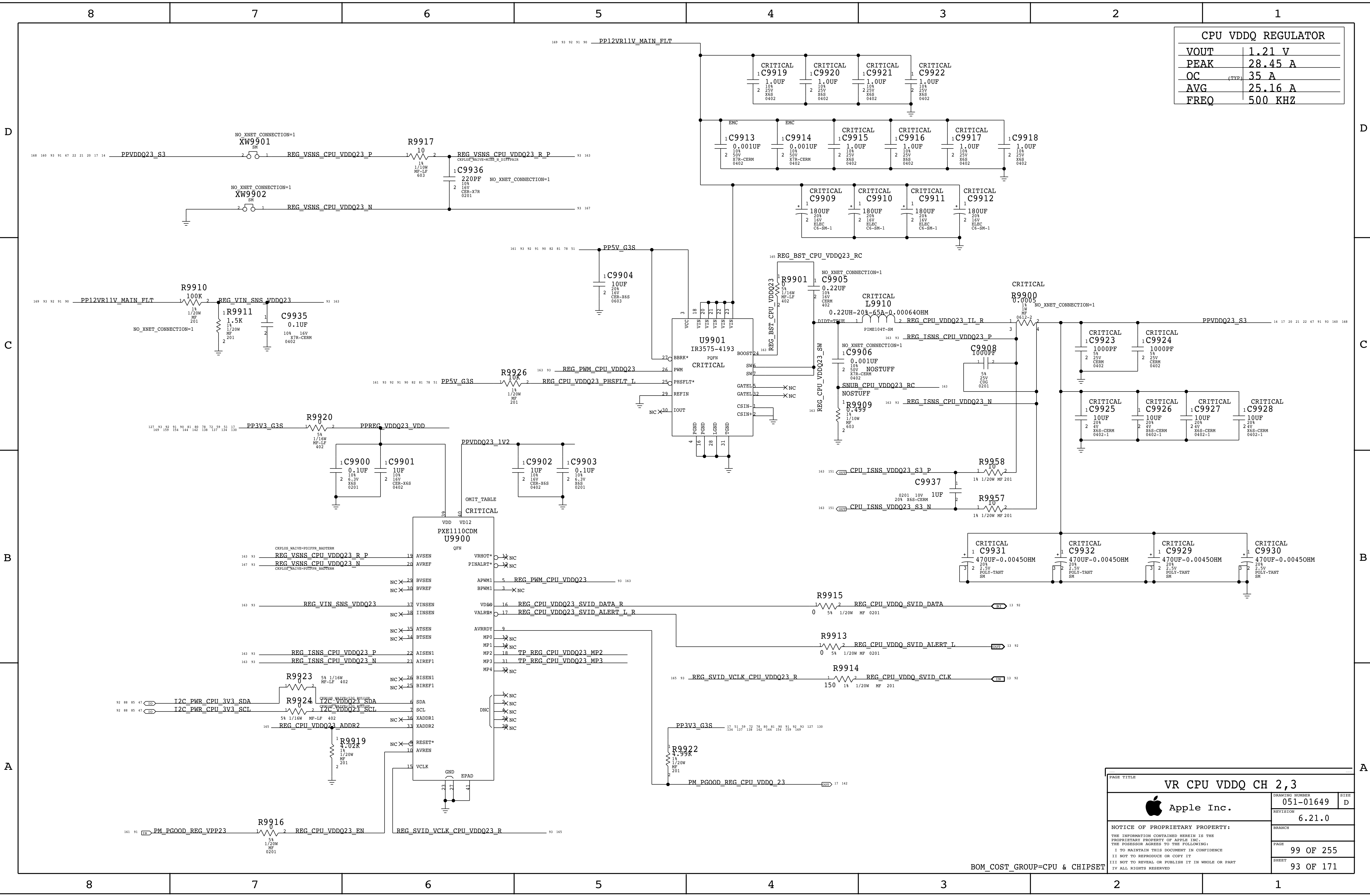
BOM_COST_GROUP=CPU & CHIPSET



CPU VPP23 REGULATOR	
VOUT	2.57 V
PEAK	5.92 A
OC (TYP)	18.75 A
AVG	4.4 A
FREQ	500 KHZ

CPU VTT23 REGULATOR	
VOUT	0.606 V
PEAK	1.5 A
OC (TYP)	3 A
AVG	1.5A A

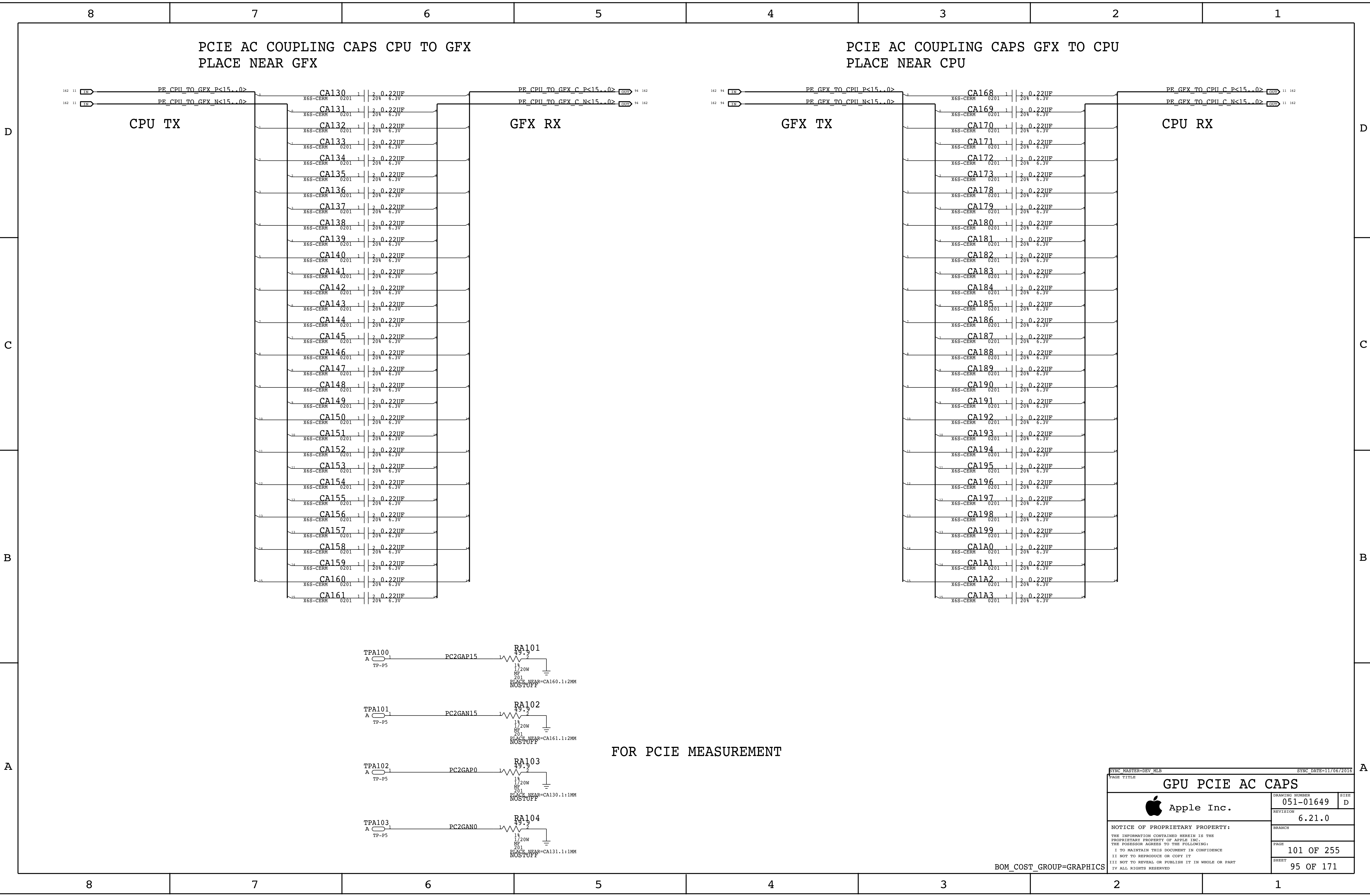


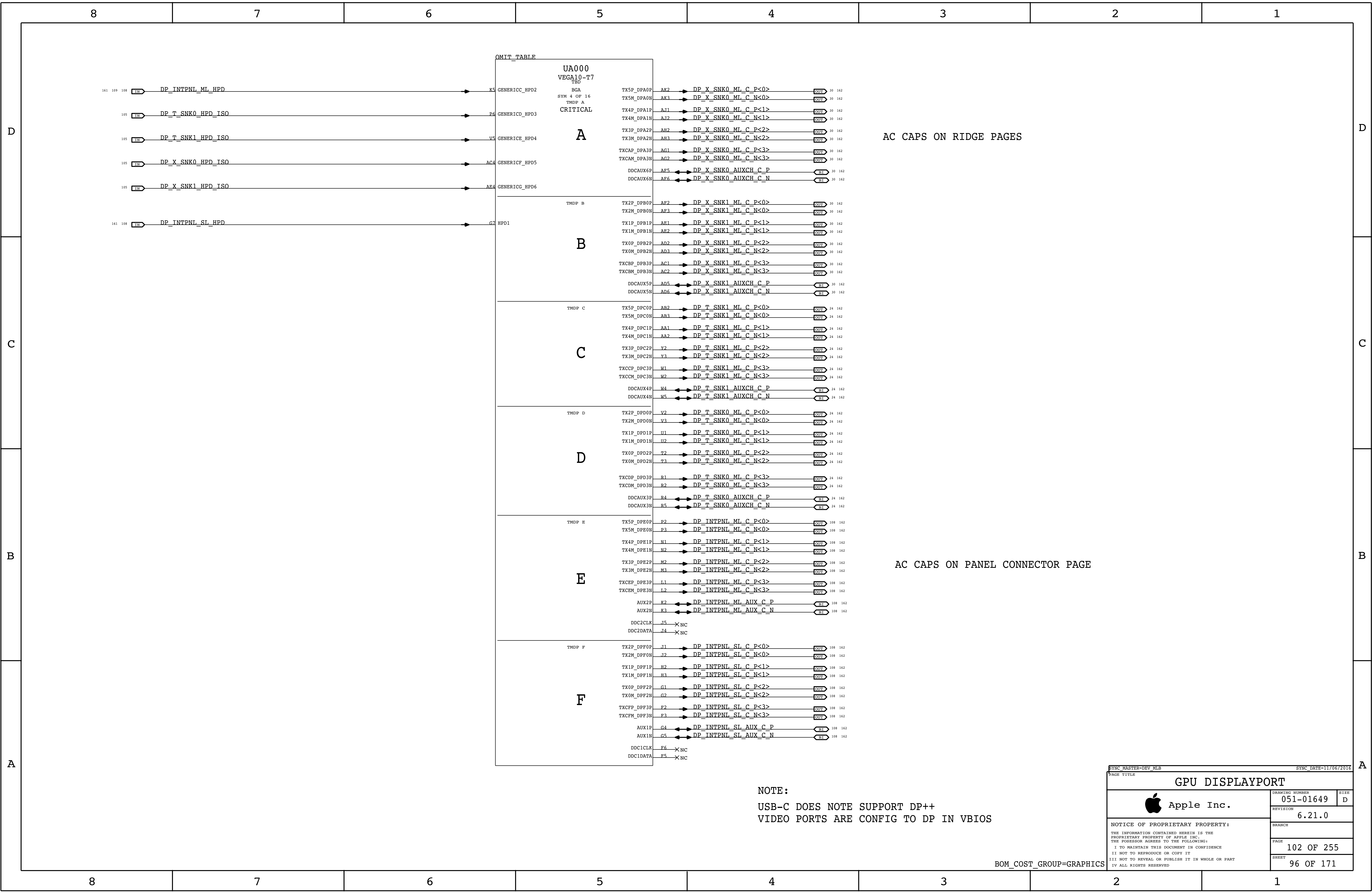


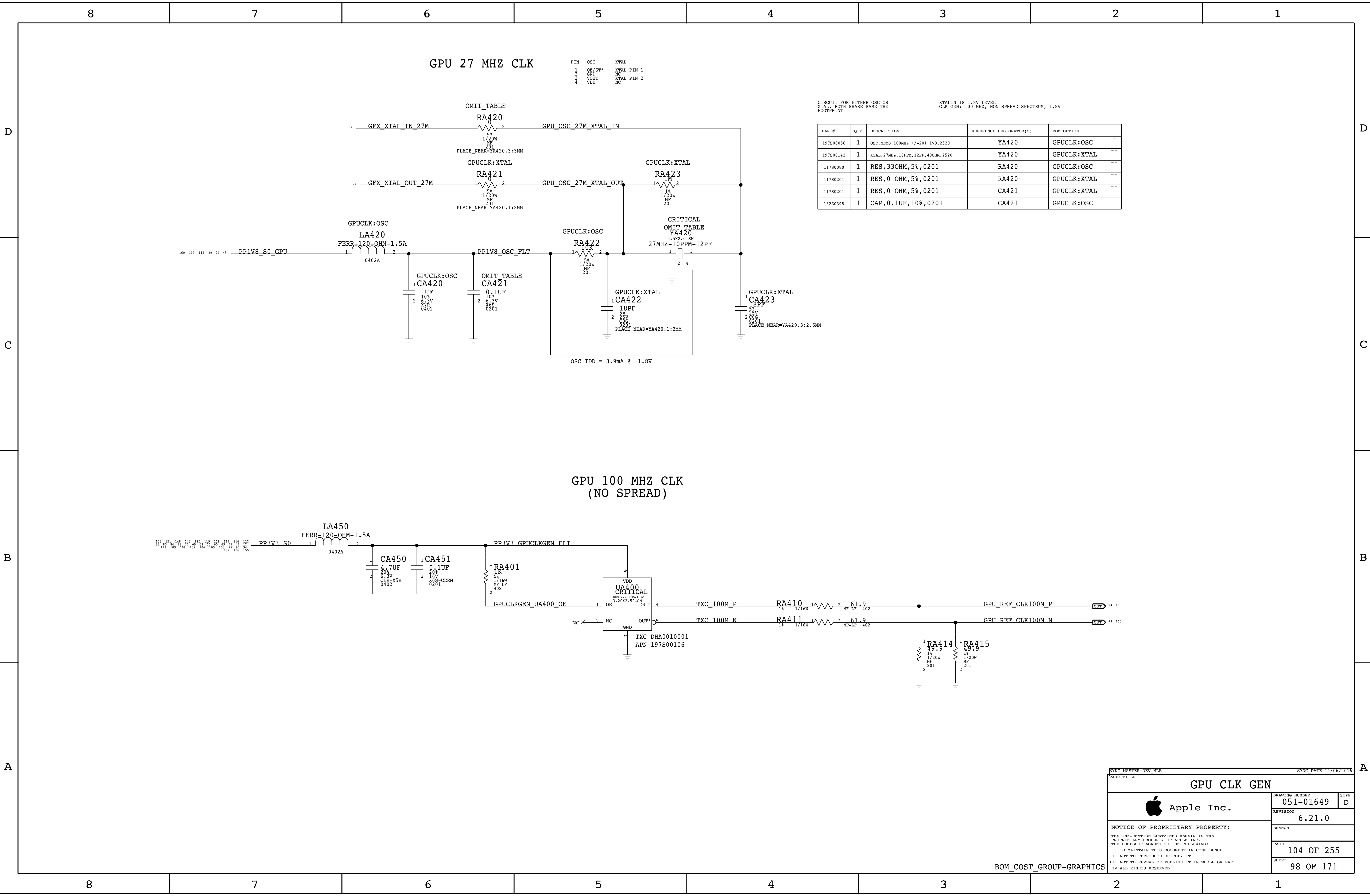
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AVG	25.16 A
FREQ	500 KHZ

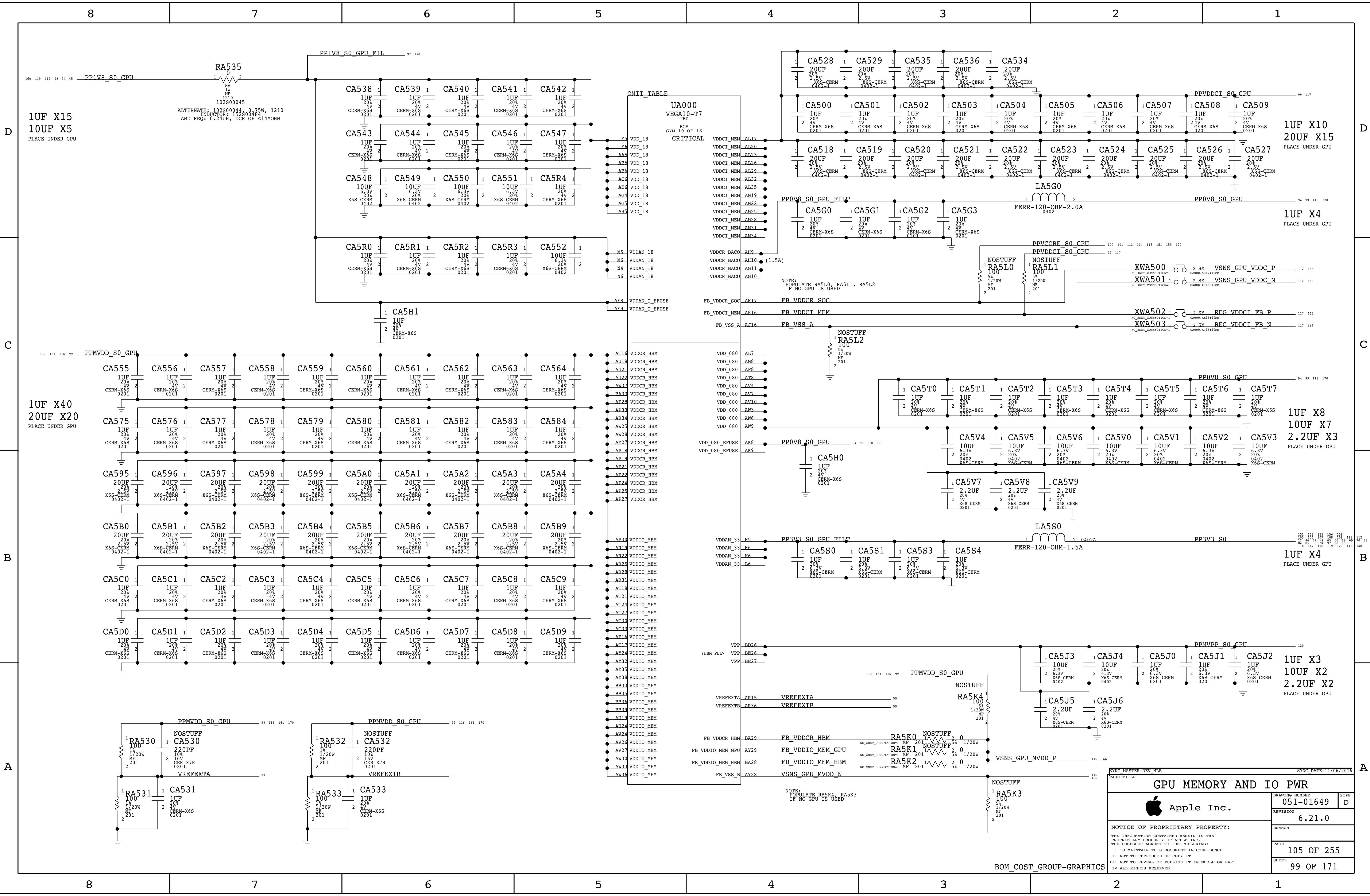
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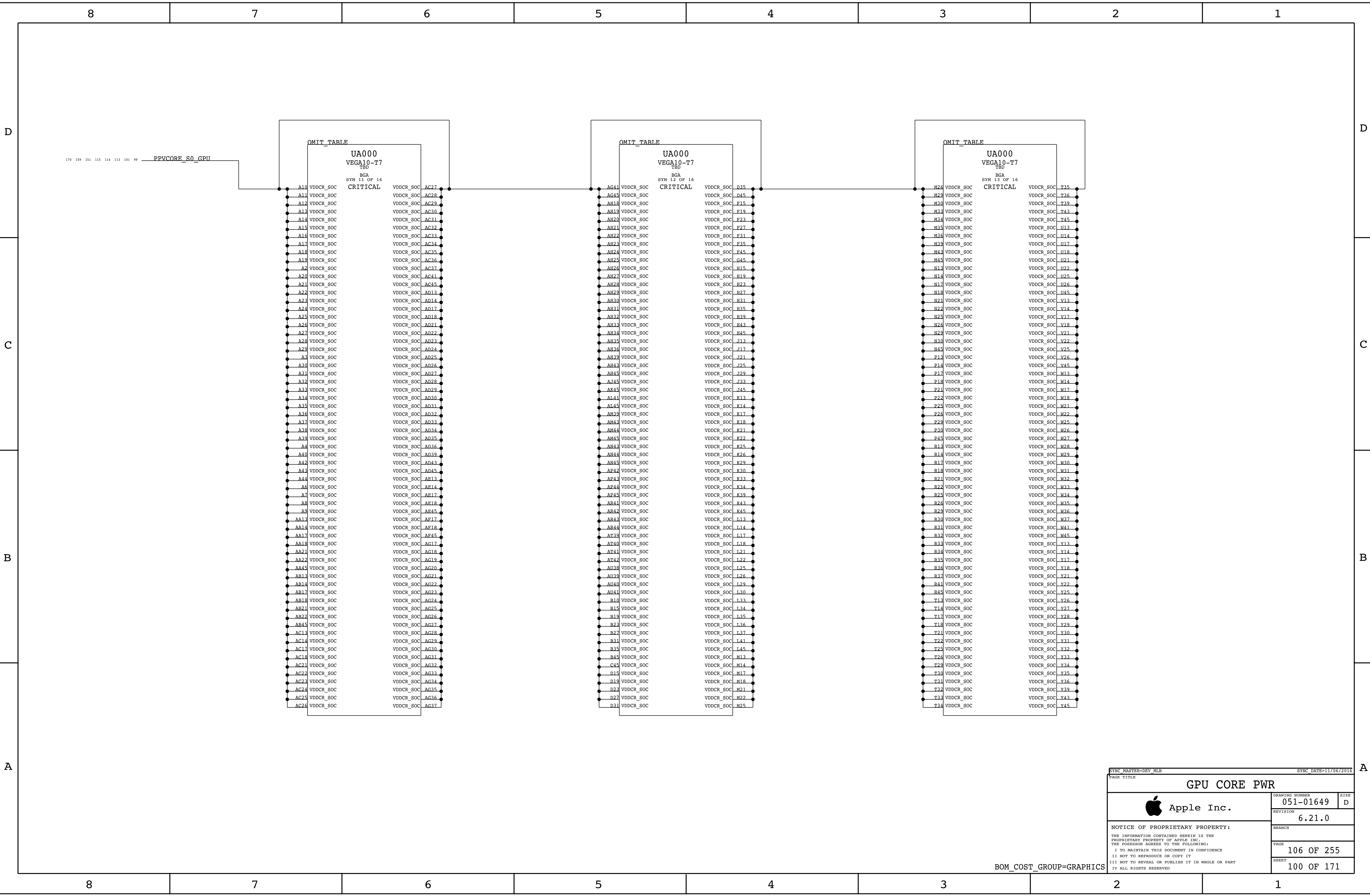


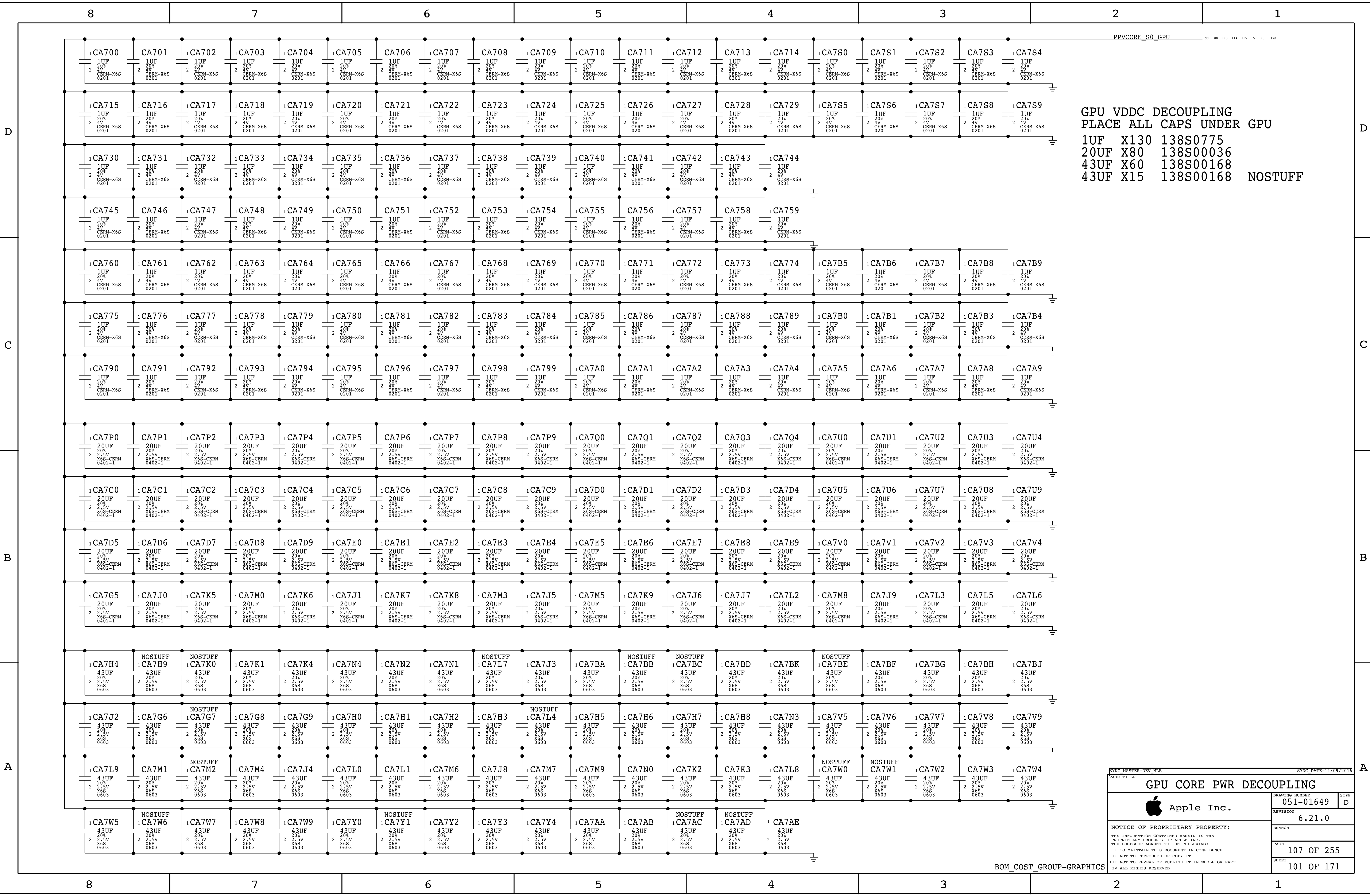




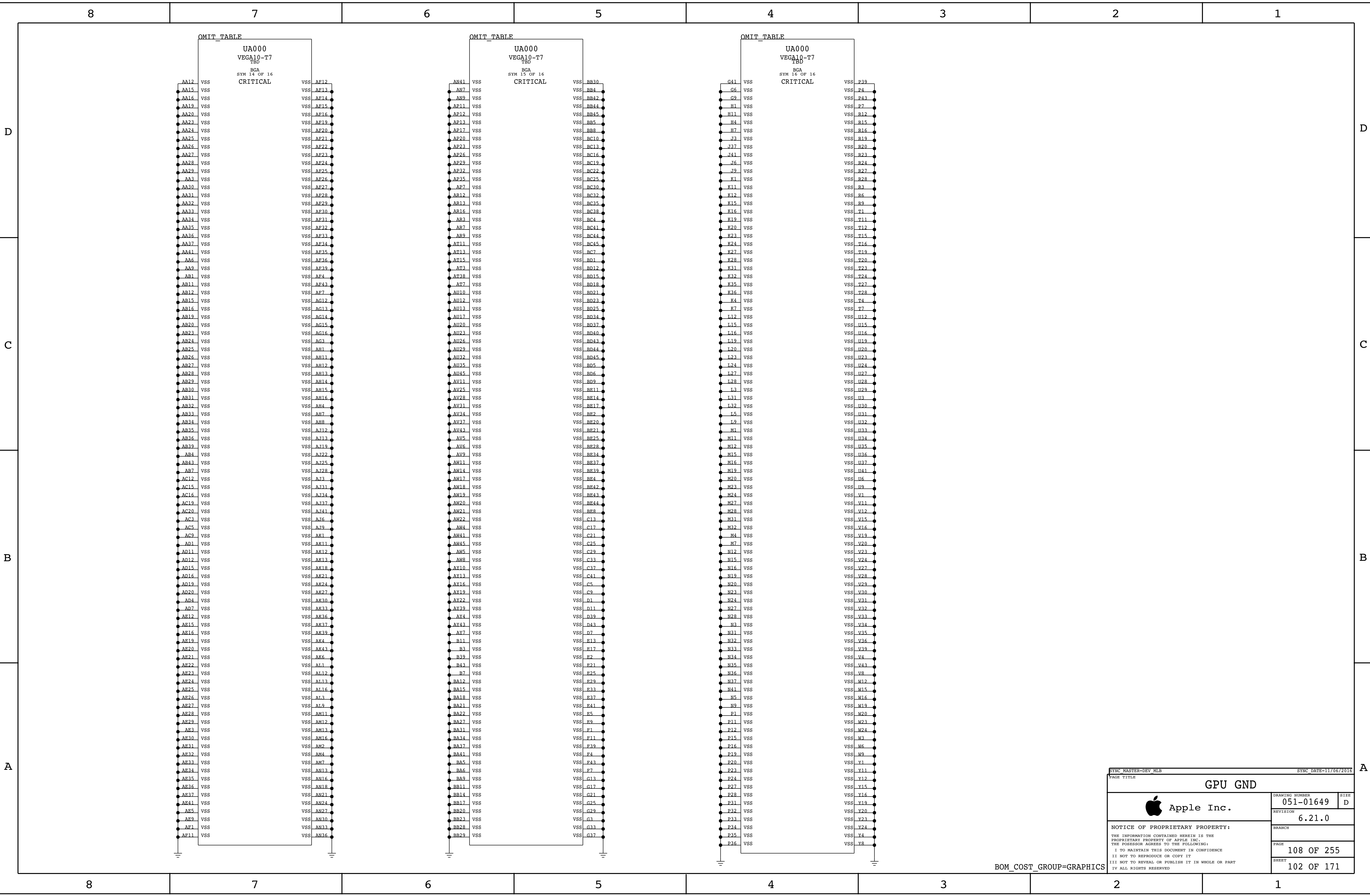


GPU MEMORY AND IO PWR		
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SYNC MASTER=DEV_RLB			SYNC DATE=11/09/2016		
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 Apple Inc.			DRAWING NUMBER	051-01649	
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SYNC MASTER=DEV MLB

SYNC DATE=11/06/2016

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GPU GND

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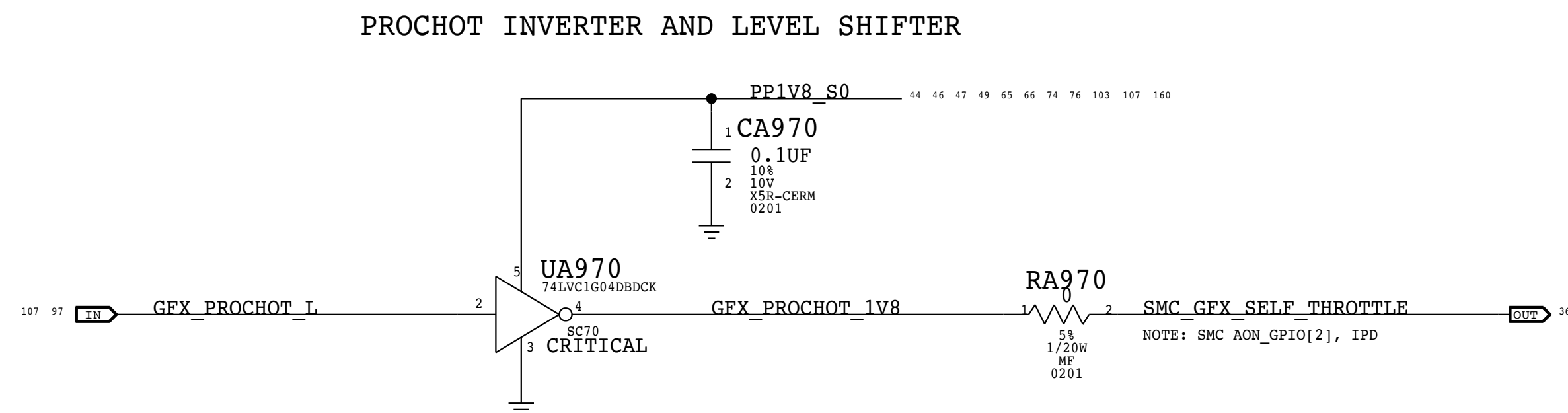
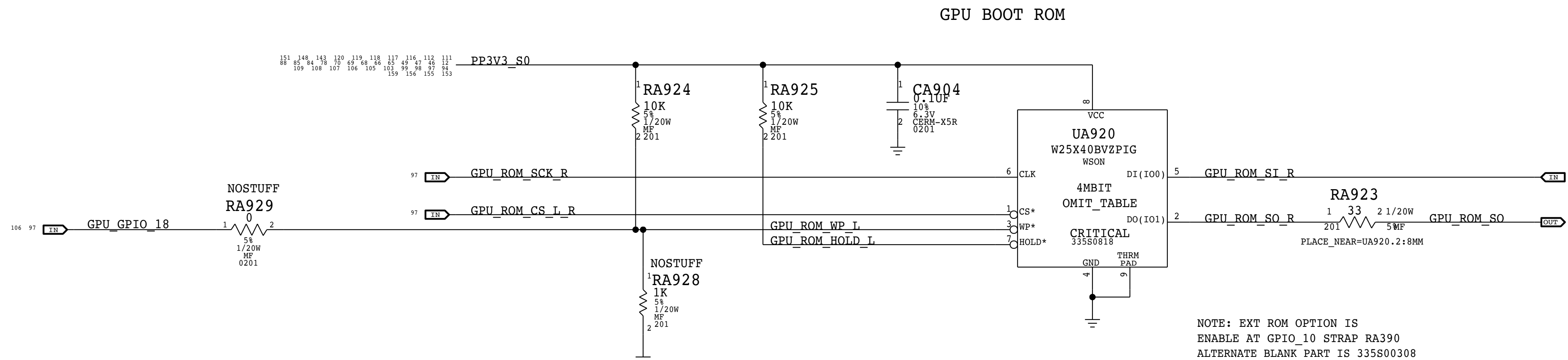
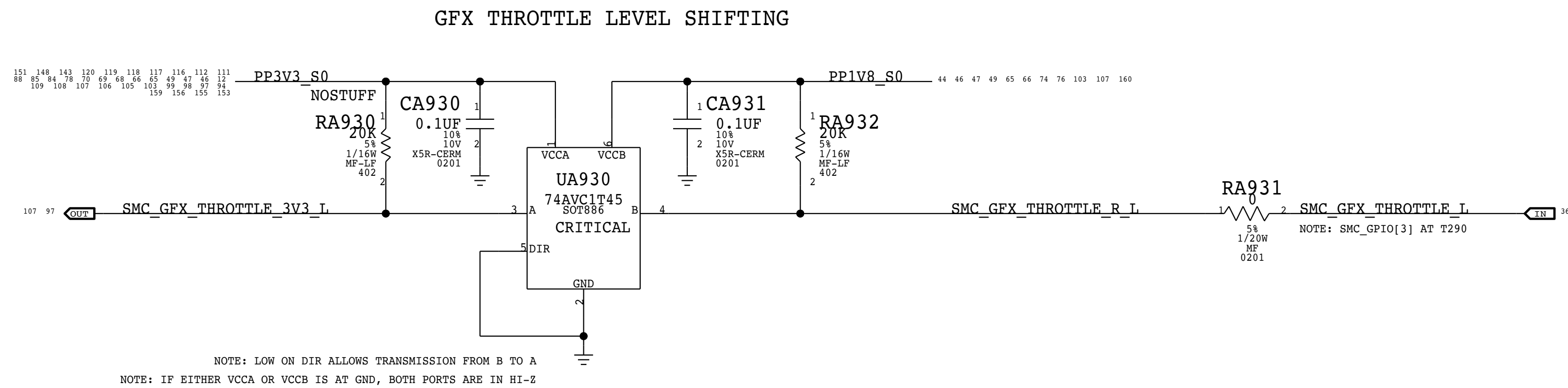
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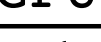
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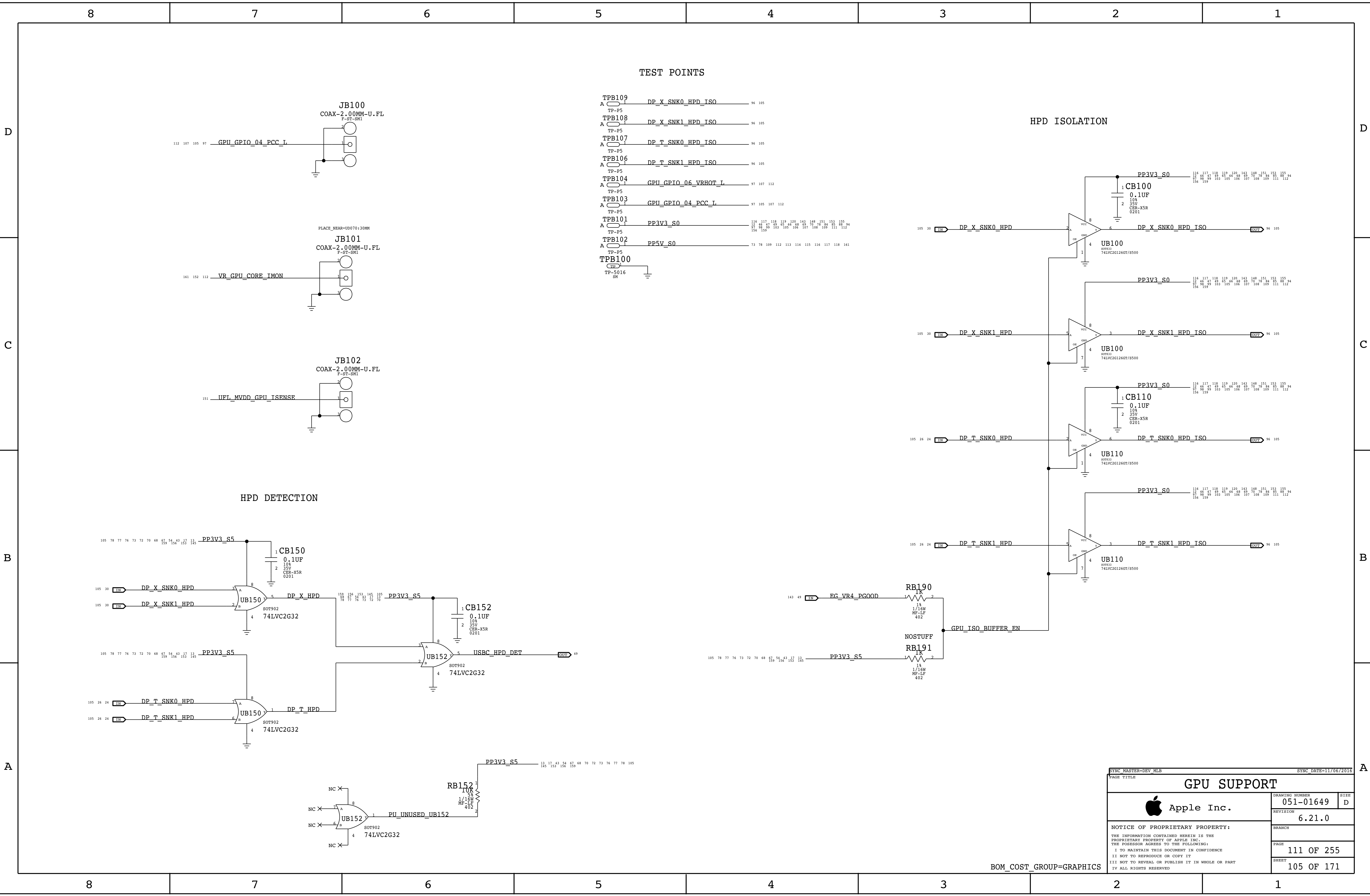
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SYNC MASTER=DEV_MLB		SYNC DATE=11/06/2016	
PAGE TITLE			
GPU ROM, TEMP, LVL SHIFT			
 Apple Inc.		GRANTING NUMBER	SIZE
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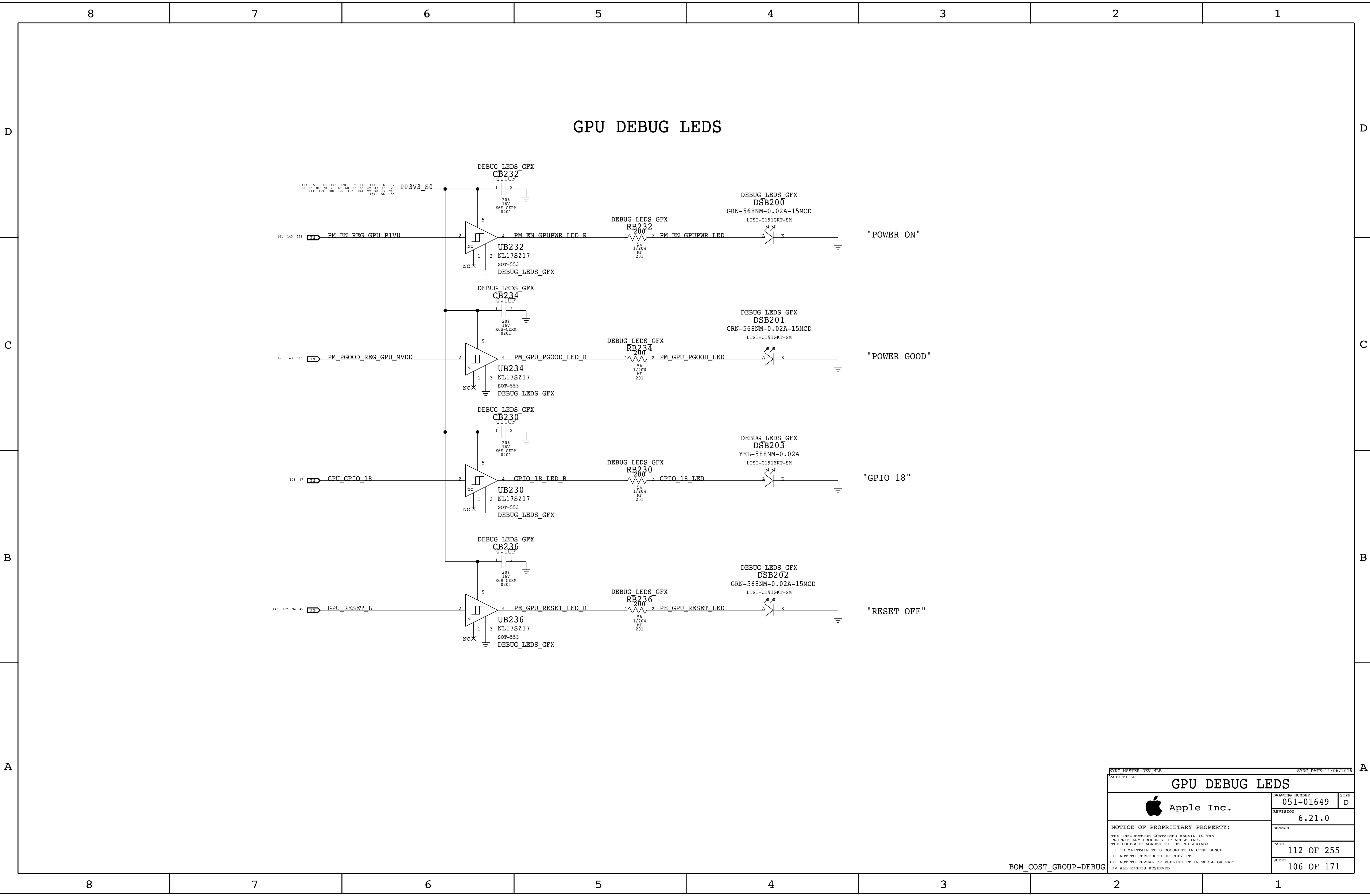
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UA000				UA000				UA000		UA000		UA000			
VEGA10-T7				VEGA10-T7				VEGA10-T7		VEGA10-T7		VEGA10-T7			
TBD				TBD				TBD		TBD		TBD			
BGA				BGA				BGA		BGA		BGA			
SYM 5 OF 16				SYM 6 OF 16				SYM 8 OF 16		SYM 8 OF 16		SYM 9 OF 16			
CRITICAL				CRITICAL				CRITICAL		CRITICAL		CRITICAL			
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NC X L11 DFTTIO_1		DFTTIO_97 B16 X NC		NC X J35 DFTTIO_193		DFTTIO_289 L39 X NC		NC X AK26 HBM2_DAP_1		HBM2_DAP_1 AJ36 X NC		NC X BC26 RSVD		RSVD AG43 X NC	
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NC X F13 DFTTIO_50		DFTTIO_146 D26 X NC		NC X J42 DFTTIO_242		DFTTIO_338 AT45 X NC		NC X AJ24 HBM2_DAP_50		HBM2_DAP_50 AL30 X NC		NC X BE40 RSVD		RSVD AL50 X NC	
NC X J16 DFTTIO_51		DFTTIO_147 C32 X NC		NC X F37 DFTTIO_243		DFTTIO_339 AM42 X NC		NC X AJ24 HBM2_DAP_51		HBM2_DAP_51 AV29 X NC		NC X BE40 RSVD		RSVD AL51 X NC	
NC X C8 DFTTIO_52		DFTTIO_148 E30 X NC		NC X G40 DFTTIO_244		DFTTIO_340 AT44 X NC		NC X AJ24 HBM2_DAP_52		HBM2_DAP_52 AT29 X NC		NC X BE40 RSVD		RSVD AL52 X NC	
NC X F9 DFTTIO_53		DFTTIO_149 D36 X NC		NC X G39 DFTTIO_245		DFTTIO_341 AB45 X NC		NC X AJ24 HBM2_DAP_53		HBM2_DAP_53 AN29 X NC		NC X BE40 RSVD		RSVD AL53 X NC	
NC X J15 DFTTIO_54		DFTTIO_150 J23 X NC		NC X K40 DFTTIO_246		DFTTIO_342 AJ40 X NC		NC X AJ24 HBM2_DAP_54		HBM2_DAP_54 AR28 X NC		NC X BE40 RSVD		RSVD AL54 X NC	
NC X E12 DFTTIO_55		DFTTIO_151 C34 X NC		NC X J34 DFTTIO_247		DFTTIO_343 AY45 X NC		NC X AJ24 HBM2_DAP_55		HBM2_DAP_55 AK29 X NC		NC X BE40 RSVD		RSVD AL55 X NC	
NC X F11 DFTTIO_56		DFTTIO_152 B28 X NC		NC X L42 DFTTIO_248		DFTTIO_344 AV40 X NC		NC X AJ24 HBM2_DAP_56		HBM2_DAP_56 AU28 X NC		NC X BE40 RSVD		RSVD AL56 X NC	
NC X G11 DFTTIO_57		DFTTIO_153 C30 X NC		NC X H41 DFTTIO_249		DFTTIO_345 AM43 X NC		NC X AJ24 HBM2_DAP_57		HBM2_DAP_57 AT28 X NC		NC X BE40 RSVD		RSVD AL57 X NC	
NC X B8 DFTTIO_58		DFTTIO_154 D30 X NC		NC X N43 DFTTIO_250		DFTTIO_346 AY42 X NC		NC X AJ24 HBM2_DAP_58		HBM2_DAP_58 AL28 X NC		NC X BE40 RSVD		RSVD AL58 X NC	
NC X H12 DFTTIO_59		DFTTIO_155 H24 X NC		NC X N44 DFTTIO_251		DFTTIO_347 AM42 X NC		NC X AJ24 HBM2_DAP_59		HBM2_DAP_59 AK28 X NC		NC X BE40 RSVD		RSVD AL59 X NC	
NC X H16 DFTTIO_60		DFTTIO_156 G31 X NC		NC X G34 DFTTIO_252		DFTTIO_348 BA43 X NC									
NC X F14 DFTTIO_61		DFTTIO_157 F28 X NC		NC X H42 DFTTIO_253		DFTTIO_349 AM44 X NC									
NC X E14 DFTTIO_62		DFTTIO_158 B29 X NC		NC X F36 DFTTIO_254		DFTTIO_350 BA44 X NC									
NC X G14 DFTTIO_63		DFTTIO_159 F26 X NC		NC X J36 DFTTIO_255		DFTTIO_351 AY41 X NC									
NC X F10 DFTTIO_64		DFTTIO_160 J24 X NC		NC X J35 DFTTIO_256		DFTTIO_352 AM40 X NC									
NC X D13 DFTTIO_65		DFTTIO_161 B30 X NC		NC X G38 DFTTIO_257		DFTTIO_353 AY44 X NC									
NC X H20 DFTTIO_66		DFTTIO_162 D32 X NC		NC X F44 DFTTIO_258		DFTTIO_354 BA40 X NC									
NC X H18 DFTTIO_67		DFTTIO_163 E31 X NC		NC X J40 DFTTIO_259		DFTTIO_355 AT43 X NC									
NC X B9 DFTTIO_68		DFTTIO_164 D38 X NC		NC X N42 DFTTIO_260		DFTTIO_356 AV44 X NC									
NC X H13 DFTTIO_69		DFTTIO_165 B32 X NC		NC X M41 DFTTIO_261		DFTTIO_357 AU42 X NC									
NC X D10 DFTTIO_70		DFTTIO_166 B33 X NC		NC X R43 DFTTIO_262		DFTTIO_358 AU43 X NC									
NC X H17 DFTTIO_71		DFTTIO_167 C35 X NC		NC X R44 DFTTIO_263		DFTTIO_359 AY40 X NC									
NC X H21 DFTTIO_72		DFTTIO_168 F28 X NC		NC X J38 DFTTIO_264		DFTTIO_360 AK42 X NC									
NC X H14 DFTTIO_73		DFTTIO_169 C38 X NC		NC X K37 DFTTIO_265		DFTTIO_361 BA42 X NC									
NC X C14 DFTTIO_74		DFTTIO_170 H25 X NC		NC X Y44 DFTTIO_266		DFTTIO_362 AK44 X NC									
NC X C10 DFTTIO_75		DFTTIO_171 G30 X NC		NC X P42 DFTTIO_267		DFTTIO_363 AN39 X NC									
NC X G18 DFTTIO_76		DFTTIO_172 D40 X NC		NC X T42 DFTTIO_268		DFTTIO_364 AR39 X NC									
NC X D14 DFTTIO_77		DFTTIO_173 E32 X NC		NC X H38 DFTTIO_269		DFTTIO_365 AR40 X NC									
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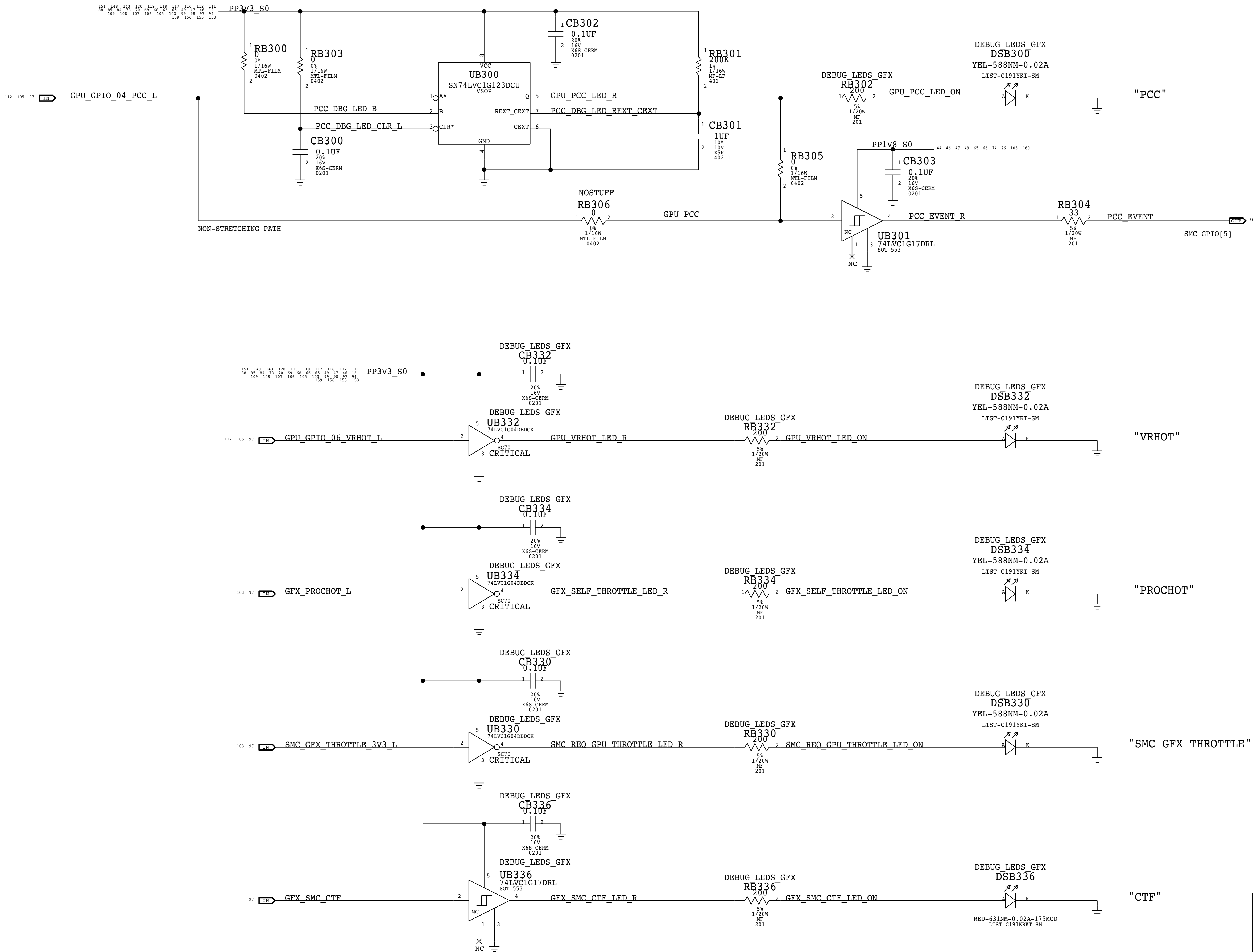


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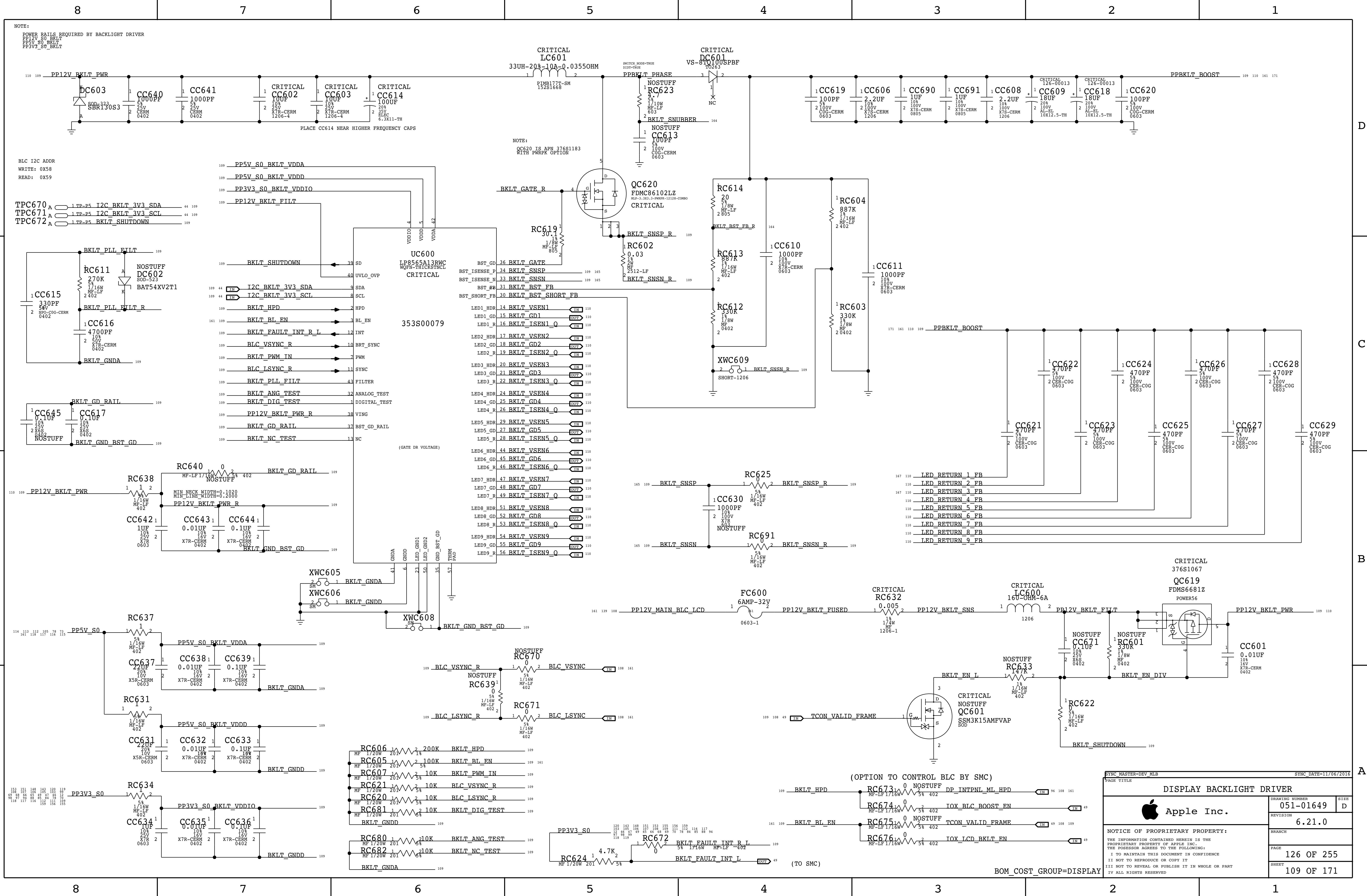


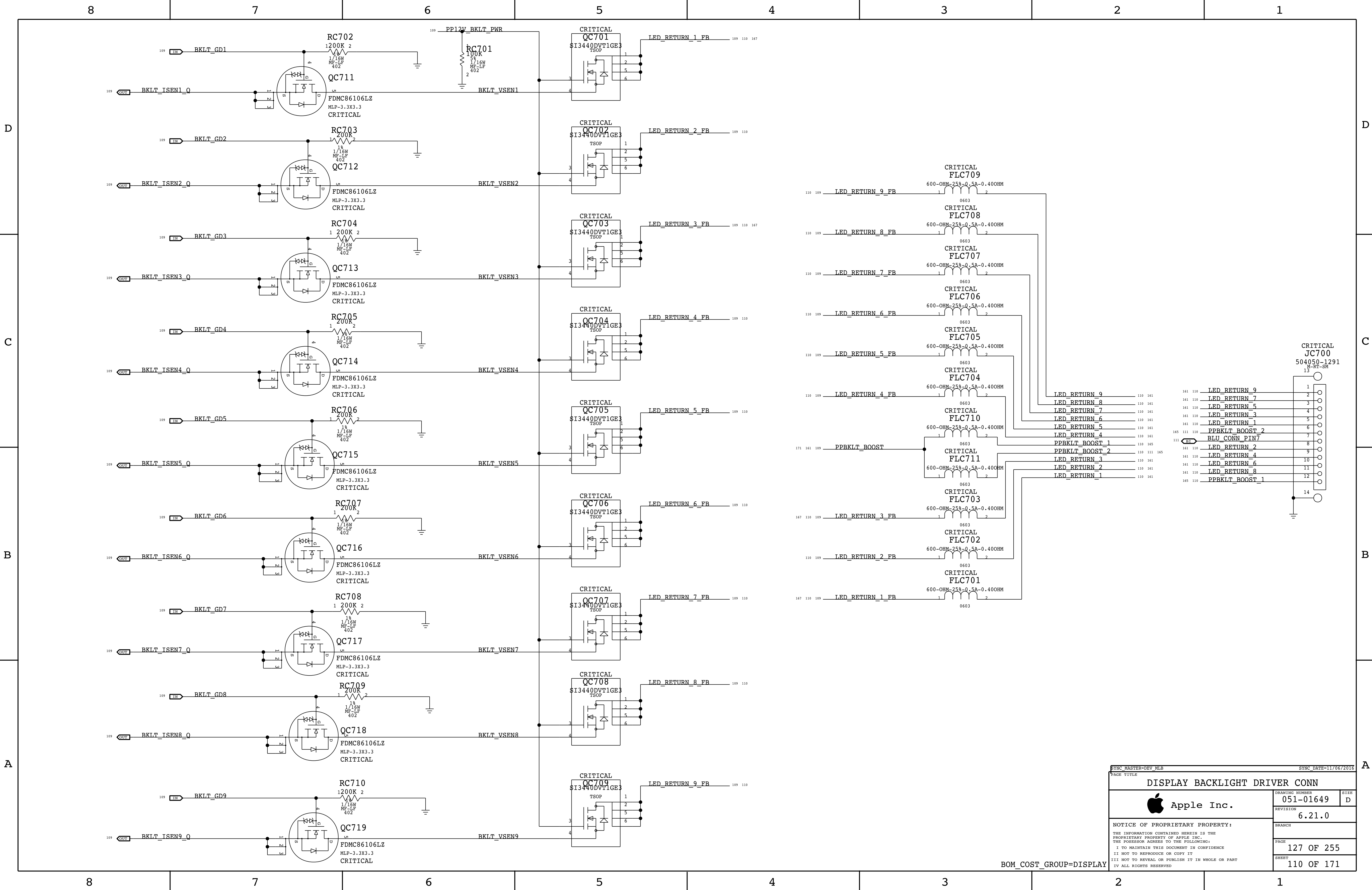
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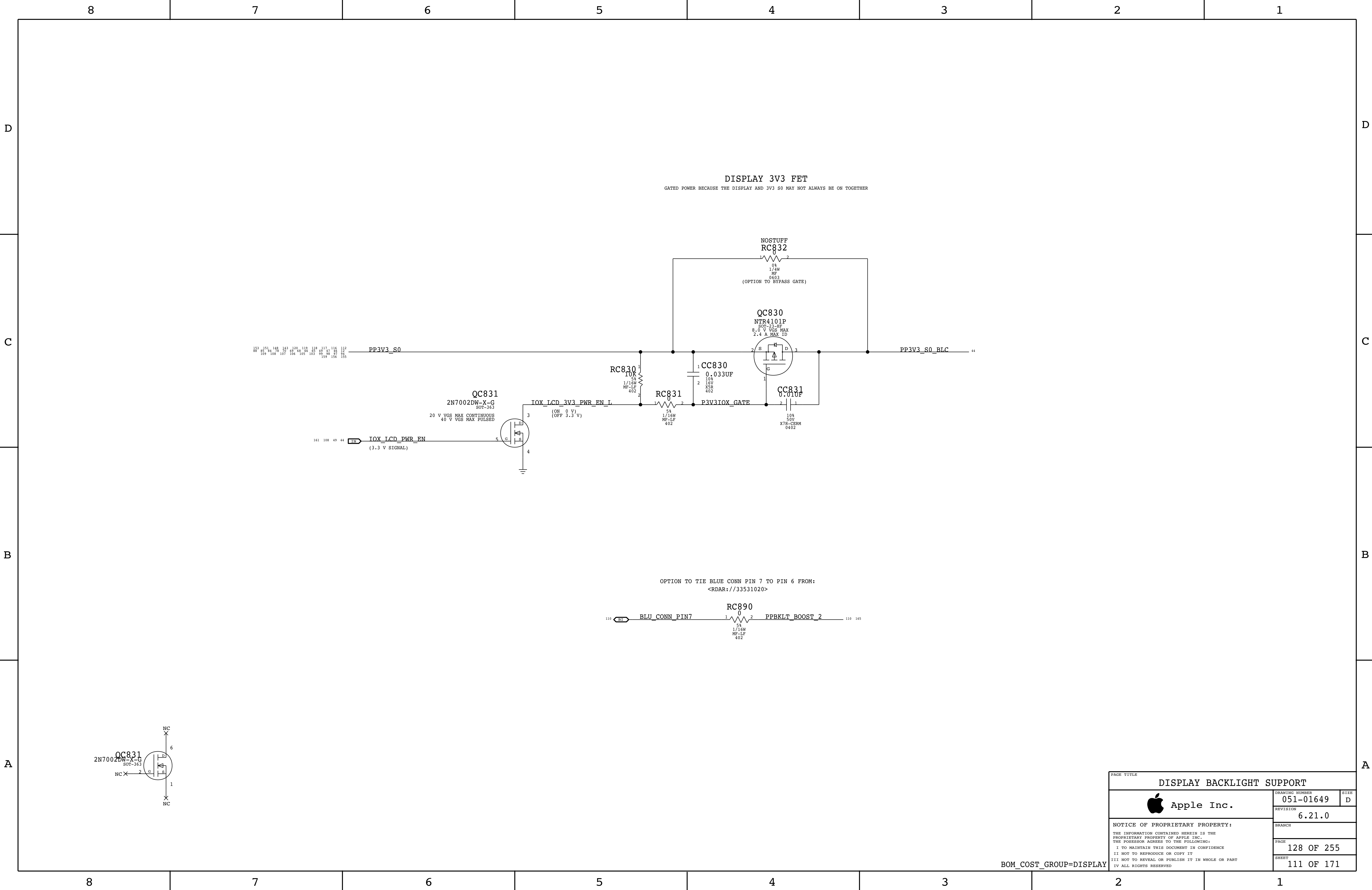
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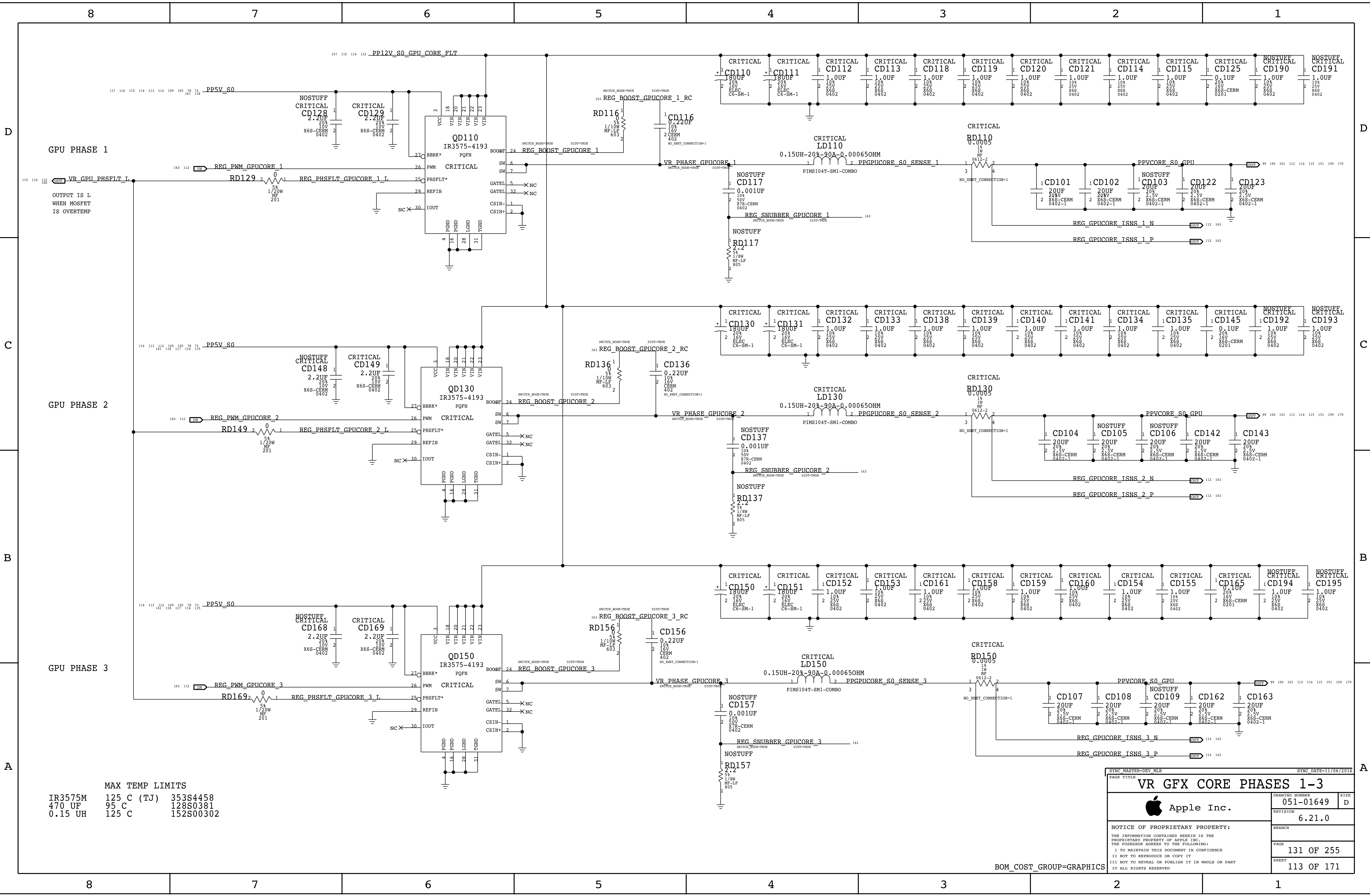




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GPU PHASE 1

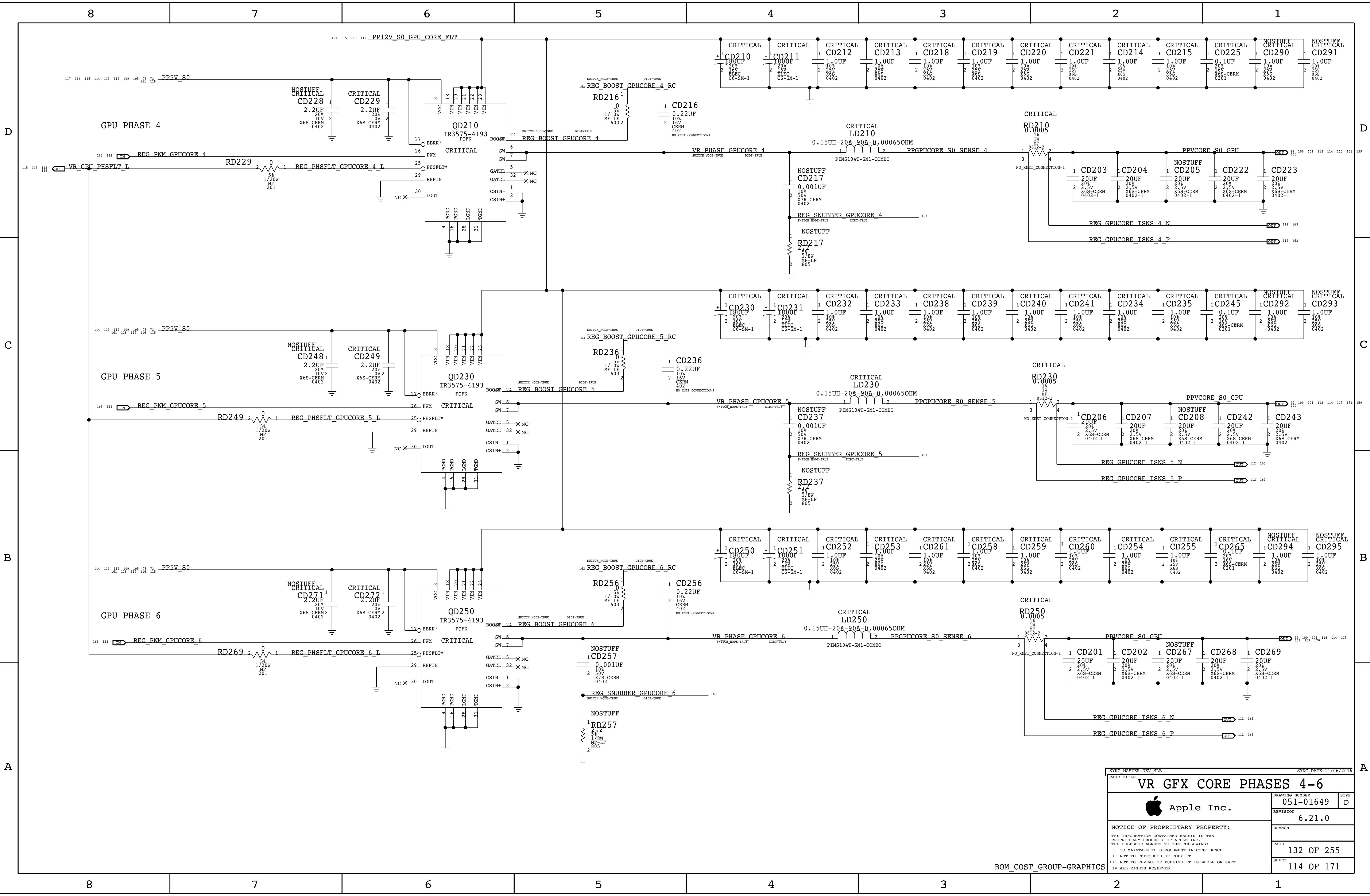
GPU PHASE 2

GPU PHASE 3

MAX TEMP LIMITS		
IR3575M	125 C (TJ)	353S4458
470 UF	95 C	128S0381
0.15 UH	125 C	152S00302

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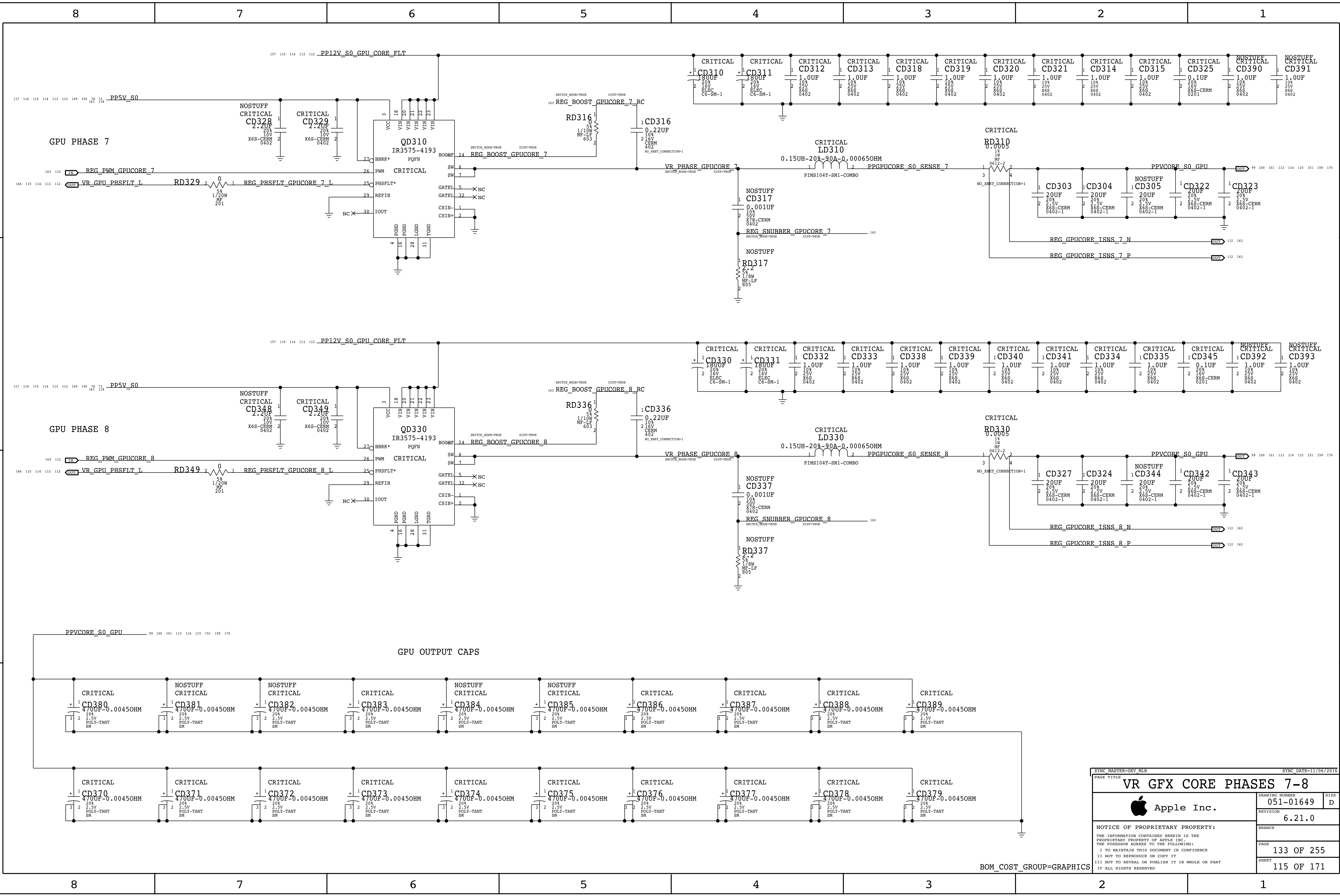
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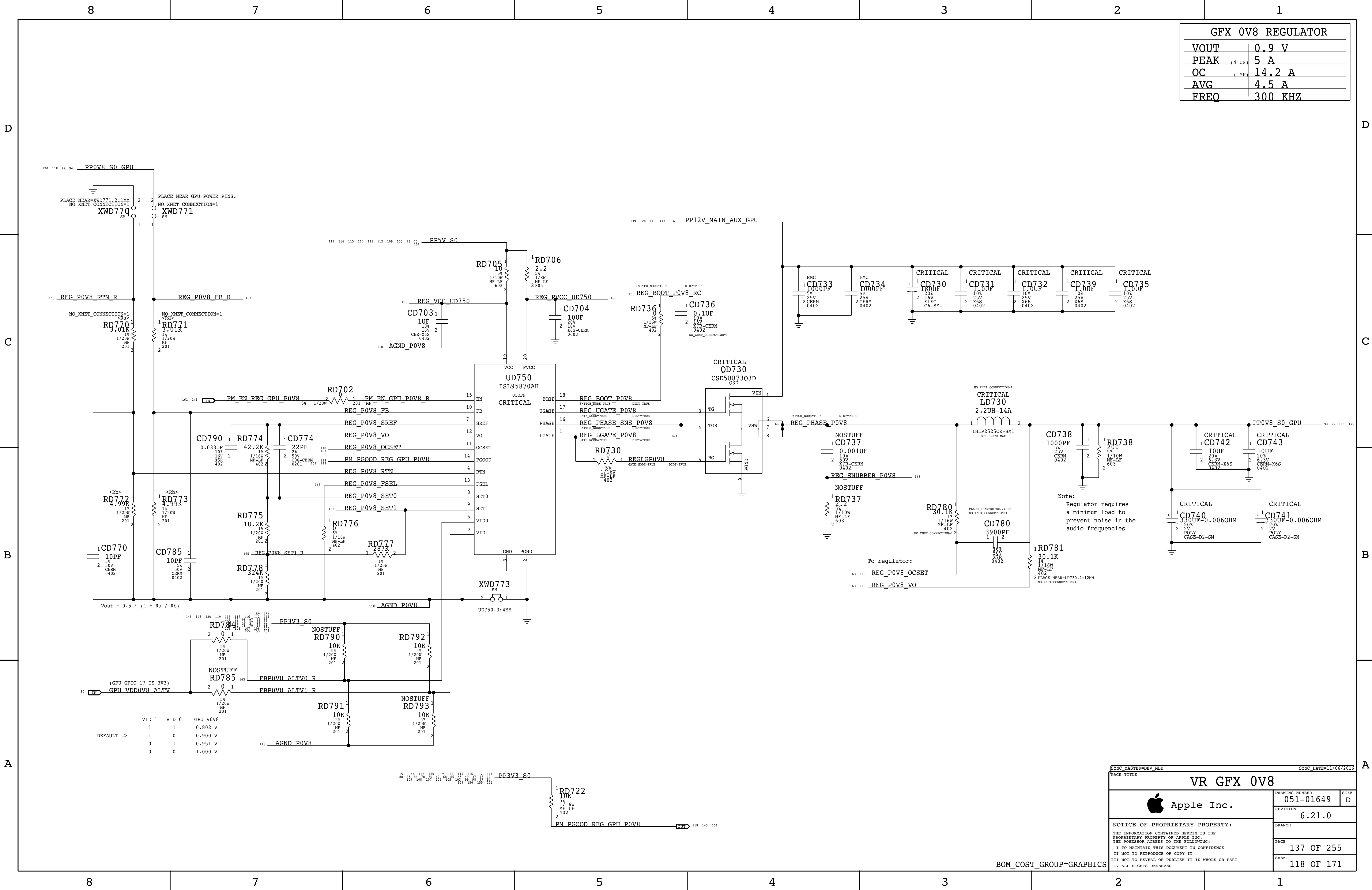
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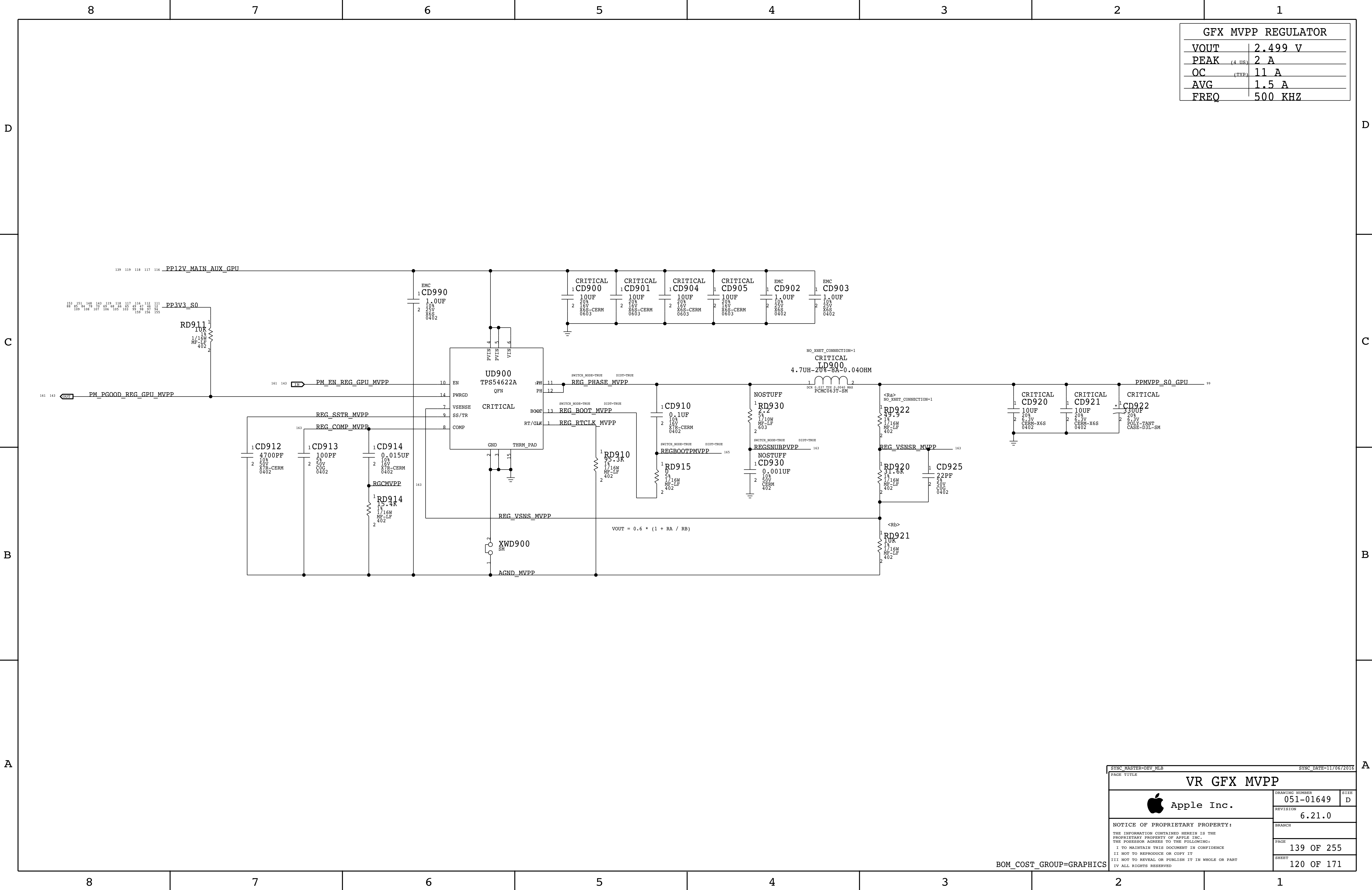
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GFX 0V8 REGULATOR	
VOUT	0.9 V
PEAK	5 A
OC	14.2 A
AVG	4.5 A
FREQ	300 KHZ

PAGE TITLE		VR GFX 0V8	
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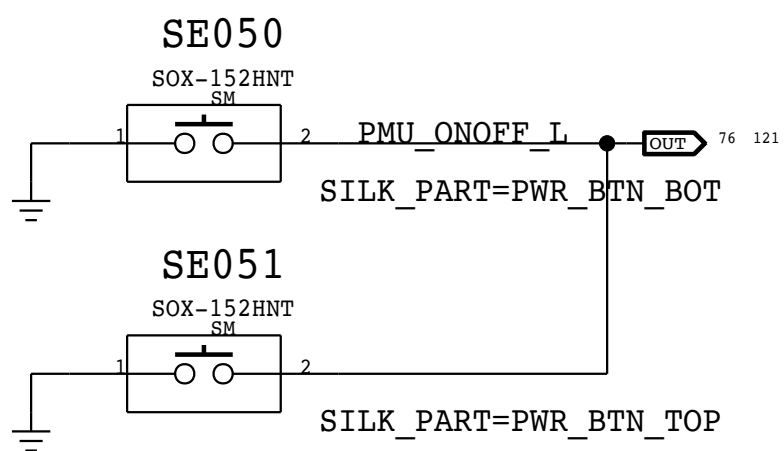


GFX MVPP REGULATOR	
VOUT	2.499 V
PEAK	2 A
OC	11 A
AVG	1.5 A
FREQ	500 KHZ

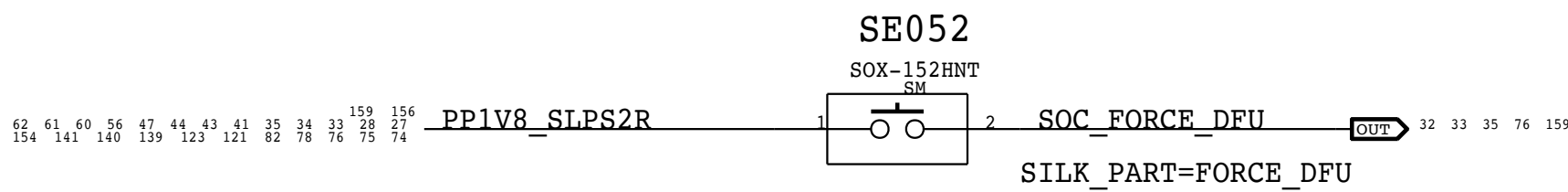
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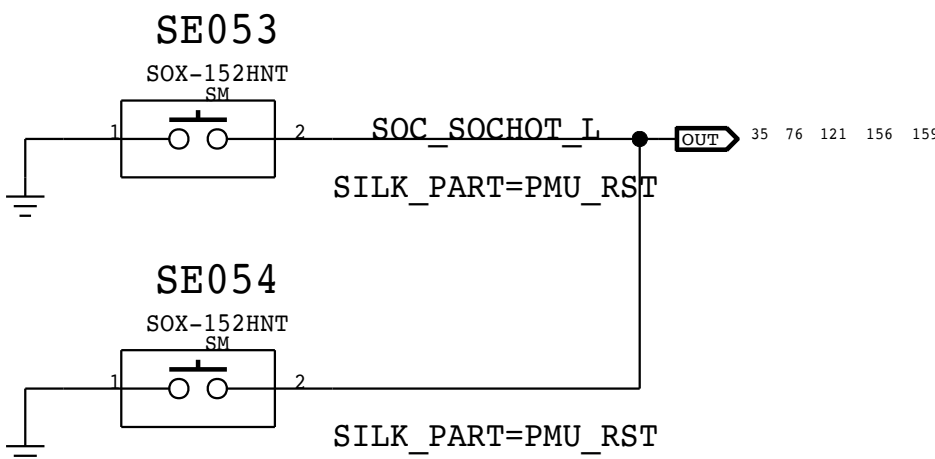
Power Button



Force DFU

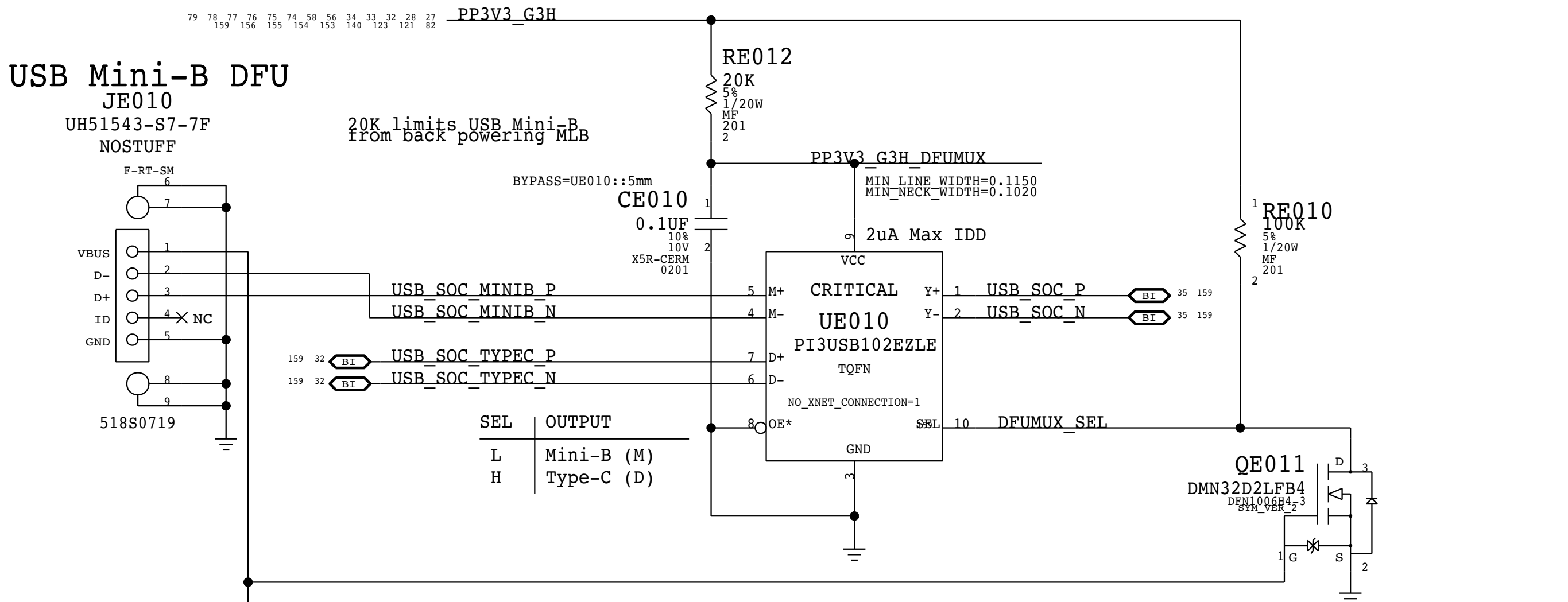


PMU Reset



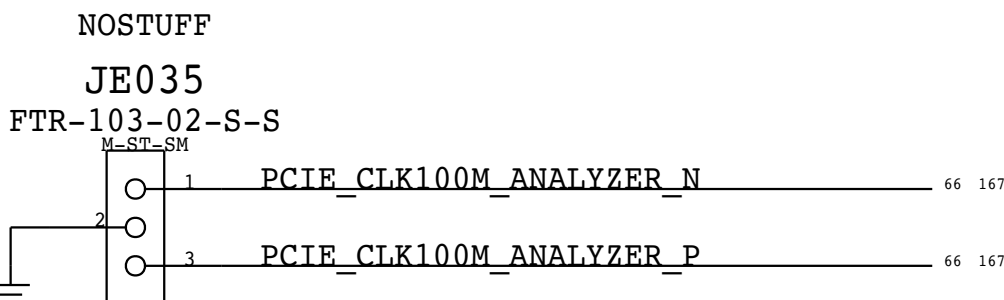
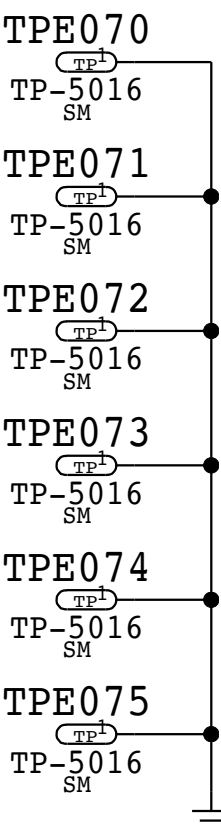
UE010: pin7 and pin8 Signal name must match the Debug Mux 2 page

SoC USB DFU Mux

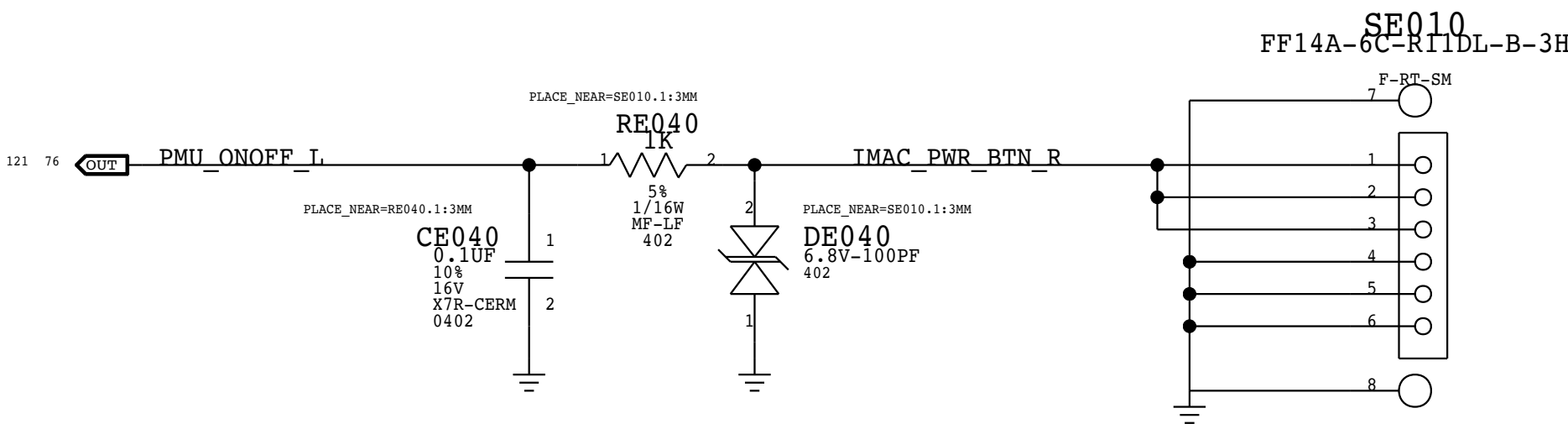


Ground Clips

Distribute between top and bottom of board

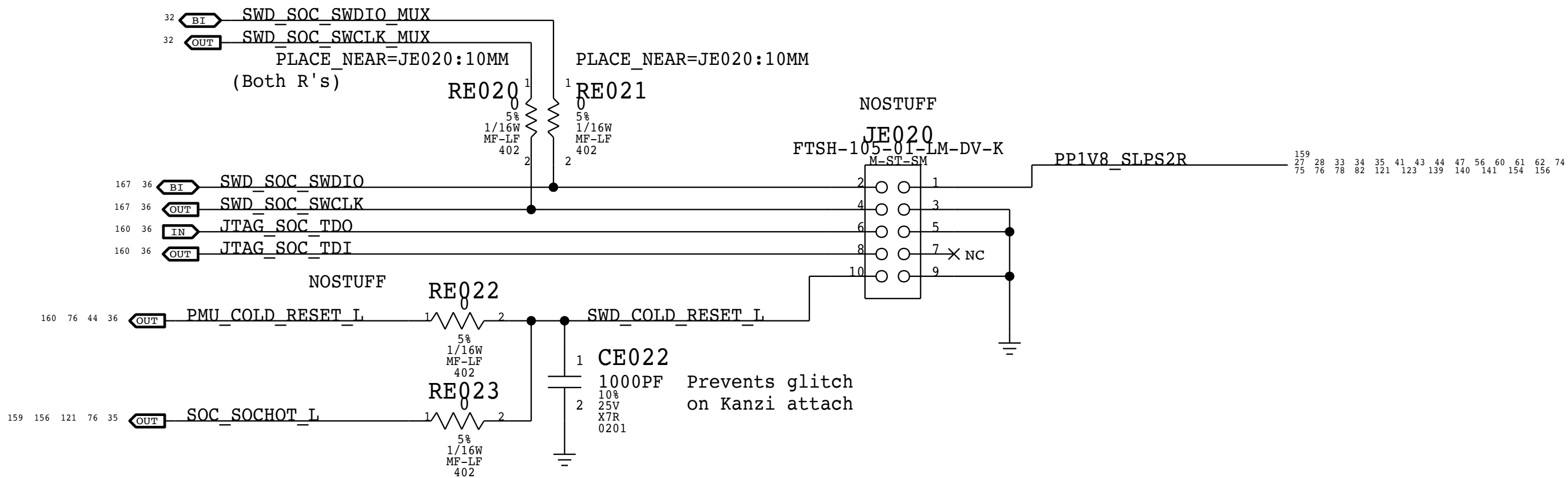


FOR PWR BUTTON FLEX

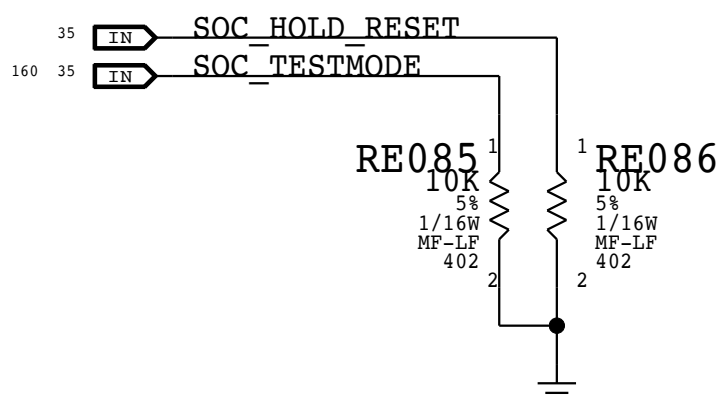


Kanzi SWD Header

Default SWD path
(Connects to the Debug Mux)



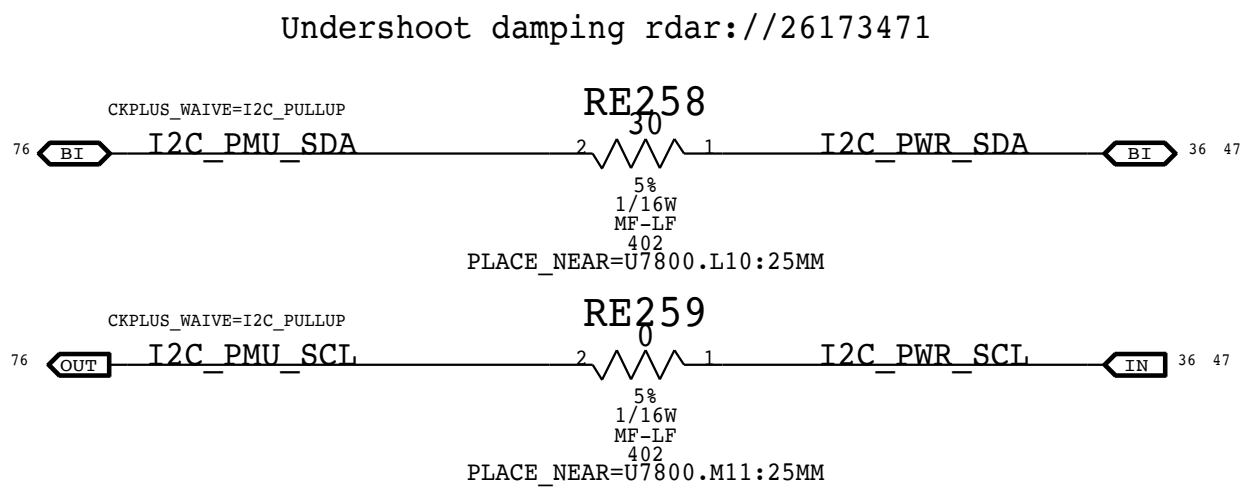
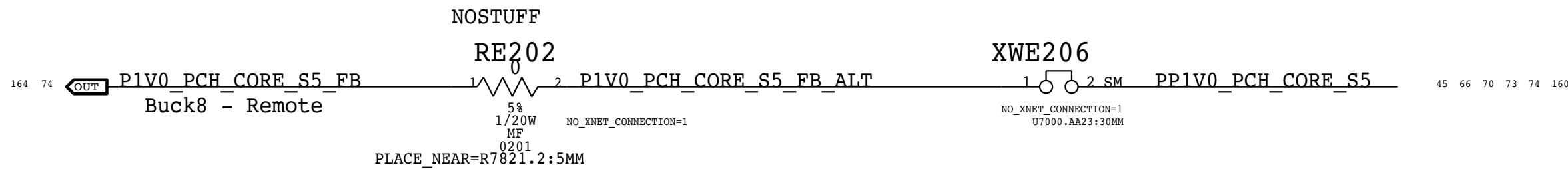
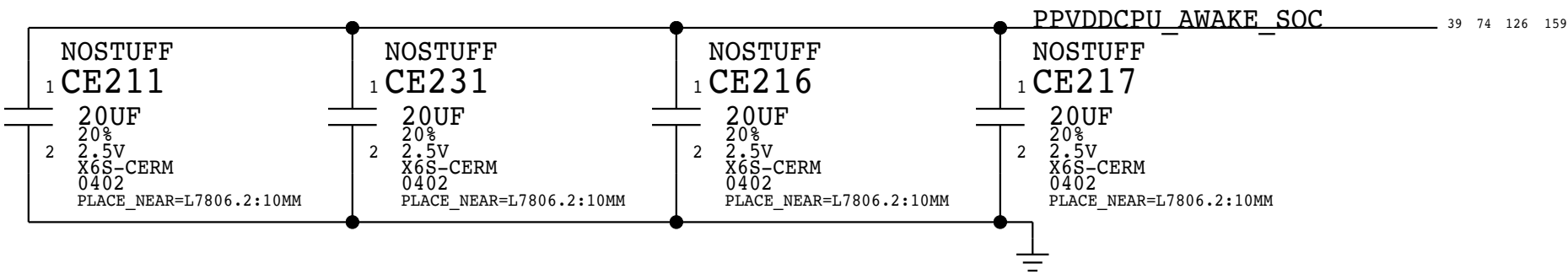
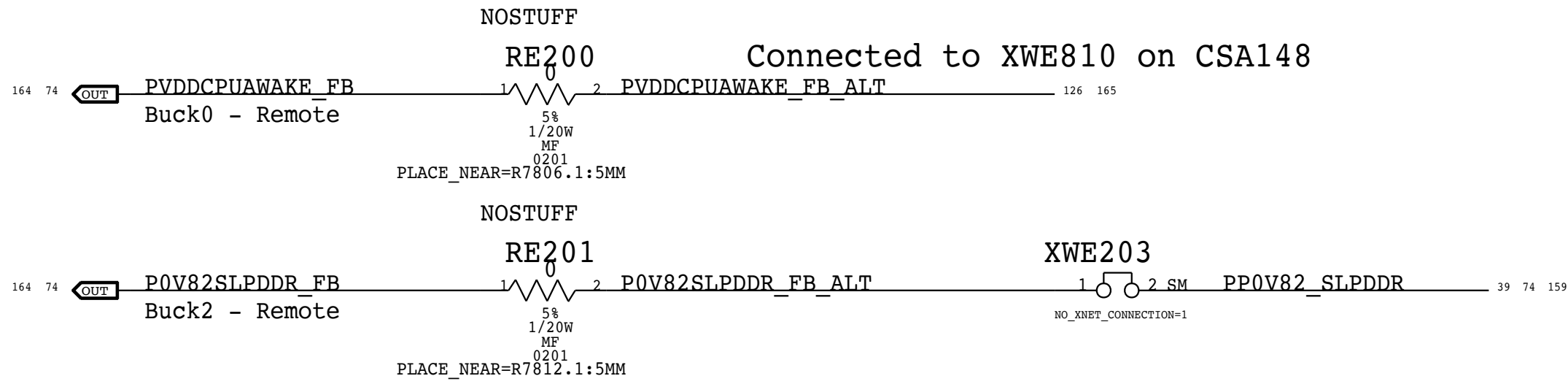
Needed for test mode purposes



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	PAGE	140 OF 255
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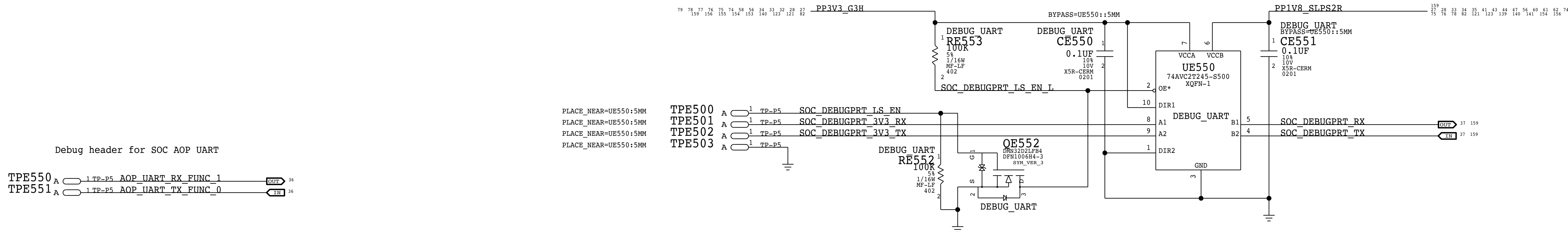
Alternate Feedback Sense



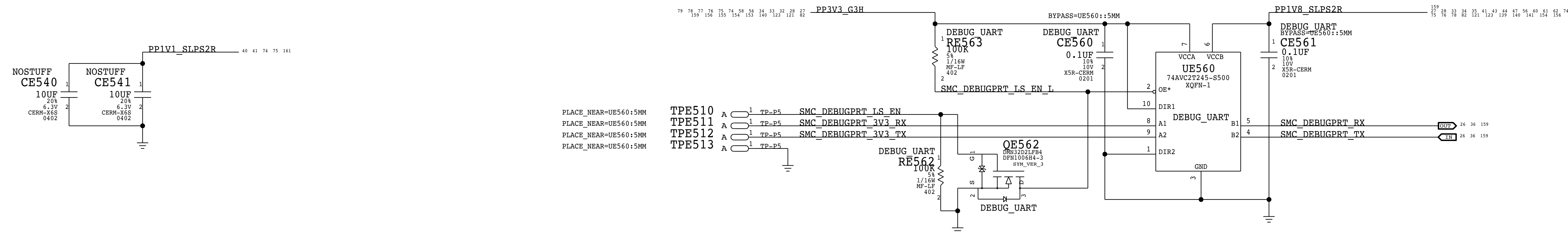
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SOC UART0 Header



SMC UART0 Header



XDP_PRESENT and PCIE_UP_CLKREQ PU are moved to their pages

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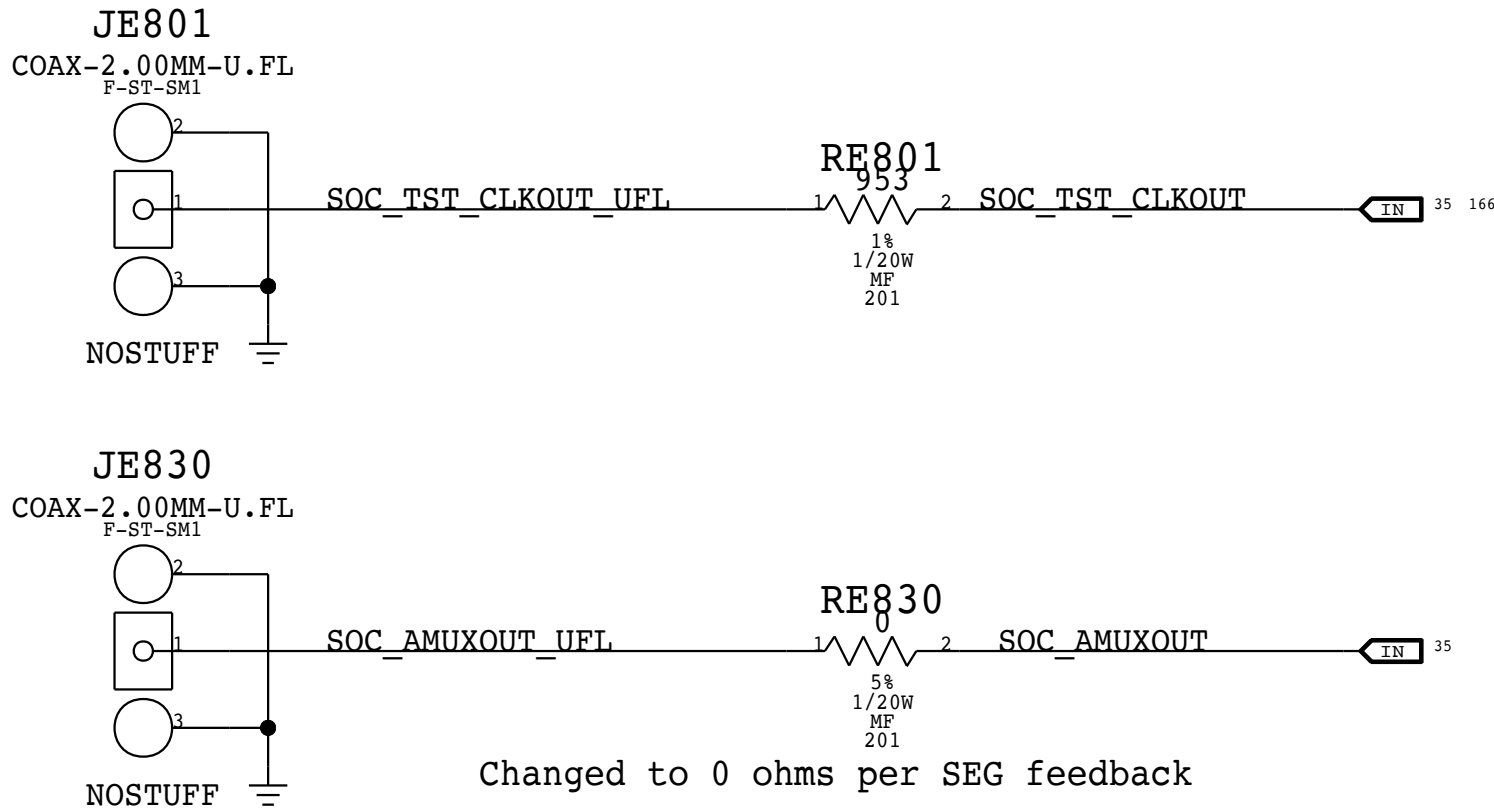
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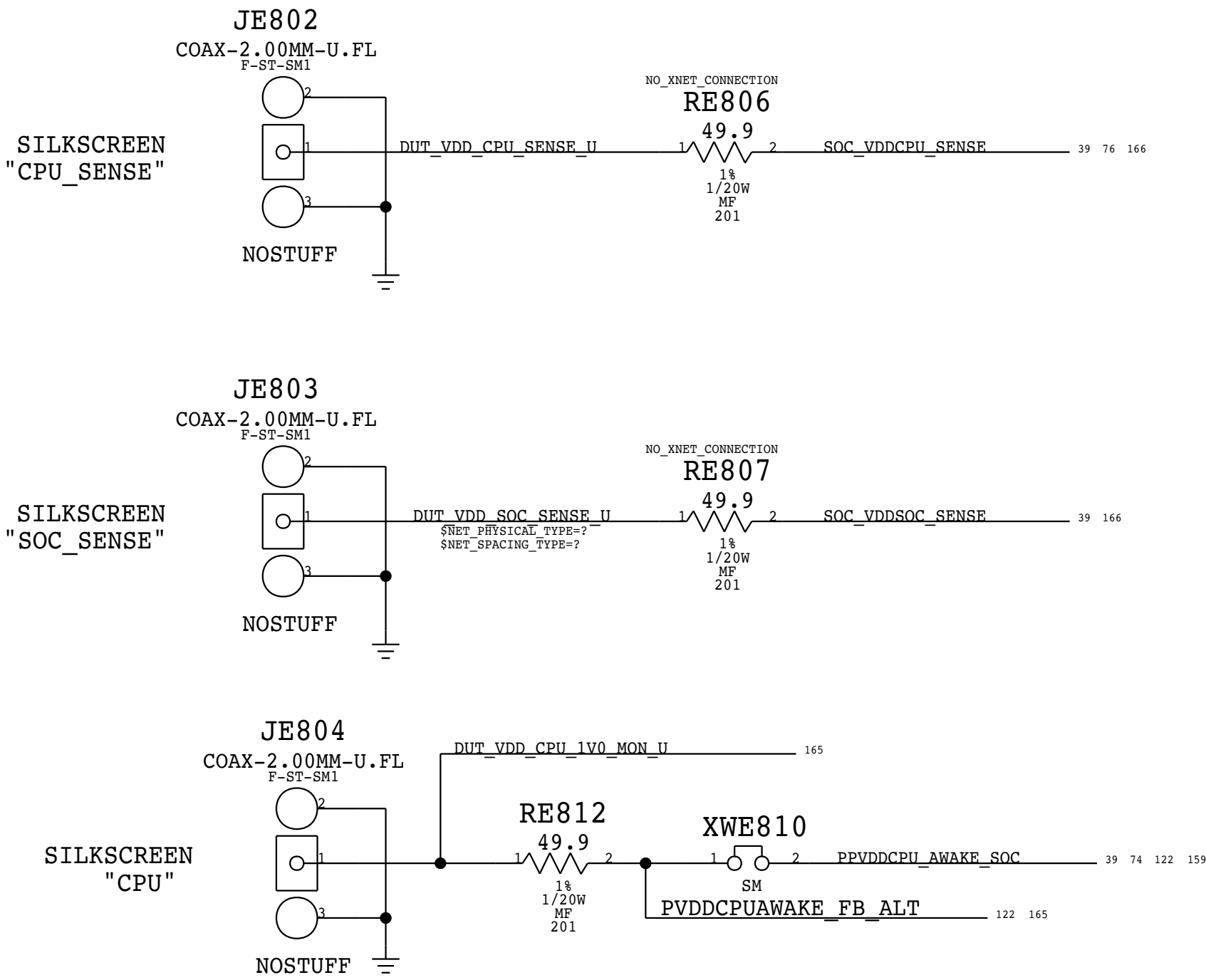
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U.FL CONNECTOR FOR SOC CLOCKS

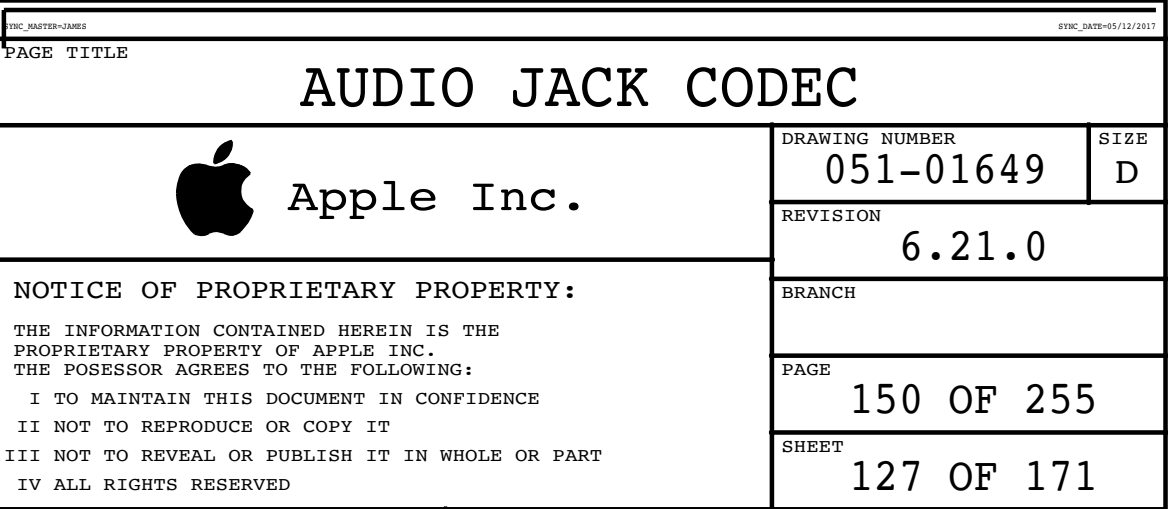


U.FL CONNECTOR FOR SOC POWER



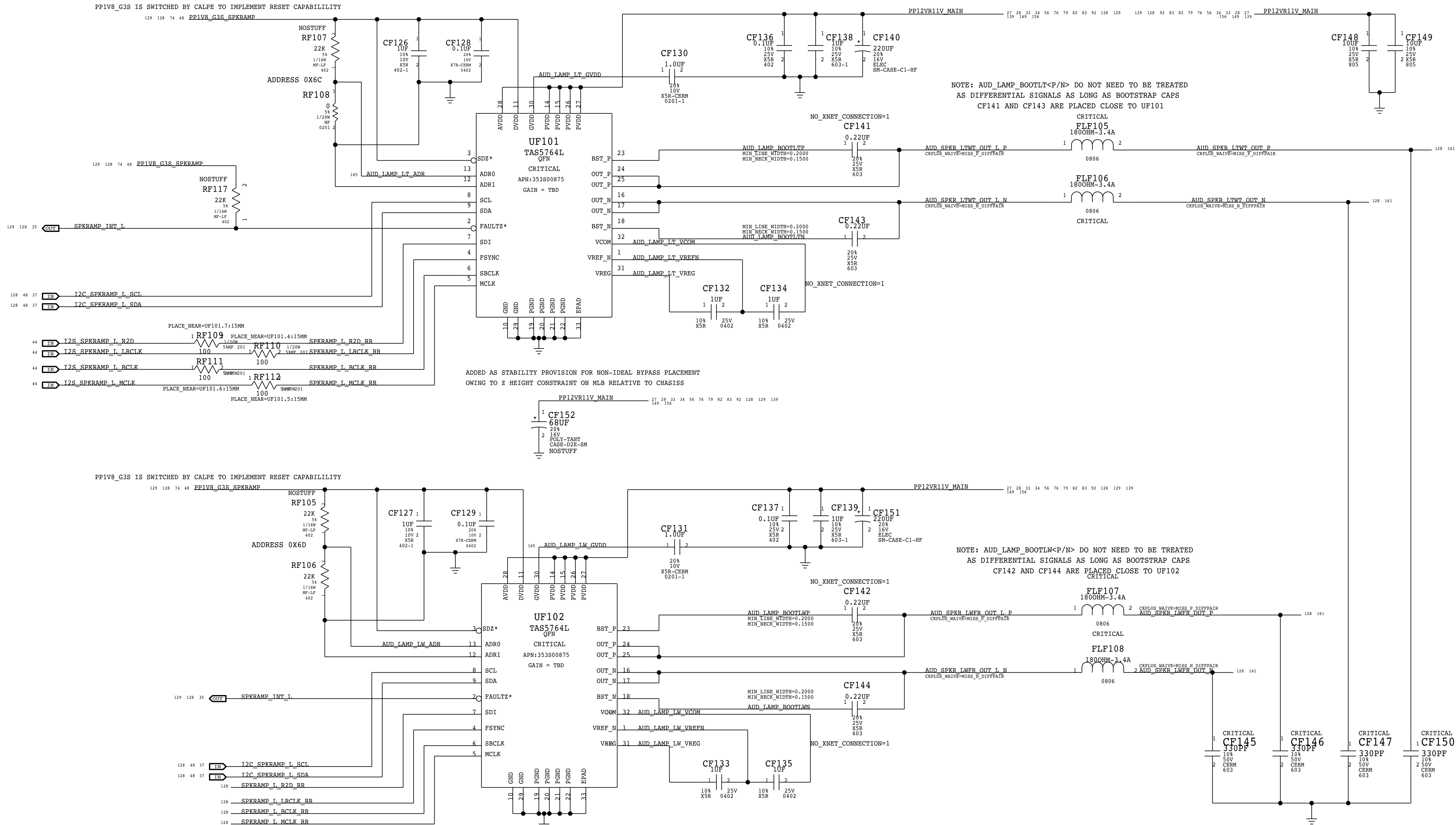
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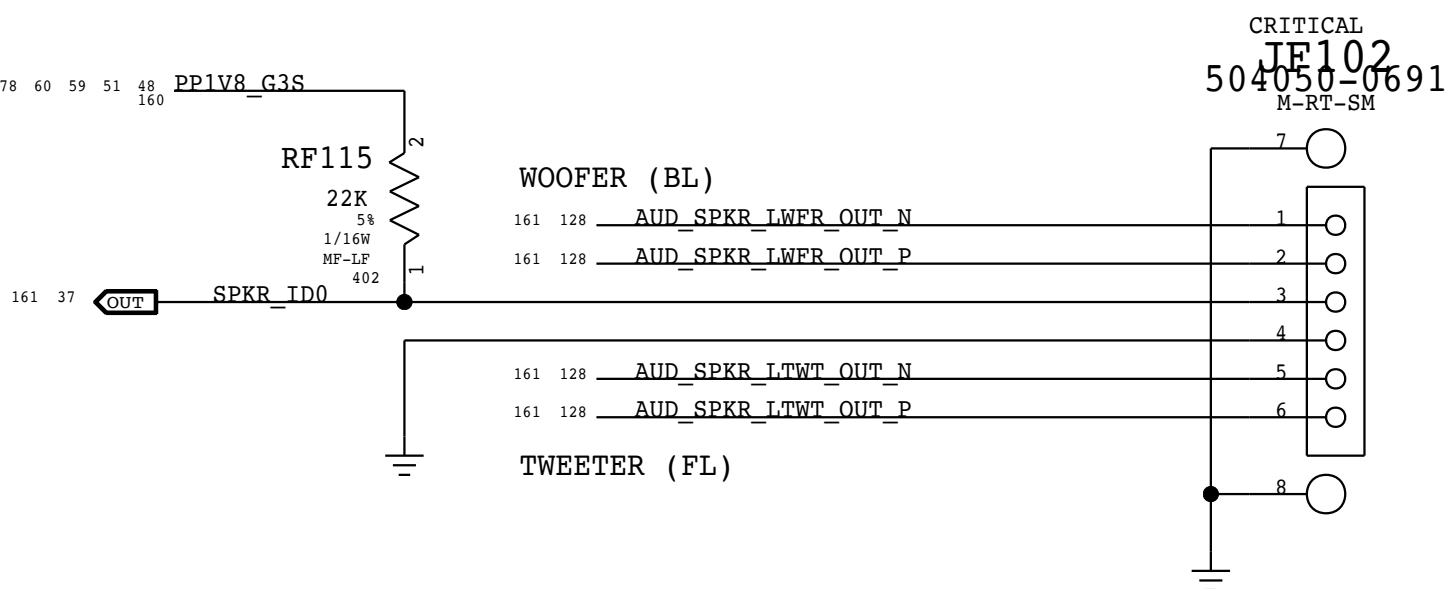
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SPEAKER AMPLIFIERS - LEFT CHANNEL
2X MONO SPEAKER AMPLIFIERS (TAS5764L)



SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862



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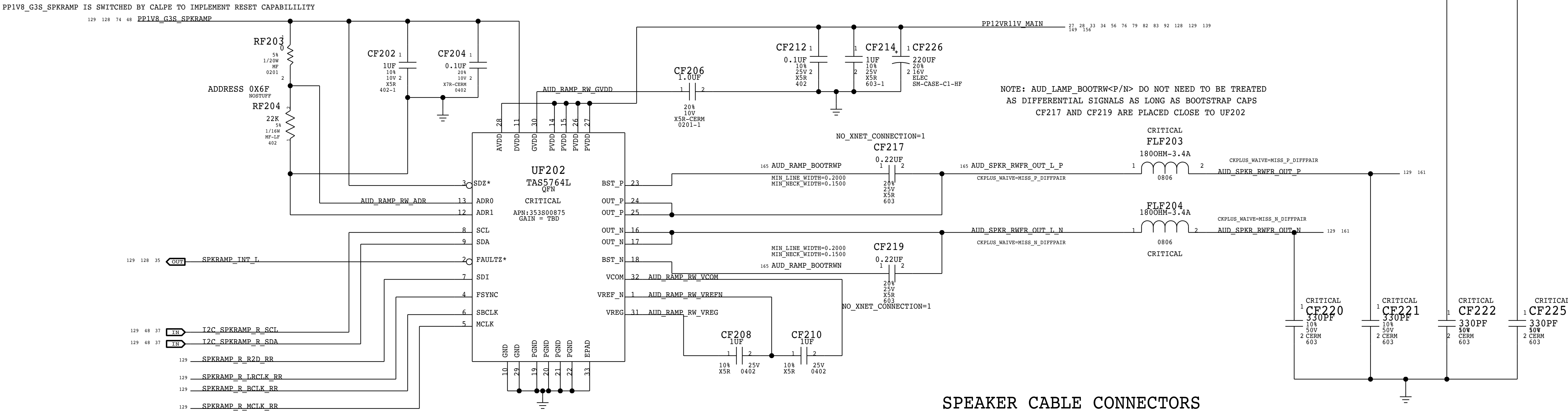
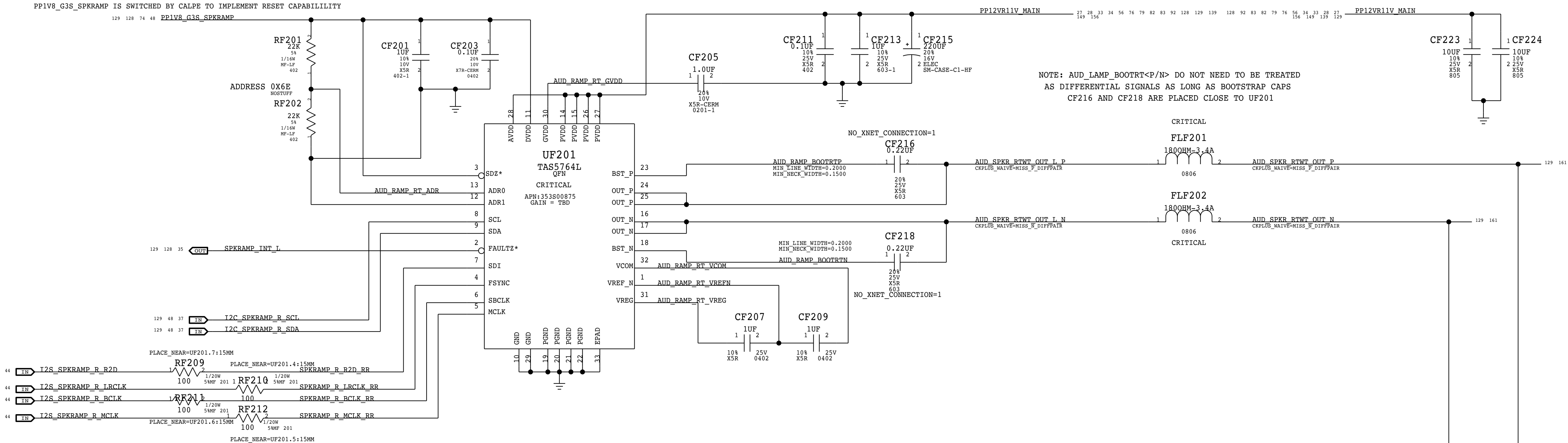
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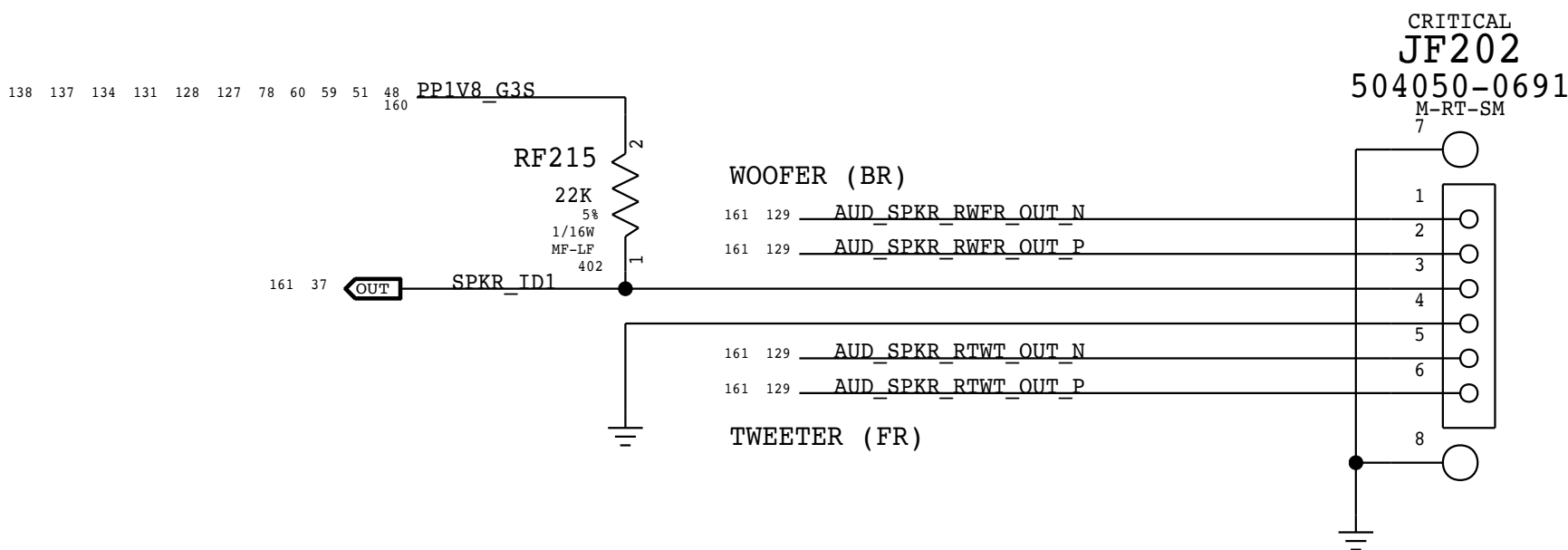
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2X MONO SPEAKER AMPLIFIERS (TAS5764L)

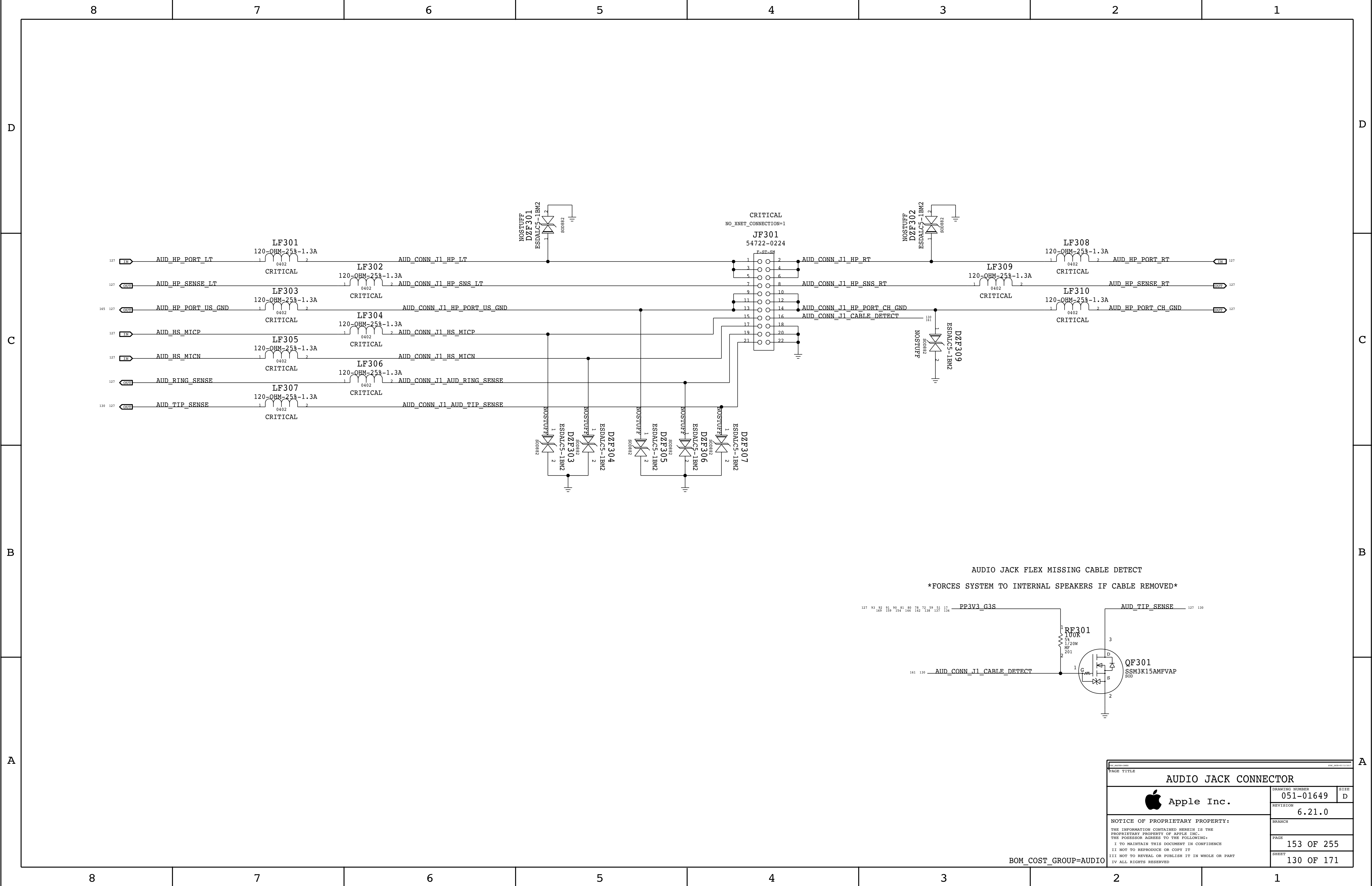


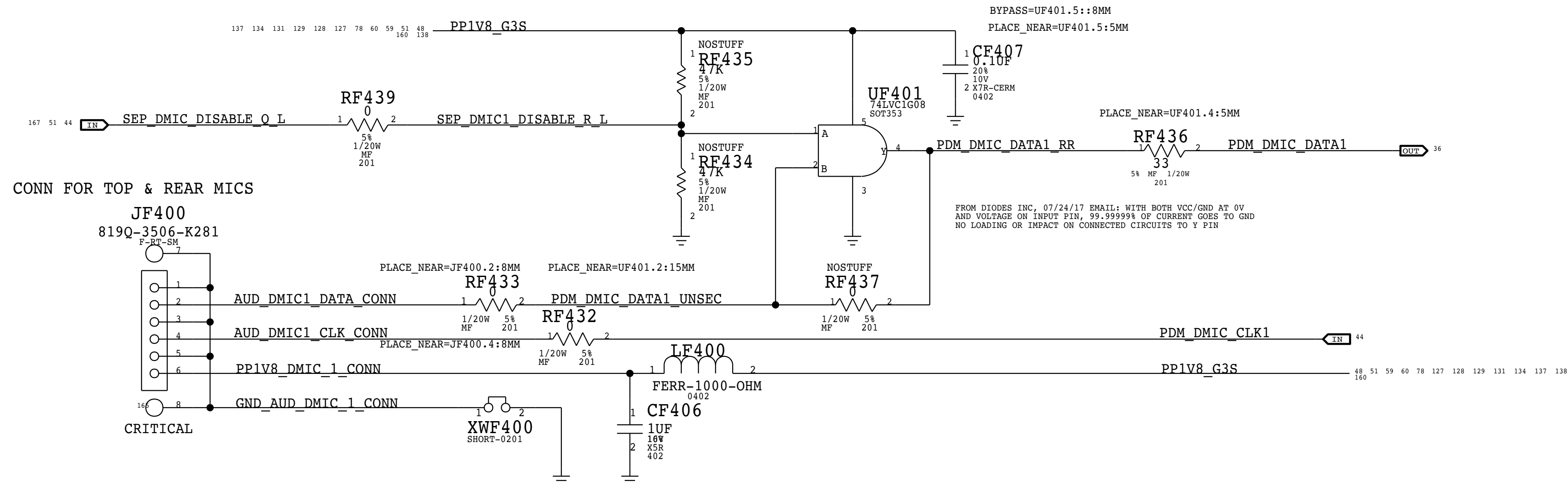
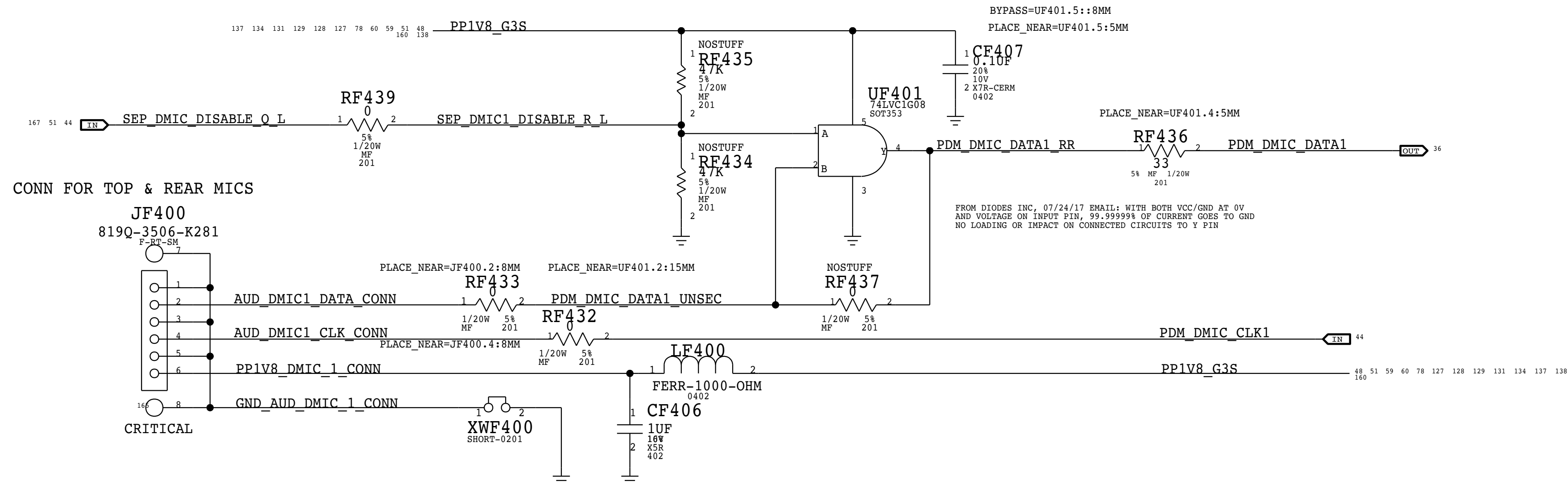
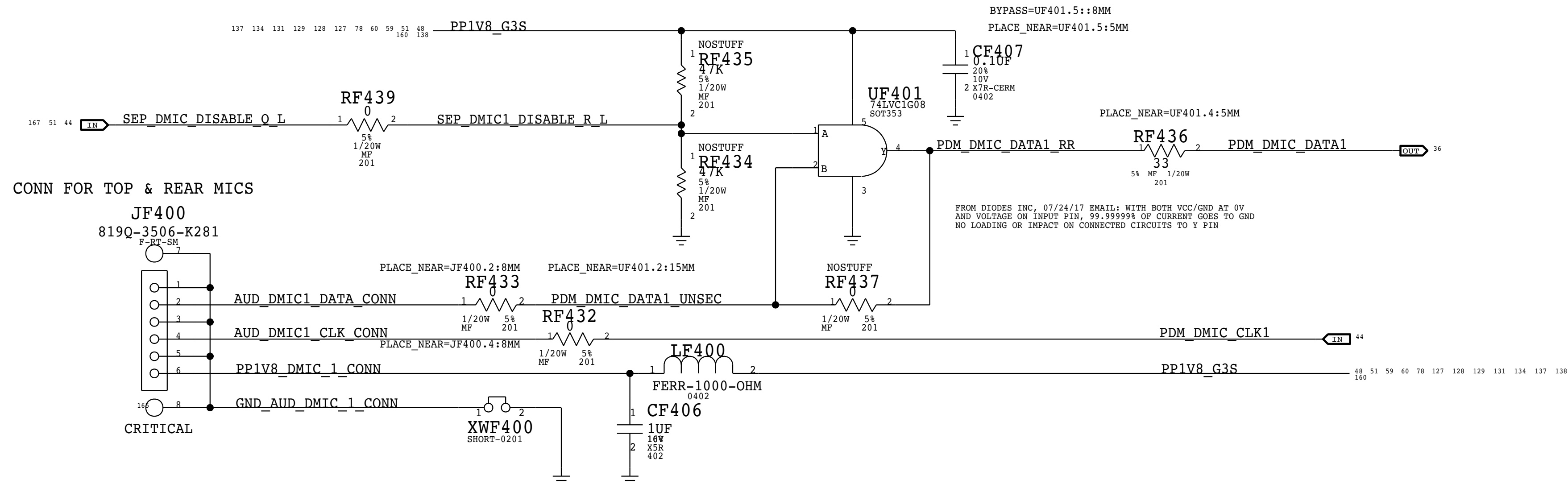
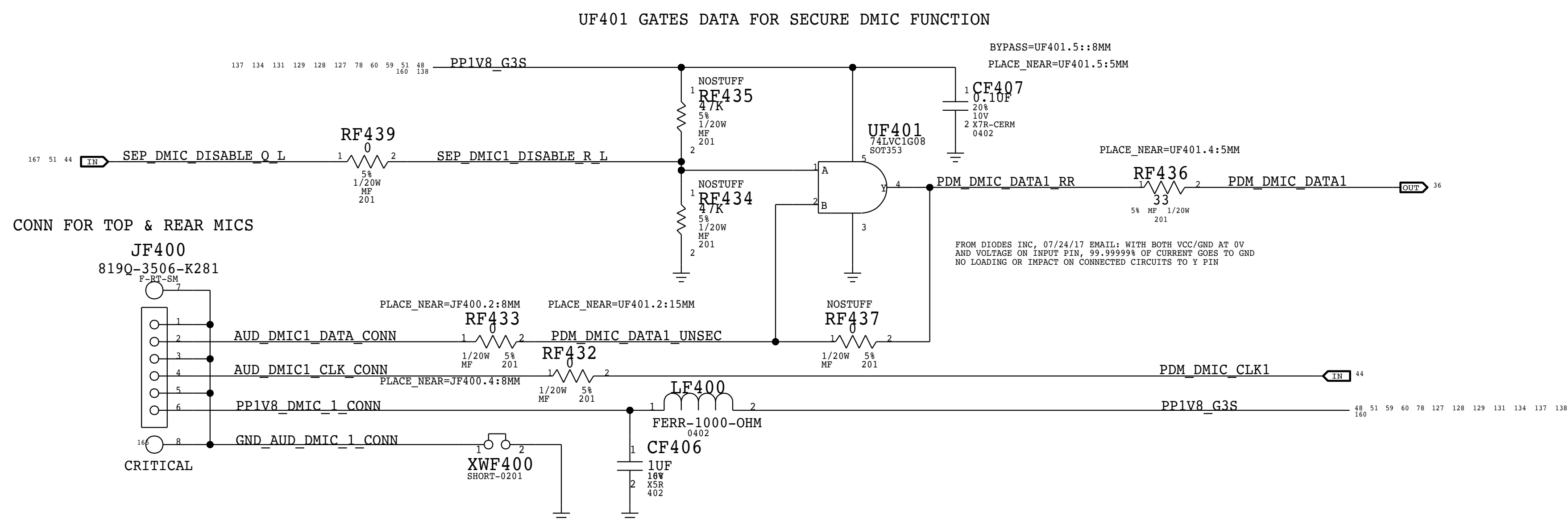
SPEAKER CABLE CONNECTORS
APPLE P/N 518S0862



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BOM_COST_GROUP=AUDIO





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BOM_COST_GROUP=AUDIO

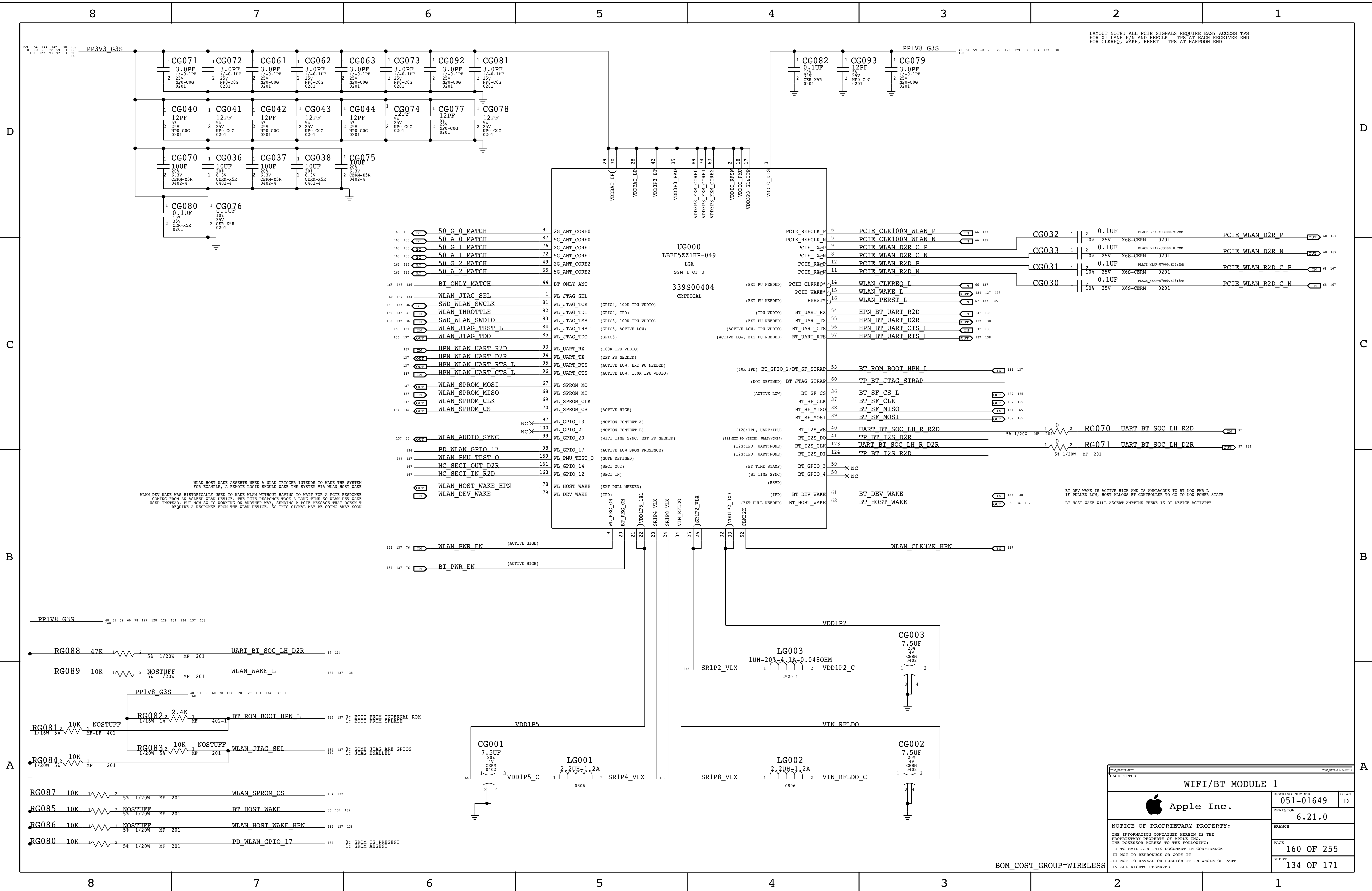
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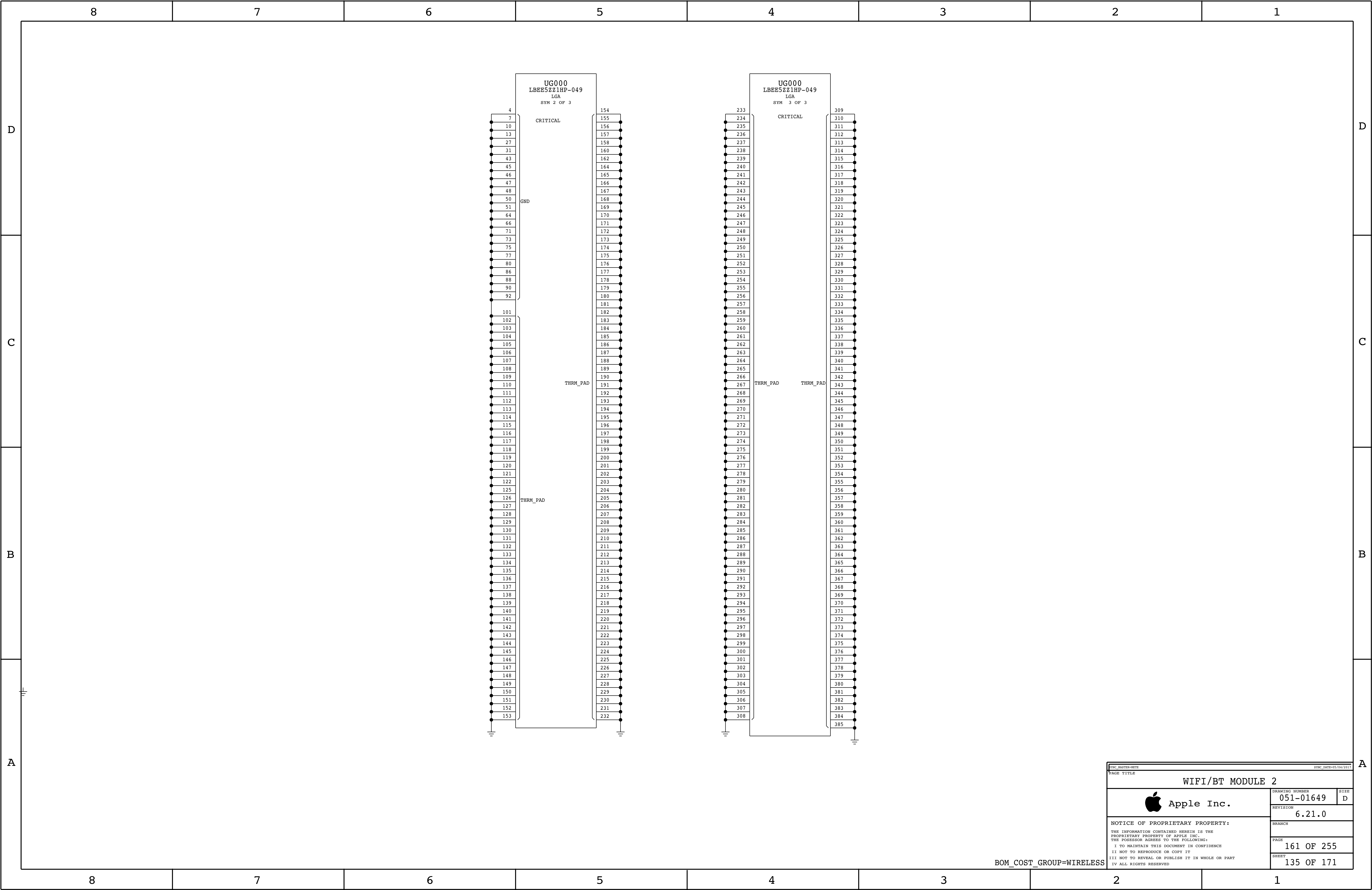
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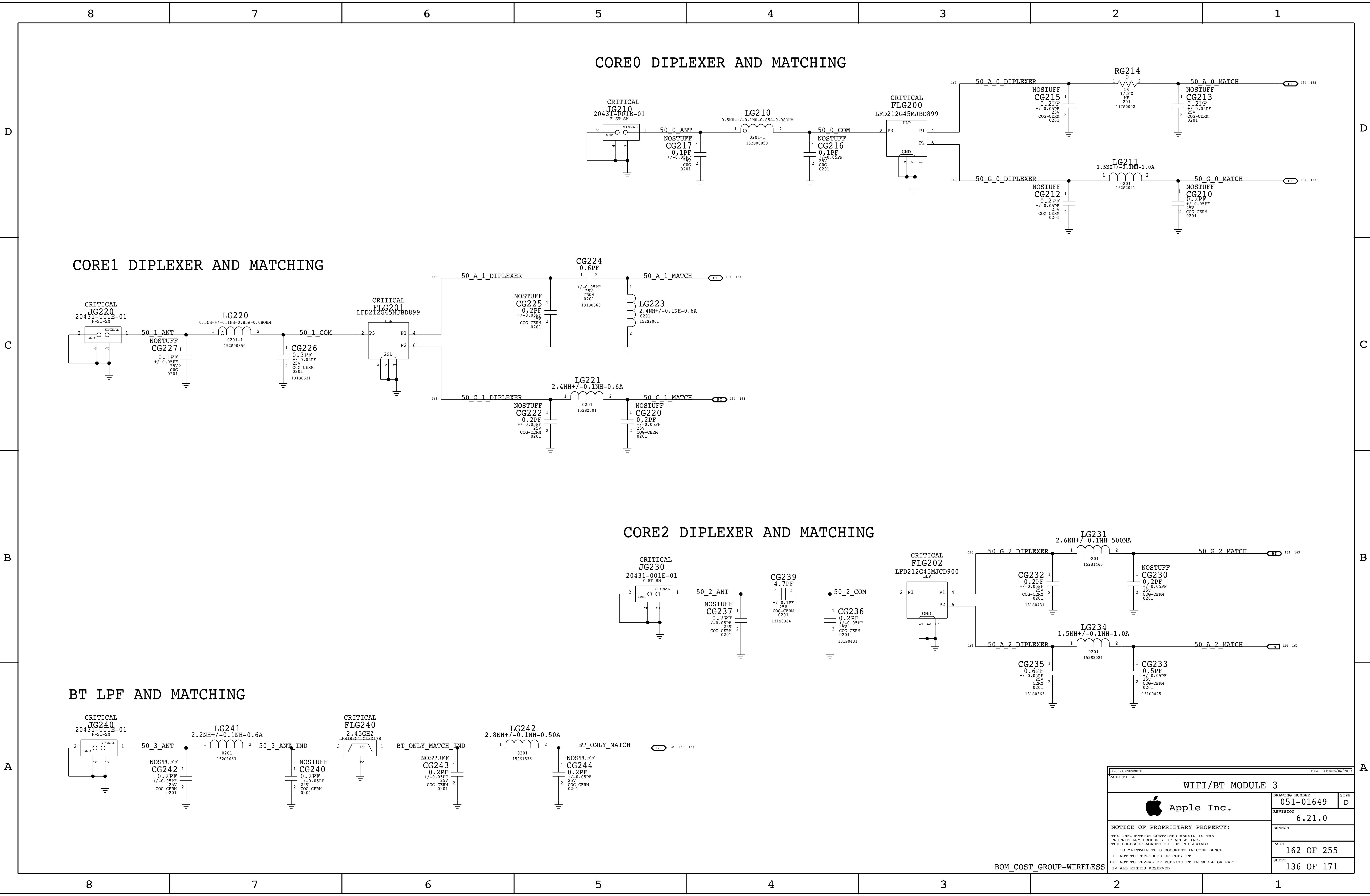
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BOM_COST_GROUP=AUDIO







CORE0 DIPLEXER AND MATCHING

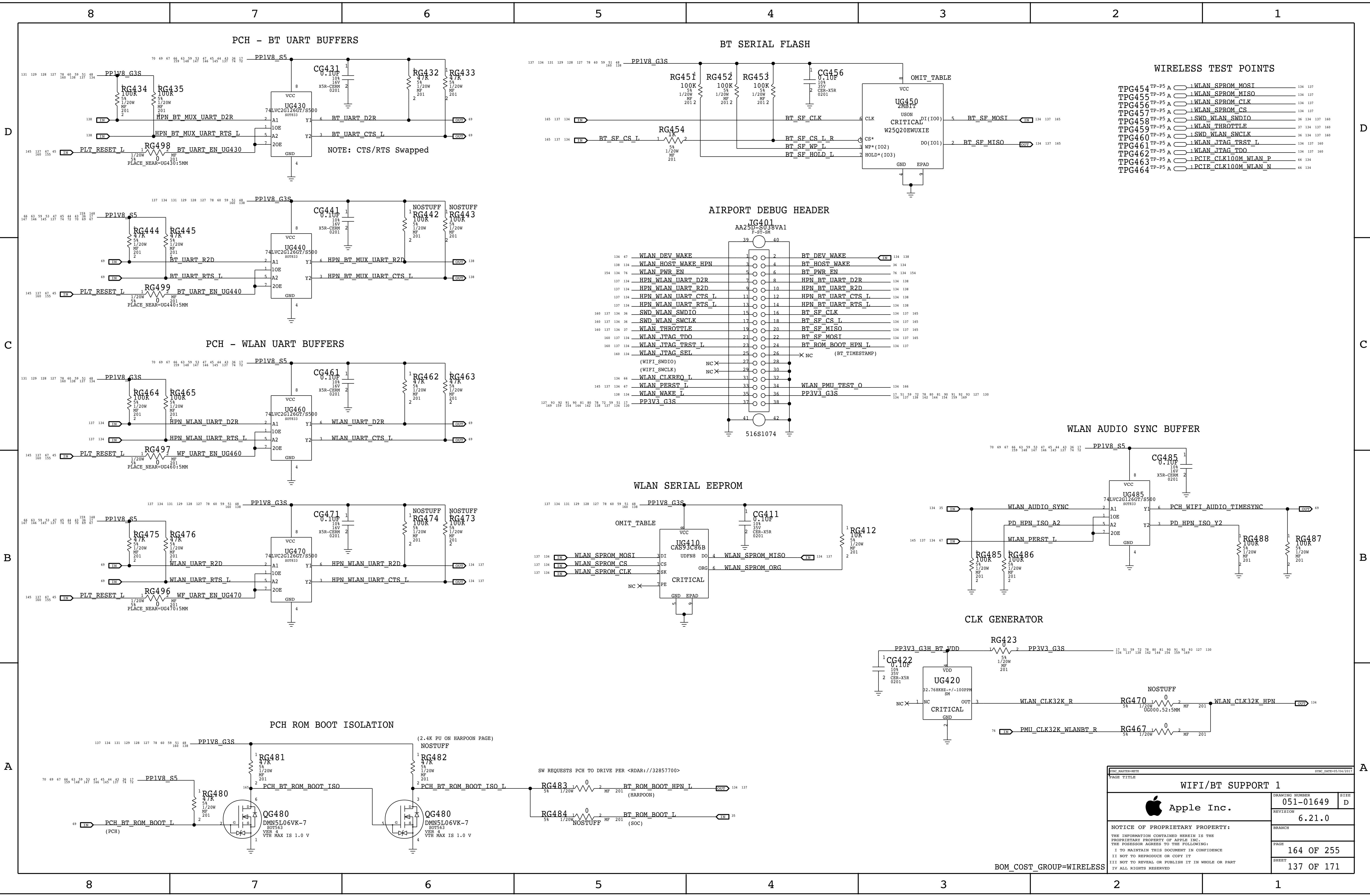
CORE1 DIPLEXER AND MATCHING

CORE2 DIPLEXER AND MATCHING

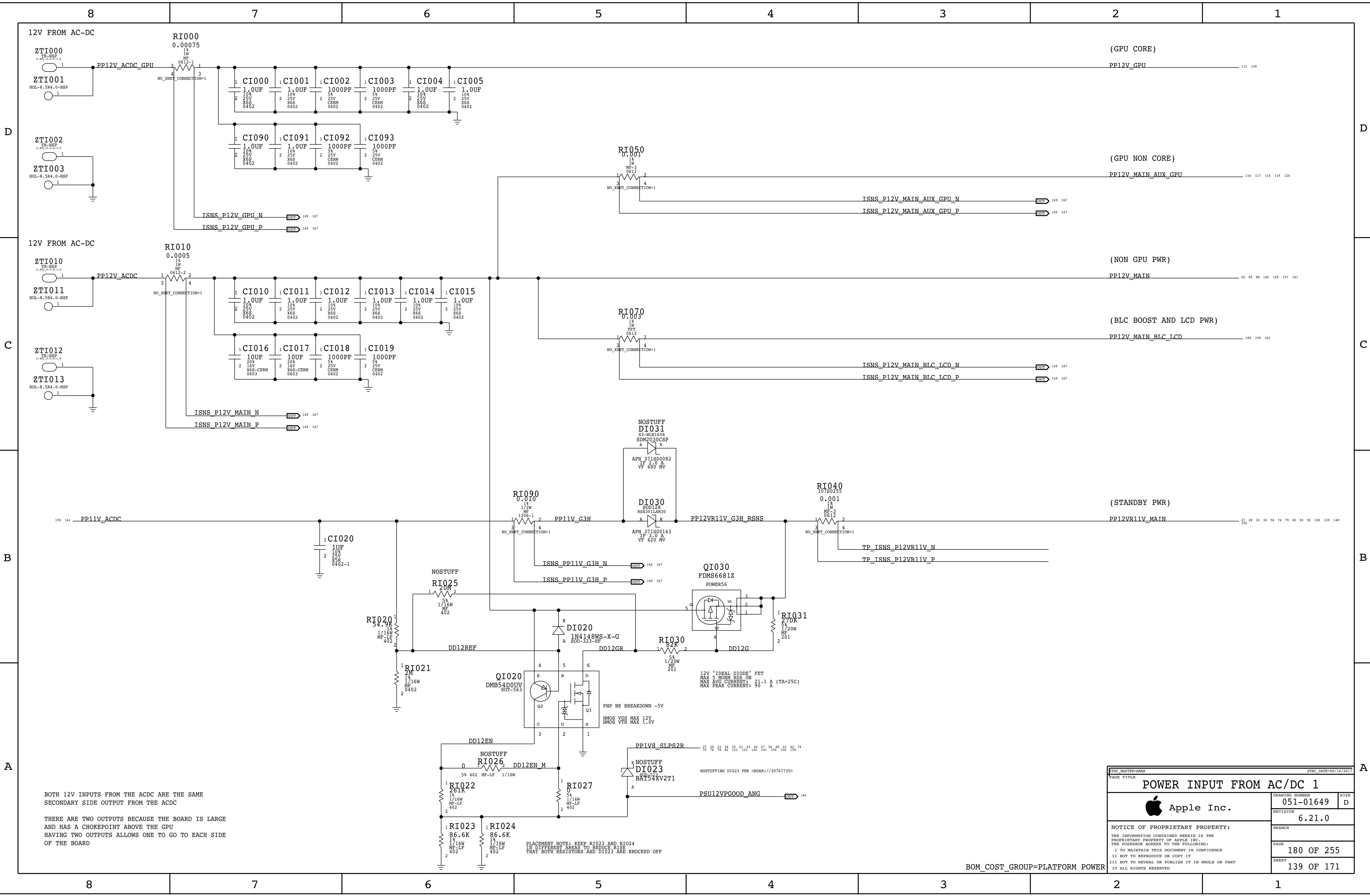
BT LPF AND MATCHING

WIFI/BT MODULE 3		
	DRAWING NUMBER	051-01649
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BOM_COST_GROUP=WIRELESS



WIFI/BT SUPPORT 1		DRAWING NUMBER	051-01649	SIZE	D
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BOTH 12V INPUTS FROM THE ACDC ARE THE SAME
SECONDARY SIDE OUTPUT FROM THE ACDC

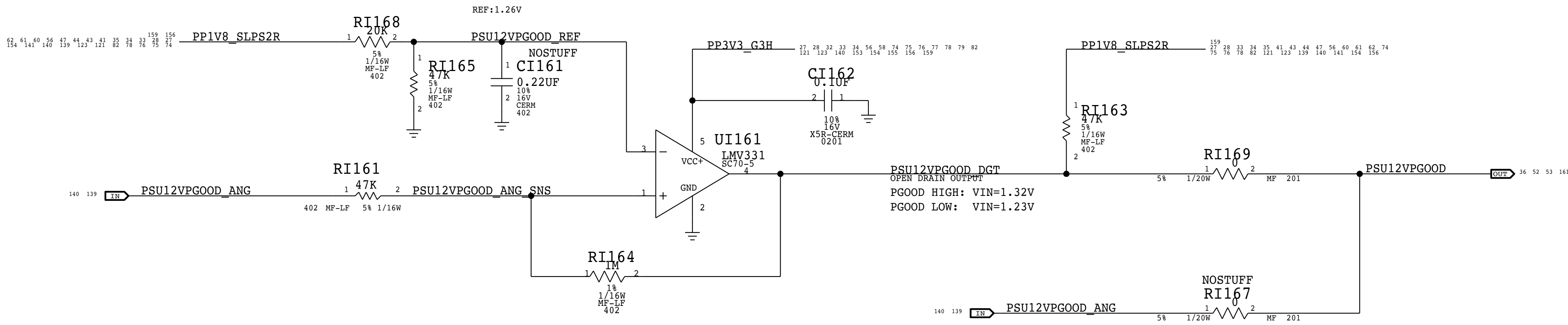
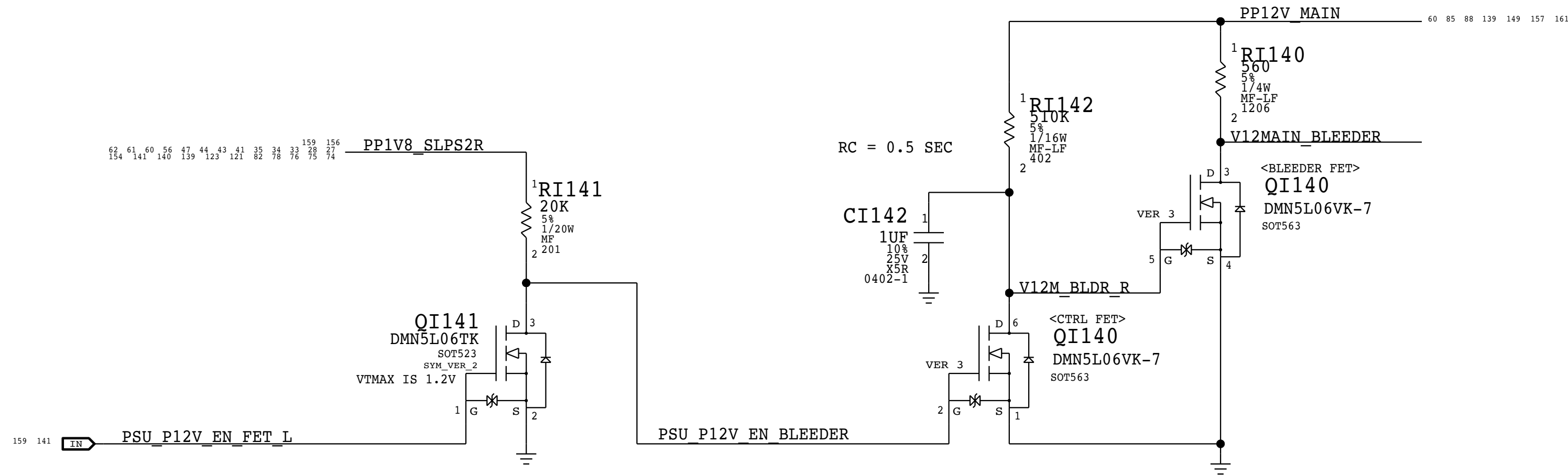
THERE ARE TWO OUTPUTS BECAUSE THE BOARD IS LARGE
AND HAS A CHOKEPOINT ABOVE THE GPU
HAVING TWO OUTPUTS ALLOWS ONE TO GO TO EACH SIDE
OF THE BOARD

PLACEMENT NOTE: KEEP RI023 AND RI024
IN DIFFERENT AREAS TO REDUCE RISK
THAT BOTH RESISTORS AND DI023 ARE KNOCKED OFF

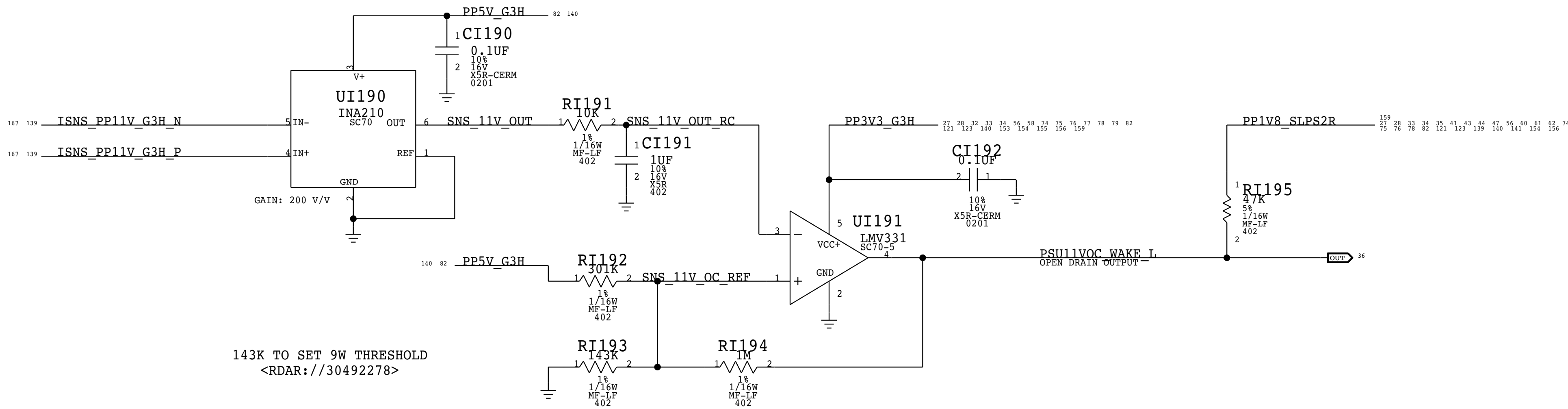
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POWER INPUT FROM AC/DC 1		DRAWING NUMBER		051-01649	
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BOM_COST_GROUP=PLATFORM POWER

PP12_MAIN BLEEDER CIRCUIT



PP11V_STBY OVERCURRENT DETECTION



143K TO SET 9W THRESHOLD
<RDAR://30492278>

EYEC MASTER=AMAN		SYNC DATE=06/16/2017	
PAGE TITLE			
POWER INPUT FROM AC/DC 2			
 Apple Inc.	DRAWING NUMBER		SHEET
	051-01649		D
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BOM_COST_GROUP=PLATFORM POWER

D

C

B

A

D

C

B

A

Power Sequencing requirements

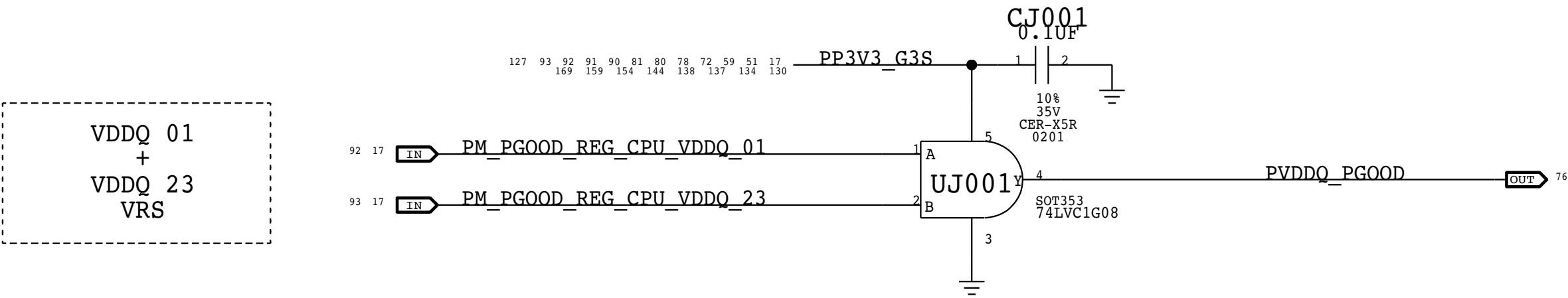
System:

1. 12V_ACDC -> 3V42G3H -> 12V_S5/12V_S5_SSD -> 3V3_S5 -> 1V0_S5 -> 5V_S4/3V3_S4 -> 1V0_S3

Intel CPU:

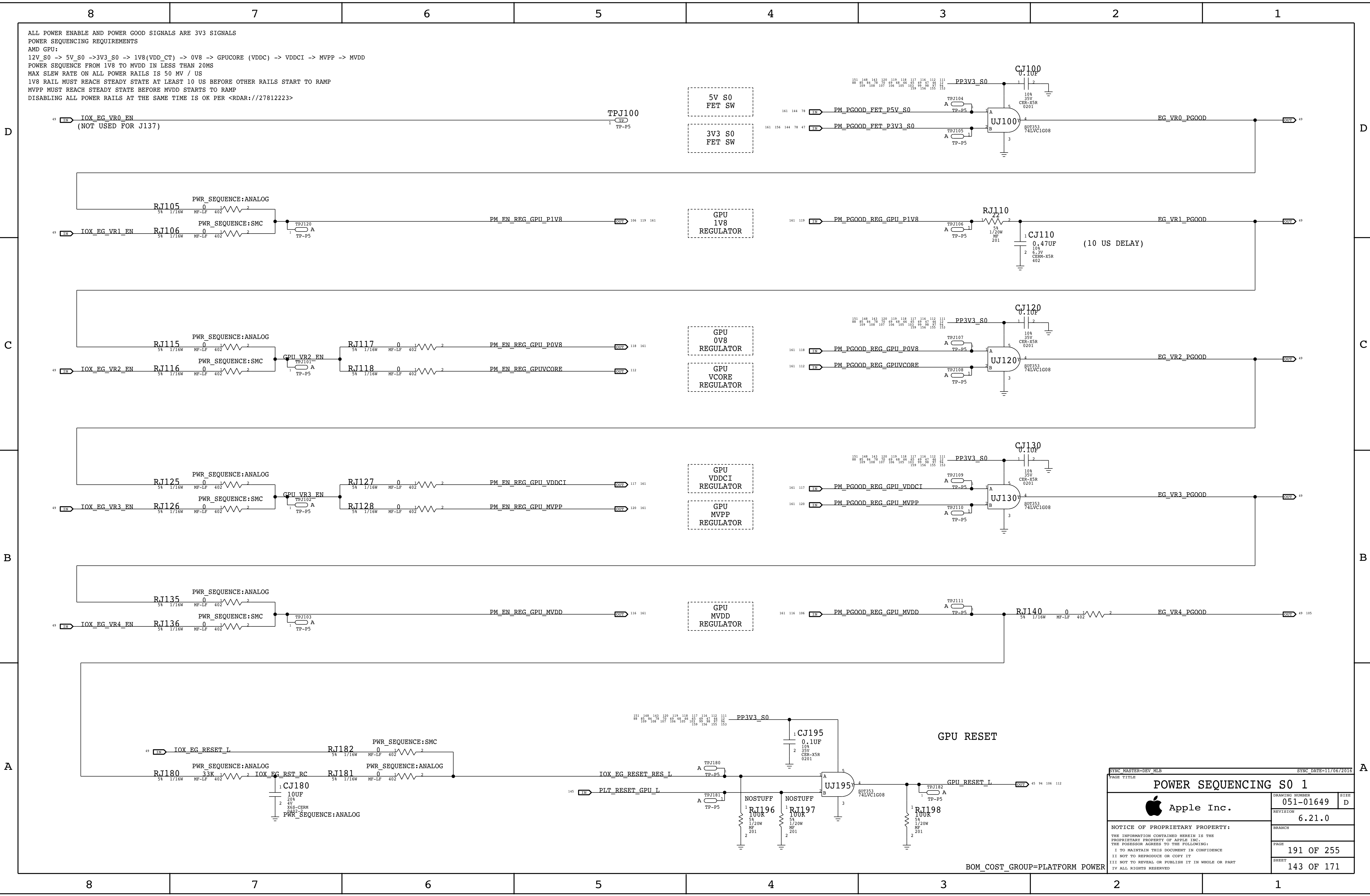
2. 1V0_S3 -> 2V5_S3 -> VDDQ_S3 -> 12V_S0 -> 5V_S0 -> 3V3_S0 -> GPU Rails -> VCCIO_S0-> VCC_CPU_S0 (VCCGT & VCCSA)

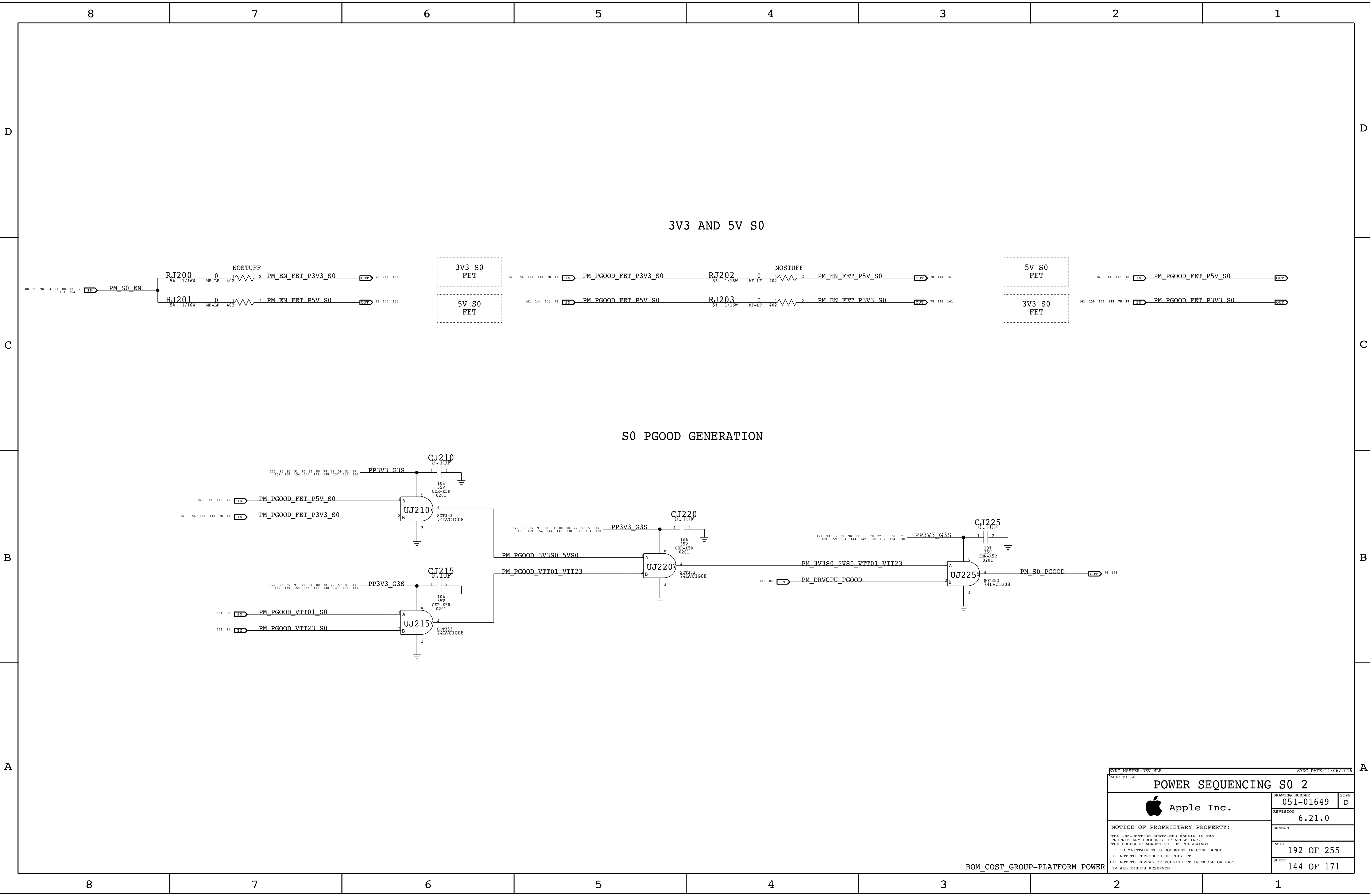
MEMORY POWER PGOOD

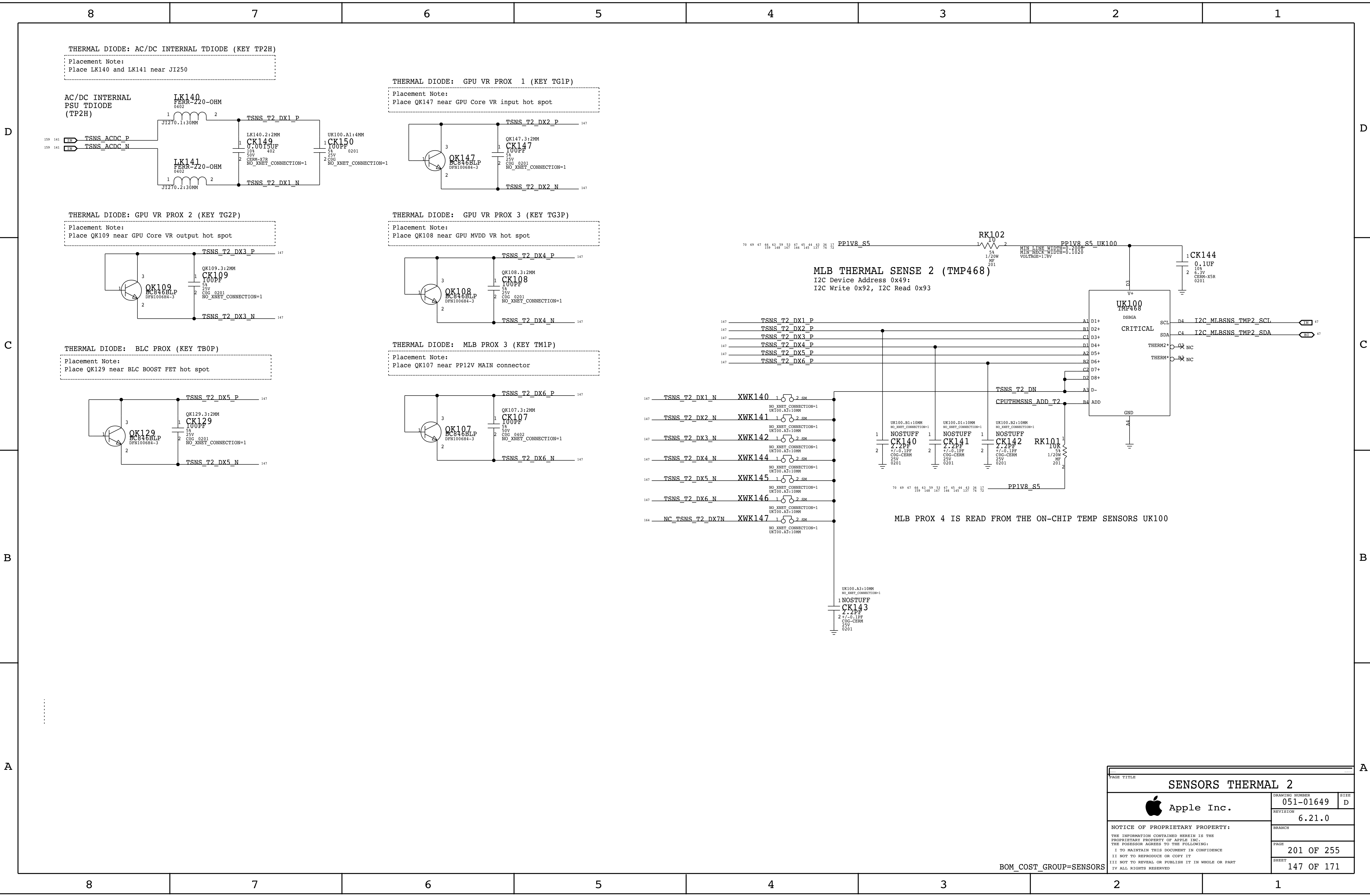


BOM_COST_GROUP=PLATFORM POWER

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POWER SEQUENCING S3		051-01649		6.21.0	
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BOM_COST_GROUP=SENSORS

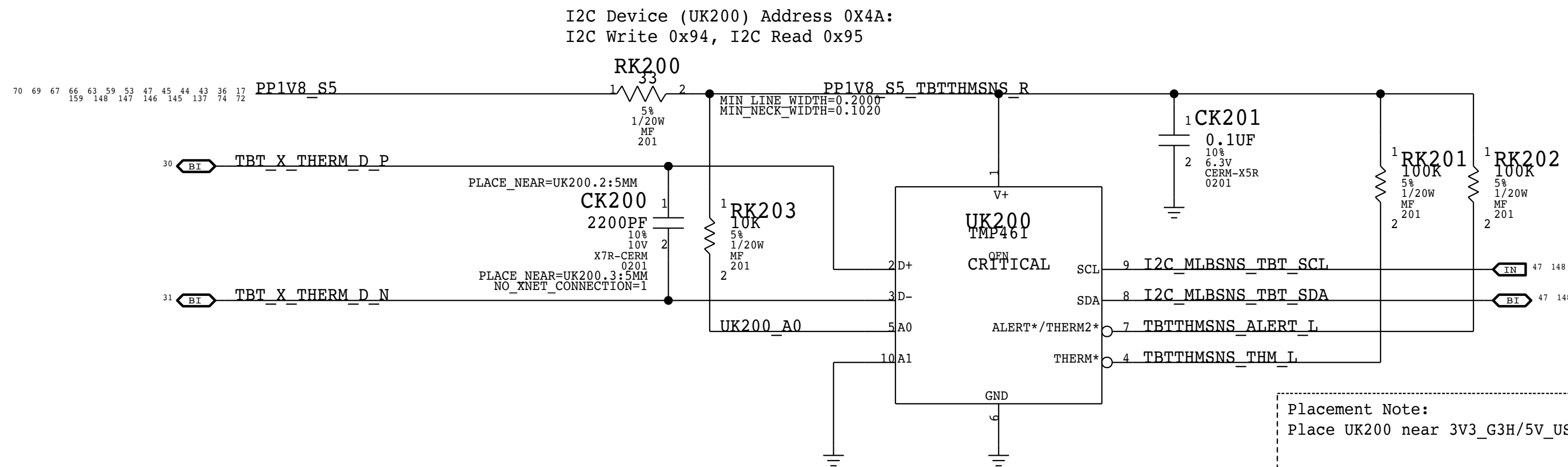
Thermal Sensor A:
Thunderbolt Die

I2C Write: 0xD8, I2C Read: 0xD9

THERMAL DIODE: TBT DIE (TBT1 - KEY TI0D)

Placement Note:

The P leg connects to THERMDA pin of the TBT chip, the N leg connect to TitanRidge pin V9.



Placement Note:
Place UK200 near 3V3_G3H/5V_USBC VR

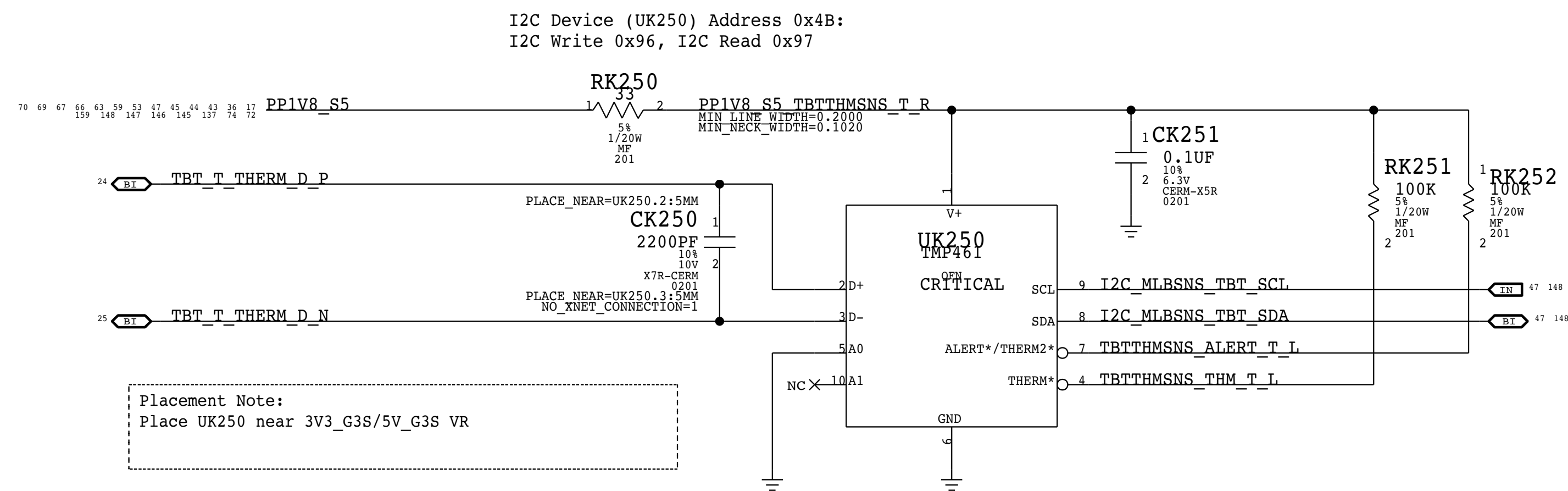
Thermal Sensor B:
Thunderbolt Die

I2C Write: 0xB8, I2C Read: 0xB9

THERMAL DIODE: TBT DIE (TBT2 - KEY TI1D)

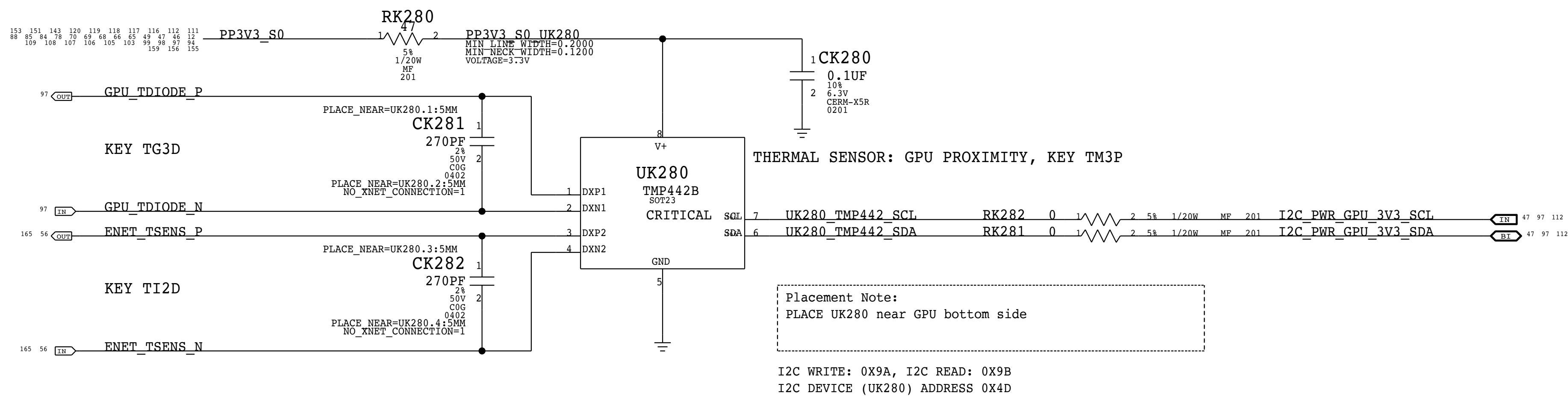
Placement Note:

The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.



Placement Note:
Place UK250 near 3V3_G3S/5V_G3S VR

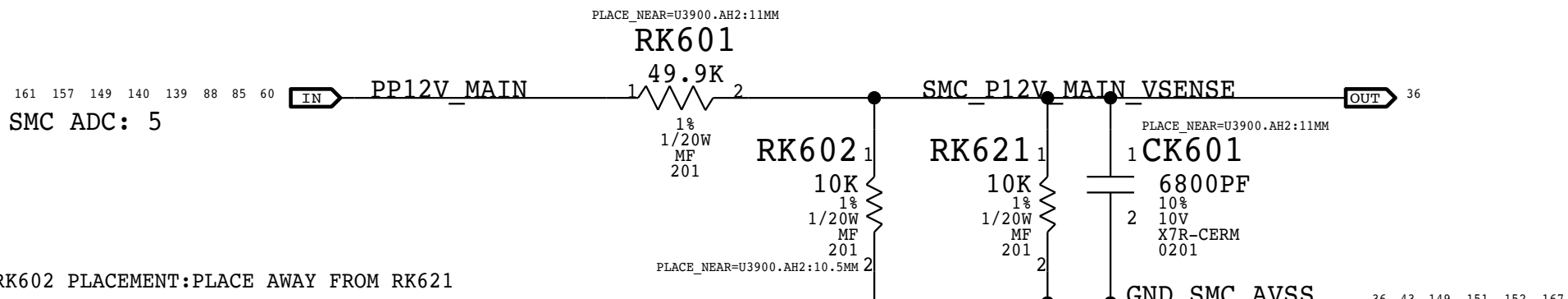
GPU THERMAL - KEY TM3P



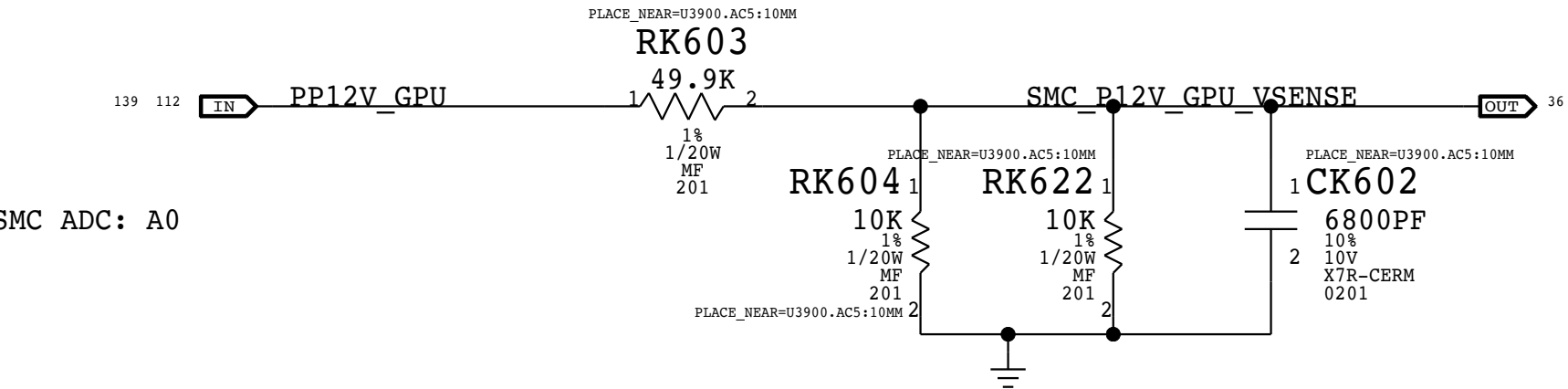
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I2C WRITE: 0X9A, I2C READ: 0X9B
I2C DEVICE (UK280) ADDRESS 0X4D
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SENSORS THERMAL 3			
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AC/DC 12V MAIN LOW SIDE VOLTAGE SENSE (KEY VD2R)

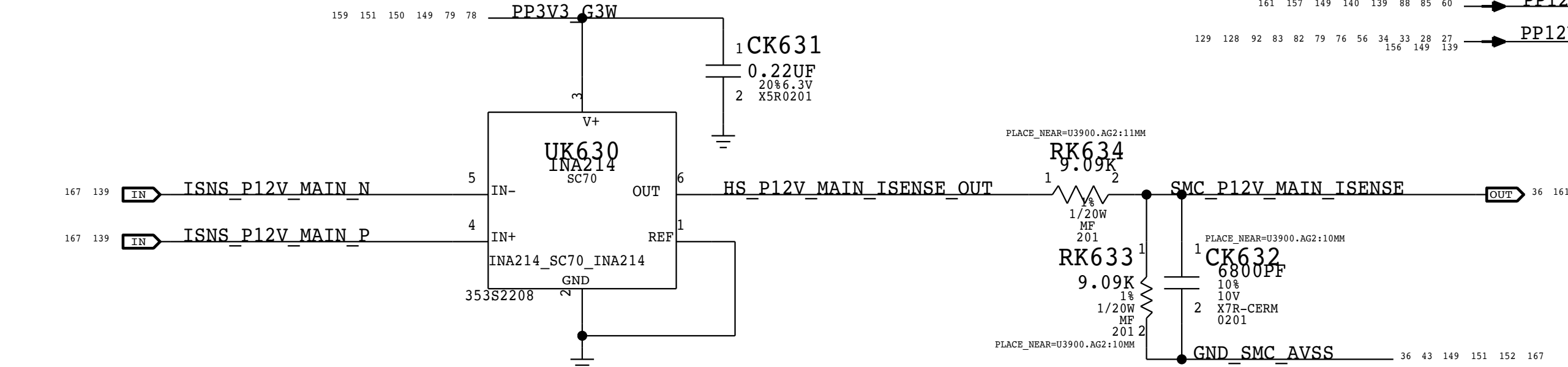


AC/DC 12V GPU HIGH SIDE VOLTAGE SENSE (KEY VG2R)



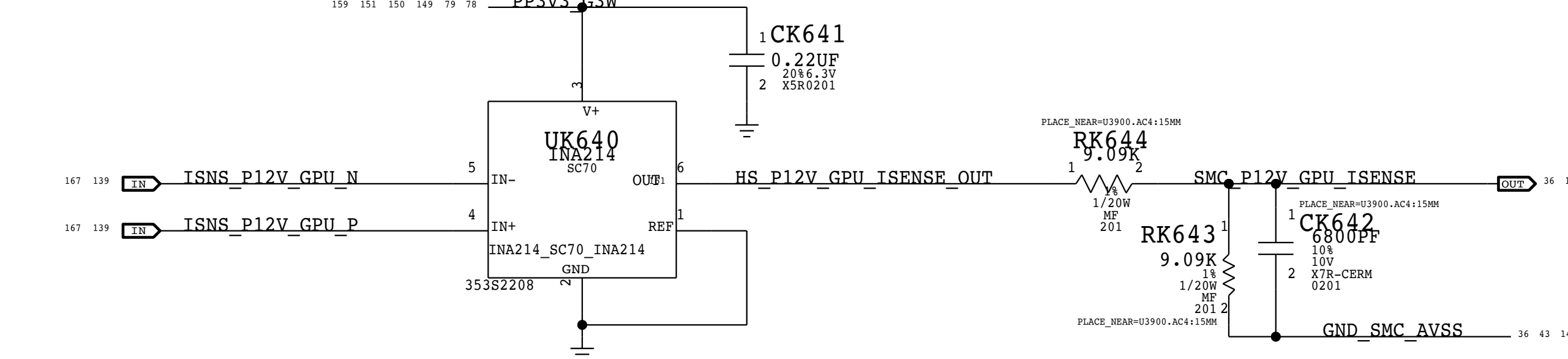
PP12V MAIN HIGH SIDE CURRENT SENSE (KEY ID2M)

GAIN: 100X, EDP: 30 A
RSENSE: 0.00075 (RI010) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 33.33 A
SMC ADC: 0



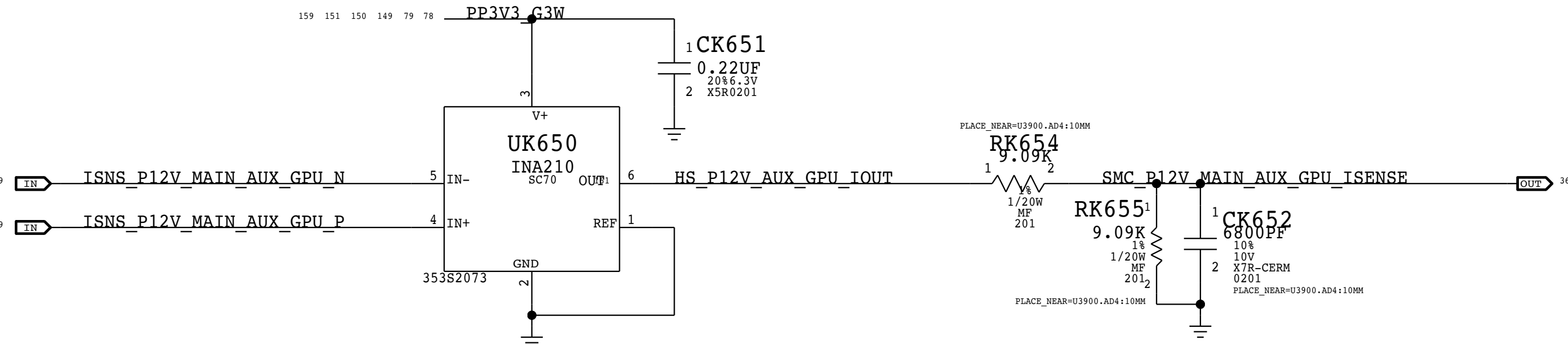
PP12V GPU HIGH SIDE CURRENT SENSE (KEY IG2R)

GAIN: 100X, EDP: 30 A
RSENSE: 0.00075 (RI000) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 33.33 A
SMC ADC: 1



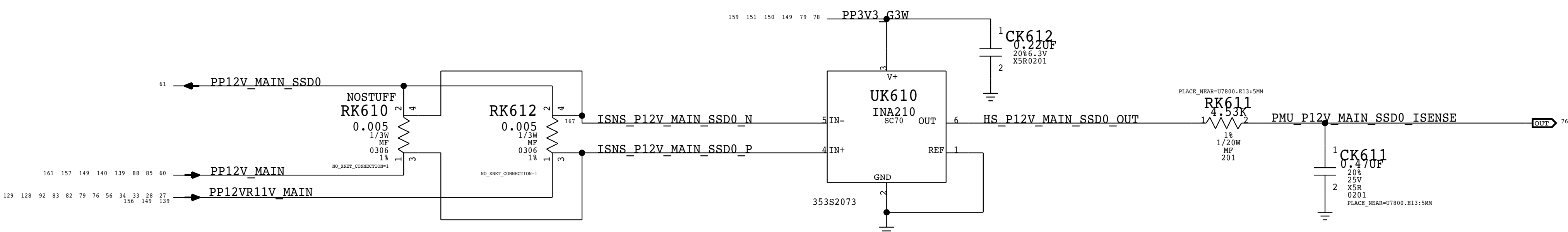
PP12V MAIN AUXILIARY GPU RAIL HIGH SIDE CURRENT SENSE (KEY IG2A)

GAIN: 100X, EDP: 6 A
RSENSE: 0.003 (RI050) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 8.3 A
PMU ADC: A5



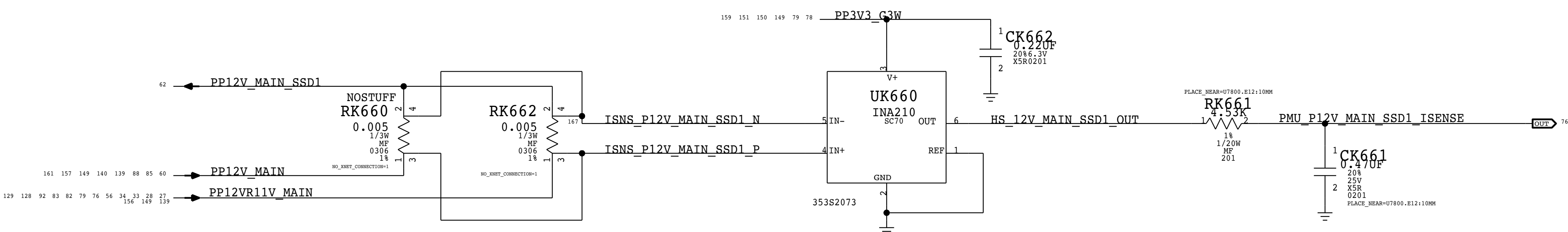
PP12V SSD0 HIGH SIDE CURRENT SENSE (KEY IH02)

GAIN: 200X, EDP: 0.5 A
RSENSE: 0.005 (RK610) OR RSENSE SHORT
VSENSE: 13 MV, RANGE: 2.5 A
PMU ADC: B1



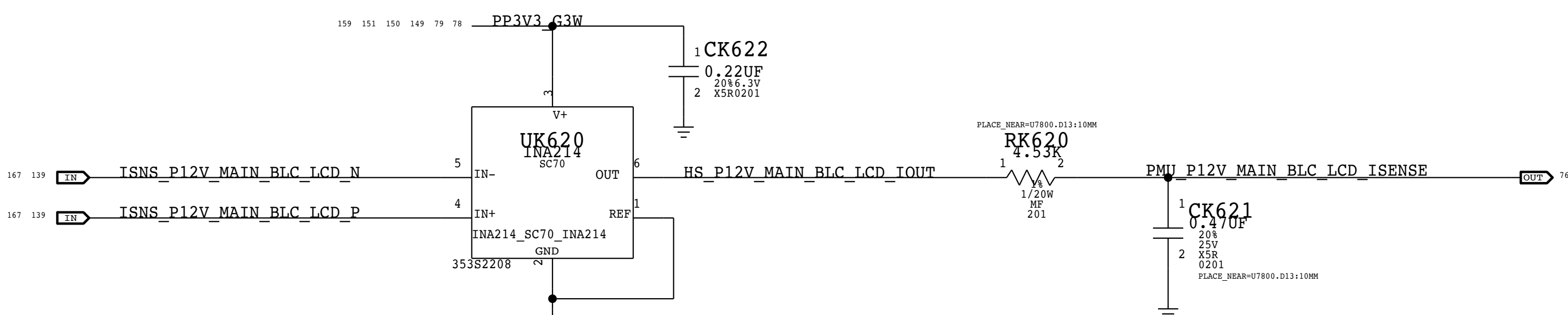
PP12V SSD1 HIGH SIDE CURRENT SENSE (KEY IH12)

GAIN: 200X, EDP: 0.5 A
RSENSE: 0.005 (RK660) OR RSENSE SHORT
VSENSE: 13 MV, RANGE: 2.5 A
PMU ADC: B2



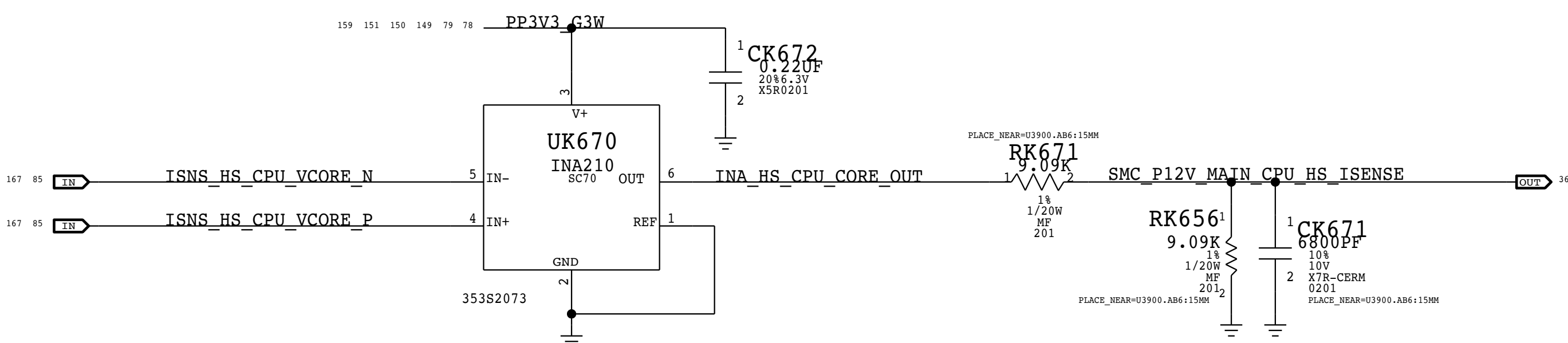
PP12V BACKLIGHT AND LCD HIGH SIDE CURRENT SENSE (KEY IB2R)

GAIN: 100X, EDP: 6 A
RSENSE: 0.003 (RI070) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 8.3 A
PMU ADC: B0



PP12V MAIN CPU HIGH SIDE CURRENT SENSE (KEY IC2R)

GAIN: 100X, EDP: 15 A
RSENSE: 0.005 (R9090) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 25 A
PMU ADC: B7

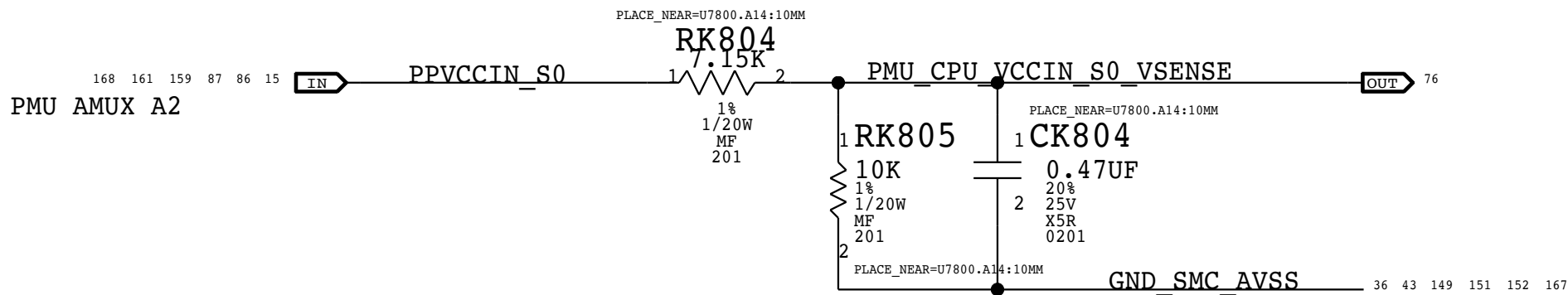


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BOM_COST_GROUP=SENSORS

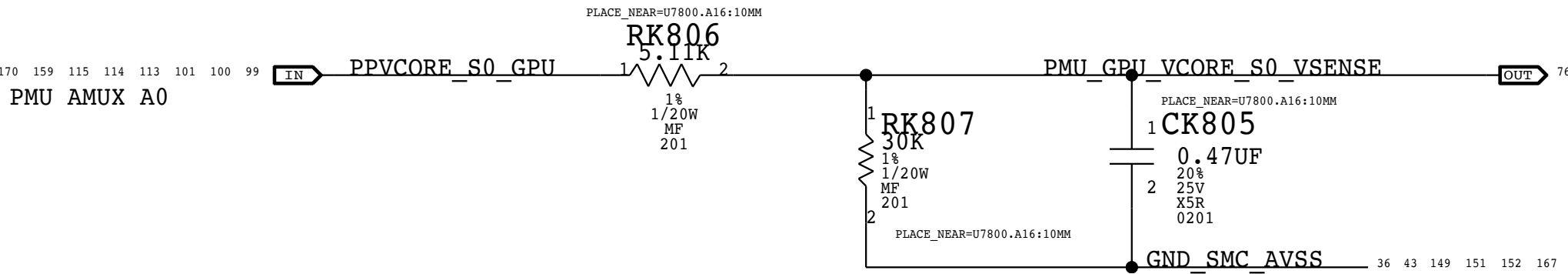
PPVCCIN CPU CORE VOLTAGE SENSE (KEY VC0C)

PPVCCIN RANGE: 1.5V - 2V



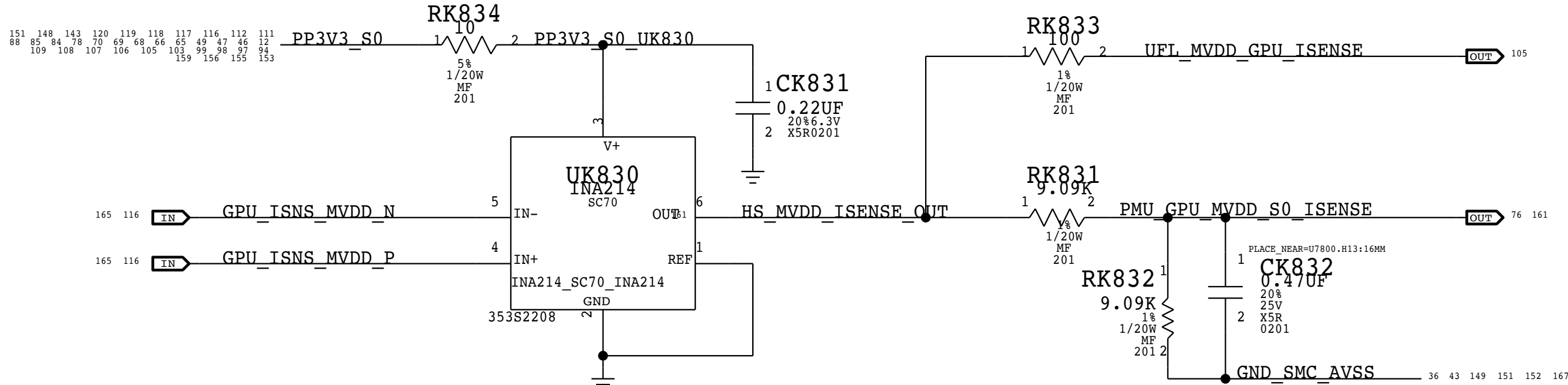
PPVCORE S0 GPU CORE VOLTAGE SENSE (KEY VG0C)

PPVCORE S0 GPU RANGE: 0.6V - 1.2V



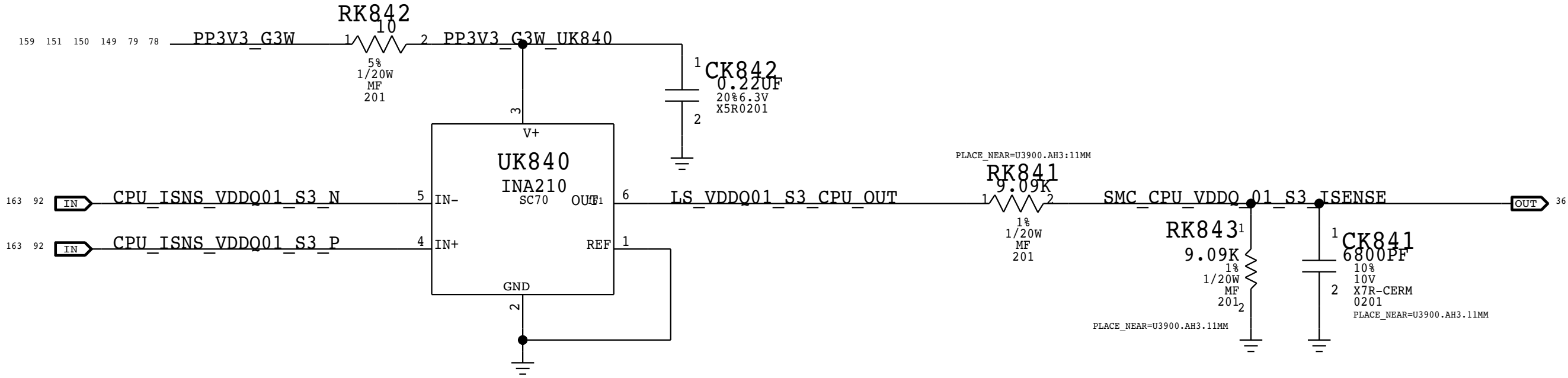
MVDD GPU LOWSIDE CURRENT SENSE (KEY IG0M)

GAIN: 100X, EDP: 30 A
RSENSE: 0.0005 (RD400) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 50 A
SMC ADC: 4



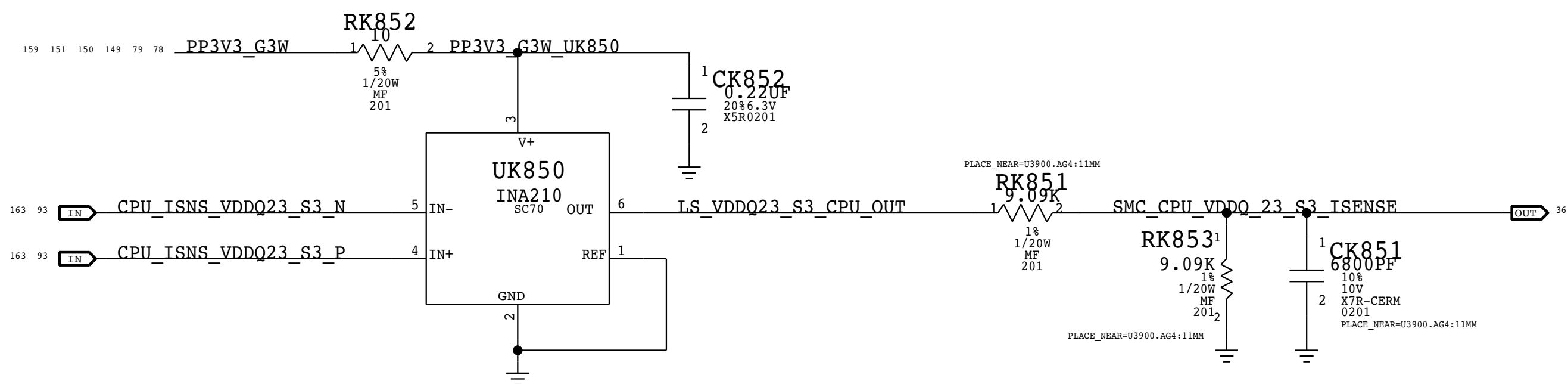
PPVDDQ01 S3 LOWSIDE CURRENT SENSE (KEY IR01)

GAIN: 100X, EDP: 25 A
RSENSE: 0.0005 (R9800) OR RSENSE SHORT
VSENSE: 25 MV, RANGE:50 A
PMU ADC: A1



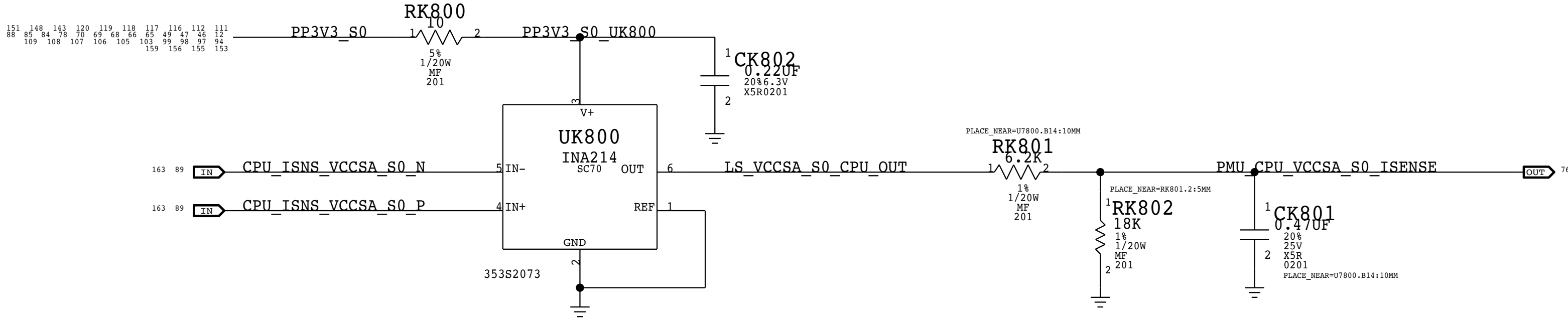
PPVDDQ23 S3 LOWSIDE CURRENT SENSE (KEY IC23)

GAIN: 100X, EDP: 25 A
RSENSE: 0.0005 (R9900) OR RSENSE SHORT
VSENSE: 25 MV, RANGE:50 A
PMU ADC: A2



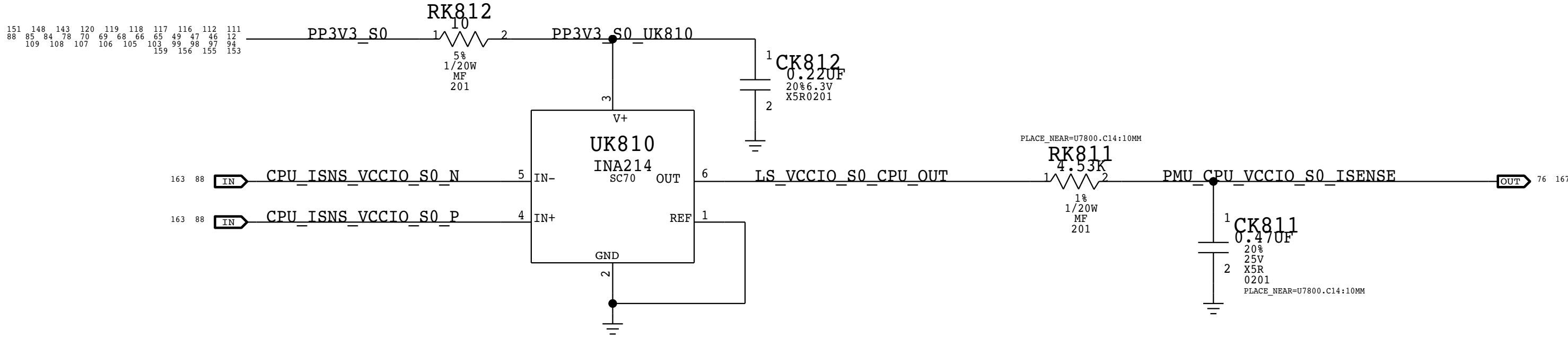
PPVCCSA S0 LOWSIDE CURRENT SENSE (KEY IC0S)

GAIN: 100X, EDP: 15 A
RSENSE: 0.001 (R9525) OR RSENSE SHORT
VSENSE: 25 MV, RANGE:25 A
PMU ADC: A3



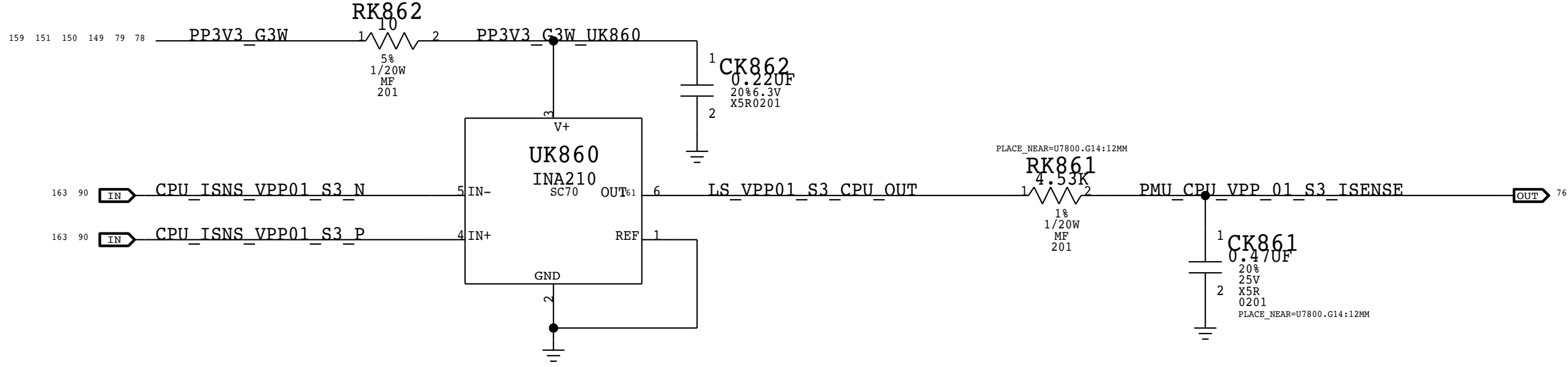
PPVCCIO S0 LOWSIDE CURRENT SENSE (KEY IC0I)

GAIN: 200X, EDP: 12 A
RSENSE: 0.001 (R9400) OR RSENSE SHORT
VSENSE: 13 MV, RANGE:12.5 A
PMU ADC: A4



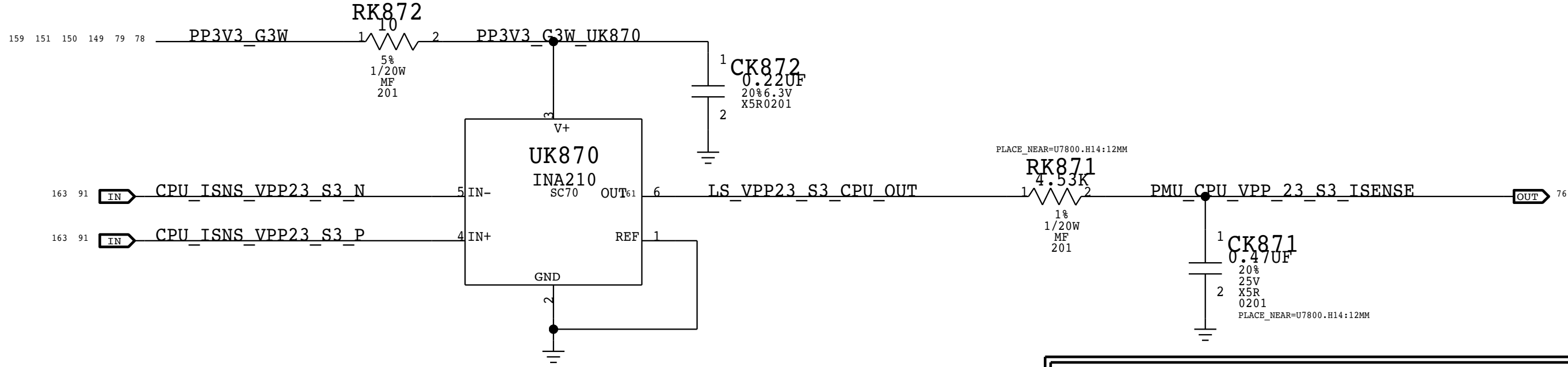
PPVPP01 S3 LOWSIDE CURRENT SENSE (KEY IM0P)

GAIN: 200X, EDP: 4 A
RSENSE: 0.001 (R9610) OR RSENSE SHORT
VSENSE: 13 MV, RANGE:12.5 A
PMU ADC: B5



PPVPP23 S3 LOWSIDE CURRENT SENSE (KEY IM1P)

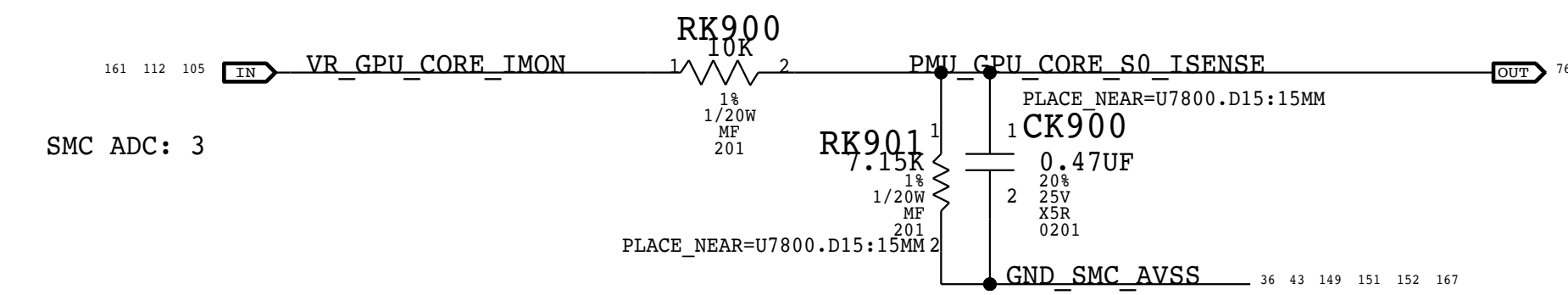
GAIN: 200X, EDP: 4 A
RSENSE: 0.001 (R9710) OR RSENSE SHORT
VSENSE: 13 MV, RANGE:12.5 A
PMU ADC: B6



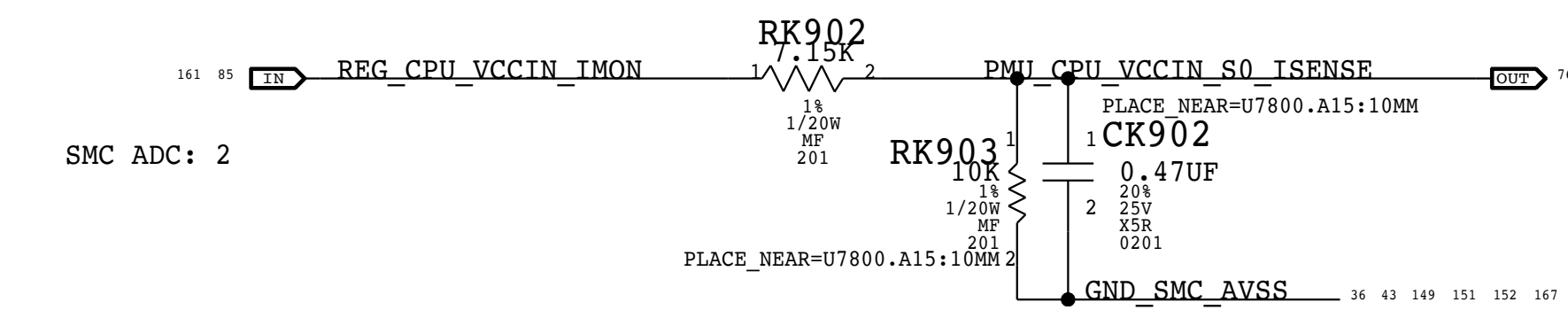
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SENSORS POWER LOWSIDE 1		
	DRAWING NUMBER	051-01649
	REVISION	6.21.0
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BOM_COST_GROUP=SENSORS

CPU CORE CURRENT RANGE: 432A
VOLTAGE RANGE: INPUT 2.16V/432A - OUTPUT $2.16 \times 7.15 / (10 + 7.15) = 2.2\text{mV}$
VSENSE: 2.2mV/A

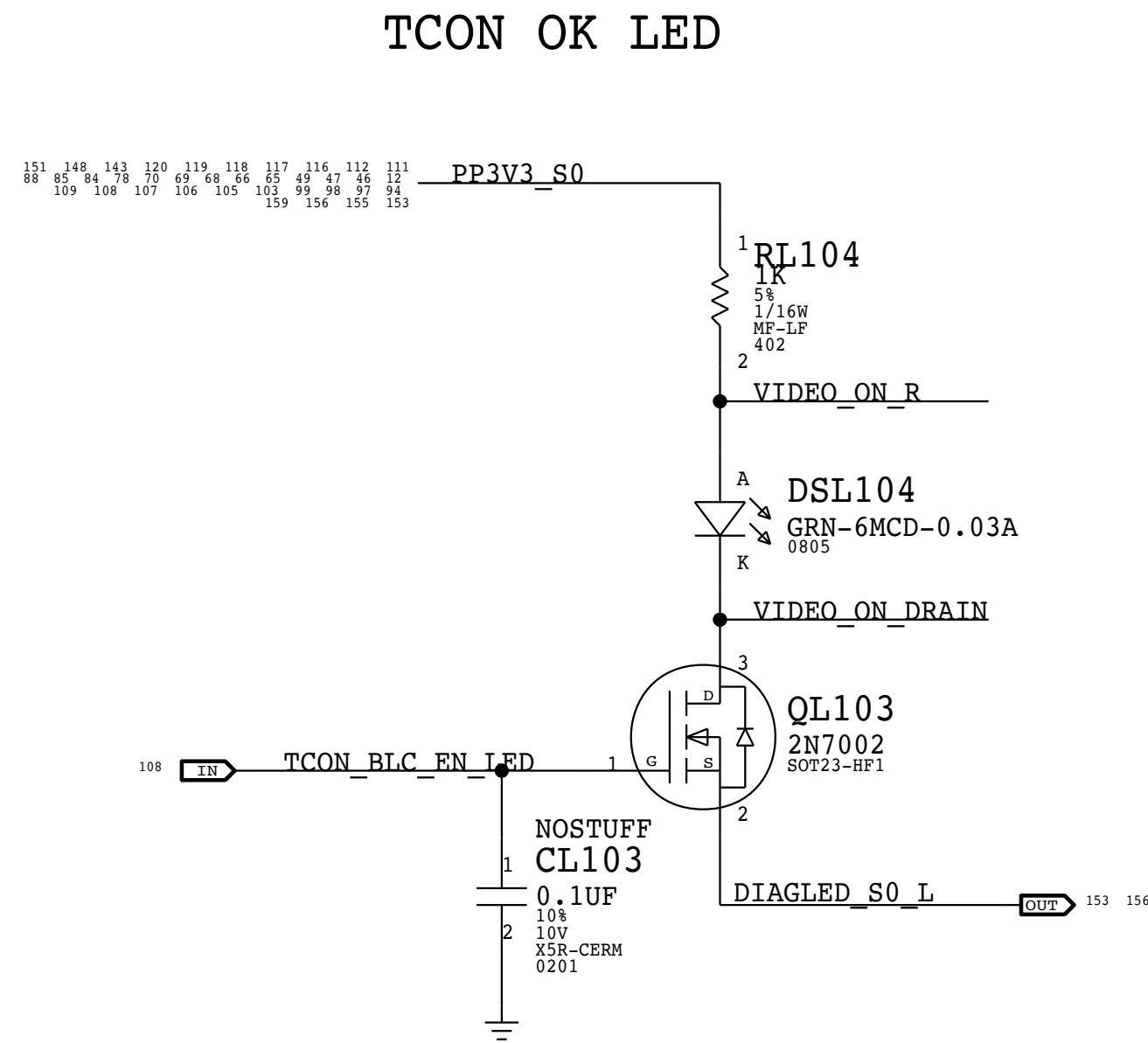
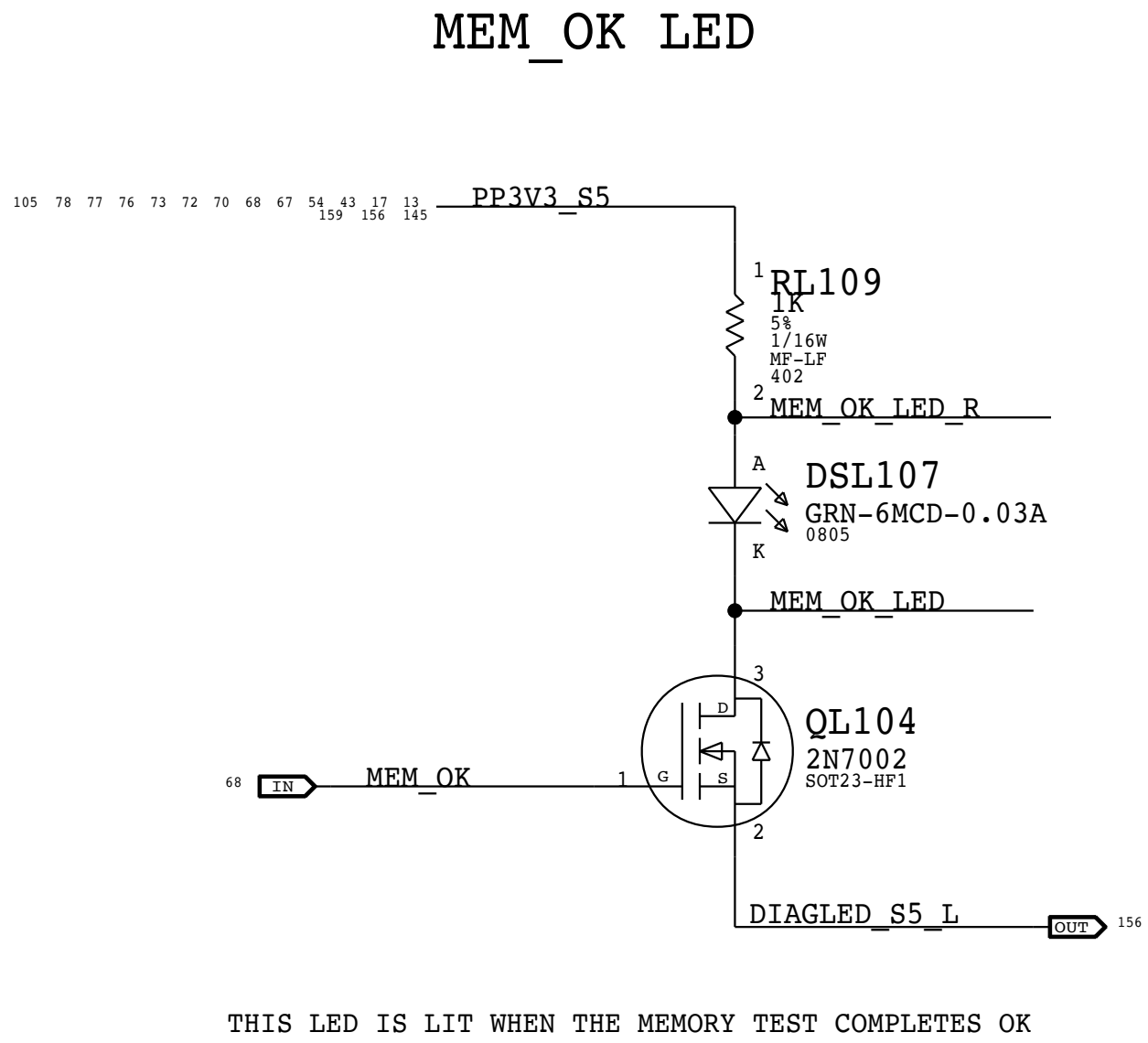
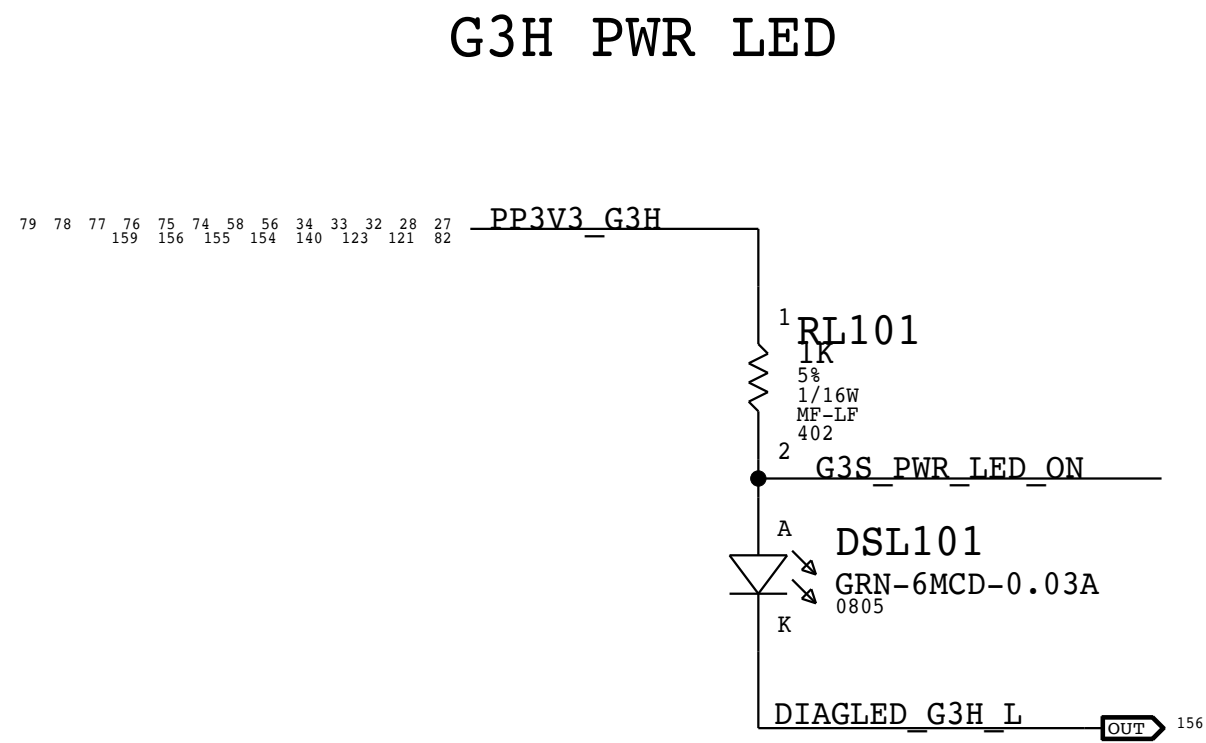
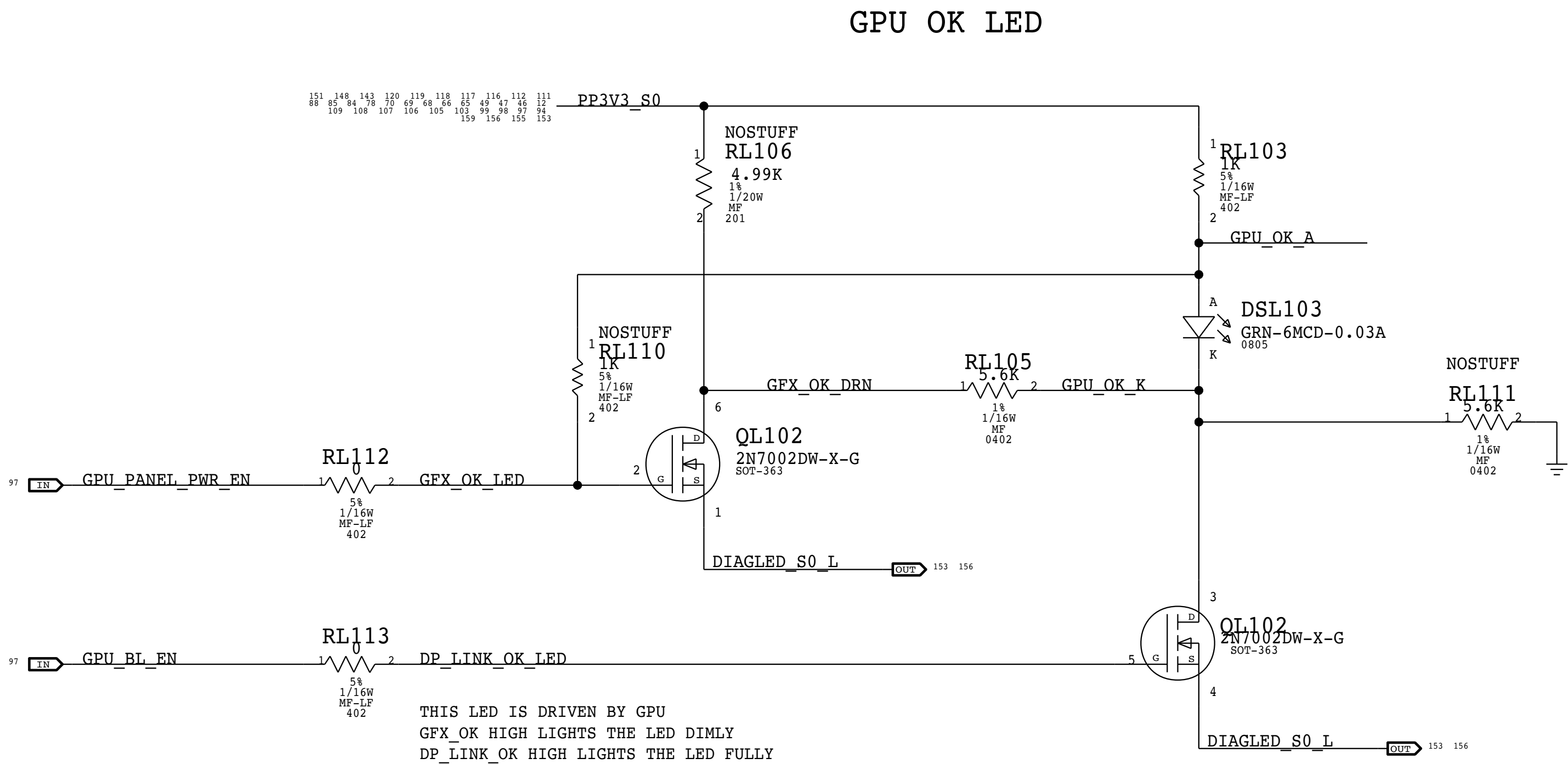
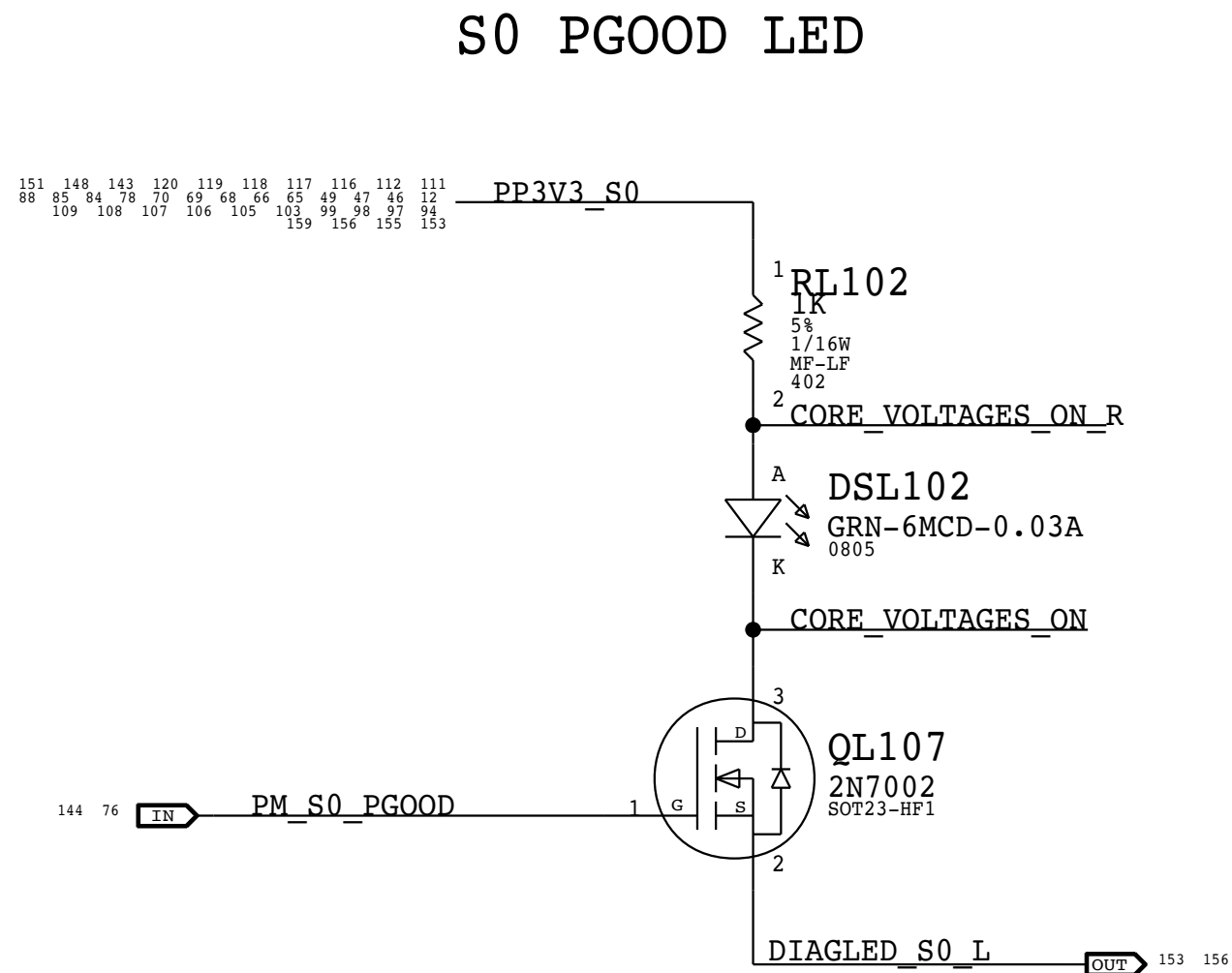


CPU CORE CURRENT RANGE: 200A
VSENSE: 1.25 MV/A



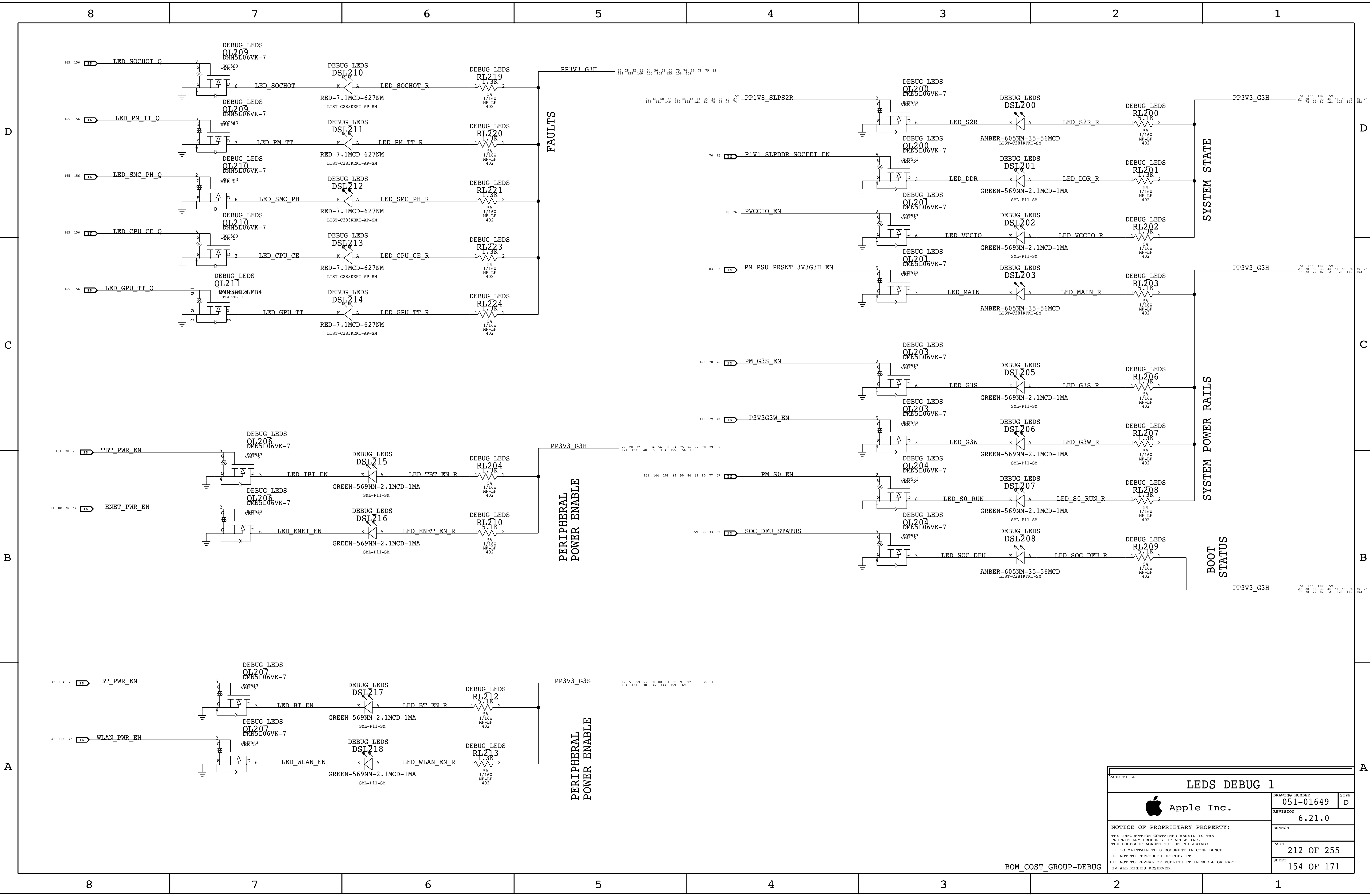
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	SHEET	152 OF 171	

POR LED LIST TRACKED AT <RDAR://30685511>
NOTE, NONE OF THESE WILL TURN ON BY DEFAULT UNLESS SWITCH SL550 IS PUSHED

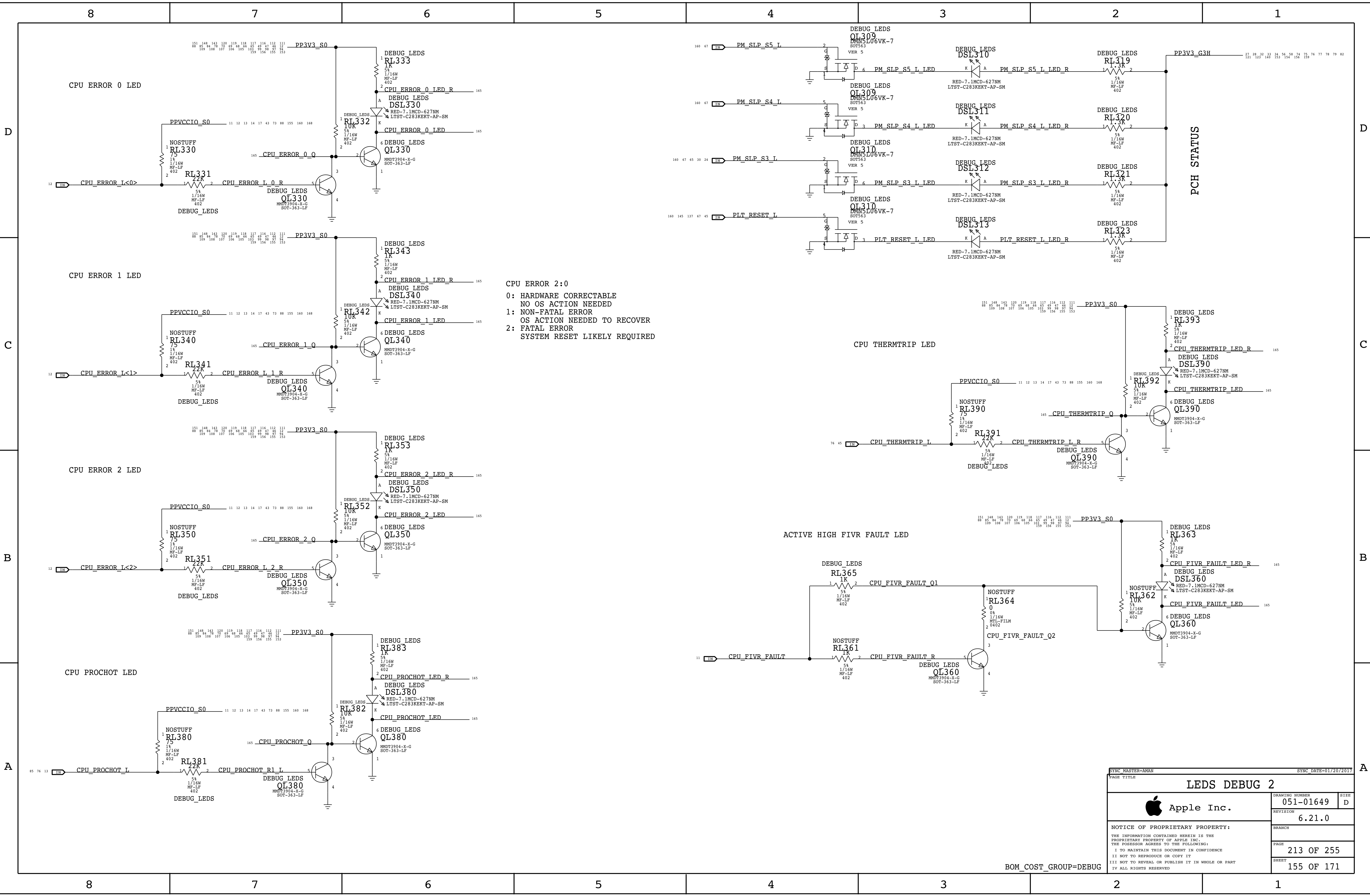


SYNC_MASTER=DEV_RLB		SYNC_DATE=11/06/2016	
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LEDS POR			
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		PAGE	211 OF 255
		SHEET	153 OF 171

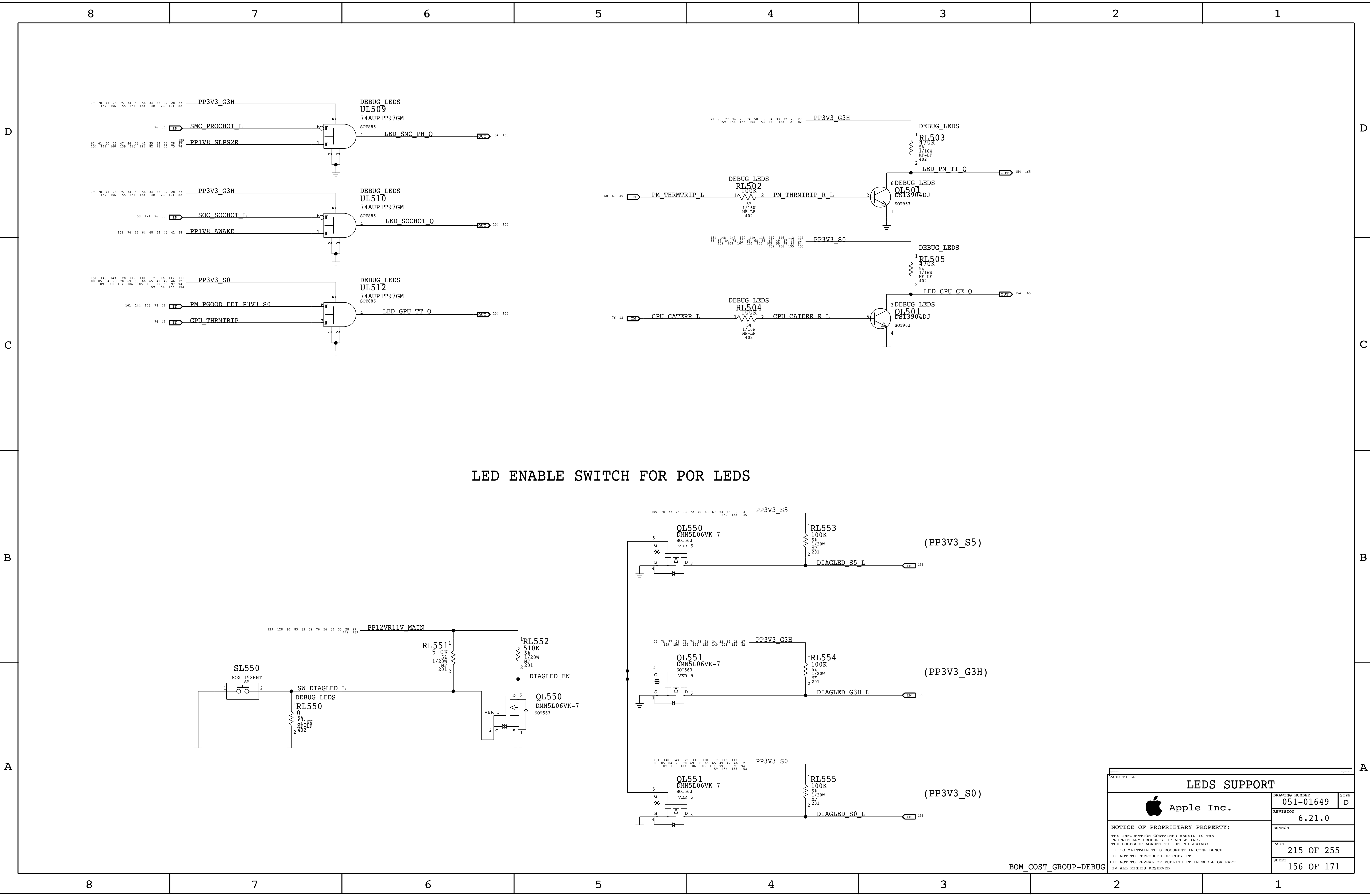
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LEDS DEBUG 1		
 Apple Inc.	DRAWING NUMBER	051-01649
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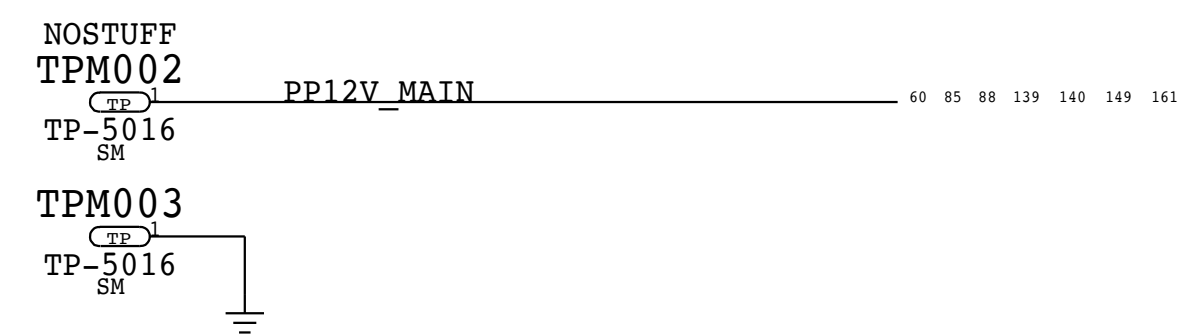
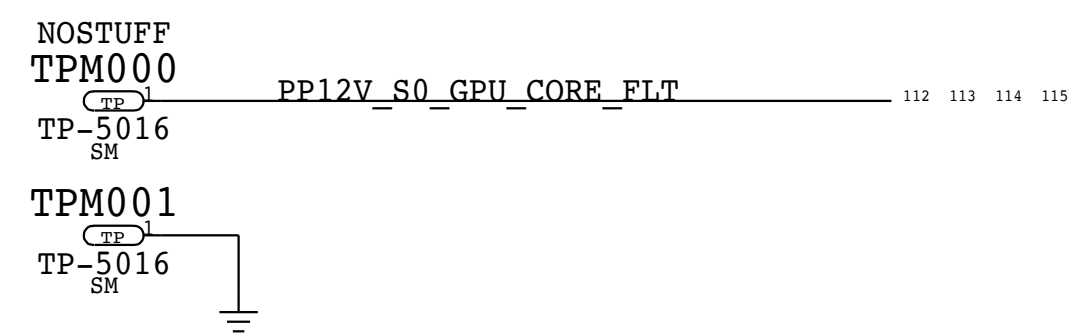
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LED ENABLE SWITCH FOR POR LEDs

BOM_COST_GROUP=DEBUG

PAGE TITLE		
LEDS SUPPORT		
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PAGE TITLE			
PSU RESISTANCE MEASUREMENT			
 Apple Inc.	DRAWING NUMBER		SIZE
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BOM_COST_GROUP=PLATFORM POWER

D

C

B

A

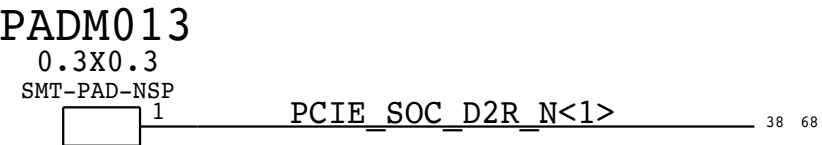
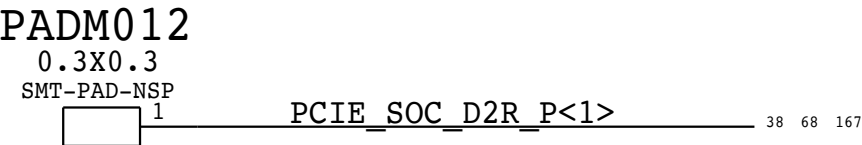
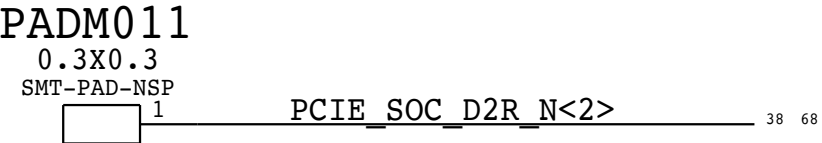
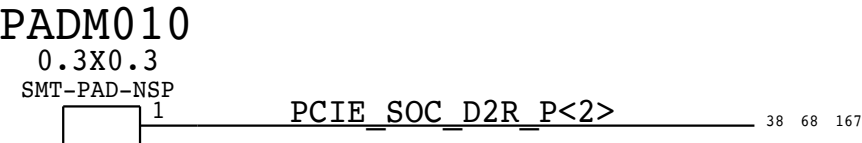
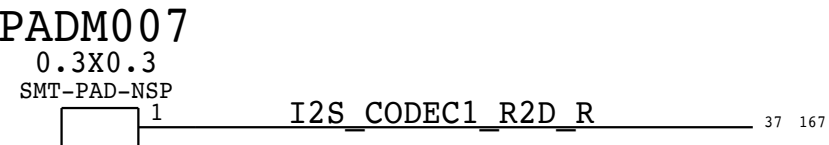
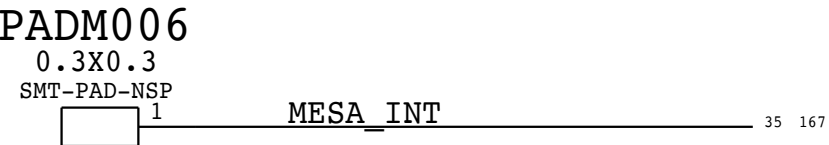
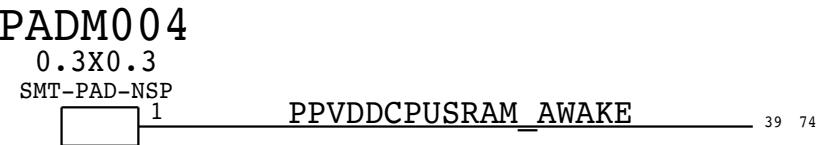
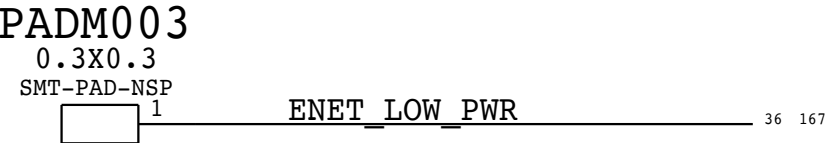
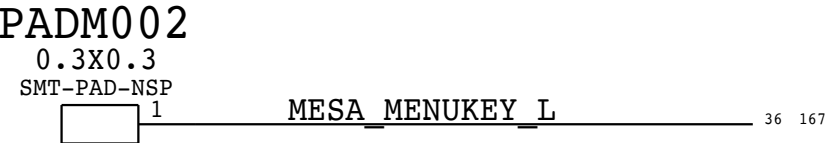
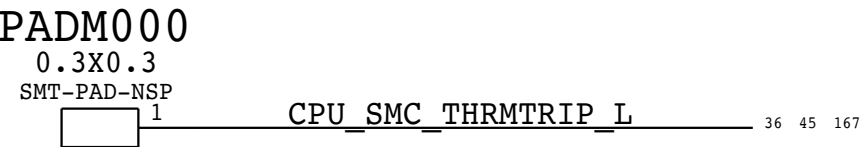
D

C

B

A

SMALL PADS TO EXPOSE NETS IN SOC AREA WHERE TP WON'T FIT



BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE		
TEST PADS FOR SOC AREA		
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DFU TEST POINTS <RDAR://30758642>

POWER SEQUENCE

TPN070	A	1	TP-P5	PP3V3_G3H	27	28	29	33	34	56	58	74	75	76	77	78	79	82
TPN071	A	1	TP-P5	PP3V3_G3W	121	122	140	153	154	155	156							
TPN072	A	1	TP-P5	PP3V3_G3S	78	79	149	150	151									
TPN073	A	1	TP-P5	PP1V8_SLPS2R	17	51	59	72	78	80	81	89	91	92	93	127	130	
TPN074	A	1	TP-P5	PP0V82_SLDDR	27	28	33	34	35	41	43	44	47	56	60	61	62	74
TPN075	A	1	TP-P5	PP1V2_AWAKE	75	76	78	82	121	123	139	140	141	154	156			
TPN076	A	1	TP-P5	PPVDDCRU_AWAKE_SOC	39	74	122	126										
TPN077	A	1	TP-P5	PP1V8_S5	116	117	118	119	120	143	148	151	153	155				
TPN078	A	1	TP-P5	PP1V0_S0	13	45	74	76										
TPN079	A	1	TP-P5	PP3V3_S0	116	117	118	119	120	143	148	151	153	155				
TPN090	A	1	TP-P5	PPVCCIN_S0	15	86	87	151	161	168	156							
TPN091	A	1	TP-P5	PPVCORE_S0_GPU	99	100	101	113	114	115	151	170						

POWER

TPN030	A	1	TP-P5	PP12V_ACDC_GPU	139													
TPN031	A	1	TP-P5															
TPN032	A	1	TP-P5	PP12V_ACDC	139													
TPN033	A	1	TP-P5															
TPN034	A	1	TP-P5	PP11V_ACDC	139	141												
TPN035	A	1	TP-P5	TSNS_ACDC_P	141	147												
TPN036	A	1	TP-P5	TSNS_ACDC_N	141	147												
TPN037	A	1	TP-P5	PSU_P12V_EN_FET_L	140	141												
TPN038	A	1	TP-P5	ACDC_ID_R	141													
TPN039	A	1	TP-P5															
TPN080	A	1	TP-P5	IMAC_PWR_BTN_R	121													
TPN081	A	1	TP-P5	PP5V_G3S_USBC	26	27	28	32	33	34	79							
TPN082	A	1	TP-P5	PP5V_G3S_USBA	53	53	82											
TPN024	A	1	TP-P5	PP3V3_S5	13	17	43	54	67	68	70	72	73	76	77	78	105	

DEBUG MOJO

TPN040	A	1	TP-P5	SMC_DEBUGPRT_RX	26	36	123											
TPN041	A	1	TP-P5	SMC_DEBUGPRT_TX	26	36	123											
TPN042	A	1	TP-P5	PCH_DEBUG_UART_R2D	34	69	72											
TPN043	A	1	TP-P5	PCH_DEBUG_UART_D2R	34	69	72											
TPN044	A	1	TP-P5															

DFU

TPN045	A	1	TP-P5	SOC_USB_VBUS	32	35	121											
TPN046	A	1	TP-P5	USB_SOC_N	35	121												
TPN047	A	1	TP-P5	USB_SOC_P	35	121												
TPN049	A	1	TP-P5	PMU_ACTIVE_READY	32	33	35	76										
TPN050	A	1	TP-P5	SOC_FORCE_DFU	32	33	35	76	121									
TPN051	A	1	TP-P5	SOC_DFU_STATUS	32	33	35	154										
TPN054	A	1	TP-P5	SOC_DOCK_CONNECT	33	34	36											
TPN055	A	1	TP-P5	SOC_DEBUGPRT_TX	37	123												
TPN056	A	1	TP-P5	SOC_DEBUGPRT_RX	37	123												
TPN057	A	1	TP-P5															
TPN062	A	1	TP-P5	USBC_XA_SBU1	29	33	161											
TPN063	A	1	TP-P5	USBC_XA_SBU2	29	33	161											
TPN064	A	1	TP-P5	USBC_XA_CC1	29	33	161											
TPN065	A	1	TP-P5	USBC_XA_CC2	29	33	161											
TPN066	A	1	TP-P5	PPVBUS_USBC_XA	29	32	33	161										
TPN067	A	1	TP-P5															
TPN068	A	1	TP-P5															
TPN093	A	1	TP-P5	SOC_SWD_MUX_SEL	32	69												
TPN069	A	1	TP-P5	SOC_SOCHOT_L	35	76	121	156										
TPN083	A	1	TP-P5	DEMUX_SEL	121													

(USBC_XA_USB_DBG_TOP_P)
(USBC_XA_USB_DBG_TOP_N)
(USBC_XA_USB_DBG_BOT_P)
(USBC_XA_USB_DBG_BOT_N)
(USBC_XB_USB_TOP_P)
(USBC_XB_USB_TOP_N)
(USBC_XB_USB_BOT_P)
(USBC_XB_USB_BOT_N)

ICP TEST VIMS DUE TO
PACKAGE KEEPCUT CONSTRAINTS
AT USB-C AREA

FAN

TPN099	A	1	TP-P5	SMC_FAN_0_PWM_RES	60													
TPN098	A	1	TP-P5	SMC_FAN_0_TACH	36	60												
TPN0A0	A	1	TP-P5	SMC_FAN_1_PWM_RES	60													
TPN0A1	A	1	TP-P5	SMC_FAN_1_TACH	36	60												
TPN097	A	1	TP-P5	PP12V_S0_FAN_0_FILTER	60	161												
TPN096	A	1	TP-P5	PP12V_S0_FAN_1_FILTER	60	161												
TPN095	A	1	TP-P5															
TPN094	A	1	TP-P5															

MISC / BACK UP

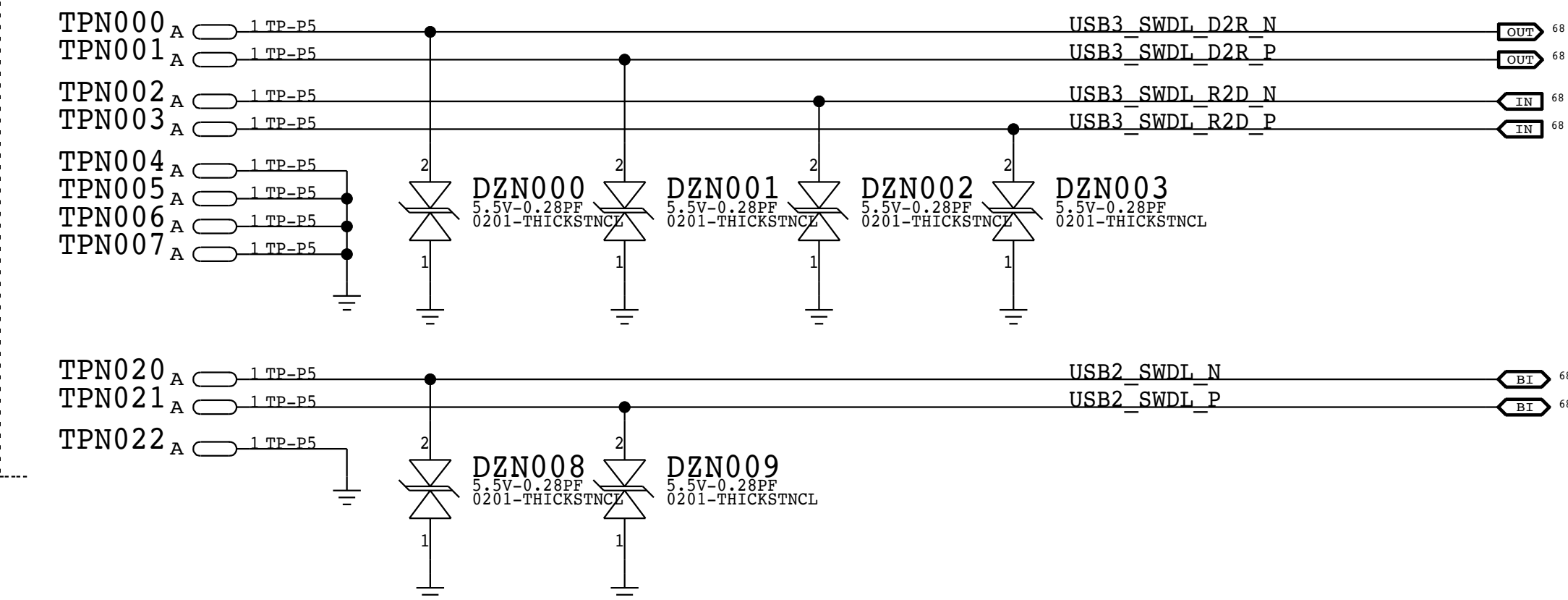
TPN060	A	1	TP-P5	USB_SOC_TYPEC_P	32	121												
TPN061	A	1	TP-P5	USB_SOC_TYPEC_N	32	121												
TPN052	A	1	TP-P5	UPC_XB_DBG3	34													
TPN053	A	1	TP-P5	UPC_XB_DBG4	34													

PLACED ON TOP

SWDL

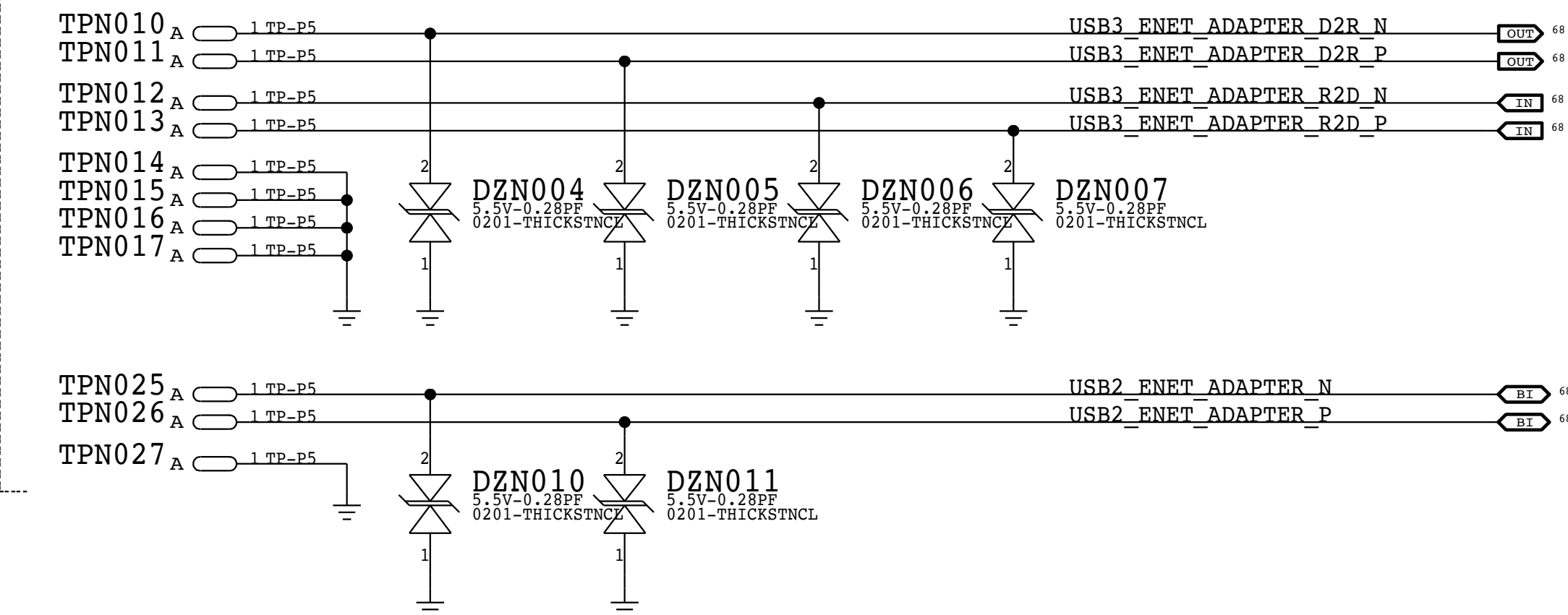
PLACE IN ACCESSIBLE BOARD AREA

NON-CUSTOMER-FACING USB PORTS



ENET ADAPTER

PLACE IN ACCESSIBLE BOARD AREA



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		BRANCH	
		PAGE	230 OF 255
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		159 OF 171	

BOM_COST_GROUP=PCB

	8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
SVT TEST POINTS <RDAR://30758642>																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
D	ITP-XDP			BSCAN			BSCAN COMPLIANCE			VOLTAGE MONITORING																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
	73	13	BT	XDP_CPU_PREQ_I	FUNC_TEST=TRUE	73	66	BT	XDP_PCH_TMS	FUNC_TEST=TRUE	67	BT	PM_SLP_SUS_I	FUNC_TEST=TRUE	88	76	73	BT	PVCCIO_PGOOD	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
	73	13	BT	XDP_CPU_PRDY_I	FUNC_TEST=TRUE	73	66	BT	XDP_PCH_TDI	FUNC_TEST=TRUE	76	72	36	BT	PM_SLP_S0_I	FUNC_TEST=TRUE	168	89	14	BT	PPVCCSA_S0	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
	73	66	13	BT	XDP_CPU_TCK	FUNC_TEST=TRUE	73	66	BT	XDP_PCH_TDO	FUNC_TEST=TRUE	155	67	65	30	24	BT	PM_SLP_S3_I	FUNC_TEST=TRUE	107	103	76	74	66	65	49	47	46	44	BT	PP1V8_S0	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
	73	13	BT	XDP_CPU_TDI	FUNC_TEST=TRUE	73		BT	XDP_JTAG_CPU_ISOL_I	FUNC_TEST=TRUE	155	67	BT	PM_SLP_S4_I	FUNC_TEST=TRUE	155	67	BT	PM_SLP_S5_I	FUNC_TEST=TRUE	168	92	90	22	19	18	17	14	BT	PPVDDQ01_S3	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	73	13	BT	XDP_CPU_TDO	FUNC_TEST=TRUE	72	30	24	BT	JTAG_TBT_TDI	FUNC_TEST=TRUE	155	67	BT	PM_SLP_S5_I	FUNC_TEST=TRUE	155	67	BT	PCH_PM_SLP_S0_I	FUNC_TEST=TRUE	168	93	91	67	22	21	20	17	14	BT	PPVDDQ23_S3	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	73	13	BT	XDP_CPU_TMS	FUNC_TEST=TRUE	72	30	24	BT	JTAG_TBT_TDO	FUNC_TEST=TRUE	72	67	BT	PM_THRMTRIP_I	FUNC_TEST=TRUE	72	67	BT	PM_THRMTRIP_I	FUNC_TEST=TRUE	119	112	99	98	94	45	BT	PP1V8_S0_GPU	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
	73	66	BT	PCH_IOTP_PMODE	FUNC_TEST=TRUE	72	30	24	BT	JTAG_TBT_TCK	FUNC_TEST=TRUE	156	67	45	BT	PLT_RESET_I	FUNC_TEST=TRUE	156	67	45	BT	PLT_RESET_I	FUNC_TEST=TRUE	78	65	57	56	55	54	BT	PP3V3_G3H_ENET_FET	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
	74	73	70	66	45	122	BT	PPIV0_PCH_CORE_S5	FUNC_TEST=TRUE	72	24	BT	JTAG_TBT_T_TMS	FUNC_TEST=TRUE	155	145	137	67	45	BT	PLT_RESET_I	FUNC_TEST=TRUE	168	155	88	73	43	17	14	13	12	11	BT	PPVCCIO_S0	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
	79	73	67	53	43	BT	PM_RSMRST_PCH_I	FUNC_TEST=TRUE	72	30	BT	JTAG_TBT_X_TMS	FUNC_TEST=TRUE	43	36	BT	SMC_PCH_SYS_PWROK	FUNC_TEST=TRUE	43	36	BT	SMC_PCH_PWROK	FUNC_TEST=TRUE	137	134	131	129	128	127	78	60	59	51	48	BT	PP1V8_G3S	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
76	73	67	BT	PM_PWRBTN_I	FUNC_TEST=TRUE	121	36	BT	JTAG_SOC_TDO	FUNC_TEST=TRUE	43	36	BT	SMC_PCH_PWROK	FUNC_TEST=TRUE	43	36	BT	SMC_PCH_PWROK	FUNC_TEST=TRUE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

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SYNC MASTER=DEV_MLB		SYNC DATE=11/06/2016	
PAGE TITLE			
SVT TEST PROPERTIES			
 Apple Inc.	DRAWING NUMBER		SIZE
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8	7	6	5	4	3	2	1
FCT TEST POINTS <RDAR://30758642>							
VOLTAGE MONITORING		POWER ENABLES AND PGOODS		SENSE RESISTOR MEASUREMENTS		DISPLAY	
59 58 54 78 69 32 26 113 112 110 105 114 71 79 80 55 81 55 92 91 90 82 81 78 89 88 87 86 84 171 90 22 19 18 171 91 22 21 20 168 90 19 18 17 12 168 91 21 20 17 108 171 110 109 170 116 99 149 140 139 88 85 153 139 109 108 25 26 25 24 31 31 32 31 34 33 32 31 30 32 31 75 41 40 74 40 74 40 123 75 74 41 75 41	PP3V3_S0SW_SD PP1V2_SD PP3V3_S0_TBT PP5V_S0 PP5V_G3S_USBC_REG PP1V2_ENET PP2V1_ENET PP0V85_ENET PP5V_G3S PPDRV_CPUVR PPDDRVTT01_S0 PPDDRVTT23_S0 PPVPP01_S3 PPVPP23_S3 PP12V_LCD_FET PPBKLT_BOOST PPMVDD_S0_GPU PP12V_MAIN PP12V_MAIN_BLC_LCD PP0V9_TBT_T_SVR PP3V3_TBT_T_S0 PP3V3_TBT_T_F PP3V3_TBT_T_SX PP0V9_TBT_X_SVR PP0V9_TBT_X_LVR PP3V3_TBT_X_S0 PP3V3_TBT_X_SX PP3V3_TBT_X_LC PP1V1_SLPDDR PP0V9_SLPDDR PP1V8_AWAKE PP1V1_SLP2SR PP3V3_AWAKE	FUNC_TEST=TRUE FUNC_TEST					

GRAPHICS VOLTAGE REGULATORS

113	REG_BOOST_GPUICORE_1	NO_TEST=1
113	REG_BOOST_GPUICORE_1_RC	NO_TEST=1
113	REG_BOOST_GPUICORE_2	NO_TEST=1
113	REG_BOOST_GPUICORE_2_RC	NO_TEST=1
113	REG_BOOST_GPUICORE_3	NO_TEST=1
113	REG_BOOST_GPUICORE_3_RC	NO_TEST=1
114	REG_BOOST_GPUICORE_4	NO_TEST=1
114	REG_BOOST_GPUICORE_4_RC	NO_TEST=1
114	REG_BOOST_GPUICORE_5	NO_TEST=1
114	REG_BOOST_GPUICORE_5_RC	NO_TEST=1
114	REG_BOOST_GPUICORE_6	NO_TEST=1
114	REG_BOOST_GPUICORE_6_RC	NO_TEST=1
115	REG_BOOST_GPUICORE_7	NO_TEST=1
115	REG_BOOST_GPUICORE_7_RC	NO_TEST=1
115	REG_BOOST_GPUICORE_8	NO_TEST=1
115	REG_BOOST_GPUICORE_8_RC	NO_TEST=1
163	VR_PHASE_GPUICORE_1	NO_TEST=1
163	VR_PHASE_GPUICORE_2	NO_TEST=1
163	VR_PHASE_GPUICORE_3	NO_TEST=1
163	VR_PHASE_GPUICORE_4	NO_TEST=1
163	VR_PHASE_GPUICORE_5	NO_TEST=1
163	VR_PHASE_GPUICORE_6	NO_TEST=1
163	VR_PHASE_GPUICORE_7	NO_TEST=1
163	VR_PHASE_GPUICORE_8	NO_TEST=1
118	REG_LGPUVR	NO_TEST=1
163	REGSNUBPVDDR	NO_TEST=1
163	REGSNUBPVPP	NO_TEST=1
116	REG_BOOT_MVDD	NO_TEST=1
120	REG_BOOT_MVPP	NO_TEST=1
118	REG_BOOT_POVR	NO_TEST=1
118	REG_BOOT_POVR_RC	NO_TEST=1
163	REG_COMP_MVPP	NO_TEST=1
119	REG_COMP_PVDDR	NO_TEST=1
113	REG_GPUICORE_ISNS_1_N	NO_TEST=1
113	REG_GPUICORE_ISNS_1_P	NO_TEST=1
112	REG_GPUICORE_ISNS_1_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_1_R_P	NO_TEST=1
113	REG_GPUICORE_ISNS_2_N	NO_TEST=1
112	REG_GPUICORE_ISNS_2_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_2_R_P	NO_TEST=1
113	REG_GPUICORE_ISNS_3_N	NO_TEST=1
113	REG_GPUICORE_ISNS_3_P	NO_TEST=1
112	REG_GPUICORE_ISNS_3_R_N	NO_TEST=1
165	REG_GPUICORE_ISNS_3_R_P	NO_TEST=1
114	REG_GPUICORE_ISNS_4_N	NO_TEST=1
114	REG_GPUICORE_ISNS_4_P	NO_TEST=1
112	REG_GPUICORE_ISNS_4_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_4_R_P	NO_TEST=1
114	REG_GPUICORE_ISNS_5_N	NO_TEST=1
114	REG_GPUICORE_ISNS_5_P	NO_TEST=1
112	REG_GPUICORE_ISNS_5_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_5_R_P	NO_TEST=1
114	REG_GPUICORE_ISNS_6_N	NO_TEST=1
114	REG_GPUICORE_ISNS_6_P	NO_TEST=1
112	REG_GPUICORE_ISNS_6_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_6_R_P	NO_TEST=1
115	REG_GPUICORE_ISNS_7_N	NO_TEST=1
115	REG_GPUICORE_ISNS_7_P	NO_TEST=1
112	REG_GPUICORE_ISNS_7_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_7_R_P	NO_TEST=1
115	REG_GPUICORE_ISNS_8_N	NO_TEST=1
115	REG_GPUICORE_ISNS_8_P	NO_TEST=1
112	REG_GPUICORE_ISNS_8_R_N	NO_TEST=1
112	REG_GPUICORE_ISNS_8_R_P	NO_TEST=1
116	REG_LGATE_MVDD	NO_TEST=1
116	REG_LGATE_POVR	NO_TEST=1
118	REG_MVDD_FB	NO_TEST=1
116	REG_MVDD_FB_R	NO_TEST=1
116	REG_MVDD_FSEL	NO_TEST=1
116	REG_MVDD_OCSET	NO_TEST=1
116	REG_MVDD_RTN	NO_TEST=1
116	REG_MVDD_RTN_R	NO_TEST=1
116	REG_MVDD_SET0	NO_TEST=1
116	REG_MVDD_SET1	NO_TEST=1
116	REG_MVDD_SET1_R	NO_TEST=1
116	REG_MVDD_SREF	NO_TEST=1

GRAPHICS VOLTAGE REGULATORS

118	REG_POVR_FB	NO_TEST=1
118	REG_POVR_FB_R	NO_TEST=1
118	REG_POVR_FSEL	NO_TEST=1
118	REG_POVR_OCSET	NO_TEST=1
118	REG_POVR_RTN	NO_TEST=1
118	REG_POVR_RTN_R	NO_TEST=1
118	REG_POVR_SET0	NO_TEST=1
118	REG_POVR_SET1	NO_TEST=1
118	REG_POVR_SREF	NO_TEST=1
118	REG_POVR_VO	NO_TEST=1
118	REG_PHASE_MVDD	NO_TEST=1
120	REG_PHASE_MVPP	NO_TEST=1
118	REG_PHASE_POVR	NO_TEST=1
118	REG_PHASE_SNS_POVR	NO_TEST=1
113	REG_PWM_GPUICORE_1	NO_TEST=1
113	REG_PWM_GPUICORE_2	NO_TEST=1
113	REG_PWM_GPUICORE_3	NO_TEST=1
114	REG_PWM_GPUICORE_4	NO_TEST=1
114	REG_PWM_GPUICORE_5	NO_TEST=1
114	REG_PWM_GPUICORE_6	NO_TEST=1
115	REG_PWM_GPUICORE_7	NO_TEST=1
115	REG_PWM_GPUICORE_8	NO_TEST=1
120	REG_RTCLK_MVPP	NO_TEST=1
113	REG_SNUBBER_GPUICORE_1	NO_TEST=1
113	REG_SNUBBER_GPUICORE_2	NO_TEST=1
113	REG_SNUBBER_GPUICORE_3	NO_TEST=1
114	REG_SNUBBER_GPUICORE_4	NO_TEST=1
114	REG_SNUBBER_GPUICORE_5	NO_TEST=1
114	REG_SNUBBER_GPUICORE_6	NO_TEST=1
115	REG_SNUBBER_GPUICORE_7	NO_TEST=1
115	REG_SNUBBER_GPUICORE_8	NO_TEST=1
114	REG_SNUBBER_MVDD	NO_TEST=1
120	REG_SNUBBER_POVR	NO_TEST=1
120	REG_SSTR_MVPP	NO_TEST=1
114	REG_UGATE_MVDD	NO_TEST=1
120	REG_VSNSR_MVPP	NO_TEST=1
120	REG_VSNS_MVPP	NO_TEST=1
163	REG_CPU_VDDQ01_SW	NO_TEST=1
120	RCGMVPP	NO_TEST=1
163	VR_PHASE_GPUICORE_1	NO_TEST=1
163	VR_PHASE_GPUICORE_2	NO_TEST=1
163	VR_PHASE_GPUICORE_3	NO_TEST=1
163	VR_PHASE_GPUICORE_4	NO_TEST=1
163	VR_PHASE_GPUICORE_5	NO_TEST=1
163	VR_PHASE_GPUICORE_6	NO_TEST=1
163	VR_PHASE_GPUICORE_7	NO_TEST=1
163	VR_PHASE_GPUICORE_8	NO_TEST=1
114	VR_PHASE_MVDD	NO_TEST=1

ETHERNET VOLTAGE REGULATORS

78	P3V3ENET_SS	NO_TEST=1
163	REG_OVR85_ENET_FB	NO_TEST=1
163	REG_OVR85_ENET_OCSET	NO_TEST=1
163	REG_OVR85_ENET_RTN	NO_TEST=1
163	REG_OVR85_ENET_FB_N	NO_TEST=1
81	REG_OVR85_ENET_FB_P	NO_TEST=1
81	REG_OVR85_ENET_FSEL	NO_TEST=1
163	REG_OVR85_ENET_OCSET	NO_TEST=1
163	REG_OVR85_ENET_RTN	NO_TEST=1
81	REG_OVR85_ENET_SENSE_N	NO_TEST=1
81	REG_OVR85_ENET_SENSE_P	NO_TEST=1
81	REG_OVR85_ENET_SREF	NO_TEST=1
81	REG_BOOT_OVR85_ENET	NO_TEST=1
81	REG_BOOT_OVR85_ENET_RC	NO_TEST=1
81	REG_LGATE_OVR85_ENET	NO_TEST=1
81	REG_PHASE_OVR85_ENET	NO_TEST=1
81	REG_PHASE_OVR85_ENET_TGR	NO_TEST=1
81	REG_PVCC_U8500	NO_TEST=1
81	REG_SNUBBER_OVR85_ENET	NO_TEST=1
81	REG_UGATE_OVR85_ENET	NO_TEST=1
81	REG_UGATE_OVR85_ENET_R	NO_TEST=1
81	REG_VCC_U8500	NO_TEST=1
81	REG_FB_I1V2_ENET	NO_TEST=1
81	REG_FB_I1V2_ENET_R	NO_TEST=1
81	REG_FB_2V1_ENET	NO_TEST=1
81	REG_FB_2V1_ENET_R	NO_TEST=1

CPU VOLTAGE REGULATORS

86	REG_BOOST_CPUICORE_1	NO_TEST=1
86	REG_BOOST_CPUICORE_1_RC	NO_TEST=1
86	REG_BOOST_CPUICORE_2	NO_TEST=1
86	REG_BOOST_CPUICORE_2_RC	NO_TEST=1
86	REG_BOOST_CPUICORE_3	NO_TEST=1
86	REG_BOOST_CPUICORE_3_RC	NO_TEST=1
87	REG_BOOST_CPUICORE_4	NO_TEST=1
87	REG_BOOST_CPUICORE_4_RC	NO_TEST=1
87	REG_BOOST_CPUICORE_5	NO_TEST=1
87	REG_BOOST_CPUICORE_5_RC	NO_TEST=1
163	REGSNUBPVDDR	NO_TEST=1
163	REGSNUBPVPP	NO_TEST=1
163	REG_6VR_FSW	NO_TEST=1
163	REG_6VR_SS	NO_TEST=1
151	CPU_ISNS_VCCIO_S0_N	NO_TEST=1
151	CPU_ISNS_VCCIO_S0_P	NO_TEST=1
151	CPU_ISNS_VCCSA_S0_N	NO_TEST=1
151	CPU_ISNS_VCCSA_S0_P	NO_TEST=1
151	CPU_ISNS_VDDQ01_S3_N	NO_TEST=1
151	CPU_ISNS_VDDQ01_S3_P	NO_TEST=1
151	CPU_ISNS_VDDQ23_S3_N	NO_TEST=1
151	CPU_ISNS_VDDQ23_S3_P	NO_TEST=1
151	CPU_ISNS_VPP01_S3_N	NO_TEST=1
151	CPU_ISNS_VPP01_S3_P	NO_TEST=1
151	CPU_ISNS_VPP23_S3_N	NO_TEST=1
151	CPU_ISNS_VPP23_S3_P	NO_TEST=1
163	REG_6VR_FSW	NO_TEST=1
163	REG_6VR_SS	NO_TEST=1
88	REG_BOOST_VCCIO	NO_TEST=1
119	REG_BOOT_PVDDR	NO_TEST=1
117	REG_BOOT_VDDCT	NO_TEST=1
117	REG_BOOT_VDDCT_RC	NO_TEST=1
90	REG_BOOT_VPP01	NO_TEST=1
90	REG_BOOT_VPP23	NO_TEST=1
92	REG_BST_CPU_VDDQ01	NO_TEST=1
93	REG_BST_CPU_VDDQ23	NO_TEST=1
89	REG_BST_VCCSA	NO_TEST=1
92	REG_CPU_VDDQ01_SW	NO_TEST=1
93	REG_CPU_VDDQ23_SW	NO_TEST=1
85	REG_CPU_VINSNS	NO_TEST=1
85	REG_DRVCPUI_VSNS	NO_TEST=1
84	REG_DRVCPUI_VSNS_R	NO_TEST=1
84	REG_DRV_VR_AGN0	NO_TEST=1
91	REG_FSEL_VPP23	NO_TEST=1
92	REG_ISNS_CPU_VDDQ01_N	NO_TEST=1
92	REG_ISNS_CPU_VDDQ01_P	NO_TEST=1
92	REG_ISNS_CPU_VDDQ23_N	NO_TEST=1
93	REG_ISNS_CPU_VDDQ23_P	NO_TEST=1
88	REG_ISNS_REF_VCCIO	NO_TEST=1
88	REG_ISNS_VCCIO	NO_TEST=1
117	REG_LGATE_VDDCT	NO_TEST=1
90	REG_LG_VPP01	NO_TEST=1
91	REG_LG_VPP23	NO_TEST=1
86	REG_PHASE_CPUICORE_1	NO_TEST=1
86	REG_PHASE_CPUICORE_2	NO_TEST=1
86	REG_PHASE_CPUICORE_3	NO_TEST=1
86	REG_PHASE_CPUICORE_4	NO_TEST=1
87	REG_PHASE_CPUICORE_5	NO_TEST=1
119	REG_PHASE_PVDDR	NO_TEST=1
117	REG_PHASE_VDDCT	NO_TEST=1
117	REG_PHASE_VDDCT_I	NO_TEST=1
90	REG_PHASE_VPP01	NO_TEST=1
90	REG_PHASE_VPP01_I	NO_TEST=1
91	REG_PHASE_VPP23	NO_TEST=1
91	REG_PHASE_VPP23_I	NO_TEST=1
85	REG_PWM_CPUICORE_1	NO_TEST=1
86	REG_PWM_CPUICORE_2	NO_TEST=1
86	REG_PWM_CPUICORE_3	NO_TEST=1
87	REG_PWM_CPUICORE_4	NO_TEST=1
89	REG_PWM_CPUVCCSA	NO_TEST=1
92	REG_PWM_CPU_VDDQ01	NO_TEST=1
93	REG_PWM_CPU_VDDQ23	NO_TEST=1
88	REG_PWM_VCCIO	NO_TEST=1
119	REG_RTCLK_PVDDR	NO_TEST=1
86	REG_SNUBBER_CPUICORE_1	NO_TEST=1
86	REG_SNUBBER_CPUICORE_2	NO_TEST=1
86	REG_SNUBBER_CPUICORE_3	NO_TEST=1
87	REG_SNUBBER_CPUICORE_4	NO_TEST=1
87	REG_SNUBBER_CPUICORE_5	NO_TEST=1
117	REG_SNUBBER_VDDCT	NO_TEST=1
90	REG_SNUB_VPP01_RC	NO_TEST=1
91	REG_SNUB_VPP23_RC	NO_TEST=1
119	REG_SSTR_PVDDR	NO_TEST=1

CPU VOLTAGE REGULATORS

117	REG_UGATE_VDDCT	NO_TEST=1
117	REG_UGATE_P3V3G3H_R	NO_TEST=1
90	REG_UG_VPP01	NO_TEST=1
90	REG_UG_VPP01_R	NO_TEST=1
91	REG_UG_VPP23	NO_TEST=1
91	REG_UG_VPP23_R	NO_TEST=1
86	REG_VCCIN_ISNS_1_N	NO_TEST=1
86	REG_VCCIN_ISNS_1_P	NO_TEST=1
86	REG_VCCIN_ISNS_2_N	NO_TEST=1
86	REG_VCCIN_ISNS_2_P	NO_TEST=1
86	REG_VCCIN_ISNS_3_N	NO_TEST=1
86	REG_VCCIN_ISNS_3_P	NO_TEST=1
87	REG_VCCIN_ISNS_4_N	NO_TEST=1
87	REG_VCCIN_ISNS_4_P	NO_TEST=1
87	REG_VCCIN_ISNS_5_N	NO_TEST=1
87	REG_VCCIN_ISNS_5_P	NO_TEST=1
85	REG_VCCIN_TSNS_A	NO_TEST=1
88	REG_VCCIO_SW	NO_TEST=1
163	REG_VCCSA_PH	NO_TEST=1
163	REG_VCCSA_PH	NO_TEST=1
89	REG_VCCSA_SNUB	NO_TEST=1
85	REG_VCCSA_TSNS_R	NO_TEST=1
117	REG_VDDCT_FB	NO_TEST=1
117	REG_VDDCT_FB_P	NO_TEST=1
117	REG_VDDCT_FSEL	NO_TEST=1
117	REG_VDDCT_OCSET	NO_TEST=1
117	REG_VDDCT_RTN	NO_TEST=1
117	REG_VDDCT_SET0	NO_TEST=1
117	REG_VDDCT_SET1	NO_TEST=1
117	REG_VDDCT_SREF	NO_TEST=1
92	REG_VIN_SNS_CPU_VDDQ01	NO_TEST=1
88	REG_VIN_SNS_VCCIO	NO_TEST=1
163	REG_VIN_SNS_VDDQ23	NO_TEST=1
163	REG_VIN_SNS_VDDQ23	NO_TEST=1
90	REG_VPP01	NO_TEST=1
90	REG_VPP01_FB	NO_TEST=1
90	REG_VPP01_FB_XW_N	NO_TEST=1
90	REG_VPP01_FB_XW_P	NO_TEST=1
90	REG_VPP01_OCSET	NO_TEST=1
90	REG_VPP01_RTN	NO_TEST=1
90	REG_VPP01_SREF	NO_TEST=1
90	REG_VPP01_VO	NO_TEST=1
91	REG_VPP23_FB	NO_TEST=1
91	REG_VPP23_FB_XW_N	NO_TEST=1
91	REG_VPP23_FB_XW_P	NO_TEST=1
91	REG_VPP23_OCSET	NO_TEST=1
91	REG_VPP23_RTN	NO_TEST=1
91	REG_VPP23_SREF	NO_TEST=1
91	REG_VPP23_VO	NO_TEST=1
119	REG_VSNSR_PVDDR	NO_TEST=1
85	REG_VSNS_CPU_VCCIN_N	NO_TEST=1
85	REG_VSNS_CPU_VCCIN_P_R	NO_TEST=1
88	REG_VSNS_CPU_VCCIO_N	NO_TEST=1
88	REG_VSNS_CPU_VCCIO_P_R	NO_TEST=1
85	REG_VSNS_CPU_VCCSA_N	NO_TEST=1
85	REG_VSNS_CPU_VCCSA_P_R	NO_TEST=1
92	REG_VSNS_CPU_VDDQ01_N	NO_TEST=1
92	REG_VSNS_CPU_VDDQ01_R_P	NO_TEST=1
92	REG_VSNS_CPU_VDDQ23_R_P	NO_TEST=1
119	REG_VSNS_PVDDR	NO_TEST=1
90	REG_VTT01_REFIN	NO_TEST=1
91	REG_VTT23_REFIN	NO_TEST=1
93	SNUB_CPU_VDDQ23_RC	NO_TEST=1
92	SNUB_CPU_VDDQ_01	NO_TEST=1

WIRELESS ANTENNAS

136	50_0_ANT	NO_TEST=1
136	50_0_COM	NO_TEST=1
136	50_A_0_DIPLEXER	NO_TEST=1
136	50_G_0_DIPLEXER	NO_TEST=1
136	50_A_0_MATCH	NO_TEST=1
136	50_G_0_MATCH	NO_TEST=1
136	50_1_ANT	NO_TEST=1
136	50_1_COM	NO_TEST=1
136	50_A_1_DIPLEXER	NO_TEST=1
136	50_G_1_DIPLEXER	NO_TEST=1
136	50_A_1_MATCH	NO_TEST=1
136	50_G_1_MATCH	NO_TEST=1
136	50_2_ANT	NO_TEST=1
136	50_2_COM	NO_TEST=1
136	50_A_2_DIPLEXER	NO_TEST=1
136	50_G_2_DIPLEXER	NO_TEST=1
136	50_A_2_MATCH	NO_TEST=1
136	50_G_2_MATCH	NO_TEST=1
136	50_3_ANT	NO_TEST=1
136	50_3_ANT_IND	NO_TEST=1
136	BT_ONLY_MATCH_IND	NO_TEST=1
165	BT_ONLY_MATCH	NO_TEST=1

SYSTEM VOLTAGE REGULATORS

87	P5V_G3S_SNS_P	NO_TEST=1
87	REG_BOOT_P3V3G3H	NO_TEST=1
87	REG_BOOT_P3V3G3H_RC	NO_TEST=1
87	REG_BOOT_P5VG3S	NO_TEST=1
87	REG_BOOT_P5VG3S_RC	NO_TEST=1
87	REG_LGATE_P3V3G3H	NO_TEST=1
87	REG_LGATE_P5VG3S	NO_TEST=1
87	REG_P3V3G3H_DCR_ISNS_N	NO_TEST=1
87	REG_P3V3G3H_DCR_ISNS_P	NO_TEST=1
87	REG_P3V3G3H_FSEL	NO_TEST=1
87	REG_P3V3G3H_ISEN	NO_TEST=1
87	REG_P3V3G3H_OCSET	NO_TEST=1
87	REG_P3V3G3H_VOUT	NO_TEST=1
87	REG_P3V3G3H_VOUT_R	NO_TEST=1
87	REG_P5VG3S_FB	NO_TEST=1
87	REG_P5VG3S_FSET	NO_TEST=1
87	REG_P5VG3S_ISEN	NO_TEST=1
87	REG_P5VG3S_OCSET	NO_TEST=1
87	REG_P5VG3S_VOUT_R	NO_TEST=1
87	REG_PHASE_P3V3G3H	NO_TEST=1
87	REG_PHASE_P3V3G3H_TGR	NO_TEST=1
87	REG_PHASE_P5VG3S	NO_TEST=1
87	REG_SNUBBER_P3V3G3H	NO_TEST=1
87	REG_SNUBBER_P5VG3S	NO_TEST=1
87	REG_UGATE_P3V3G3H	NO_TEST=1
87	REG_UGATE_P5VG3S	NO_TEST=1
87	REG_VIN_U8600	NO_TEST=1
87	REG_SNUBBER_P3V3G3W	NO_TEST=1
79	REG_P3V3G3W_VOUT_R	NO_TEST=1
79	REG_VIN_U8300	NO_TEST=1
79	REG_VCC_I_U8300	NO_TEST=1
79	REG_UGATE_P3V3G3W	NO_TEST=1
79	REG_LGATE_P3V3G3W	NO_TEST=1
79	REG_BOOT_P3V3G3W	NO_TEST=1
79	REG_P3V3G3W_ISEN	NO_TEST=1
163	REG_P3V3G3W_VOUT	NO

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D	127	BT	AUD_HP_PORT_US_GND	NO_TEST=1
	128	BT	AUD_LAMP_BOOTLTN	NO_TEST=1
	129	BT	AUD_LAMP_BOOTLTP	NO_TEST=1
	129	BT	AUD_LAMP_BOOTLWN	NO_TEST=1
	129	BT	AUD_LAMP_BOOTLWP	NO_TEST=1
	129	BT	AUD_LAMP_LT_ADDR	NO_TEST=1
	129	BT	AUD_LAMP_LW_GVDD	NO_TEST=1
	129	BT	AUD_LAMP_LW_VCOM	NO_TEST=1
	129	BT	AUD_LAMP_LW_VREG	NO_TEST=1
	127 76	BT	AUD_PWR_EN_	NO_TEST=1
C	129	BT	AUD_RAMP_BOOTRTN	NO_TEST=1
	129	BT	AUD_RAMP_BOOTRTP	NO_TEST=1
	129	BT	AUD_RAMP_BOOTRWN	NO_TEST=1
	129	BT	AUD_RAMP_BOOTRWP	NO_TEST=1
	129	BT	AUD_SPKR_RTWT_OUT_I_N	NO_TEST=1
	129	BT	AUD_SPKR_RTWT_OUT_I_P	NO_TEST=1
	129	BT	AUD_SPKR_RWER_OUT_I_N	NO_TEST=1
	129	BT	AUD_SPKR_RWER_OUT_I_P	NO_TEST=1
	109	BT	BKLT_SNSH	NO_TEST=1
	109	BT	BKLT_SNSP	NO_TEST=1
B	43 35	BT	BOARD_REV0	NO_TEST=1
	43 35	BT	BOARD_REV1	NO_TEST=1
	43 35	BT	BOARD_REV2	NO_TEST=1
	43 35	BT	BOOT_CONFIG0	NO_TEST=1
	43 35	BT	BOOT_CONFIG1	NO_TEST=1
	43 35	BT	BOOT_CONFIG2	NO_TEST=1
	163 136	BT	BT_ONLY_MATCH	NO_TEST=1
	137 134	BT	BT_SF_CLK	NO_TEST=1
	137 134	BT	BT_SF_CS_L	NO_TEST=1
	137	BT	BT_SF_CS_L_R	NO_TEST=1
A	137	BT	BT_SF_HOLD_I	NO_TEST=1
	137 134	BT	BT_SF_MISO	NO_TEST=1
	137 134	BT	BT_SF_MOSI	NO_TEST=1
	137	BT	BT_SF_WP_L	NO_TEST=1
	154	BT	CPU_CATERR_R_I	NO_TEST=1
	154	BT	CPU_ERROR_0_LED	NO_TEST=1
	154	BT	CPU_ERROR_0_LED_R	NO_TEST=1
	154	BT	CPU_ERROR_0_Q	NO_TEST=1
	154	BT	CPU_ERROR_1_LED	NO_TEST=1
	154	BT	CPU_ERROR_1_LED_R	NO_TEST=1

59	BT	ENET_A_CMC_P	NO_TEST=1
59	BT	ENET_A_CT_CONN	NO_TEST=1
59	BT	ENET_A_CT_PHY	NO_TEST=1
59	BT	ENET_BG_REXT1_R	NO_TEST=1
59	BT	ENET_B_CMC_N	NO_TEST=1
59	BT	ENET_B_CMC_P	NO_TEST=1
59	BT	ENET_B_CT_CONN	NO_TEST=1
59	BT	ENET_B_CT_PHY	NO_TEST=1
60	BT	ENET_CLKREQ_PCH_I	NO_TEST=1
59	BT	ENET_C_CMC_N	NO_TEST=1
59	BT	ENET_C_CMC_P	NO_TEST=1
59	BT	ENET_C_CT_CONN	NO_TEST=1
59	BT	ENET_C_CT_PHY	NO_TEST=1
59	BT	ENET_D_CMC_N	NO_TEST=1
59	BT	ENET_D_CMC_P	NO_TEST=1
59	BT	ENET_D_CT_CONN	NO_TEST=1
59	BT	ENET_D_CT_PHY	NO_TEST=1
59	BT	ENET_EFUSE_EN	NO_TEST=1
59	BT	ENET_MISO_R	NO_TEST=1
59	BT	ENET_RG_K12	NO_TEST=1
59	BT	ENET_RNC_N9	NO_TEST=1
59	BT	ENET_SMB_CLK_R	NO_TEST=1
59	BT	ENET_SMB_DAT_R	NO_TEST=1
72 54	BT	ENET_SYNC_1588	NO_TEST=1
72 61	BT	ENET_SYNC_1588_IN	NO_TEST=1
72 61	BT	ENET_SYNC_1588_OUT	NO_TEST=1
59	BT	ENET_TSENS_A_N	NO_TEST=1
59	BT	ENET_TSENS_A_P	NO_TEST=1
59	BT	ENET_TSENS_E_N	NO_TEST=1
148 56	BT	ENET_TSENS_N	NO_TEST=1
148 56	BT	ENET_TSENS_P	NO_TEST=1
59	BT	ENET_XTALI	NO_TEST=1
59	BT	ENET_XTALO	NO_TEST=1
59	BT	ENET_XTALO_R	NO_TEST=1
59	BT	ENET_XTAL_SELECT_I	NO_TEST=1
124 66 30	BT	ESPI_CLK	NO_TEST=1
60	BT	ESPI_CLK60M_R	NO_TEST=1
124	BT	ESPI_IO_HDR<0>	NO_TEST=1
124	BT	ESPI_IO_HDR<1>	NO_TEST=1
124	BT	ESPI_IO_HDR<2>	NO_TEST=1
124	BT	ESPI_IO_HDR<3>	NO_TEST=1
30	BT	ESPI_IO_R<0>	NO_TEST=1
30	BT	ESPI_IO_R<1>	NO_TEST=1
30	BT	ESPI_IO_R<2>	NO_TEST=1
30	BT	ESPI_IO_R<3>	NO_TEST=1
114	BT	FBMVDD_ALTV0_R	NO_TEST=1
114	BT	FBMVDD_ALTV1_R	NO_TEST=1
114	BT	FBP0V8_ALTV0_R	NO_TEST=1
114	BT	FBP0V8_ALTV1_R	NO_TEST=1
117	BT	FBVDDCI_ALTV0_R	NO_TEST=1
117	BT	FBVDDCI_ALTV1_R	NO_TEST=1
99	BT	FB_VDDCI_MEM	NO_TEST=1
99	BT	FB_VDDCR_HBM	NO_TEST=1
99	BT	FB_VDDCR_SOC	NO_TEST=1
99	BT	FB_VDDIO_MEM_GPU	NO_TEST=1
99	BT	FB_VDDIO_MEM_HBM	NO_TEST=1
99	BT	FB_VSS_A	NO_TEST=1
131	BT	GND_AUD_DMIC_1_CONN	NO_TEST=1
99	BT	GPU_GPIO_14	NO_TEST=1
151 116	BT	GPU_ISNS_MVDD_N	NO_TEST=1
151 116	BT	GPU_ISNS_MVDD_P	NO_TEST=1
99	BT	GPU_REF_CLK100M_N	NO_TEST=1
99	BT	GPU_REF_CLK100M_P	NO_TEST=1
127	BT	HS_MICN	NO_TEST=1
127	BT	HS_MICP	NO_TEST=1
150	BT	HS_P3V3_G3W_SSD0_OUT	NO_TEST=1
85	BT	I2C_CPU_VCCIN_SCL	NO_TEST=1
85	BT	I2C_CPU_VCCIN_SDA	NO_TEST=1
43 30	BT	I2C_SEP_SCL	NO_TEST=1
43 30	BT	I2C_SEP_SDA	NO_TEST=1
26	BT	I2C_TBT_T_RIDGE_SCL	NO_TEST=1
32 30	BT	I2C_TBT_X_RIDGE_SCL	NO_TEST=1
32 30	BT	I2C_TBT_X_RIDGE_SDA	NO_TEST=1
86	BT	I2C_VCCIO_SCL	NO_TEST=1
86	BT	I2C_VCCIO_SDA	NO_TEST=1
86	BT	I2C_VDDQ01_SCL	NO_TEST=1
90	BT	I2C_VDDQ01_SDA	NO_TEST=1
90	BT	I2C_VDDQ23_SCL	NO_TEST=1
90	BT	I2C_VDDQ23_SDA	NO_TEST=1

44 30	BT	I2S_CODEC_BCLK_R	NO_TEST=1
127	BT	I2S_CODEC_BCLK_RR	NO_TEST=1
127	BT	I2S_CODEC_D2R_R	NO_TEST=1
127	BT	I2S_CODEC_LRCLK_RR	NO_TEST=1
44 30	BT	I2S_CODEC_R2D_R	NO_TEST=1
44 30	BT	I2S_CODEC_R2D_RR	NO_TEST=1
44 30	BT	I2S_SPKRAMP_I_BCLK_R	NO_TEST=1
44 30	BT	I2S_SPKRAMP_I_MCLK_R	NO_TEST=1
44 30	BT	I2S_SPKRAMP_I_R2D_R	NO_TEST=1
44 30	BT	I2S_SPKRAMP_R_BCLK_R	NO_TEST=1
44 30	BT	I2S_SPKRAMP_R_R2D_R	NO_TEST=1
50	BT	J5500_PIN_5	NO_TEST=1
127	BT	L83_FLYC	NO_TEST=1
127	BT	L83_FLYN	NO_TEST=1
127	BT	L83_FLYP	NO_TEST=1
127	BT	L83_HSBIAS_FILT	NO_TEST=1
127	BT	L83_HSBIAS_FILT_REF	NO_TEST=1
127	BT	L83_VCP_FILTIN	NO_TEST=1
70	BT	LDO_CORE	NO_TEST=1
70	BT	LDO_RTC	NO_TEST=1
154	BT	LED_CPU_CE	NO_TEST=1
156	BT	LED_CPU_CE_Q	NO_TEST=1
154	BT	LED_CPU_CE_R	NO_TEST=1
154	BT	LED_DDR	NO_TEST=1
154	BT	LED_DDR_R	NO_TEST=1
154	BT	LED_ENET_EN_R	NO_TEST=1
154	BT	LED_G3S	NO_TEST=1
154	BT	LED_G3S_R	NO_TEST=1
154	BT	LED_G3W	NO_TEST=1
154	BT	LED_G3W_R	NO_TEST=1
154	BT	LED_GPU_TT	NO_TEST=1
156	BT	LED_GPU_TT_Q	NO_TEST=1
154	BT	LED_GPU_TT_R	NO_TEST=1
154	BT	LED_MAIN	NO_TEST=1
154	BT	LED_MAIN_R	NO_TEST=1
154	BT	LED_PM_TT	NO_TEST=1
156	BT	LED_PM_TT_Q	NO_TEST=1
154	BT	PIT_RST_BUF_TBT_T_I	NO_TEST=1
154	BT	LED_S0_RUN	NO_TEST=1
154	BT	LED_S0_RUN_R	NO_TEST=1
154	BT	LED_S2R	NO_TEST=1
154	BT	LED_S2R_R	NO_TEST=1
154	BT	LED_SMC_PH	NO_TEST=1
156	BT	LED_SMC_PH_Q	NO_TEST=1
154	BT	LED_SMC_PH_R	NO_TEST=1
154	BT	LED_SOCHOT	NO_TEST=1
156	BT	LED_SOCHOT_Q	NO_TEST=1
154	BT	LED_SOCHOT_R	NO_TEST=1
154	BT	LED_SOC_DFU	NO_TEST=1
154	BT	LED_SOC_DFU_R	NO_TEST=1
154	BT	LED_TBT_EN_R	NO_TEST=1
154	BT	LED_VCCIO	NO_TEST=1
154	BT	LED_VCCIO_R	NO_TEST=1
50	BT	MIPI_AIN	NO_TEST=1
50	BT	MIPI_AIP	NO_TEST=1
50	BT	MIPI_BIN	NO_TEST=1
50	BT	MIPI_BIP	NO_TEST=1
51 30	BT	MIPI_FTCAM_CLK_N	NO_TEST=1
51 30	BT	MIPI_FTCAM_CLK_P	NO_TEST=1
50	BT	MIPI_FTCAM_C_CLK_N	NO_TEST=1
50	BT	MIPI_FTCAM_C_CLK_P	NO_TEST=1
50	BT	MIPI_FTCAM_C_DATA_N<0>	NO_TEST=1
50	BT	MIPI_FTCAM_C_DATA_N<1>	NO_TEST=1
50	BT	MIPI_FTCAM_C_DATA_P<0>	NO_TEST=1
50	BT	MIPI_FTCAM_C_DATA_P<1>	NO_TEST=1
51 30	BT	MIPI_FTCAM_DATA_N<0>	NO_TEST=1
51 30	BT	MIPI_FTCAM_DATA_N<1>	NO_TEST=1
51 30	BT	MIPI_FTCAM_DATA_P<0>	NO_TEST=1
51 30	BT	MIPI_FTCAM_DATA_P<1>	NO_TEST=1
50	BT	MIPI_LED_A	NO_TEST=1
50	BT	MIPI_LED_K	NO_TEST=1
50	BT	MIPI_SW_BYP0	NO_TEST=1
50	BT	MIPI_SW_BYP1	NO_TEST=1
50	BT	MIPI_SW_BYP2	NO_TEST=1
50	BT	MIPI_SW_BYP3	NO_TEST=1
50	BT	MIPI_SW_BYP4	NO_TEST=1
50	BT	MIPI_SW_BYP5	NO_TEST=1
50	BT	MIPI_SW_G0	NO_TEST=1
50	BT	MIPI_SW_G1	NO_TEST=1

50	BT	MIPI_SW_G2	NO_TEST=1
50	BT	MIPI_SW_G3	NO_TEST=1
50	BT	MIPI_SW_G4	NO_TEST=1
50	BT	MIPI_SW_G5	NO_TEST=1
120	BT	P0V82SLPDDR_FB_ALT	NO_TEST=1
120	BT	P1V0_PCH_CORE_S5_FB_ALT	NO_TEST=1
70	BT	P1V1SLPDDR_RAMP	NO_TEST=1
70	BT	P3V3S5_RAMP	NO_TEST=1
70	BT	P3V3_G3S_CT	NO_TEST=1
137	BT	PCH_BT_ROM_BOOT_ISO	NO_TEST=1
70	BT	PCH_CLK32K_RTCX1_OSC	NO_TEST=1
70	BT	PCH_CLK32K_RTCX1_R	NO_TEST=1
70	BT	PCH_CLK32K_RTCX2_OSC	NO_TEST=1
70	BT	PCH_CLK32K_RTCX2_R	NO_TEST=1
60	BT	PCH_DIFFCLK_BIASREF	NO_TEST=1
72	BT	PCH_GPD_0_FET	NO_TEST=1
60	BT	PCH_INTRUDER_I	NO_TEST=1
60	BT	PCH_SUSACK_I	NO_TEST=1
60	BT	PCH_UART2_CTS_I	NO_TEST=1
60	BT	PCH_USB2_ID	NO_TEST=1
60	BT	PCH_USB_VBUSSENSE	NO_TEST=1
60	BT	PCH_XTAL24_IN	NO_TEST=1
60	BT	PCH_XTAL24_IN_R	NO_TEST=1
60	BT	PCH_XTAL24_OUT	NO_TEST=1
60	BT	PCH_XTAL24_OUT_R	NO_TEST=1
40	BT	PDM_DMIC_CLK0_R	NO_TEST=1
40	BT	PDM_DMIC_CLK1_R	NO_TEST=1
60	BT	PD_PCH_GPP_I0	NO_TEST=1
60	BT	PD_PCH_GPP_I1	NO_TEST=1
60	BT	PD_PCH_GPP_I2	NO_TEST=1
60	BT	PD_PCH_GPP_I4	NO_TEST=1
60	BT	PEG_CLKREQ_PCH_I	NO_TEST=1
155	BT	PIT_RESET_I_LED	NO_TEST=1
155	BT	PIT_RESET_I_LED_R	NO_TEST=1
145	BT	PIT_RST_BUF_ENET_I	NO_TEST=1
145	BT	PIT_RST_BUF_CPU_I	NO_TEST=1
145	BT	PIT_RST_BUF_SD_I	NO_TEST=1
145	BT	PIT_RST_BUF_TBT_T_I	NO_TEST=1
145	BT	PIT_RST_BUF_TBT_X_I	NO_TEST=1
145	BT	PIT_RST_BUF_WLAN_I	NO_TEST=1
76	BT	PMU_CLK32K_PCH_R	NO_TEST=1
76	BT	PMU_CLK32K_SOC_R	NO_TEST=1
76	BT	PMU_INT_I	NO_TEST=1
76	BT	PMU_IREF	NO_TEST=1
76	BT	PMU_IDO3_OUT	NO_TEST=1
76	BT	PMU_VOUT_RTC	NO_TEST=1
76	BT	PMU_VPUMP	NO_TEST=1
76	BT	PMU_VREF	NO_TEST=1
76	BT	PMU_XTAL1	NO_TEST=1
76	BT	PMU_XTAL1_R	NO_TEST=1
76	BT	PMU_XTAL2	NO_TEST=1
155	BT	PM_SLP_S3_L_LED	NO_TEST=1
155	BT	PM_SLP_S3_L_LED_R	NO_TEST=1
155	BT	PM_SLP_S4_L_LED	NO_TEST=1
155	BT	PM_SLP_S4_L_LED_R	NO_TEST=1
155	BT	PM_SLP_S5_L_LED	NO_TEST=1
155	BT	PM_SLP_S5_L_LED_R	NO_TEST=1
154	BT	PM_THRMTRIP_R_I	NO_TEST=1
90	BT	PM_VTT01_EN	NO_TEST=1
90	BT	PM_VTT23_EN	NO_TEST=1
131	BT	PP1V8_DMIC_1_CONN	NO_TEST=1
50	BT	PP3V3_G3H_ENET_EFUSE	NO_TEST=1
110	BT	PPBKLT_BOOST_1	NO_TEST=1
110	BT	PPBKLT_BOOST_2	NO_TEST=1
60	BT	PU_CLKREQ_ANALYZER_I	NO_TEST=1
60	BT	PU_CLKREQ_ANA_PCH_I	NO_TEST=1
126 120	BT	PVDDCPUAWAKE_FB_ALT	NO_TEST=1
120	BT	REGBOOTPMVPP	NO_TEST=1
119	BT	REGBOOTPVDDRR	NO_TEST=1

80	BT	REG_IV2_ENET_SKIP	NO_TEST=1
80	BT	REG_2V1_ENET_SKIP	NO_TEST=1
116	BT	REG_BOOT_MVDD_RC	NO_TEST=1
90	BT	REG_BOOT_VPP23_RC	NO_TEST=1
90	BT	REG_BST_CPU_VDDQ23_RC	NO_TEST=1
89	BT	REG_BST_VCCSA_RC	NO_TEST=1
90	BT	REG_BST_VPP01_RC	NO_TEST=1
90	BT	REG_CPUCORE_VRHOT_I	NO_TEST=1
85	BT	REG_CPU_ADDR2	NO_TEST=1
13	BT	REG_CPU_SVID_ALERT_SKI_I	NO_TEST=1
13	BT	REG_CPU_SVID_CLK_SKI	NO_TEST=1
13	BT	REG_CPU_SVID_DATA_SKI	NO_TEST=1
85	BT	REG_CPU_VALRT_I_R	NO_TEST=1
89	BT	REG_CPU_VCCSA_PHSFLT_I	NO_TEST=1
90	BT	REG_CPU_VDDQ01_ADDR2	NO_TEST=1
92	BT	REG_CPU_VDDQ01_ISNS_R	NO_TEST=1
92	BT	REG_CPU_VDDQ01_RC	NO_TEST=1
92	BT	REG_CPU_VDDQ01_SVID_ALERT_R	NO_TEST=1
92	BT	REG_CPU_VDDQ01_SVID_DATA_R	NO_TEST=1
92	BT	REG_CPU_VDDQ23_ADDR2	NO_TEST=1
93	BT	REG_CPU_VDDQ23_IL_R	NO_TEST=1
93	BT	REG_CPU_VDDQ23_PHSFLT_I	NO_TEST=1
93	BT	REG_CPU_VDDQ23_SVID_ALERT_I_R	NO_TEST=1
93	BT	REG_CPU_VDDQ23_SVID_DATA_R	NO_TEST=1
13	BT	REG_CPU_VDDQ_SVID_ALERT_SKI_I	NO_TEST=1
12	BT	REG_CPU_VDDQ_SVID_CLK_SKI	NO_TEST=1
13	BT	REG_CPU_VDDQ_SVID_DATA_SKI	NO_TEST=1
85	BT	REG_CPU_VDIO_R	NO_TEST=1
84	BT	REG_DRVCPUI_AVIN	NO_TEST=1
112	BT	REG_GPUCORE_CFIT	NO_TEST=1
163 112	BT	REG_GPUCORE_ISNS_3_R_P	NO_TEST=1
116	BT	REG_MVDD_OCSET_R	NO_TEST=1
116	BT	REG_MVDD_VO_R	NO_TEST=1
118	BT	REG_P0V8_SET1_R	NO_TEST=1
80	BT	REG_P3V3G3H_FB	NO_TEST=1
79	BT	REG_P3V3G3W_FB	NO_TEST=1
79	BT	REG_P3V3G3W_OCSET	NO_TEST=1
113	BT	REG_PHSFLT_GPUCORE_1_I	NO_TEST=1
113	BT	REG_PHSFLT_GPUCORE_2_I	NO_TEST=1
113	BT	REG_PHSFLT_GPUCORE_3_I	NO_TEST=1
114	BT	REG_PHSFLT_GPUCORE_4_I	NO_TEST=1

8		7		6		5		4		3		2		1	
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D	85	14	BT	REG_VSNS_CPU_VCCSA_P	NO_TEST=1	59	BT	USB2_EXTC_N	NO_TEST=1	D					
	59	59	BT	SDCONN_CLK	NO_TEST=1	59	BT	USB2_EXTC_P	NO_TEST=1						
	59	59	BT	SDCONN_CMD	NO_TEST=1	59	BT	USB2_EXTD_N	NO_TEST=1						
	59	59	BT	SDCONN_D0	NO_TEST=1	59	BT	USB2_EXTD_P	NO_TEST=1						
	59	59	BT	SDCONN_D1	NO_TEST=1	32	BT	USB_DBG_XA_F_N	NO_TEST=1						
	59	59	BT	SDCONN_R_WP	NO_TEST=1	32	BT	USB_DBG_XA_F_P	NO_TEST=1						
	59	59	BT	SDCONN_UHS2_CLK_P	NO_TEST=1	68	BT	USB_DBG_XA_N	NO_TEST=1						
	59	59	BT	SDCONN_UHS2_D0_N	NO_TEST=1	68	BT	USB_DBG_XA_P	NO_TEST=1						
	59	59	BT	SDCONN_UHS2_D0_P	NO_TEST=1	68	BT	USB_EXTA_N	NO_TEST=1						
	59	59	BT	SDCONN_UHS2_D1_N	NO_TEST=1	68	BT	USB_EXTA_P	NO_TEST=1						
C	59	59	BT	SDCONN_UHS2_D1_P	NO_TEST=1	68	BT	USB_EXTB_N	NO_TEST=1	C					
	59	59	BT	SD_SPI_CLK_R	NO_TEST=1	68	BT	USB_EXTB_P	NO_TEST=1						
	59	59	BT	SD_SPI_CS_L_R	NO_TEST=1	68	BT	USB_EXTC_N	NO_TEST=1						
	59	59	BT	SD_SPI_HOLD	NO_TEST=1	68	BT	USB_EXTC_P	NO_TEST=1						
	59	59	BT	SD_SPI_MISO_R	NO_TEST=1	68	BT	USB_EXTD_N	NO_TEST=1						
	59	59	BT	SD_SPI_MOSI_R	NO_TEST=1	68	BT	USB_EXTD_P	NO_TEST=1						
	59	59	BT	SEP_DMIC0_DISABLE_R_L	NO_TEST=1	59	BT	USB_SD_DM	NO_TEST=1						
	137	59	BT	SEP_DMIC1_DISABLE_R_L	NO_TEST=1	59	BT	USB_SD_DP	NO_TEST=1						
	73	44	BT	SMBUS_PCH_XDP_3V3_SCL	NO_TEST=1	27	BT	USB_UPC_TA_F_N	NO_TEST=1						
	73	44	BT	SMBUS_PCH_XDP_3V3_SDA	NO_TEST=1	27	BT	USB_UPC_TA_F_P	NO_TEST=1						
B	124	67	BT	SOC_AMUXOUT_UFL	NO_TEST=1	28	BT	USB_UPC_TB_F_N	NO_TEST=1	B					
	67	67	BT	SOC_CLKREQ_PCH_L	NO_TEST=1	28	BT	USB_UPC_TB_F_P	NO_TEST=1						
	36	67	BT	SOC_COLD_RESET_L	NO_TEST=1	34	BT	USB_UPC_XB_F_N	NO_TEST=1						
	37	67	BT	SOC_MIPI0C_REXT	NO_TEST=1	34	BT	USB_UPC_XB_F_P	NO_TEST=1						
	37	67	BT	SOC_MIPI1C_REXT	NO_TEST=1	97	BT	V10_SMBCLK	NO_TEST=1						
	37	67	BT	SOC_MIPI0_REXT	NO_TEST=1	97	BT	V10_SMRDAT	NO_TEST=1						
	126	37	BT	SOC_TST_CLKOUT	NO_TEST=1	134	BT	VDDIP2_C	NO_TEST=1						
	126	37	BT	SOC_TST_CLKOUT_UFL	NO_TEST=1	134	BT	VDDIP5_C	NO_TEST=1						
	126	76	37	SOC_VDDCPU_SENSE	NO_TEST=1	112	BT	VR_GPU_ADDR_OFFSET	NO_TEST=1						
	126	37	BT	SOC_VDDSOC_SENSE	NO_TEST=1	112	BT	VR_GPU_IMON_DIGITAL_FILTER	NO_TEST=1						
A	37	67	BT	SOC_XTAL24M_IN	NO_TEST=1	112	BT	VR_GPU_IMON_EN	NO_TEST=1	A					
	37	67	BT	SOC_XTAL24M_OUT	NO_TEST=1	112	BT	VR_GPU_IMON_RG	NO_TEST=1						
	37	67	BT	SOC_XTAL24M_OUT_R	NO_TEST=1	112	BT	VR_GPU_IOUT_ALERT_L	NO_TEST=1						
	37	67	BT	SPI_SOCROM_CLK_R	NO_TEST=1	112	BT	VR_GPU_PHSFLT_L	NO_TEST=1						
	37	67	BT	SPI_SOCROM_MOSI_R	NO_TEST=1	112	BT	VR_GPU_RCSM	NO_TEST=1						
	43	37	BT	SPI_TPAD_CLK	NO_TEST=1	112	BT	VR_GPU_RCSN_R	NO_TEST=1						
	43	37	BT	SPI_TPAD_MISO	NO_TEST=1	112	BT	VR_GPU_RCSP	NO_TEST=1						
	43	37	BT	SPI_TPAD_MOSI	NO_TEST=1	112	BT	VR_GPU_RCSP_R	NO_TEST=1						
	134	67	BT	SRIP2_VLX	NO_TEST=1	112	BT	VR_GPU_SM_ALERT_L	NO_TEST=1						
	134	67	BT	SRIP4_VLX	NO_TEST=1	112	BT	VR_GPU_SVC_R	NO_TEST=1						
134	67	BT	SRIP8_VLX	NO_TEST=1	112	BT	VR_GPU_SVD_R	NO_TEST=1							
BOM_COST_GROUP=PCB															
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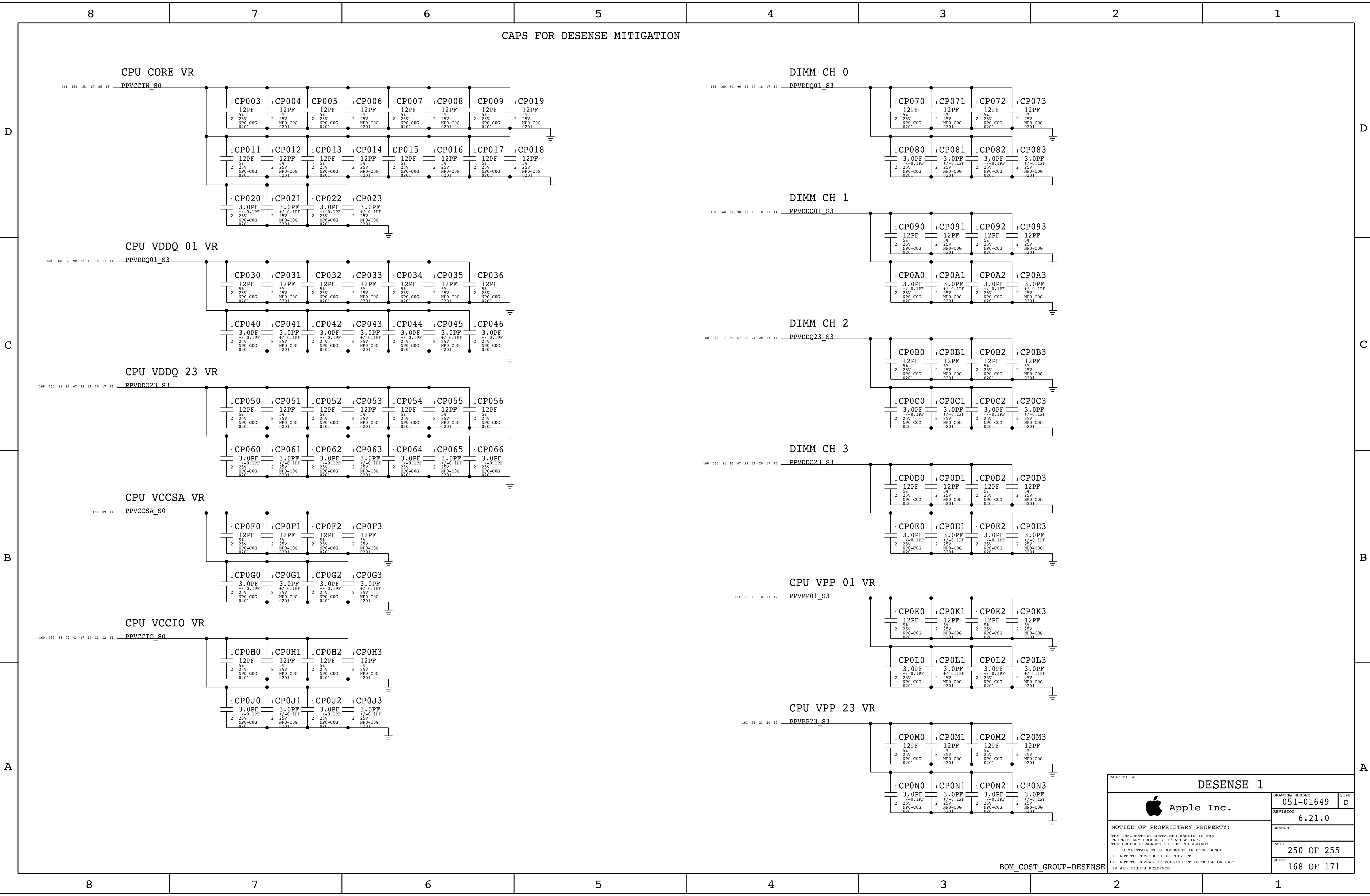
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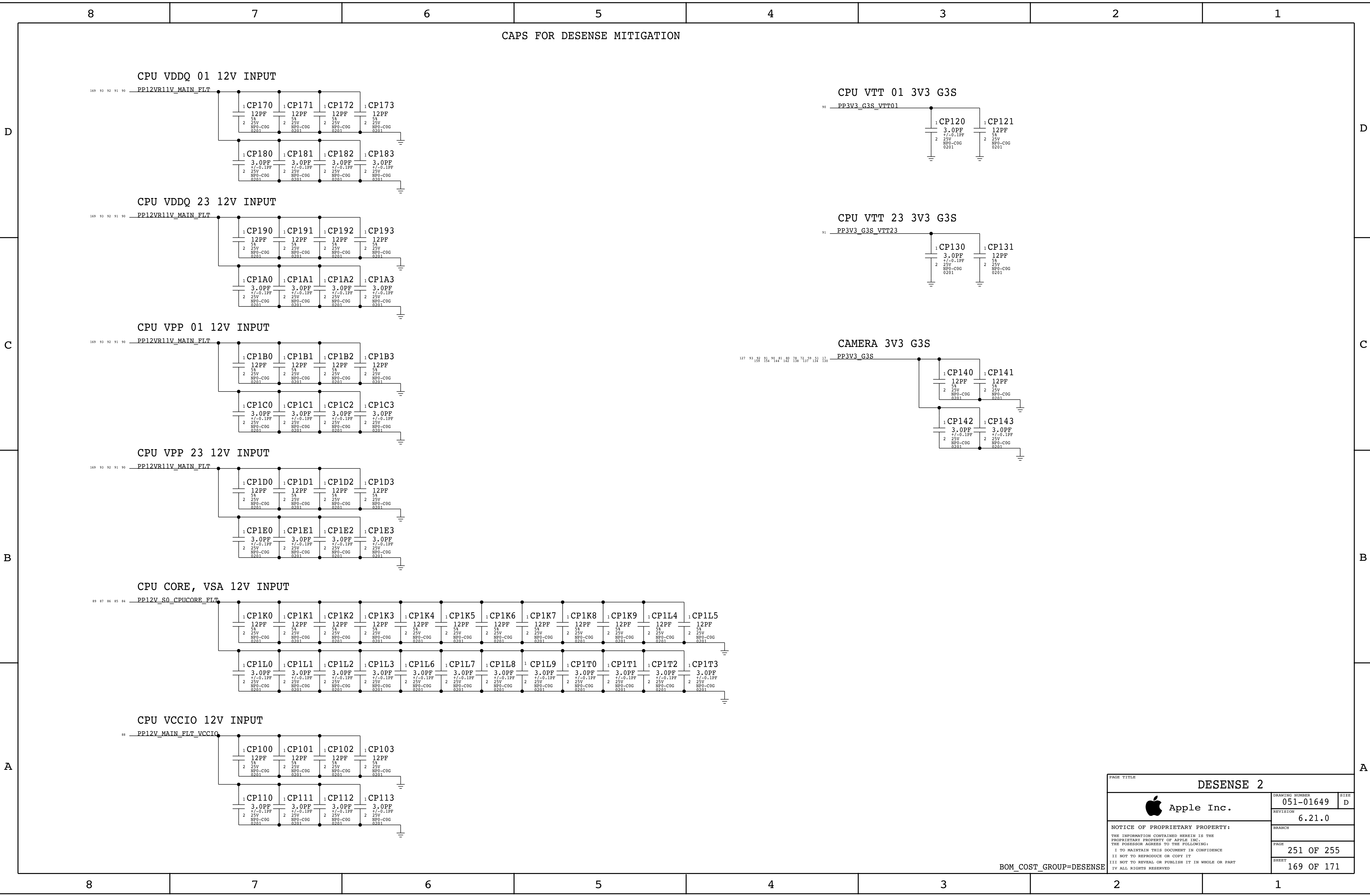
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8		7		6		5		4		3		2		1						
MISC NETS NOT REQUESETED BY ICT OR FCT <RDAR://30758642>																				
D	30	BT	TBT_X_RBIA5	NO_TEST=1	74	BT	PP1V0_PCH_CORE_S5_SW0	NO_TEST=1	11	BT	PCIE_HOT_PLUG_SDA	NO_TEST=1	32	30	BT	DP_X_SNK0_DDC_CLK	NO_TEST=1			
	30	BT	TBT_X_PCIE_BIAS	NO_TEST=1	74	BT	PP1V0_PCH_CORE_S5_SW1	NO_TEST=1	82	BT	REG_PHASE_P5VG3S_TGR	NO_TEST=1	32	30	BT	DP_X_SNK0_DDC_DATA	NO_TEST=1			
	27	23	BT	USBC_TA_USB_TOP_N	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_N<1>	NO_TEST=1	38	BT	PCIE_DN_CLKREQ1_L	NO_TEST=1	32	30	BT	DP_X_SNK1_DDC_CLK	NO_TEST=1	
	27	23	BT	USBC_TA_USB_TOP_P	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_N<2>	NO_TEST=1	149	BT	ISNS_P12V_MAIN_SSD0_N	NO_TEST=1	32	30	BT	DP_X_SNK1_DDC_DATA	NO_TEST=1	
	28	23	BT	USBC_TB_USB_BOT_N	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_N<3>	NO_TEST=1	149	BT	ISNS_P12V_MAIN_SSD0_P	NO_TEST=1	30	30	BT	DP_X_SRC_RBIA5	NO_TEST=1	
	28	23	BT	USBC_TB_USB_BOT_P	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_P<1>	NO_TEST=1	149	BT	ISNS_P12V_MAIN_SSD1_N	NO_TEST=1	32	31	BT	PP0V9_TBT_X_CIO	NO_TEST=1	
	28	23	BT	USBC_TB_USB_TOP_N	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_P<2>	NO_TEST=1	149	BT	ISNS_P12V_MAIN_SSD1_P	NO_TEST=1	31	30	BT	PP0V9_TBT_X_DP	NO_TEST=1	
	28	23	BT	USBC_TB_USB_TOP_P	NO_TEST=1	61	38	BT	PCIE_SSD0_D2R_P<3>	NO_TEST=1	84	BT	REG_DRVCPU_PH	NO_TEST=1	32	31	BT	PP0V9_TBT_X_USB	NO_TEST=1	
	97	BT	CLKTEST_C_N	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_N<1>	NO_TEST=1	149	85	BT	ISNS_HS_CPU_VCORE_N	NO_TEST=1	34	30	BT	TBT_XB_LSRX	NO_TEST=1	
	97	BT	CLKTEST_C_P	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_N<2>	NO_TEST=1	149	85	BT	ISNS_HS_CPU_VCORE_P	NO_TEST=1	34	30	BT	TBT_XB_LSTX	NO_TEST=1	
C	24	BT	DP_T_SNK_RBIA5	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_N<3>	NO_TEST=1	150	82	BT	ISNS_USBA_5V_G3S_N	NO_TEST=1	32	30	BT	TBT_X_HDMI_DDC_CLK	NO_TEST=1	
	24	BT	DP_T_SRC_RBIA5	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_P<1>	NO_TEST=1	150	82	BT	ISNS_USBA_5V_G3S_P	NO_TEST=1	32	30	BT	TBT_X_HDMI_DDC_DATA	NO_TEST=1	
	24	BT	TBT_TA_USB2_RBIA5	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_P<2>	NO_TEST=1	150	79	BT	ISNS_USBC_5V_G3S_N	NO_TEST=1	30	30	BT	TBT_X_THERM_D_POS	NO_TEST=1	
	24	BT	TBT_TB_USB2_RBIA5	NO_TEST=1	61	38	BT	PCIE_SSD0_R2D_C_P<3>	NO_TEST=1	150	79	BT	ISNS_USBC_5V_G3S_P	NO_TEST=1	56	55	BT	ENET_TSENS_F_P	NO_TEST=1	
	26	24	BT	TBT_T_HDMI_DDC_CLK	NO_TEST=1	61	BT	PCIE_SSD0_R2D_N<1>	NO_TEST=1	121	66	BT	PCIE_CLK100M_ANALYZER_N	NO_TEST=1	44	35	BT	SEP_DISABLE_STROBE	NO_TEST=1	
	26	24	BT	TBT_T_HDMI_DDC_DATA	NO_TEST=1	61	BT	PCIE_SSD0_R2D_N<2>	NO_TEST=1	121	66	BT	PCIE_CLK100M_ANALYZER_P	NO_TEST=1	51	54	BT	SEP_DMIC_DISABLE_Q_L	NO_TEST=1	
	24	BT	TBT_T_PCIE_BIAS	NO_TEST=1	61	BT	PCIE_SSD0_R2D_N<3>	NO_TEST=1	73	68	BT	XDP_OBSFN_C_0	NO_TEST=1	36	36	BT	NC_DFR_TOUCH_INT_L	NO_TEST=1		
	24	BT	TBT_T_RBIA5	NO_TEST=1	61	BT	PCIE_SSD0_R2D_P<1>	NO_TEST=1	73	68	BT	XDP_OBSFN_C_1	NO_TEST=1	56	56	BT	NC_Q6080_S	NO_TEST=1		
	26	BT	TBT_T_XTAL25M_OUT_R	NO_TEST=1	61	BT	PCIE_SSD0_R2D_P<2>	NO_TEST=1	149	139	BT	ISNS_P12V_GPU_N	NO_TEST=1	56	56	BT	NC_Q6080_G	NO_TEST=1		
	110	109	BT	LED_RETURN_1_FB	NO_TEST=1	61	BT	PCIE_SSD0_R2D_P<3>	NO_TEST=1	149	139	BT	ISNS_P12V_GPU_P	NO_TEST=1	56	56	BT	NC_Q6080_D	NO_TEST=1	
B	110	109	BT	LED_RETURN_3_FB	NO_TEST=1	62	38	BT	PCIE_SSD1_D2R_N<1>	NO_TEST=1	149	139	BT	ISNS_P12V_MAIN_AUX_GPU_N	NO_TEST=1	134	36	BT	NC_SECI_OUT_D2R	NO_TEST=1
	68	51	BT	USB3_SD_D2R_N	NO_TEST=1	149	139	BT	PCIE_SSD1_D2R_N<2>	NO_TEST=1	149	139	BT	ISNS_P12V_MAIN_AUX_GPU_P	NO_TEST=1	134	36	BT	NC_SECI_IN_R2D	NO_TEST=1
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	51	BT	USB3_SD_D2R_C_N	NO_TEST=1	62	38	BT	PCIE_SSD1_D2R_P<1>	NO_TEST=1	149	139	BT	ISNS_P12V_MAIN_BLC_LCD_P	NO_TEST=1	44	37	BT	I2S_SPKRAMP_R_LRCLK_R	NO_TEST=1	
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	51	BT	USB3_SD_R2D_P	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_N<1>	NO_TEST=1	150	82	BT	ISNS_P3V3_G3W_SSD0_N	NO_TEST=1	39	36	BT	NC_SOC_VSSSOC_SENSE	NO_TEST=1	
	86	BT	REG_PHSFLT_CPUCORE_1_L	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_N<2>	NO_TEST=1	150	82	BT	ISNS_P3V3_G3W_SSD0_P	NO_TEST=1	39	36	BT	NC_SOC_VSSCPU_SENSE	NO_TEST=1	
	86	BT	REG_PHSFLT_CPUCORE_2_L	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_N<3>	NO_TEST=1	150	82	BT	ISNS_P3V3_G3W_SSD1_N	NO_TEST=1	36	36	BT	NC_SOC_AOP_PDM_CLK4	NO_TEST=1	
	86	BT	REG_PHSFLT_CPUCORE_3_L	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_P<1>	NO_TEST=1	150	82	BT	ISNS_P3V3_G3W_SSD1_P	NO_TEST=1	76	76	BT	NC_SE_PWR_EN	NO_TEST=1	
A	87	BT	REG_PHSFLT_CPUCORE_4_L	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_P<2>	NO_TEST=1	92	BT	REG_VSNS_CPU_VDDQ01_P	NO_TEST=1	76	76	BT	NC_PMU_CLK32K_GNSS	NO_TEST=1		
	87	BT	REG_PHSFLT_CPUCORE_5_L	NO_TEST=1	62	38	BT	PCIE_SSD1_R2D_C_P<3>	NO_TEST=1	93	BT	REG_VSNS_CPU_VDDQ23_P	NO_TEST=1	76	76	BT	NC_PMU_CLK32K	NO_TEST=1		
	66	BT	WLAN_CLKREQ_PCH_L	NO_TEST=1	62	BT	PCIE_SSD1_R2D_N<1>	NO_TEST=1	73	67	BT	XDP_PCH_PRDY_L	NO_TEST=1	76	76	BT	NC_PMU_AMUX_BY	NO_TEST=1		
	67	BT	PCH_SUSWARN_L	NO_TEST=1	62	BT	PCIE_SSD1_R2D_N<2>	NO_TEST=1	73	67	BT	XDP_PCH_PREQ_L	NO_TEST=1	76	76	BT	NC_PEARL_PWR_EN	NO_TEST=1		
	68	BT	PCH_USB_RBIA5	NO_TEST=1	62	BT	PCIE_SSD1_R2D_N<3>	NO_TEST=1	85	BT	REG_VCCIN_TSNS_A_R	NO_TEST=1	76	76	BT	NC_LIDINT_L_0UARTHDD_D2R	NO_TEST=1			
	70	BT	PP1V0_S5_PCH_VCC_MPHYPCIE3_PLT	NO_TEST=1	62	BT	PCIE_SSD1_R2D_P<1>	NO_TEST=1	85	BT	REG_VCCSA_TSNS_B_R	NO_TEST=1	36	36	BT	NC_GNSS_HOST_WAKE	NO_TEST=1			
	134	61	BT	PCIE_WLAN_R2D_C_N	NO_TEST=1	62	BT	PCIE_SSD1_R2D_P<2>	NO_TEST=1	73	68	BT	XDP_OBSDATA_A_0	NO_TEST=1	74	74	BT	NC_BUCK8_SW1	NO_TEST=1	
	134	61	BT	PCIE_WLAN_R2D_C_P	NO_TEST=1	62	BT	PCIE_SSD1_R2D_P<3>	NO_TEST=1	73	68	BT	XDP_OBSDATA_A_1	NO_TEST=1	74	74	BT	NC_BUCK4_SW1	NO_TEST=1	
	61	BT	PCIE_RCOMP_N	NO_TEST=1	40	BT	PP0V9_SLPPDDR_SOC_PCIEREFBUF	NO_TEST=1	73	67	BT	XDP_OBSDATA_A_2	NO_TEST=1	36	36	BT	NC_ALTIMETER_INT	NO_TEST=1		
	61	BT	PCIE_RCOMP_P	NO_TEST=1	41	BT	PP1V2_AWAKE_SOC_PCIEREFBUF_F	NO_TEST=1	73	68	BT	XDP_OBSDATA_A_3	NO_TEST=1	158	45	BT	CPU_SMC_THRMTRIP_L	NO_TEST=1		
A	71	BT	XDP_VR_READY	NO_TEST=1	41	BT	PP1V2_AWAKE_SOC_PLICPU_F	NO_TEST=1	73	68	BT	XDP_OBSDATA_B_0	NO_TEST=1	158	36	BT	MESA_MENUKEY_L	NO_TEST=1		
	32	30	BT	TBT_X_TMU_CLK_OUT	NO_TEST=1	41														



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